

AKADEMIKA

DCS-B

**DIGITAL COMMUNICATION TRAINING
SYSTEM**

EXPERIMENTAL MANUAL

.... A MESSAGE FROM



Today's system designers are faced with tomorrow's problems. DIGITAL COMMUNICATION is one of the important subject need to teach while learning electronics. It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

Company's Motto

- Light years ahead – refers to leadership.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have A complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete comprehensive warranty.

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SAFETY RULES

- Read carefully and follow the instructions mentioned in this manual. This user manual includes all the important points about the installation, use and the maintenance of the product. Keep this manual always with you, for quick reference.
- After unpacking the product, arrange all the accessories in proper order, so that their integrity is checked with the packing list. Also, ensure that the accessories have no visible damage.
- Before connecting the power supply to the kit, be sure that the jumpers and the connecting chords are connected correctly, as per the experiment.
- In DCS-B, signals are to be monitored with an oscilloscope or with the help of indicators as explained in the manual. For observation of signals on oscilloscope, either use X10 (Attenuation Probe) or 180Ω resistance in series with normal oscilloscope probe.
- This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is inadvisable and dangerous too. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.
- In case of any fault or malfunctioning in the trainer kit, turn off the power supply. Please do not tamper the kit. If you require our service, kindly contact the service centre for technical assistance.
- The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not cover include perishable items like connecting chords, crystals, etc. and other imported items.

Conditions and Limitations

The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product will be Akademia Lab Solutions sole judgment. The defective product will be replaced with a new one or repaired, without charge or with charge.

In the warranty period if the service is needed, the purchaser should get in touch with the service center or the sales outlet. The purchaser should return the product to the service center or the sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of the kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case the warranty does not cover the product on Akademia Lab Solutions judgment, we would repair the product after obtaining prior permission from purchaser who would receive an estimate statement about the repairing charges. In such cases, Akademia Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but they have no rights to stretch the meaning of original warranty contents or to offer an additional warranty. Akademia Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademia Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or the sales outlet.

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TECHNICAL SPECIFICATIONS

Sine Wave

Frequency (Variable) : 1Hz - 30Hz
 Amplitude : 0 - 2 Vpp
 Frequency (Fix) : 250Hz, 500Hz, 1 KHz & 2 KHz.
 Amplitude : 0-4Vpp.

Carrier

Frequency : 500 KHz (0 deg.), 1MHz (0 deg.),
 1MHz (180 deg.)
 DC : 0-5Vpp.

Data Simulator : On-board 8 bit variable NRZ-L pattern.

Clock

Frequency : 2 KHz, 4KHz, 8KHz, 16KHz, 32KHz,
 64KHz & 128KHz.
 Duty Cycle : 10 - 90% Selectable in steps of 10%.
 Receiver Clock : Phase Lock Loop generates receiver
 clock and channel information.

Data Format (Encoding & Decoding)

NRZ (Level),
 NRZ (Mark),
 NRZ (Space),
 Bi-phase (Manchester),
 Bi-phase (Mark),
 Bi-phase (Space),
 URZ,
 AMI

Sampling Methods : Natural Sampling
 Sample and Hold
 Flat Top Sampling

Reconstruction : 2nd order and 4th order low pass Butterworth filters with 3.4
 KHz cut-off frequency

Digital Modulation & Demodulation

Pulse width modulation,
 Pulse position modulation,
 Pulse code modulation,
 Delta modulation,
 Sigma Delta modulation,
 Adaptive delta modulation,
 CVSD modulation

Carrier Modulation & Demodulation

ASK,
FSK,
PSK

Compander : μ -law Compander and Expander

Parity Code Facility : Even, Odd, Hamming, None Parity

Modes of Operation : Fast: 230 KHz

Error option : 4 Single bit error switch selectable

Error Correction : Hamming Code

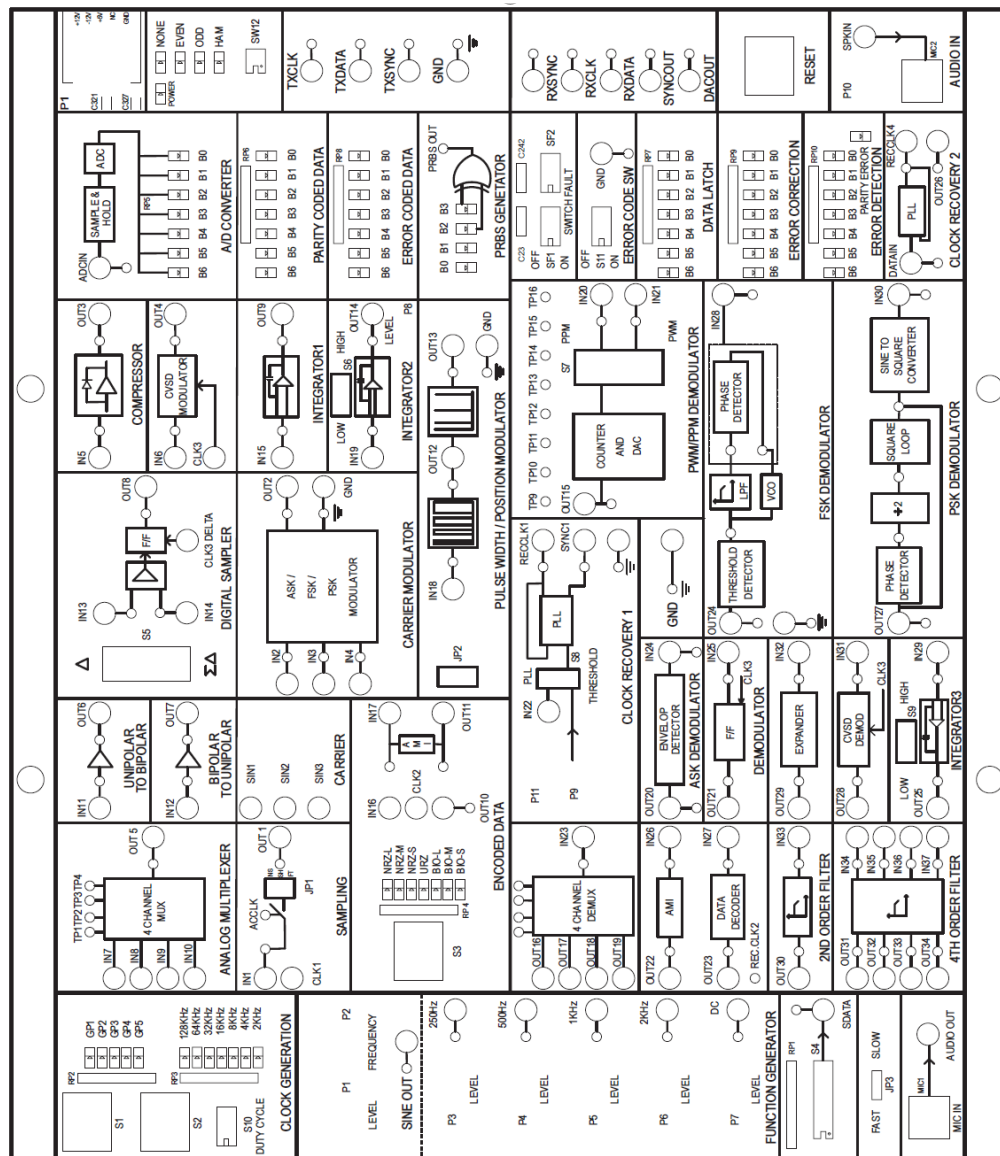
TDM : 4 Channels

Audio Amplifier Gain : 20dB

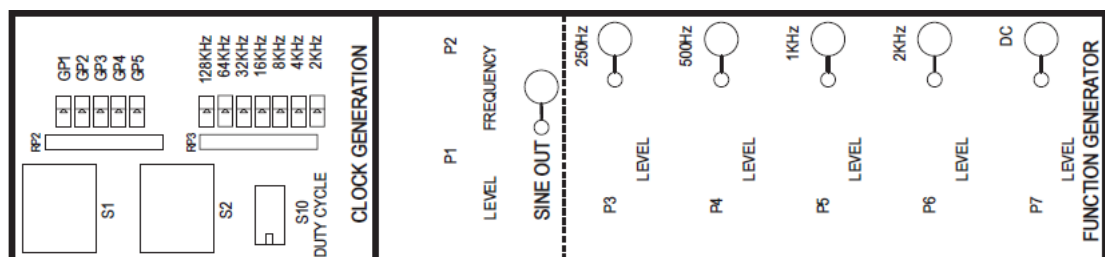
Accessories

DCS-B			Quantity
1	Short Links PCS2 Patch cord	Male To Male	15
2	Long Links PCS2 Patch cord	Male To Male	5
3	Headphone		1
4	Jumper Caps		5
5	DCS-B EXPT Manual		1
6	5 PIN DIN CONNECTOR Power Supply		1
7	Power cord		1

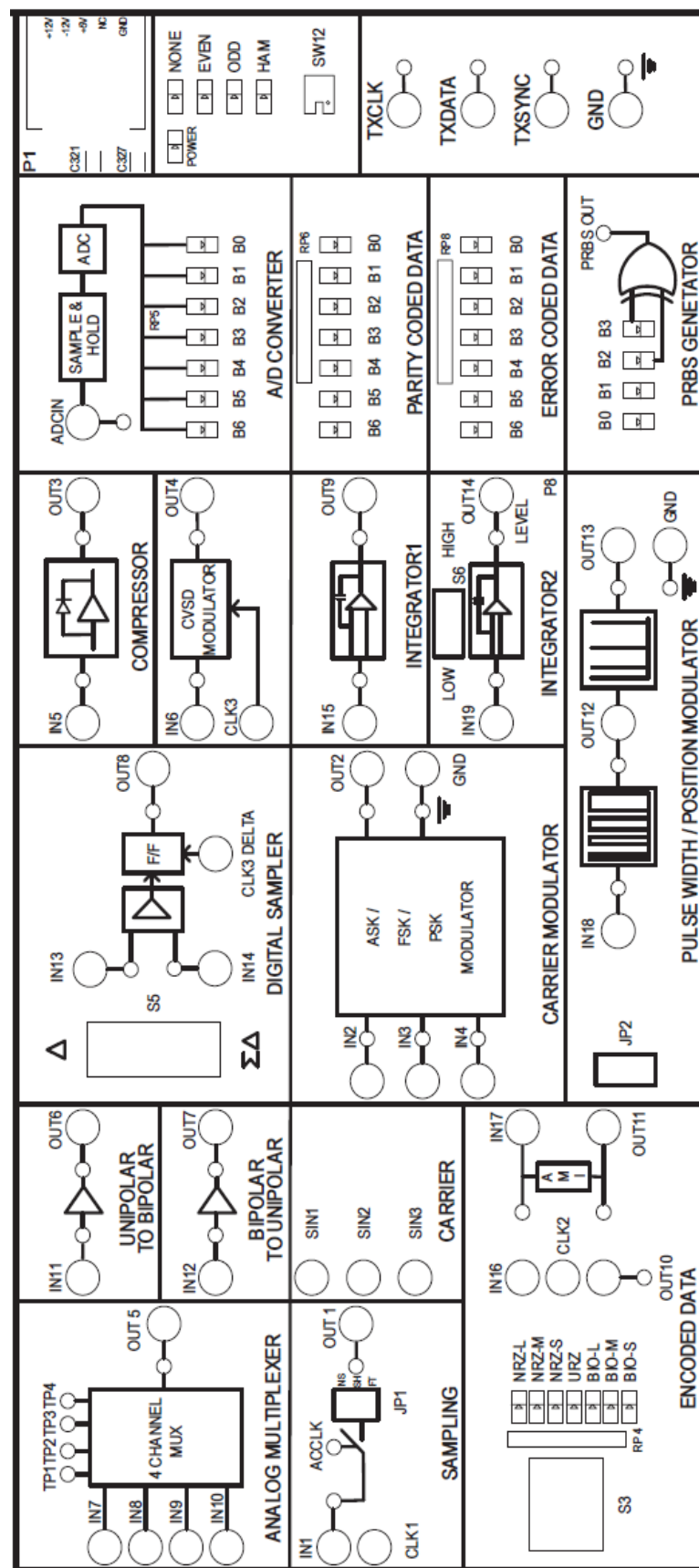
FUNCTIONAL BLOCK



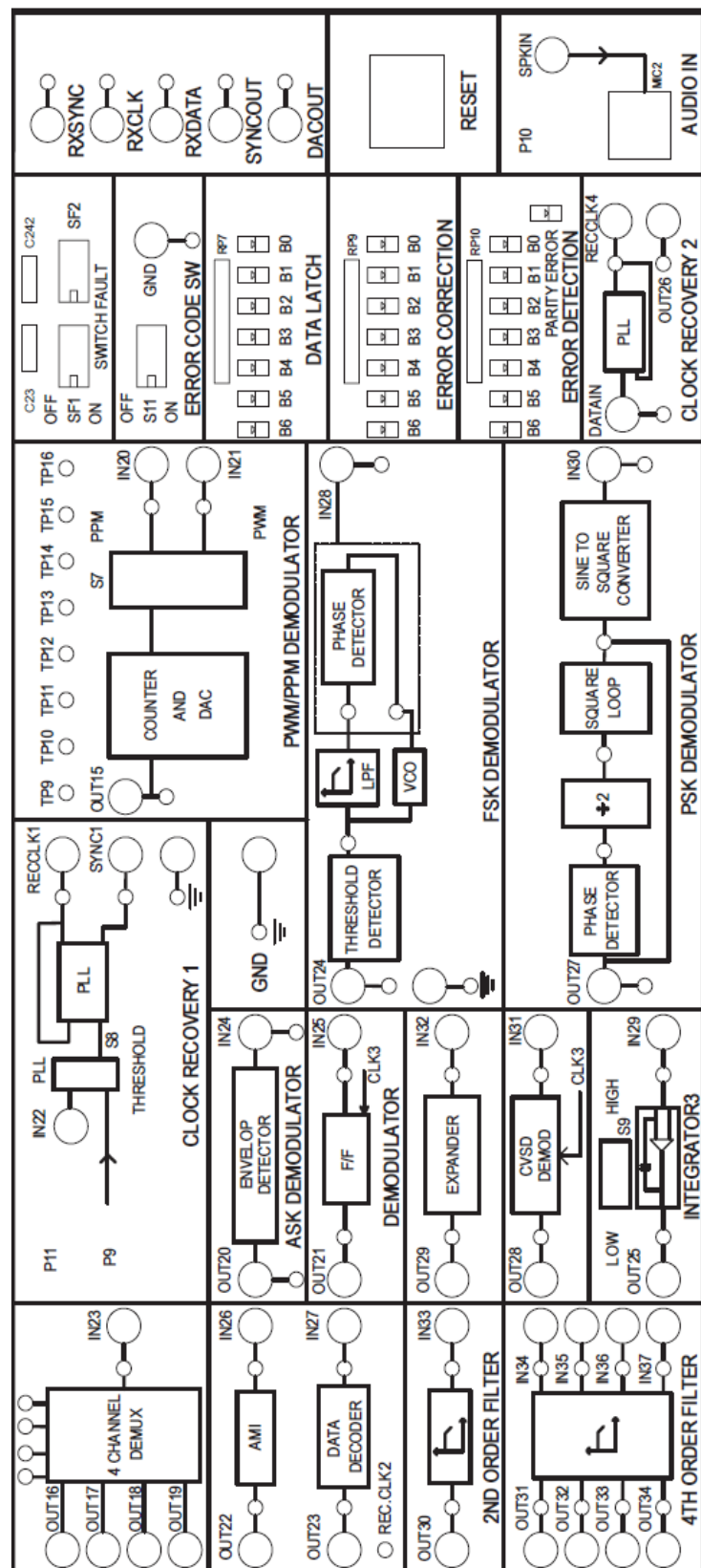
Clock Generator



Transmitter Section



Receiver Section



INTRODUCTION

“DCS-B: DIGITAL COMMUNICATION SYSTEM”: It is an introductory trainer kit to enable students to understand the basics of digital communication with the most happening technology VLSI (Very Large Scale Integration).

Digital electronic systems are increasing exponentially in their complexity.

Overtime, this fact has coupled with decreasing product's lifetime. The increasing reliability requirements have forced the system designers to dramatically increase the productivity and quality of the design.

VLSI technology is a highly specialized field that has the power of integrating millions of transistors on a single silicon chip. The designed automation tools help the designer with the design entry, hardware generation, simulation, verification and design management.

Hardware Description Language (HDL) is used to describe the hardware for the purpose of simulation, modeling, testing and designing of the digital systems. VHDL [(Very High Speed Integrated Circuit) Hardware Description Language] has been the heart of the electronic design productivity.

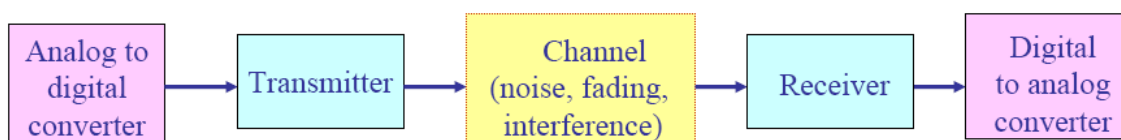
Programmable Logic Devices like FPGA (Field Programmable Gate Array) and CPLD (Complex Programmable Logic Devices) can be designed and quickly verified compared to the regular discrete ICs. These devices are software configured and field programmable.

The DCS-B board, which adopts the VLSI technology, comes in a wooden enclosure with a removable lids and it is easy to carry around in the laboratory and perform various experiments. This unit provides the user Sine waves of five different frequencies with variable amplitude. It also provides DC signal with variable amplitude, 8 Bit digital data pattern generator and has provision to select the sampling frequencies and transmitter clock frequencies from among the various sampling frequencies available, with discrete variations in the pulse width. The pulse width varies from 10% to 90% duty cycle in steps of 10%. All onboard signals can be easily connected using connecting chords. It also consists of various functional blocks to perform various studies on digital communication. All onboard signals are phase locked and enable the learner to observe a stationary signal on an Oscilloscope. Various test points are brought out onboard and identified. The trainer has a complete circuit mimic Inscribed on thePCB.

THEORY

Study of electronic transmission of information that has been encoded digitally (as for storage and processing by computers) is called Digital Communication.

In a Digital communication system, an analog (i.e., continuous-time continuous-amplitude) message signal is converted into a discrete-time discrete-amplitude digital signal by time sampling and amplitude-quantization. The resulting signal is encoded (i.e., mapped to binary codeword). The digitized information signal will be modulated back into an analog waveform for propagation through the channel.



Why Digital?

- Better resistance to noise and channel distortion and the availability of regenerative repeaters.
- Flexible/cheap hardware.
- Possibility of source and channel coding, cryptography, authentication and tradeoff bandwidth vs. SNR.

Sampling

When an analog message is conveyed over an Analog Communication System, the full message is typically used at all times. To send the same analog signal over a Digital Communication System, only its samples are required to be transmitted at periodic intervals. The receiver will receive only samples of the message. The receiver must attempt to re-construct the original message at all times from its samples only.

Various sampling techniques:

1. Natural sampling
2. Sample and hold
3. Flat-top sampling

TDM

Time-division multiplexing (TDM) is a method of putting multiple data streams in a single signal by separating the signal into many segments, each having a very short duration. Each individual data stream is reassembled at the receiving end based on the timing. The circuit that combines signals at the source (transmitting) end of a communications link is known as a multiplexer. It accepts the input from each individual end user, breaks each signal into segments, and assigns the segments to the composite signal in a rotating, repeating sequence. The composite signal thus contains data from multiple senders. At the other end of the long-distance cable, the individual signals are separated out by means of a circuit called a demultiplexer and routed to the proper end users. A two-way communications circuit requires a multiplexer/demultiplexer at each end of the long-distance, high-bandwidth cable.

If many signals must be sent along a single long-distance line, careful engineering is required to ensure that the system will perform properly. An asset of TDM is its flexibility. The scheme allows for variation in the number of signals being sent along the line, and constantly adjusts the time intervals to make optimum use of the available bandwidth

Uses of Time-Division Multiplexing

- The PDH and SDH network transmission standards.
- The GSM telephone system.
- The left-right channel splitting in use for Stereoscopic Liquid Crystal shutter glasses.
- Multichannel digital audio processing sub-systems, notably the Motorola 56000 based hardware acceleration used in Pro Tools

Time Division Multiplexing (TDM) is the means by which multiple digital signals (or analogue signals carrying digital data) can be carried on a single transmission path by interleaving portions of each signal in time ^[1]. Interleaving can be done at bits or blocks of bytes ^[1]. This enables digitally encoded speech signals to be transmitted and switched optimally within a circuit-switched network. This article consists of two sections, namely, Transmission, using TDM and Synchronous Digital Hierarchy (SDH). The first section examines the basic principles underlying TDM, while the second section discusses how SDH is used to switch TDM frames.

Pulse Code Modulation

Pulse Code Modulation is a truly digital system for transmitting analog data and is totally free from the effect of noise in the channel. This is because the communication is by the means of a group of pulses, which together represent the amplitude of the signal sample. The signals in PCM are binary; that is, there are only two possible states, represented by logic 1 (high) and logic 0 (low). This is true, no matter how complex the analog waveform happens to be. Using PCM, it is possible to digitize all forms of analog data, including full-motion video, voices, music, telemetry and virtual reality (VR).

The receiver detects the presence or absence of the pulse, irrespective of the amplitude or the shape of the pulse. It is free from channel error. The only error that remains is the quantization error. This error can be reduced at the receiver end by low pass filtering

Parity Encoder: None, Odd, Even & Hamming

Parity Bit: A bit added to a binary code that indicates parity and is used to check the integrity of data.

If Parity is none, parity checking is not performed and the parity bit is not transmitted.

An even parity generator will produce logic 1 at its output, if the data word contains an odd number of ones. If the data word contains an even number of ones, then the output of the parity generator will be low. By concatenating the Parity bit to the data word, a word will be formed which always has an even number of ones i.e. it has even parity. An Odd Parity generator is similar but it provides an odd parity.

(Hamming code) (n.) In digital data transmissions, a method of error detection and correction in which every string of four bits is replaced with a string of seven bits, the last three added bits are parity-checking bits that the receiving device uses to check for and corrects the errors.

Hamming code will detect any double errors, but it can only correct a single error. This method of error correction is best suited for situations in which randomly occurring errors are likely, not for errors that come in bursts.

Richard Hamming, a theorist with Bell Telephone Laboratories in the 1940s, developed the Hamming code method of error correction in 1949.

Data Encoding

In order to transport digital bits of data across carrier waves, encoding techniques have been developed each with their own pros and cons.

- NRZ-L
- NRZ-M
- NRZ-S
- BIO-L
- BIO-M
- BIO-S
- URZ
- AMI

Carrier Modulation

Digital Communication is the technique by which the whole information is represented in terms of binary digits, i.e., 'ones' and 'zeros'. These digits are represented by discrete voltage levels and the clock frequency of a digital communication scheme is generally low. For long distance transmission, the data is made to modulate a continuous wave (sine wave) carrier. These techniques are called Carrier Modulation Techniques. The various types of

Carrier Modulation Techniques, normally adopted in practice fall under three broad categories:

- a) Amplitude Shift Keying (ASK)
- b) Frequency Shift Keying (FSK)
- c) Phase Shift Keying (PSK)

Any form of digital modulation necessarily uses a finite number of distinct signals to represent digital data.

- In the case of PSK, a finite number of phases are used.
- In the case of FSK, a finite number of frequencies are used.
- In the case of ASK, a finite number of amplitudes are used. This is very similar to pulse code modulation.

There are three ways in which the bandwidth of the channel carrier may be altered simply. It is worth emphasizing that these methods are chosen because they are practically simple, not because they are theoretically desirable. These include altering the amplitude, frequency and phase of the carrier sine wave. These techniques give rise to amplitude-shift-keying (ASK), frequency-shift-keying (FSK) and phase-shift-keying (PSK), respectively.

Delta Modulation

It is an Analog-to-digital signal conversion in which the analog signal is approximated with a series of segments. Each segment of the approximated signal is compared to the original analog wave to determine the increase or decrease in relative amplitude. The decision process for establishing the state of successive bits is determined by this comparison, and only the change of information is sent, i.e., only an increase or decrease of the signal amplitude from the previous sample is sent whereas a no-change condition causes the modulated signal to remain at the same 0 or 1 state of the previous sample.

Adaptive Delta Modulation

ADM quantizes the difference between the current sample and the predicted value of the next sample. ADM uses a variable "step height" to adjust the predicted value of the next sample so that both slowly and rapidly changing input signals can be conveniently reproduced. One bit (i.e., "1" or "0") represents each sample in ADM.

Sigma Delta Modulation

A Sigma-Delta-Modulator (SDM) transforms a band limited input signal into a digital 1-bit output signal. The input signal modulates the output pulse density.

When low-pass-filtering the output signal, the input signal is reclaimed.

The clock frequency of the SDM (f_a) has to be much higher than the highest frequency of the input signal (U_e). To get a sufficiently high signal-to-noise-ratio at the digital output of a CD-player (16 bit/44,1kHz) over sampling by a factor of at least 32 is necessary. For a digital SDM, the input signal already has to be supplied with this high data rate. A digital interpolation filter achieves this.

In the simplest case, the loop filter is an integrator.

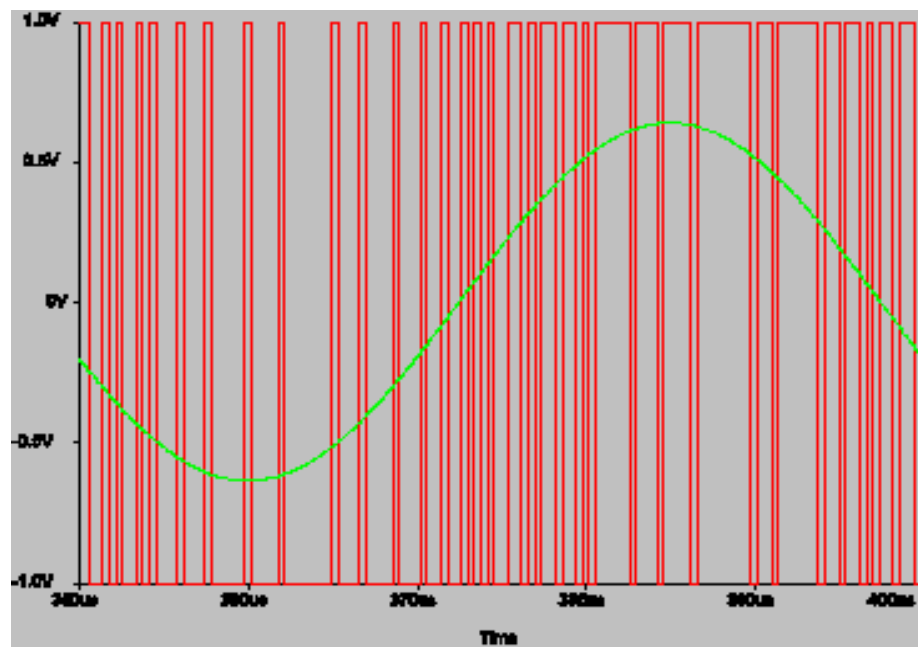


Fig. 1: SDM output signal

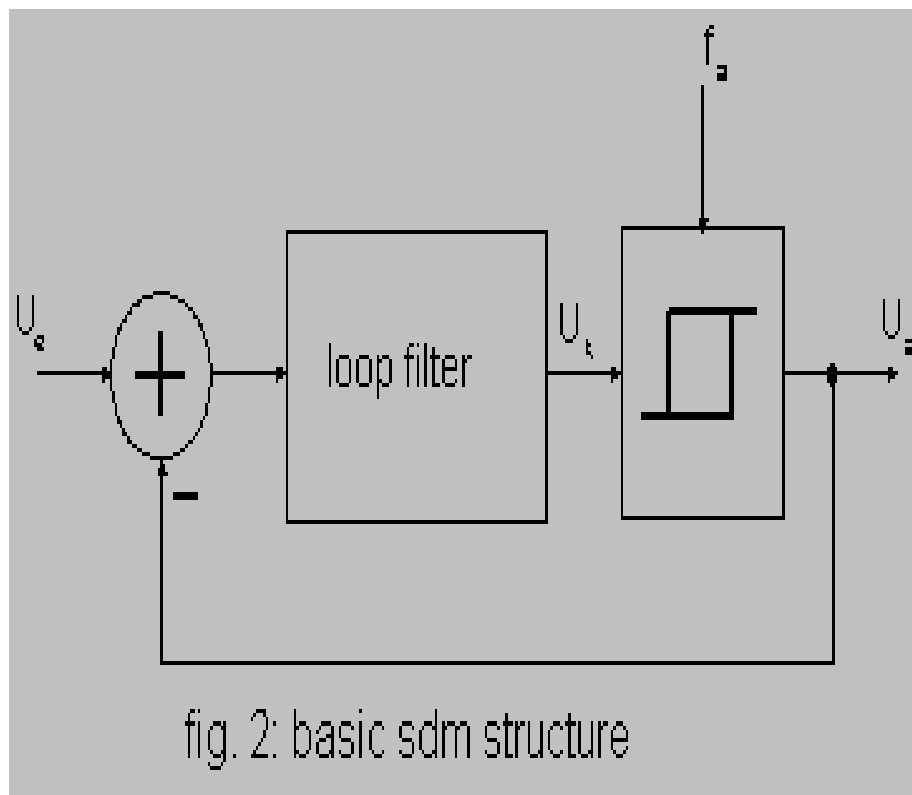
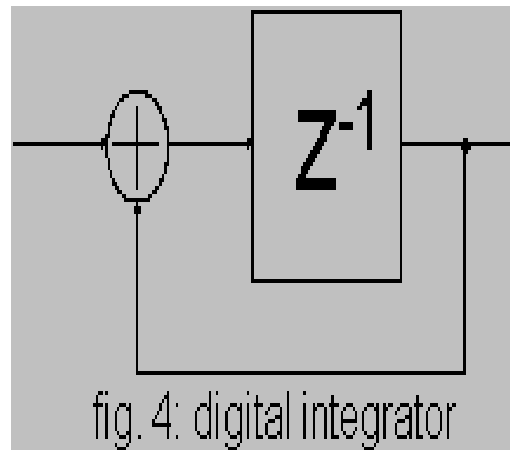
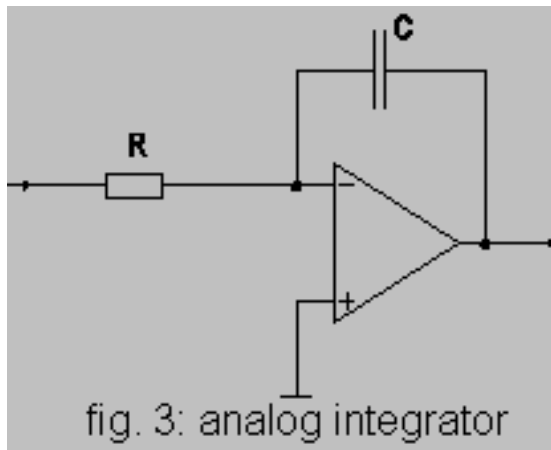


fig. 2: basic sdm structure



CVSD

Continuously variable slope delta modulation (CVSD or CVSDM) is a voice coding method. It is a delta modulation with variable step size, first proposed by Greefkes and Riemens in 1970.

The amplifier has a variable gain i.e. its gain is the function of the voltage applied at its gain control terminal. The gain increases with the increase in positive gain control voltage. Many Adaptive Delta Modulators do not change step size on a pulse-to-pulse basis, rather changes are made much more slowly. Such slow controls are referred to as syllabic. The usual implementation involves a Continuously Variable gain circuit controlled by an analog voltage from step size controller. This is often called Continuously Variable Slope Delta (CVSD). There is a variety of CVSD IC encoder and decoder in today's semiconductor market.

The CVSD is a simple alternative to the more complex conventional conversion techniques in the system requiring digital communication of analog signals. The CVSD A/D is well suited for the requirements of digital communications. A Delta Modulator consists of a comparator in the forward path and an integrator in the feedback path of a simple control loop. The input to the comparator is the simple analog signal and the integrator output. The comparator output is the difference between the input voltage and the integrator output. That sign bit is the digital output and also controls the direction of ramp in the integrator. The output of comparator is fed to the sampler. Then the sampler output is fed to the slope polarity switch and level detect algorithm. The level detect algorithm is again fed to the slope magnitude control followed by slope polarity switch. The output slope polarity switch is fed to the integrator in the control loop. With no input at the transmitter, continuous 1 and 0 alternations are transmitted. The outstanding characteristic is its ability to transmit the intelligible voice out at relatively low data rate. Companded PCM for telephone quality transmission requires about 64Kbits/sec. data rate/channel. CVSD produces equal quality at 32Kbit/sec. In CVSD Decoder, CVSD mod output is fed to the input of the comparator.

The comparator output is fed to the internal shift register. Then the output of internal shift register is fed to the digital logic followed by slope polarity switch and integrator. The output of integrator is fed to the low pass filters for the reconstruction of original signal.

Companding

Companding means compression and expansion of a signal or sample. It forms a part of the PCM (Pulse Code Modulation) process, whereby analogue signal values are logically rounded on a nonlinear scale. The decimal step number is then coded to its binary equivalent, prior to transmission. The process is reversed at the receiving terminal using the same nonlinear scale.

PWM

There are many forms of modulation used for communicating information. When a high frequency signal has amplitude varied in response to a lower frequency signal, we have AM (amplitude modulation). When the signal frequency is varied in response to the modulating signal, we have FM (frequency modulation). These signals are used for radio modulation because the high frequency carrier signal is required for efficient radiation of the signal. When communication by pulses was introduced, the amplitude, frequency and pulse width became possible modulation options. In many power electronic converters where the output voltage can be one of two values, the only option is modulation of the average conduction time.

This technique of modulation controls the variation of duty cycle of the square wave (with some fundamental frequency) according to the modulating signal.

Here the amplitude variation of the modulating signal is reflected in the ON period variation of the square wave. Hence, it is also called as the technique of V to T conversion.

Pulse-width modulation of a signal involves the modulation of its duty cycle, to convey information over a communication channel.

PPM

In this technique of modulation, the position of TTL pulse is changed on the time scale, according to the variation of input modulating signal amplitude. The pulse positions are directly proportional to the instantaneous values of the modulating signal.

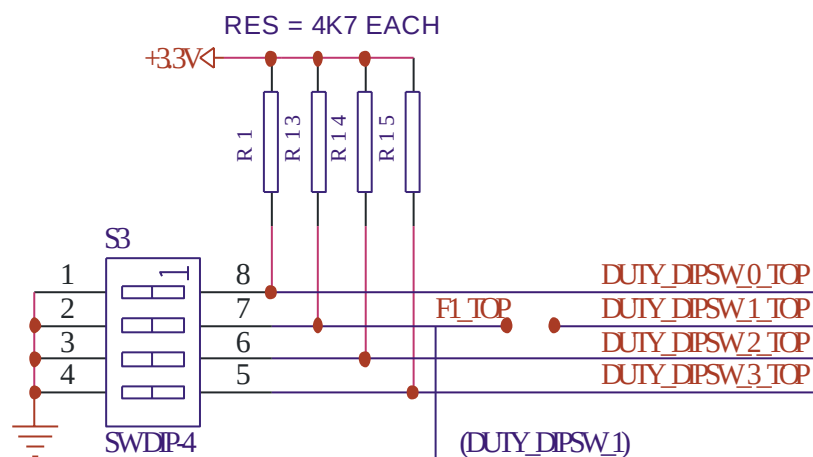
Pulse-position modulation is a form of signal modulation in which M message bits are encoded by transmitting a single pulse in one of 2^M possible time-shifts. This is repeated every T seconds, such that the transmitted bit rate is M/T bits per second. It is primarily useful for optical communications systems, where there tends to be little or no multipath interference

SWITCH FAULTS

Switch fault section on DCS-B provides 8 switches. These switches can be used to simulate fault conditions in various parts of the circuit. The faults are normally used one at a time. To ensure that the DCS-B kit is fully operational, all switches should be set to OFF position before use.

1) Effect of Switch Fault 1 in clock generation section

SCHEMATIC



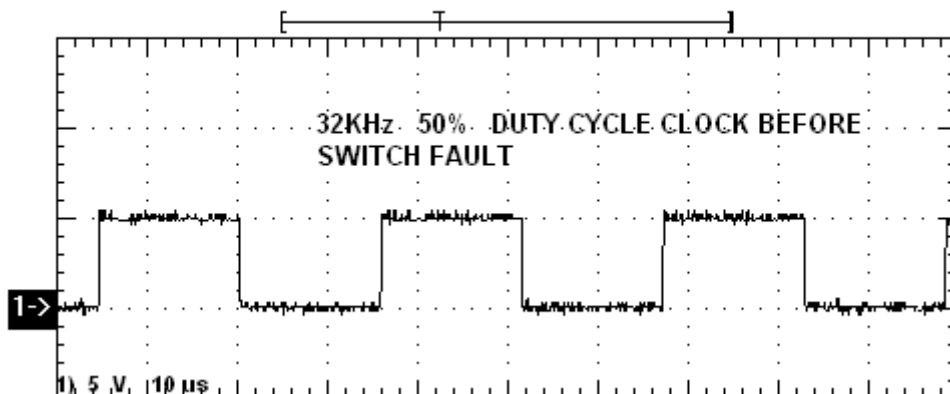
DESCRIPTION

To control the duty cycle of the selected clock The BCD counter counts from 0000 to 1001. This is connected to the A inputs of the comparator. The 4 way DIP switch S10, can be configured such that the B inputs of the comparator varies from 0001 to 1001 and the frequency of the $A < B$ output of the comparator is 10 times less than the frequency of the square wave, that is fed to the input of the BCD counter. Now, the duty cycle of the $A < B$ output of the comparator varies from 10% to 90% as the settings of the 4 way DIP switch varies from 0001 to 1001. Now, the available clock frequencies are divided by 10. Thus, we get the sampling frequencies from 64 KHz to 2 KHz.

PROCEDURE

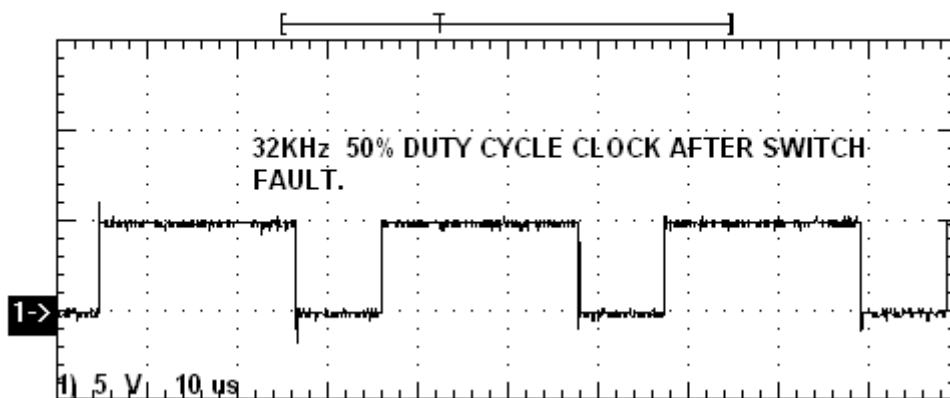
- Ensure that the group 1 (**GP1**) clock is selected in the Clock generation section. Selection is done with the help of switch **S1**. Observe the corresponding LED indication.
- Select the sampling clock of frequency 32 KHz, using the switch **S2** and observe the corresponding LED indication.
- Adjust the duty cycle of the sampling clock to 50%, using DIP switch **S10** as shown in the block diagram. Sampling clock is available at the **TX CLK** post.

Waveform Before Switch Fault



Now, put the switch 1 of (SF1) in the Switch Fault to ON position and observe its effect.

Waveform before Switch Fault

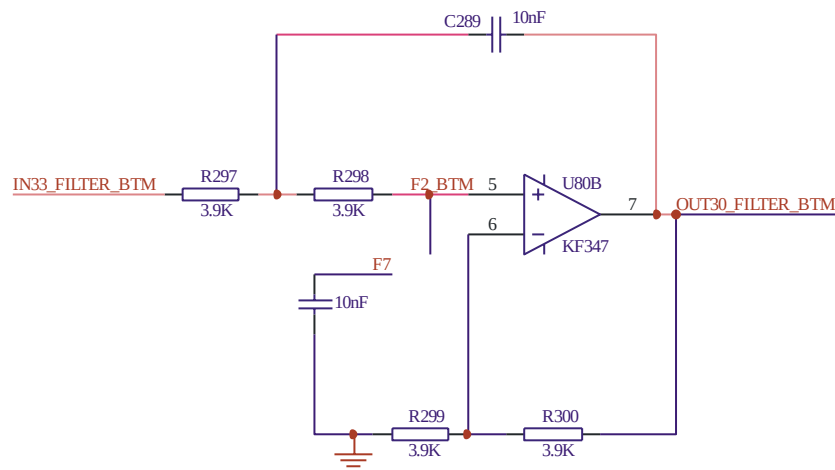


Action & Effect

When we put 1 of the SF1 in Switch Fault section to ON position, it opens the B1 bit from the B input (4-bit DIP switch output) of the comparator. This introduces the fault in the duty cycle section. With effect, change in duty cycle for settings (10%, 40%, 50%, 80% and 90%) will not be observed as expected.

2) Effect of Switch Fault 2 in filter section

SCHEMATIC



2nd ORDER BUTTERWORTH FILTER

DESCRIPTION

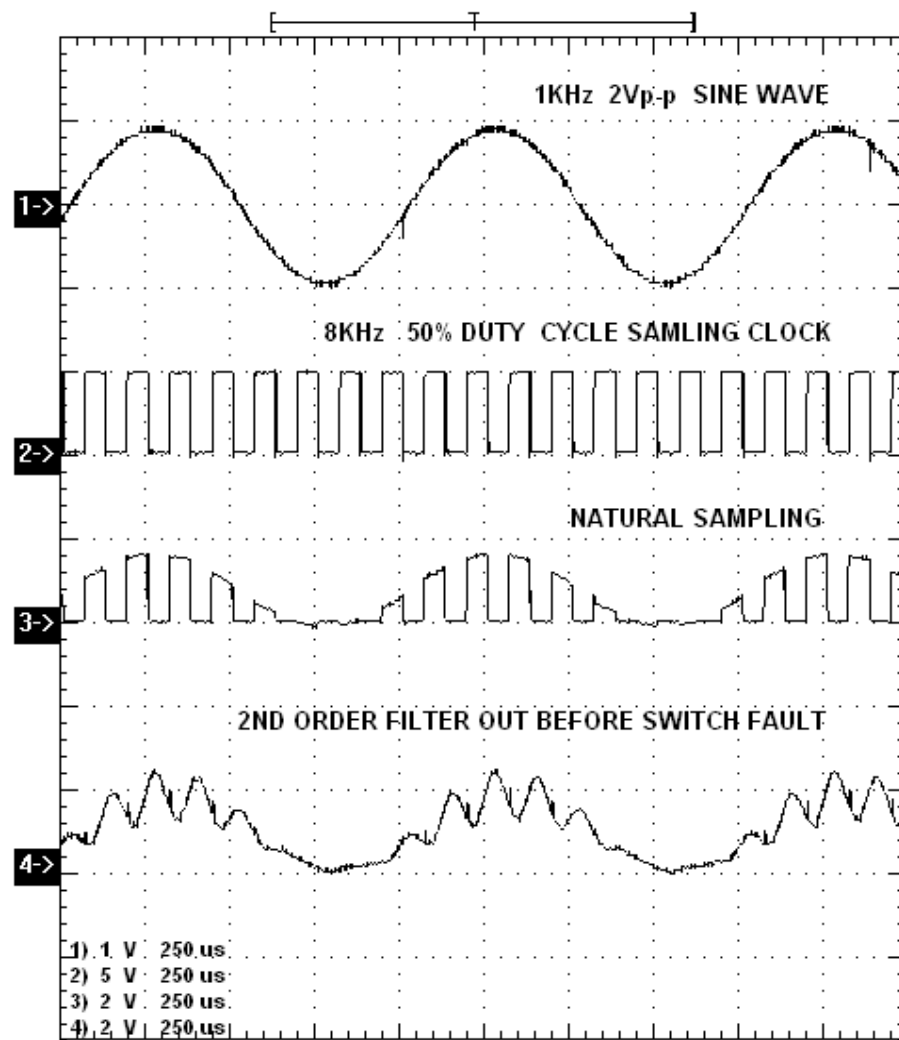
Two types of Filters are provided onboard, viz., 2nd Order Low Pass Butterworth Filter and 4th Order Low Pass Butterworth Filter. The values of the resistor and the capacitors are designed such that it gets the cut off frequency of about 3.4 KHz.

A. NATURAL SAMPLING AND ITS RECONSTRUCTION

PROCEDURE

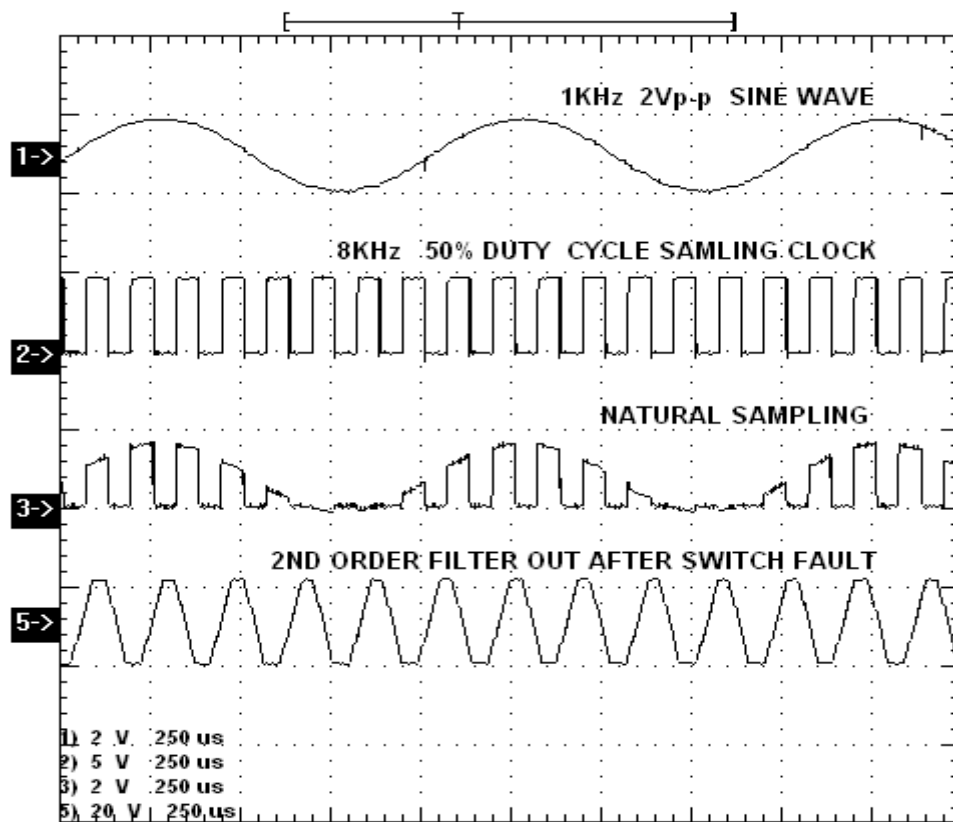
- Ensure that the group 1 (**GP1**) clock is selected in the Clock generation section. Selection is done with the help of switch **S1**. Observe the corresponding LED indication.
- Select the sampling clock of frequency 8 KHz, using the switch **S2** and observe the corresponding LED indication.
- Adjust the duty cycle of the sampling clock to 50%, using DIP switch **S10** as shown in the block diagram. Sampling clock is available at **TX CLK** post.
- The signal under test, which in our case, is a sine wave of 1 KHz, which is available at, posts **1 KHz** in the Function Generator section.
- Adjust the amplitude of the 1 KHz sine wave to 2Vpp, using post **P5**.
- Connect **TX CLK** post to **CLK 1** post and **1 KHz** sine wave to **IN1** post of the Sampling section.
- To observe the Natural sampling effect, short the jumper **JP1** to **first** position and check the signal at the **OUT1** post of the sampling section.
- Connect the 1 post to the IN 33 post of the 2nd order filter.
- Observe the recovered output at **OUT 30**

Waveform before Switch Fault



Now put the switch 2 of (SF1) in the Switch Fault to ON position and observe its effect.

Waveform before Switch Fault

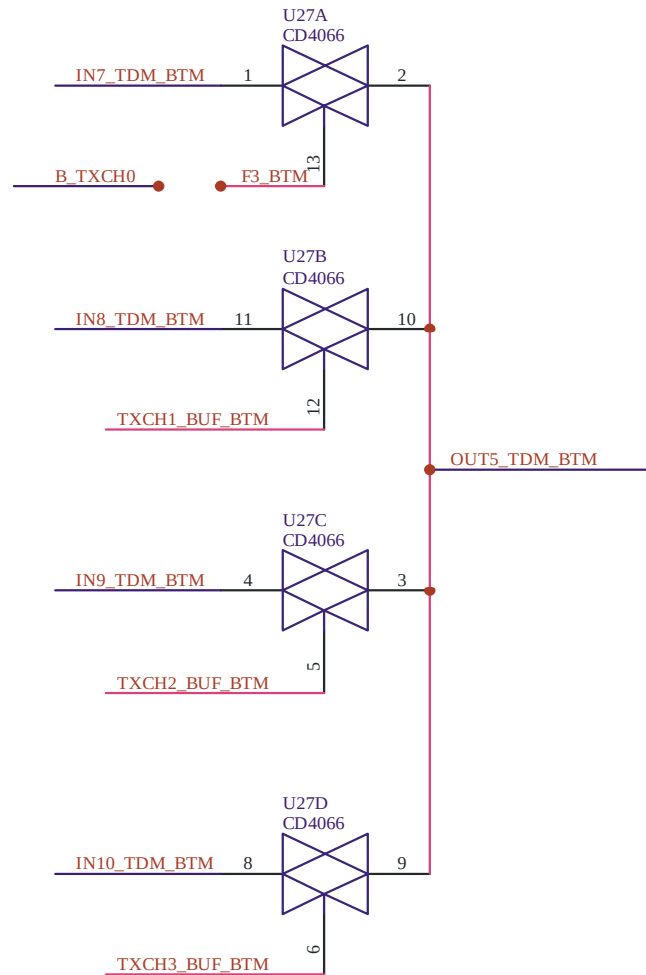


Action & Effect

When we put 2 of SF1 in Switch Fault section to ON position, it will open the bypass capacitor of the 2nd order low pass Butterworth filter, which would result in the induction of ripples or oscillation at the filter output.

3) Effect of Switch Fault 3 in TDM section

SCHEMATIC



DESCRIPTION

The Transmitter Section consists of four Analog Input Channels with 4-pole Integrated Circuit Analog Switch (IC 4066) that provides Sampling and Time Division Multiplexing of each channel, using Pulse Amplitude Modulation techniques.

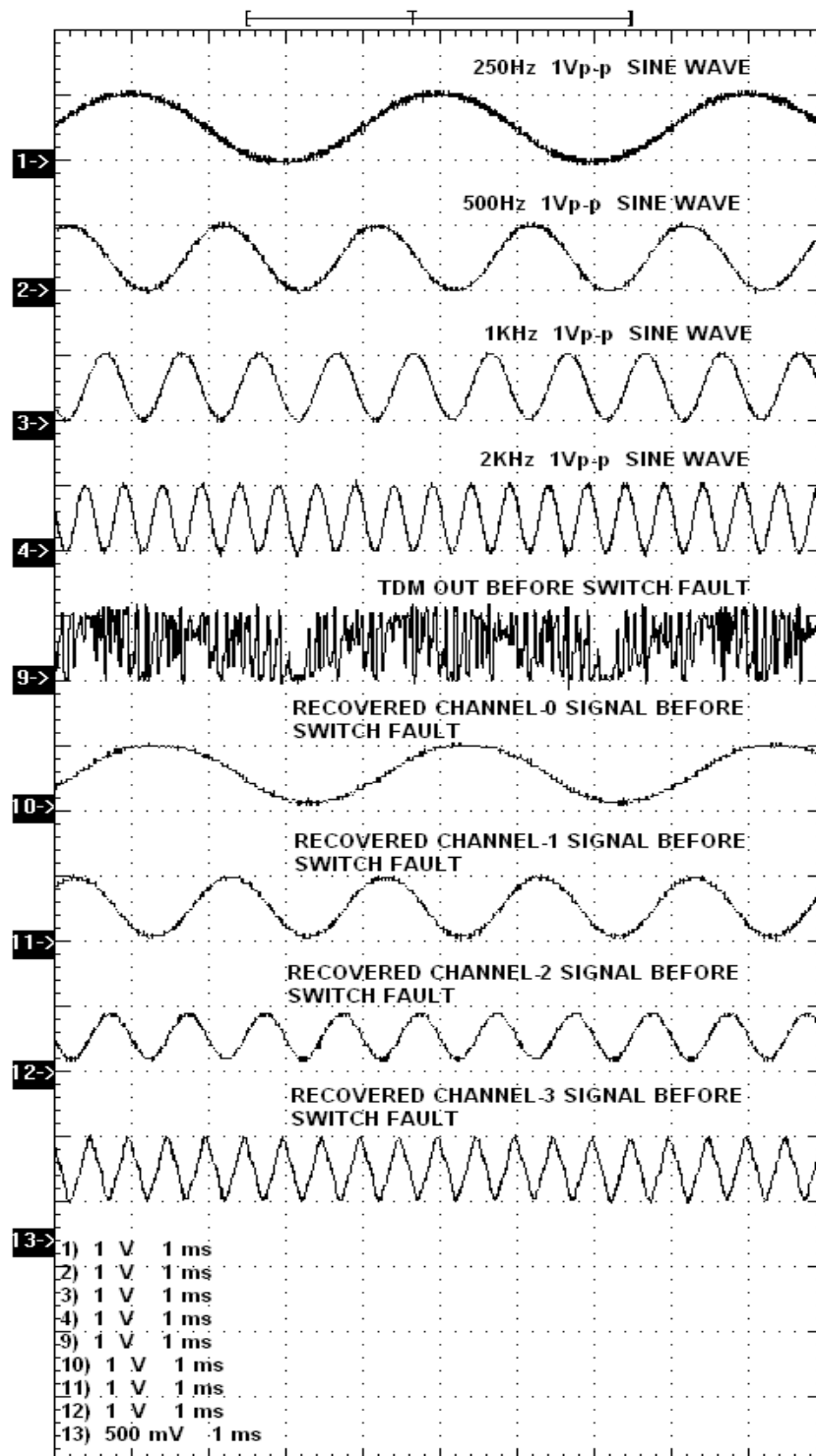
The 32 KHz sampling clock is fed to the cascade of the two counters. The two counters are configured, in such a way, that the outputs vary from 00 to 11. As the control input of the decoder varies from 00 to 11, the signal C1, C2, C3, and C4 selects the channel of the analog switch (IC 4066) and the information of that channel is transmitted. Thus, TDM is achieved effectively.

PROCEDURE

- Ensure that the group 2 (**GP2**) clock is selected in the clock generation section. Selection is done with the help of switch **S1**.
- The selected clock of frequency 32 KHz, 80% duty cycle will be available at the **TXCLK** post.
- The timing signals required for multiplexing four analog channels are derived internally from TX CLK and are available at the test points **TP1, TP2, TP3** and **TP4**.

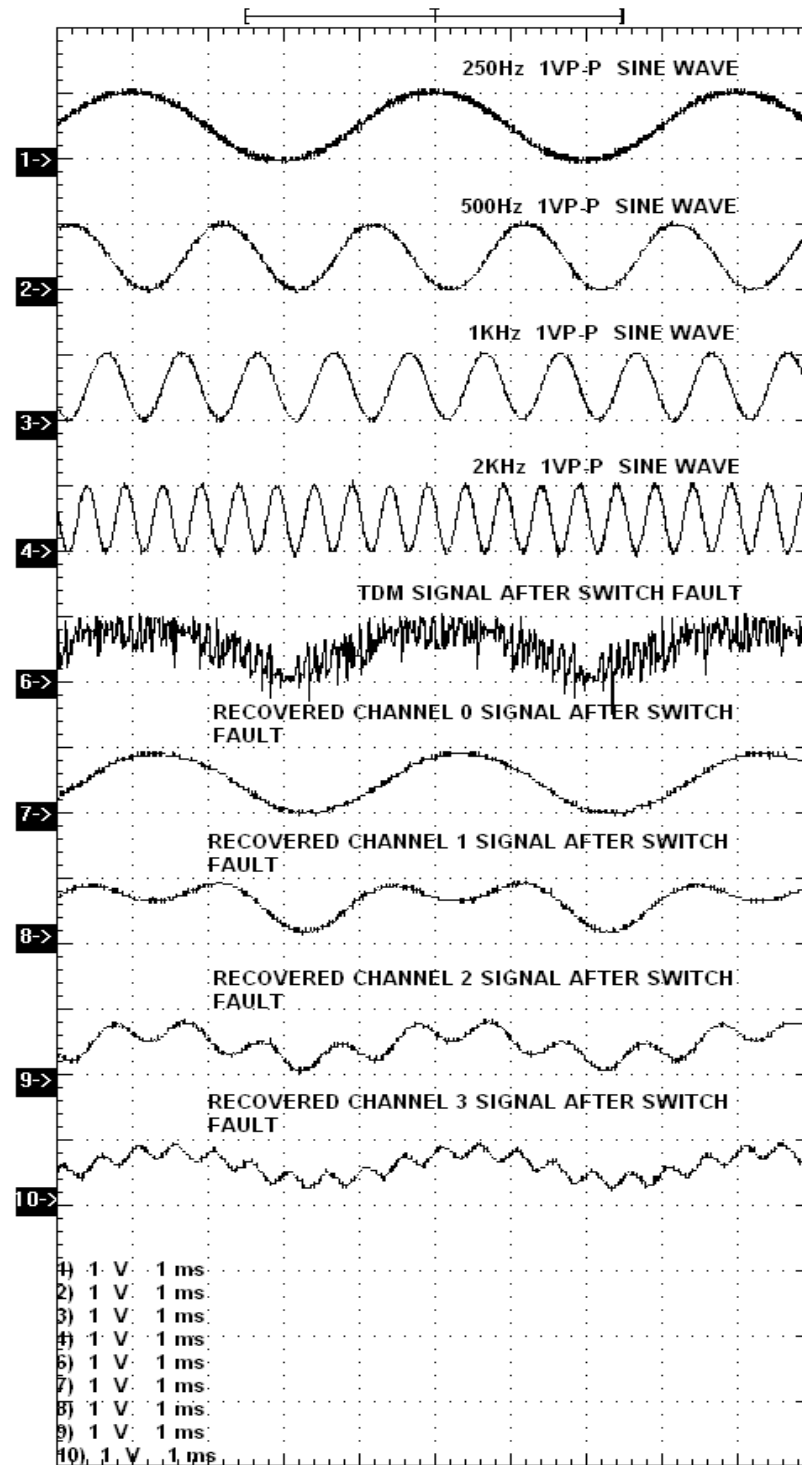
- Feed four sine waves of frequency **250Hz, 500Hz, 1 KHz** and **2 KHz** having amplitude around 1Vpp from the function generator to the four channel inputs **IN7, IN8, IN9** and **IN10** of the analog multiplexer.
- Observe the TDM output at the **OUT5** post of the 4-channel Mux section.
- To recover the individual analog signals from the multiplexed data at the **OUT5** post, connect it to **IN23** post of the 4-channel Demux section.
- To provide the timing and synchronization information to the receiver section, connect the **TX CLK** post to the **RX CLK** post and **TX SYNC** post to the **RX SYNC** post. Here, we are connecting the transmitter clock to the receiver clock and transmitter sync to the receiver sync physically.
- The timing signals required for demultiplexing the four analog channels are derived internally from the **RX CLK** and are available at the test points.
- The four demultiplexed signals are available at the post's **OUT16, OUT17, OUT18** and **OUT19** of the 4-Channel Demux section.
- The multiplexed sine waves are not pure sine waves. Therefore, it needs to be filtered. Connect **OUT16** post to **IN34** post, **OUT17** post to **IN35** post, **OUT18** post to **IN36** post and **OUT19** to **IN37** post of the 4th order LPF.
- Observe the filtered output at posts **OUT31, OUT32, OUT33** and **OUT34** respectively.

Waveform before Switch Fault



Now, put the switch 3 of (SF1) in the Switch Fault to ON position and observe its effect.

Waveform After Switch Fault

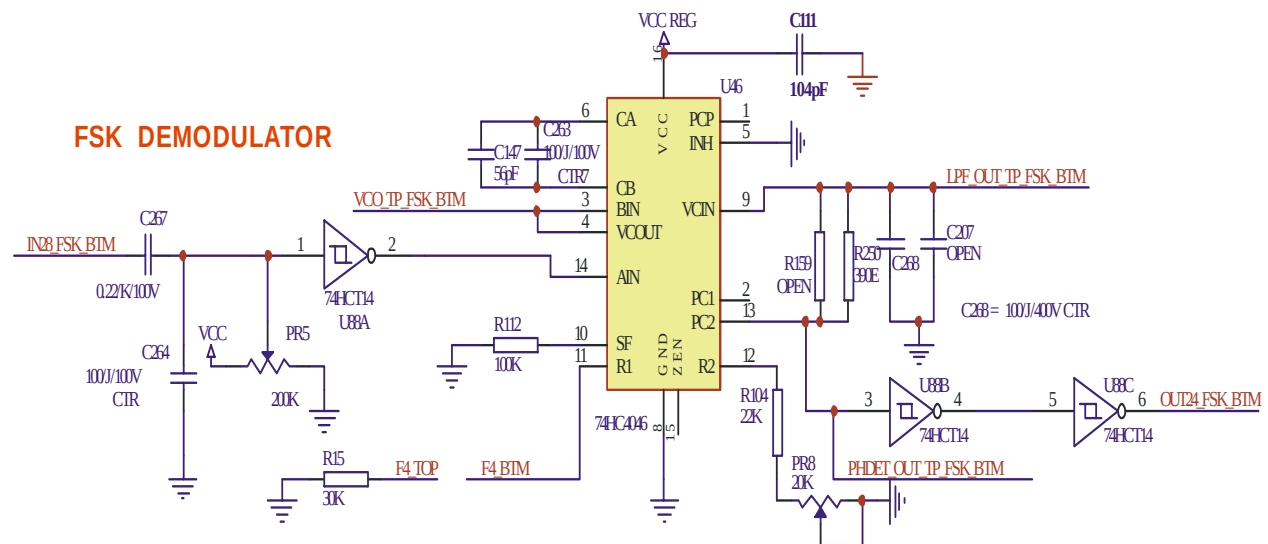


Action and Effect

When we put switch 3 of the SF1 in Switch Fault section to ON position, it will remove the TXCH0 signal. Since TXCH0 signal is used as a Synchronization signal between the transmitter and the receiver, the TDM system loses its synchronization. This will remove all the receiver-decoding pulses. The receiver outputs are disturbed.

4) Effect of Switch Fault 4 In Frequency Modulation / Demodulation Section

SCHEMATIC



DESCRIPTION

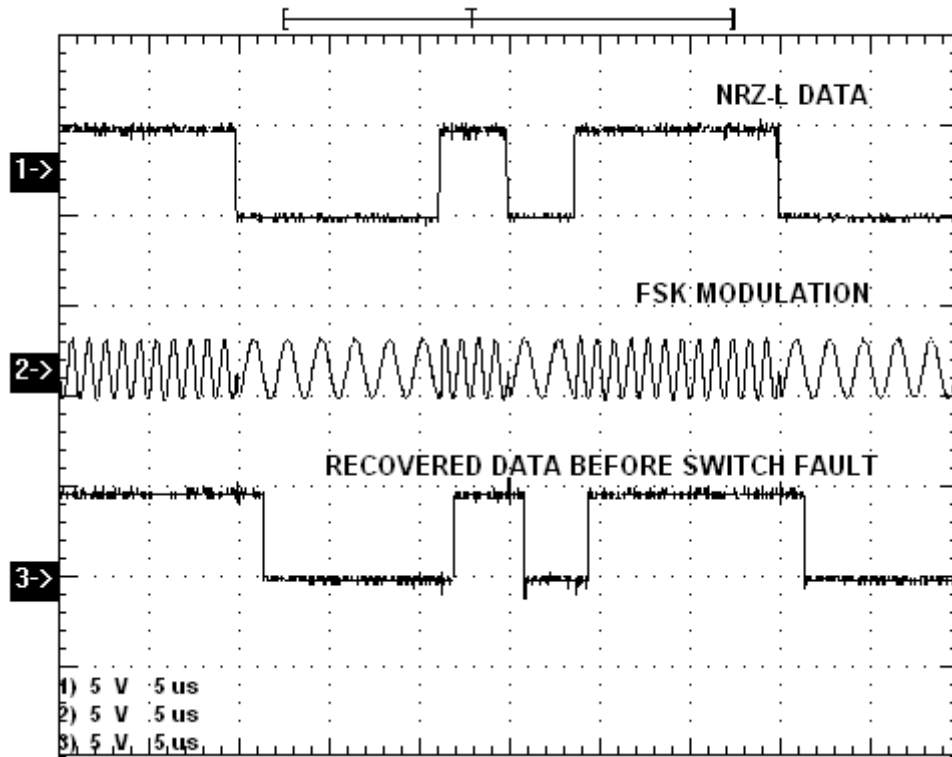
The FSK DEMODULATION SECTION employs PLL logic for the recovery of data. The digital phase locked loop (IC 74HC4046) forms the heart of this logic. The PLL center frequency and lock range are fixed around 500KHz. So, whenever the 500KHz signal is transmitted, the phase detector output of the PLL goes low. And, whenever the 1 MHz signal is transmitted, the phase detector output goes high. Thus, the phase detector output at the PLL directly gives the detected data.

PROCEDURE

- Ensure that the group 4 (**GP4**) clock is selected in the Clock Generation section. Selection is done with the help of switch **S1**. Observe the corresponding LED indication.
- Observe the transmitter clock of the frequencies **250 KHz** at **TXCLK** post.
- Set the data pattern, using switch **S4** as per the given block diagram.
- Observe the 8-bit data pattern at the **S DATA** post.
- Observe the carrier sine wave of the frequency **500 KHz** at **SIN1** post and **1MHz** at **SIN3** post in the carrier section.
- Connect the **SIN1** post to the **IN3** post and **SIN3** post to the **IN2** post of the Carrier modulator section.
- Connect the **SDATA** to the **IN16** post and the **TXCLK** to the **CLK2** post of the Encoded Data section.
- Select the **NRZ-L** data with the help of the switch **S3** and observe the corresponding LED indication in the Encoded Data section.
- Connect the **OUT10** post of the Encoded Data section to the **IN4** post as a control input for the carrier modulator section.
- Observe the FSK modulated signal at the **OUT2** post of the Carrier modulator section.

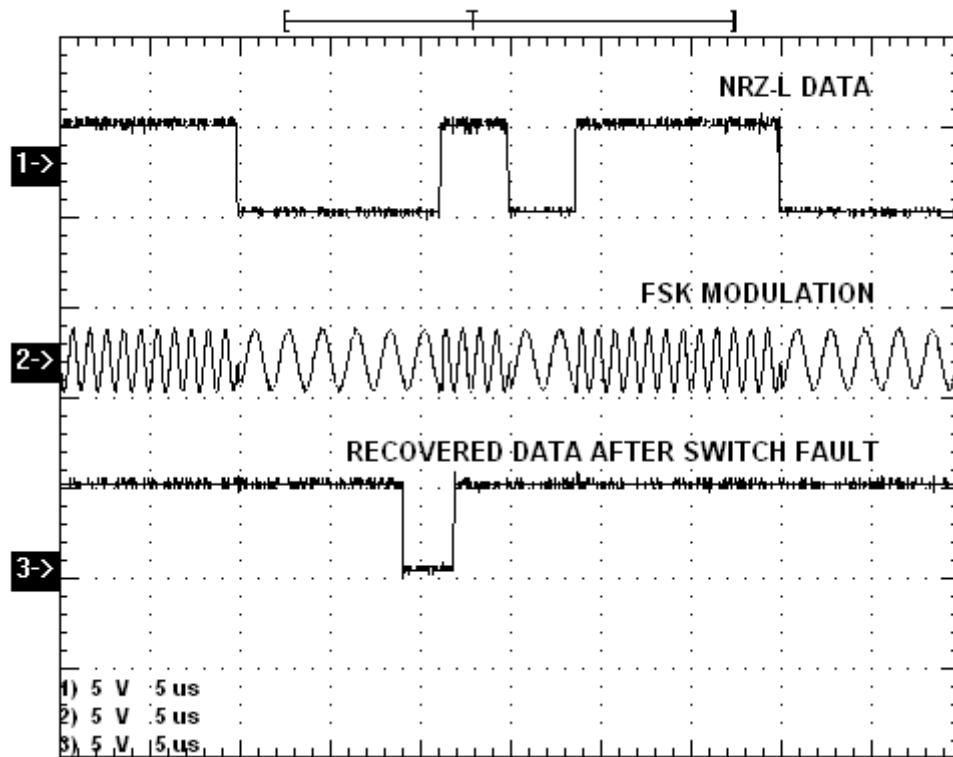
- For the demodulation of the FSK modulated data, connect the **OUT2** post of the carrier modulator to the **IN28** post of the FSK demodulator section.
- Observe the FSK demodulated data at the **OUT24** post of the FSK demodulator section.
- Verify the recovered data with the S DATA.

Waveform Before Switch Fault



Now, put the switch 4 of (SF1) in the Switch Fault to ON position and observe its effect.

Waveform after Switch Fault

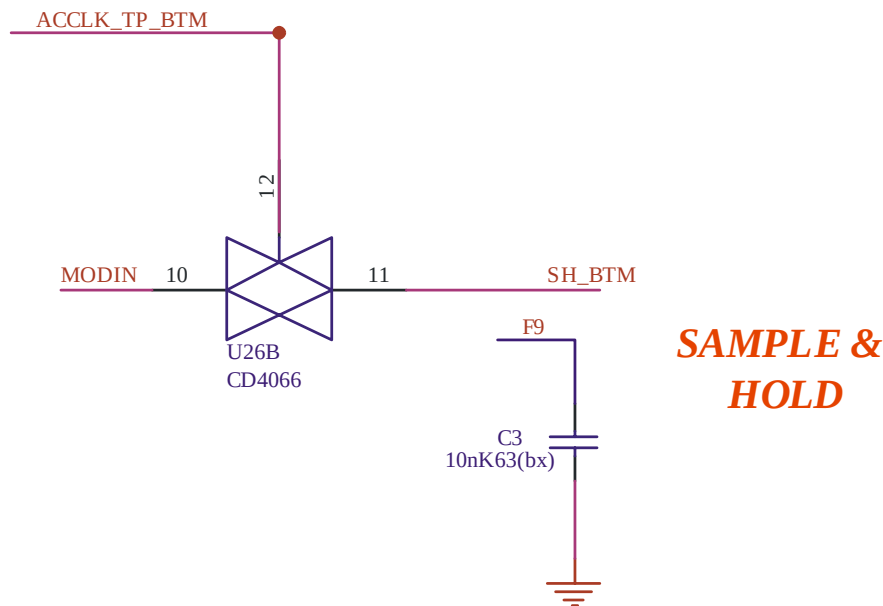


Action and Effect

When we put switch 4 of SF1 in Switch Fault section to ON position it removes the frequency determining resistor connected to PLL in FSK demodulator. Thus, the center frequency of the PLL changes and the output of FSK demodulator gets distorted.

5) Effect of Switch Fault 5 In Sample And Hold Section

Schematic



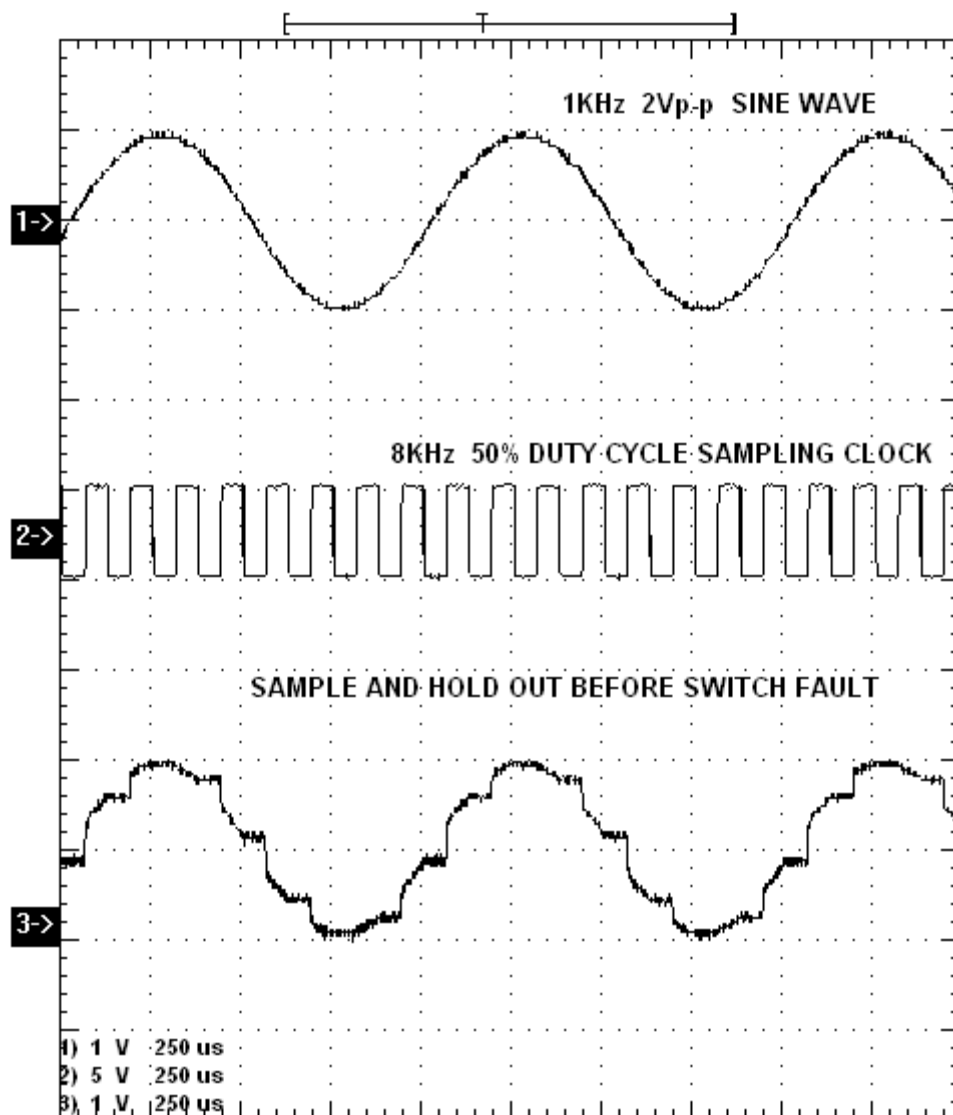
DESCRIPTION

The sine wave generated is fed as an input to the sampling switch (IC CD4066). This samples the analog input at the rate equal to the sampling frequency signal. For sample and hold circuit, the Sine wave is fed to the (IC 4066), which samples the signal and the output is taken across a capacitor, which holds the sample until the next sample arrives, which is then passed through the buffer for better output.

PROCEDURE

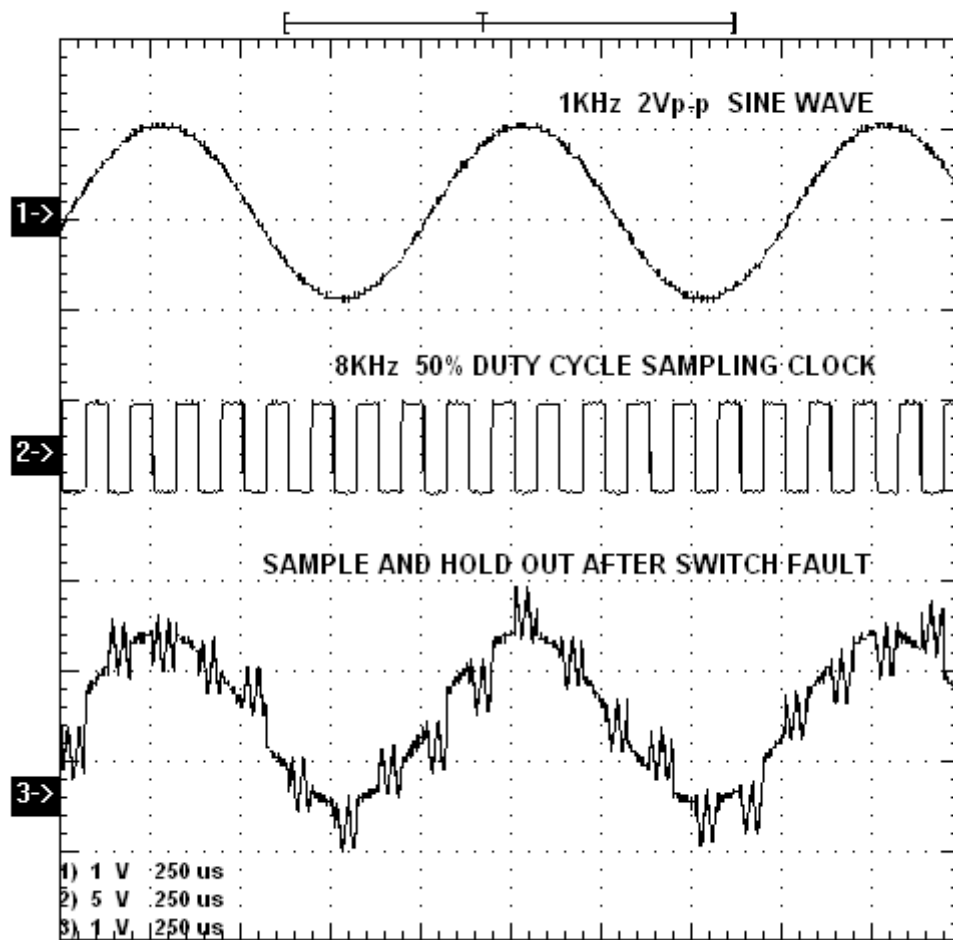
- Ensure that the group 1 (**GP1**) clock is selected in the Clock generation section. Selection is done with the help of switch **S1**, observe the corresponding LED indication.
- Select the sampling clock of frequency 8 KHz, using the switch **S2** and observe the corresponding LED indication.
- Adjust the duty cycle of the sampling clock to 50%, using DIP switch **S10** as shown in the block diagram. Sampling clock is available at the **TX CLK** post.
- The signal under test, which in our case is a sine wave of 1 KHz or 2 KHz is available at the posts **1 KHz** and **2 KHz**, in the function generator section.
- Adjust the amplitude of the 1 KHz and 2 KHz sine waves to 2Vpp, using pot **P5** and **P6** respectively.
- Connect the **TX CLK** post to the **CLK 1** post and the **1 KHz** sine wave to the **IN1** post of the Sampling section.
- To observe the Sample and Hold effect, short the jumper **JP1** to the **Second** position and check the signal at the **OUT1** post of the sampling section.

Waveform Before Switch Fault



Now, put the switch 5 of (SF2) in the Switch Fault to ON position and observe its effect.

Waveform after Switch Fault

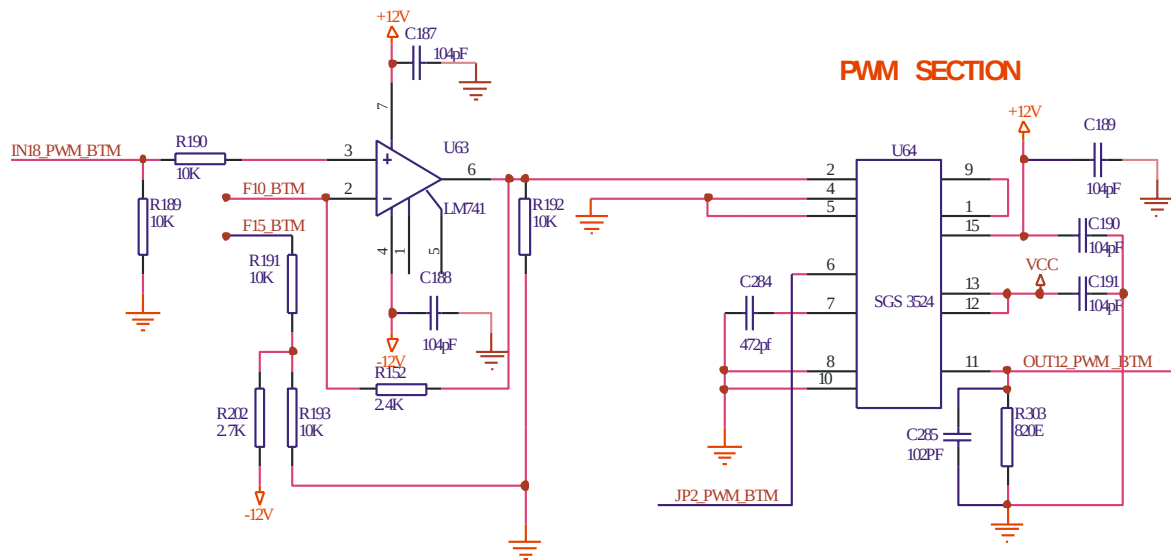


Action And Effect

When we Put switch 5 of the SF2 in Switch Fault section to ON position it removes the capacitor at the output of sample and hold section. Thus, we do not get the hold value of the sampled signal.

6) Effect of Switch Fault 6 in PWM/PPM section

SCHEMATIC



DESCRIPTION

The circuit is divided into two stages. The first stage is the circuit built around Op-Amp 741. This is unity gain non-inverting amplifier, designed as a level shifter. When there is no input signal, the output of the amplifier simply follows the voltage at inverting input terminal, with inverted polarity. This is necessary because PWM chip requires only positive polarity signal. Now, the input signal fed to non-inverting terminal Overrides this shifted voltage.

This signal is now fed to the second part of the circuit. This is specialized for regulating the Pulse Width Modulation. It primarily consists of one fixed frequency oscillator, whose frequency is decided by timing the resistor R_t and the capacitor C_t . The regulation is simply given by the formula,

$$F = 1.18 / (R_t \times C_t)$$

where, F is in KHz, R is in Ω and C is in μF . Also, the error amplifier, comparator and flip-flop are all important functioning parts within the IC. When no input signal is given to the error amplifier, the output oscillates with the same frequency decided by the external R & C network. When input is given to the error amplifier, it causes a linear charging of the output pulse. The amplifier output is compared with the linear charging voltage at C_t , resulting in a modulated pulse at the output of high gain comparator. Thus, we get the PWM. For more details please refer to the appendix SGS3524 data sheets.

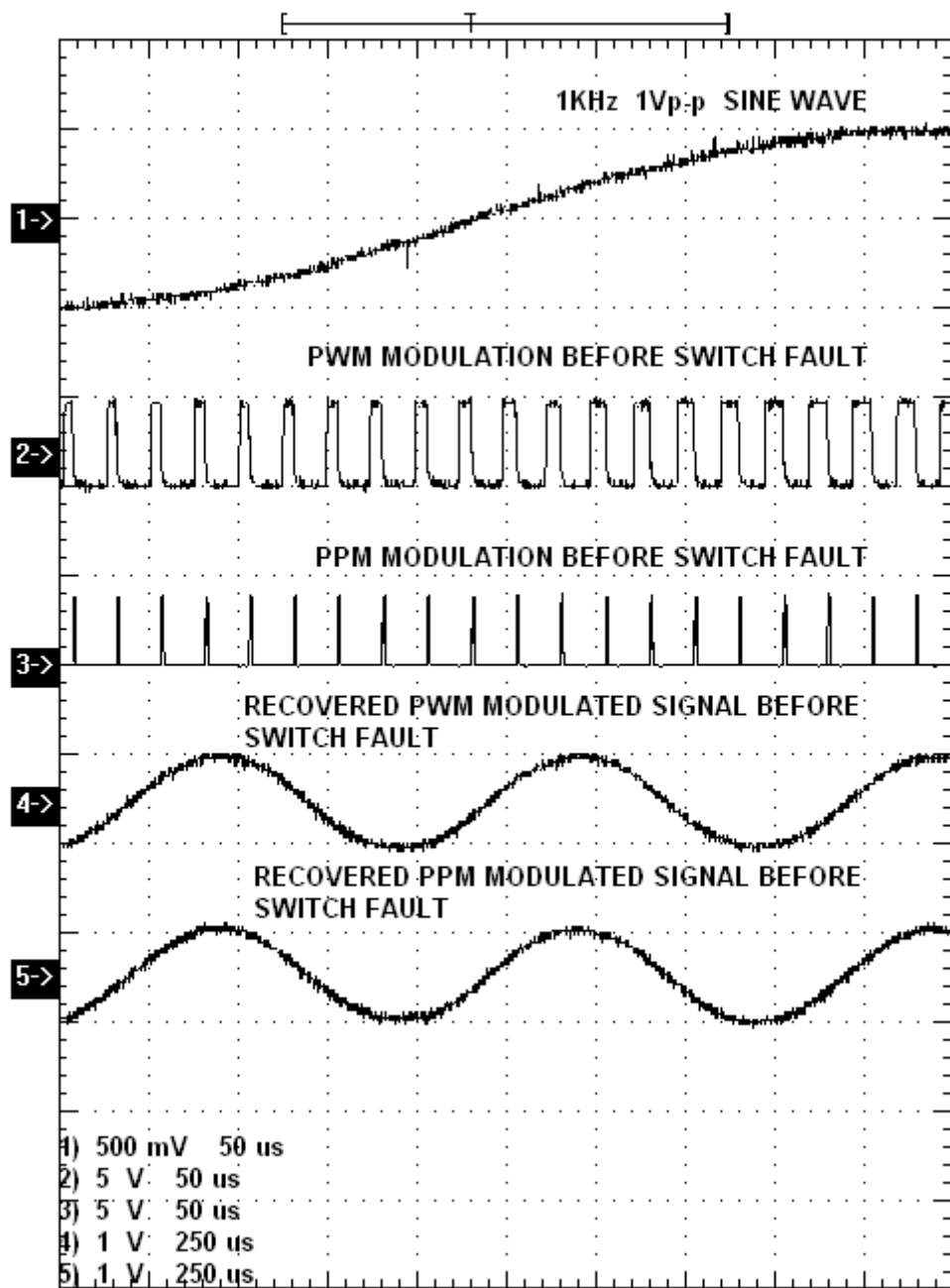
PWM MODULATION

PROCEDURE

- Connect a low frequency sine wave from the **SINE OUT** post having amplitude of 1Vpp, using pot **P2** from the Function generator section to the **IN18** post of the PWM/PPM section.
- Keep jumper **JP2** on the 2nd position.
- Observe the variation in the width of the carrier at the **OUT12** post of the PWM/PPM section. Change the frequency of the input sine wave from 1 to 30 Hz, using pot **P1** and observe the variation.
- Now, connect the **1 KHz** sine wave having amplitude of 1Vpp, using pot **P5** from the Function generator section to the **IN18** post of the PWM/PPM section. Also, observe the counter outputs at their corresponding test points (**TP9** to **TP16**).
- Observe the pulse width modulated output at **OUT12** post of the PWM/PPM section.
- Connect the **OUT12** post of the PWM/PPM to the **IN20** post of the PWM/PPM Demodulator section.
- Keep switch **S7** at PWM position.
- Observe the pulse width demodulated output at **OUT15** post of the PWM/PPM demodulator section.
- Connect the **OUT15** post of the PWM/PPM demodulator section to the **IN33** post of the 2nd order LPF.
- Connect the **OUT30** post of 2nd order LPF to the **IN 34** of the 4th order LPF.
- Observe the recovered signal at the **OUT31** post of the 4th order LPF.
- Repeat the experiment for different input signals and sampling clocks, by changing the position of the jumper **JP2**.

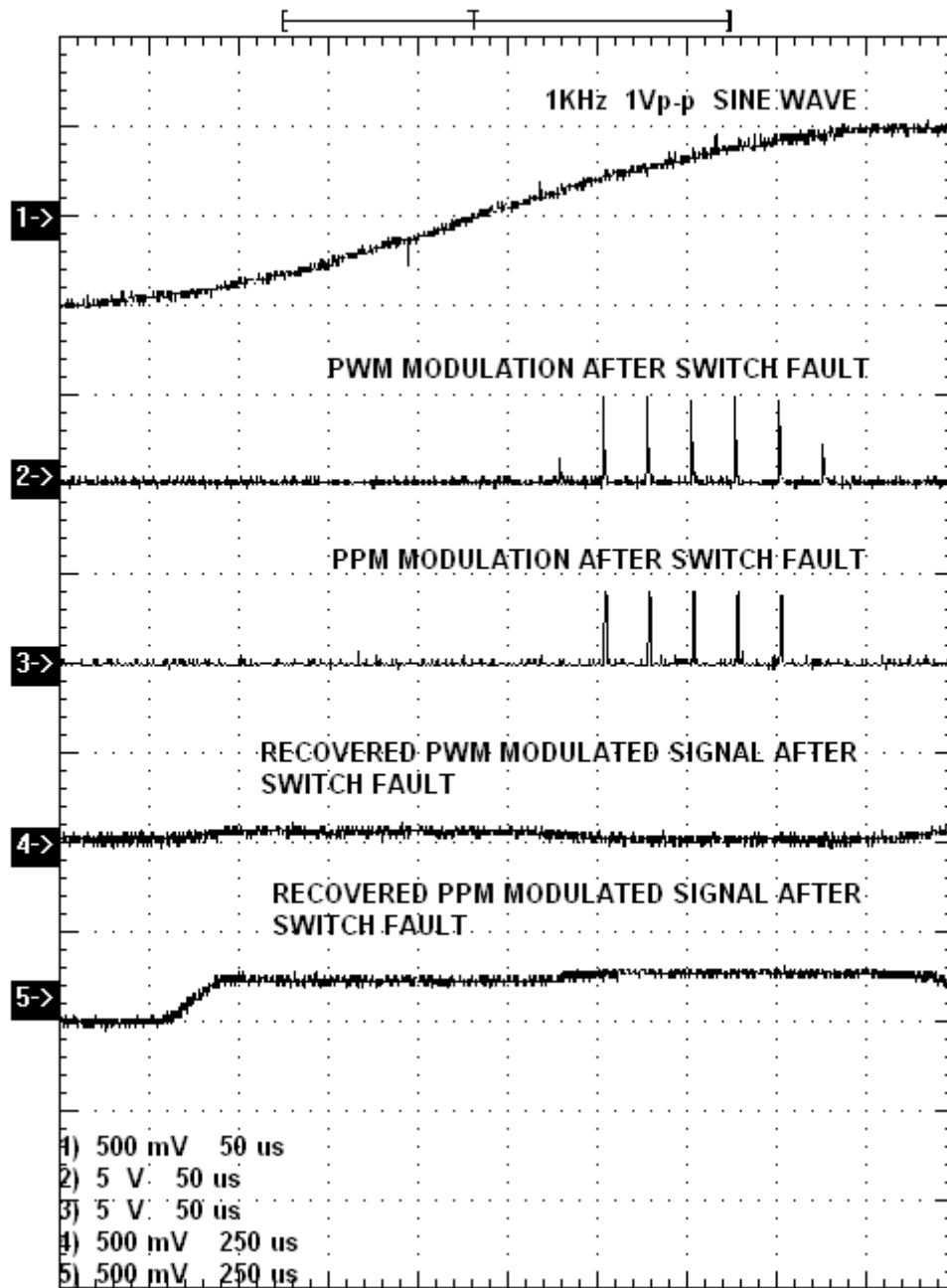
For PPM Modulation Follow The Procedure As Per Experiment 12c.

WAVEFORM BEFORE SWITCH FAULTS



Now, put the switch 6 of the SF2 in the Switch Fault to ON position and observe its effect.

Waveform After Switch Fault

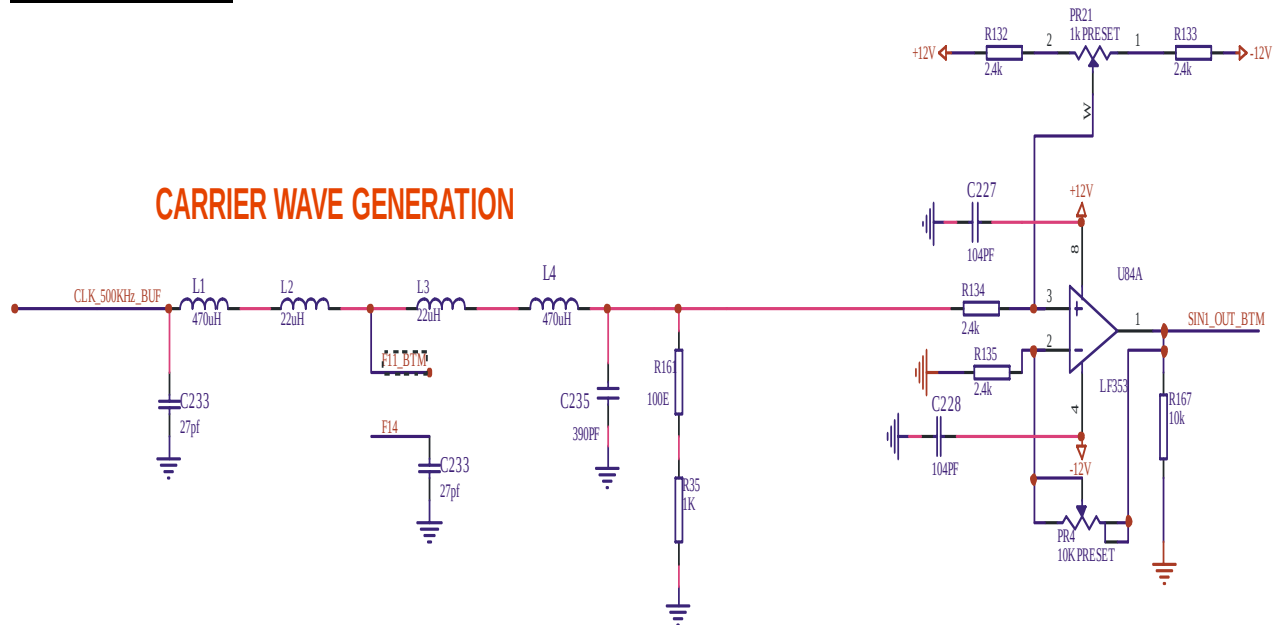


Action And Effect

When we put switch **6** of the **SF2** in Switch Fault section to **ON** position, it changes the Minimum Pulse Width to zero level. Due to this, PWM output will be absent if there is no signal at the PWM input. Minimum pulse width at the output of PWM modulator is not maintained, thus the PWM/PPM modulation is disturbed.

7) Effect of Switch Fault 7 In Carrier Generation Section

SCHEMATIC



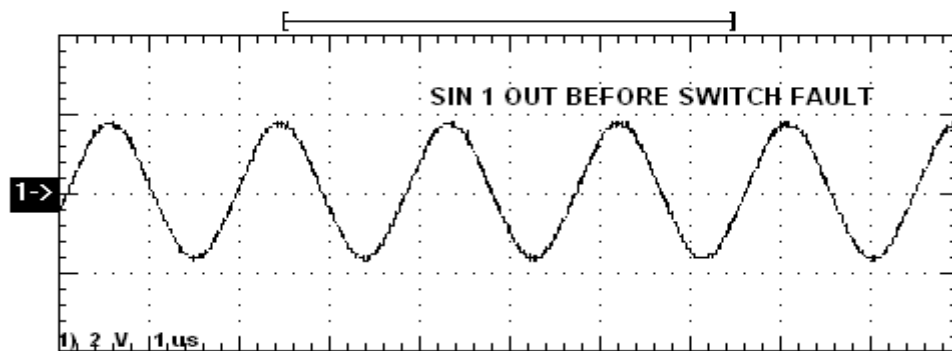
DESCRIPTION

The 500 KHz carrier is generated by filtering the 500 KHz clock. To filter the clock, LC type filter is used. The 500 KHz clock is generated by the master clock, using frequency divider logic. The 500kh clock generated then passes through the digital buffer. Output of digital buffer is then fed to the LC filter whose components (inductors and capacitors) are designed for 500 KHz frequency.

PROCEDURE

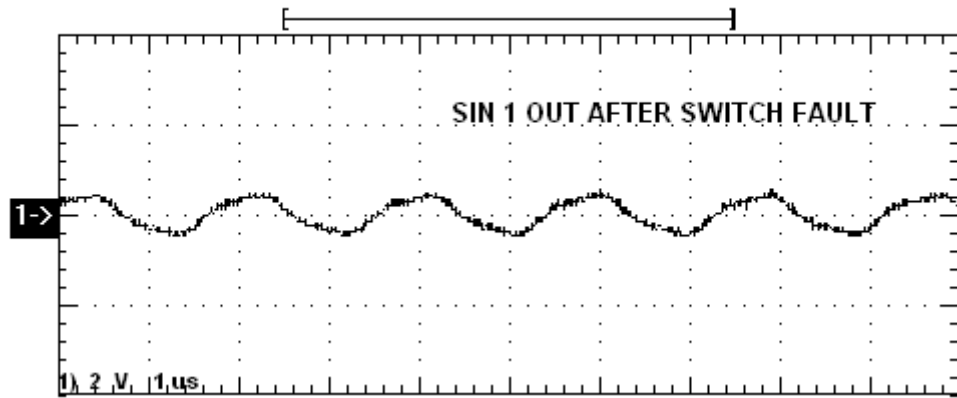
- Observe the carrier sine wave of frequency 500 KHz at the **SIN1** post.

Waveform before Switch Fault



Now, put the switch 7 of (SF2) in the Switch Fault to ON position and observe its effect.

Waveform after Switch Fault



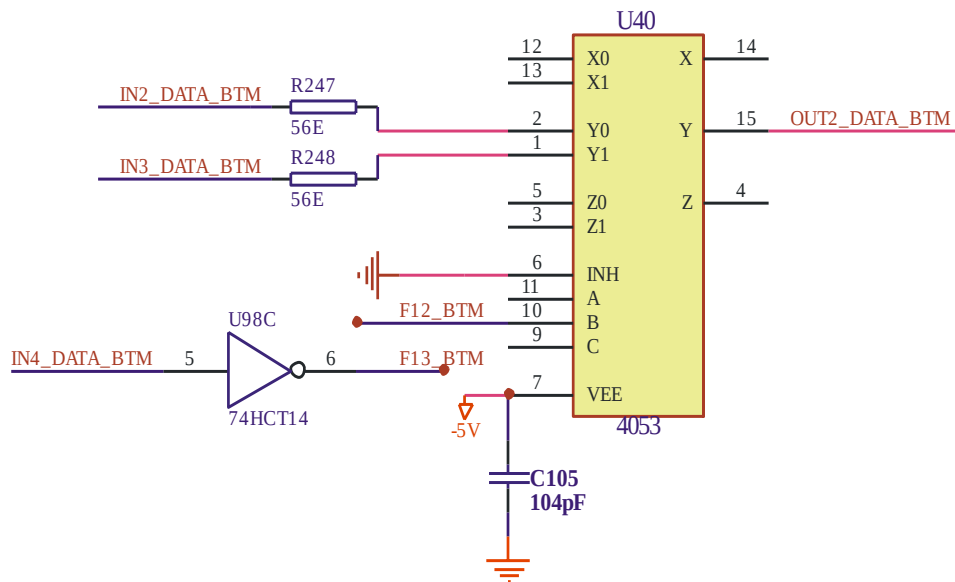
Action And Effect

When we put switch 7 of SF2 in switch fault section to ON position, it opens the capacitor to filter the carrier signal.

Due to this, amplitude of sin 1 gets reduced and also the shape of the carrier wave gets distorted.

8) Effect of Switch Fault 8 in carrier modulation section

SCHEMATIC



DESCRIPTION

The carrier-modulated section consists of (IC 4053), which is a Triple Two Channel Analog Multiplexer/ Demultiplexer. The control input for the two gates are fed from the output of (IC 74HCT14), which is an Inverter. The Control Input is fed to the data input of (IC 4053).

To obtain the ASK signal, Carrier-3 Wave is fed to the input of Gate-1, and another input of Gate-2 of (IC 4053) is grounded. We will get ASK Modulated signal at the output of (IC 4053).

To obtain the FSK signal, Carrier-1 Wave is fed to the input of Gate-1, and another Carrier-3 Wave, which is of different frequency with respect to Carrier-1, is fed to input of Gate-2. We will get FSK Modulated signal at the output.

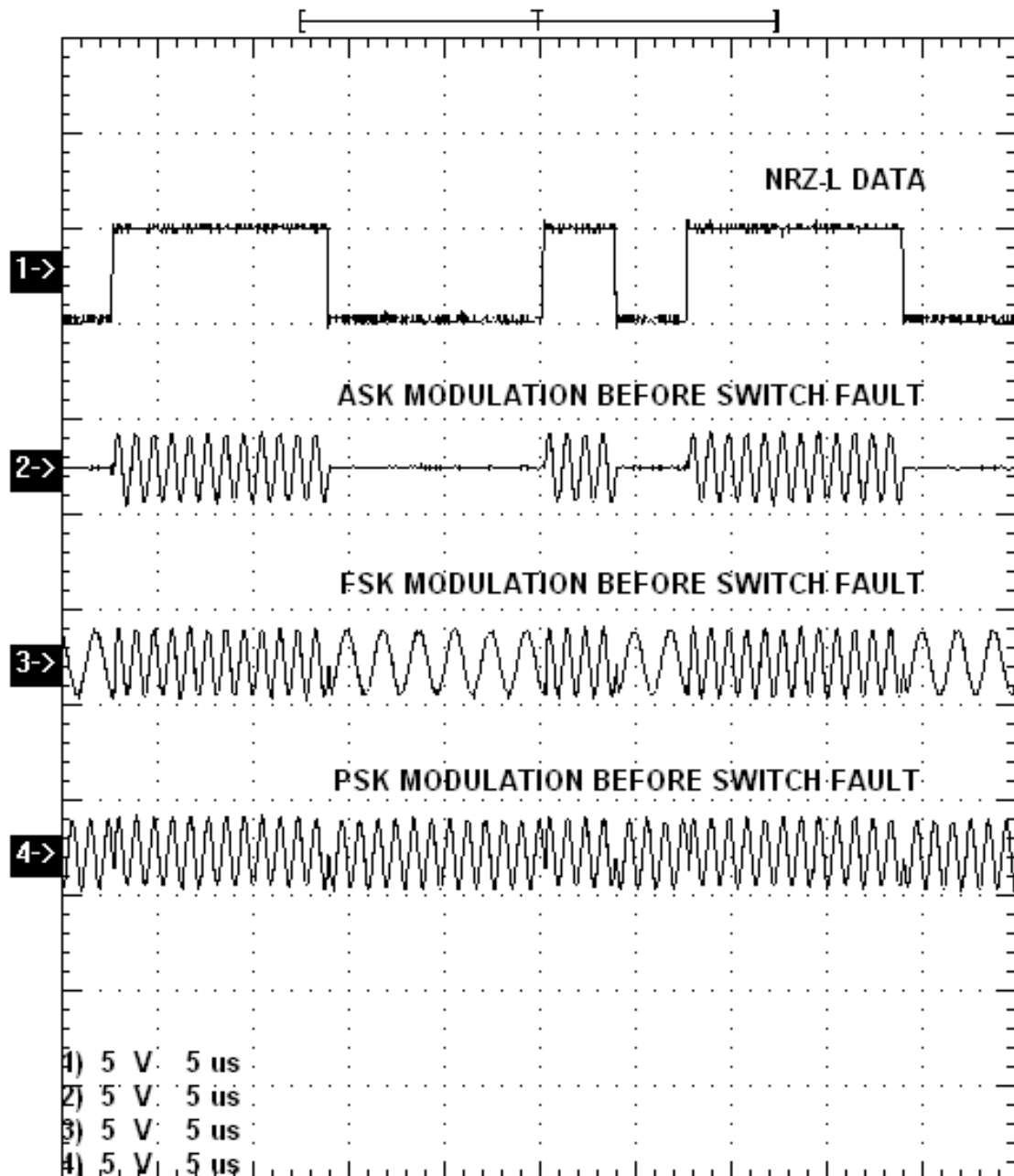
To obtain the PSK signal, Carrier-3 Wave is fed to the input of Gate-1, and another Carrier-2 Wave, which has the same frequency, but 180° out of phase with the Carrier-3 is fed to input of Gate-2 of (IC 4053). We will get the PSK Modulated signal at the output of U22.

PROCEDURE

- Ensure that the group 4 (**GP4**) clock is selected in the Clock Generation section. Selection is done with the help of switch **S1**. Observe the corresponding LED indication.
- Observe the transmitter clock of frequency **250 KHz** at the **TXCLK** post.
- Set the data pattern, using switch **S4** as per the given block diagram.
- Observe the 8-bit data pattern at the **S DATA** post.
- Observe the carrier sine wave of frequencies **500 KHz** at **SIN1** post and **1MHz** at **SIN3** post in the carrier section.
- Connect the **SIN1** post to the **IN3** post and **SIN3** post to the **IN2** post of the Carrier modulator section.
- Connect the **S DATA** to the **IN16** post and the **TXCLK** to the **CLK2** post of the Encoded Data section.
- Select the **NRZ-L** data with the help of the switch **S3** and observe the corresponding LED indication in the Encoded Data section.

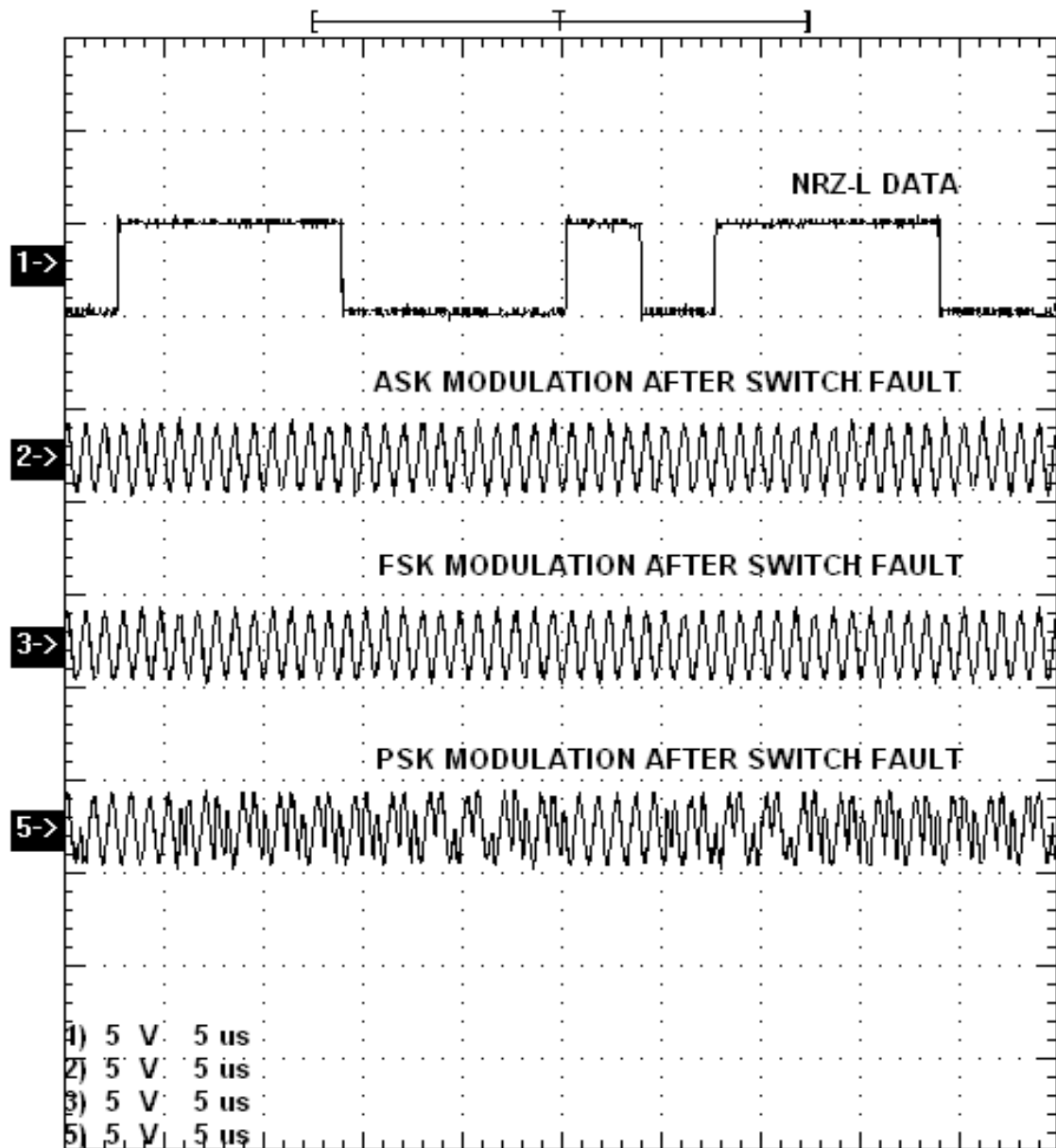
- Connect the **OUT10** post of the Encoded Data section to the **IN4** post as a control input for the carrier modulator section.
- Observe the FSK modulated signal at the **OUT2** post of the Carrier modulator section.
- For the demodulation of the FSK modulated data, connect the **OUT2** post of the carrier modulator to the **IN28** post of the FSK demodulator section.
- Observe the FSK demodulated data at the **OUT24** post of the FSK demodulator section.
- Verify the recovered data with the **S DATA**.

Waveform before Switch Faults



Now, put the switch 8 of (SF1) in the Switch Fault to ON position and observe its effect.

Waveform after Switch Fault



Action And Effect

When we Put switch 8 of SF2 in Switch Fault section to ON position, it disables the channel selection signal going to the Modulator IC. Modulator output contains only a single channel (INPUT 1) data.

**EXPERIMENT
NO. 1**

EXPERIMENT NO.1

NAME

Signal samplings and their reconstruction

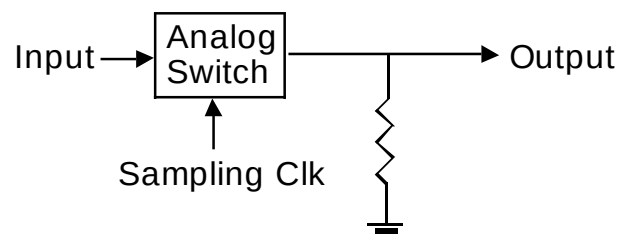
OBJECTIVES

To study different types of signal samplings and their reconstruction:

- A. Natural Sampling
- B. Sample and Hold
- C. Flat top sampling

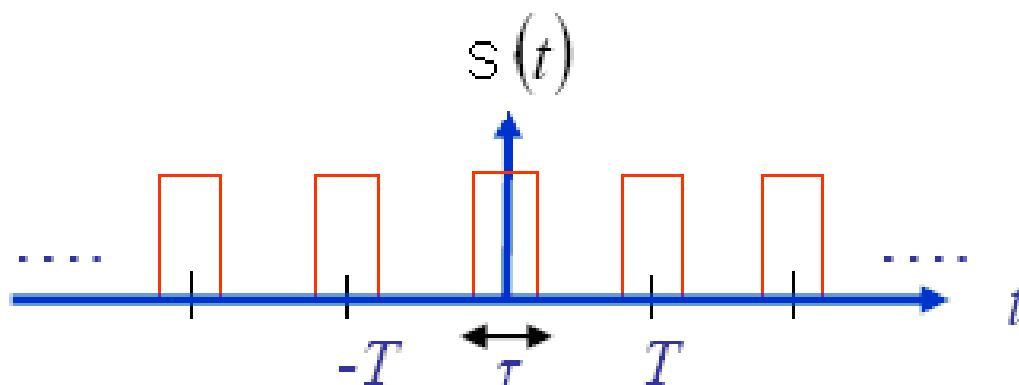
THEORY

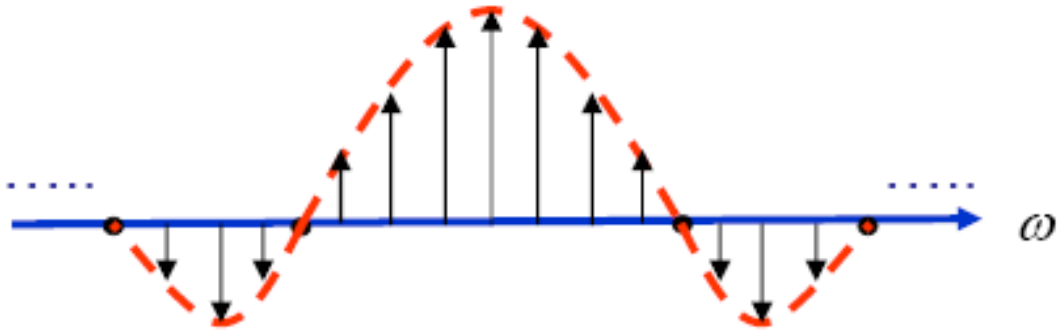
Natural Sampling



Here, the sampling waveform $S(t)$ consists of a train of pulses having duration τ and it is separated by the sampling time T_s . The baseband signal is $m(t)$, and the sampled signal $S(t)m(t)$ is shown in the figure. Observe that the sampled signal consists of a sequence of pulses varying amplitude whose tops are not flat but follow the waveform of the signal $m(t)$. With natural sampling, as with instantaneous sampling, a signal sampled at the Nyquist rate may be reconstructed exactly by passing the samples through an ideal low-pass filter with cutoff at the frequency f_m , where f_m is the highest frequency spectral component of the signal. To prove this, we note that the sampling waveform $S(t)$ is given by:

$$S(t) = \tau / T_s + 2\tau / T_s (C_1 \cos 2\pi t / T_s + C_2 \cos 2 * 2\pi t / T_s + \dots) \quad \dots\dots\dots (i)$$





With the constant C_n given by

$$C_n = \frac{\sin(n\pi\tau/T_s)}{n\pi\tau/T_s} \dots\dots\dots (ii)$$

The sampled base band signal $S(t) m(t)$ is for $T_s = 2f_M$,

$$S(t)m(t) = \tau / T_s m(t) + 2\tau / T_s [m(t) C_1 \cos 2\pi(2f_M)t + m(t) C_2 \cos 2\pi(4f_M)t + \dots] \dots (iii)$$

Therefore, as in instantaneous sampling, a low-pass filter with cutoff at f_M will deliver an output signal $s_o(t)$ given by

$$s_o(t) = \tau / T_s m(t) \dots\dots\dots (iv)$$

With samples of finite duration, it is not possible to completely eliminate the crosstalk generated in a channel, sharply band limited to a bandwidth f_c . If N signals are to be multiplexed, then the maximum sample duration is $\tau = T_s / N$. It is advantageous, for the purpose of increasing the level of the output signal, to make τ as large as possible. For (a), as it is seen in Eq. (iv), $s_o(t)$ increases with τ . However, to help suppress crosstalk, it is ordinarily required that the samples be limited to a duration much less than T_s / N . The result is a large guard time between the end of one sample and the beginning of the next.

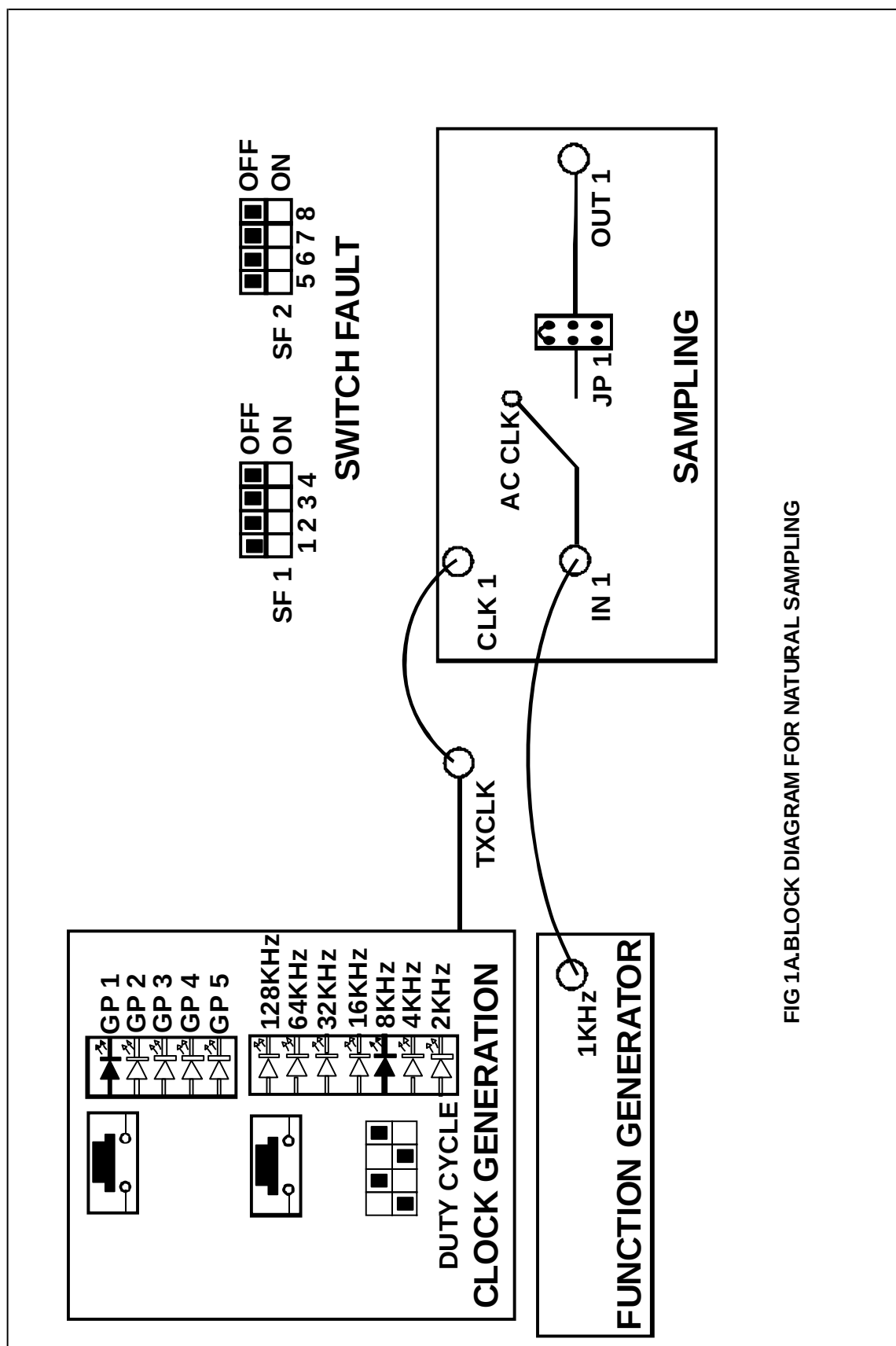
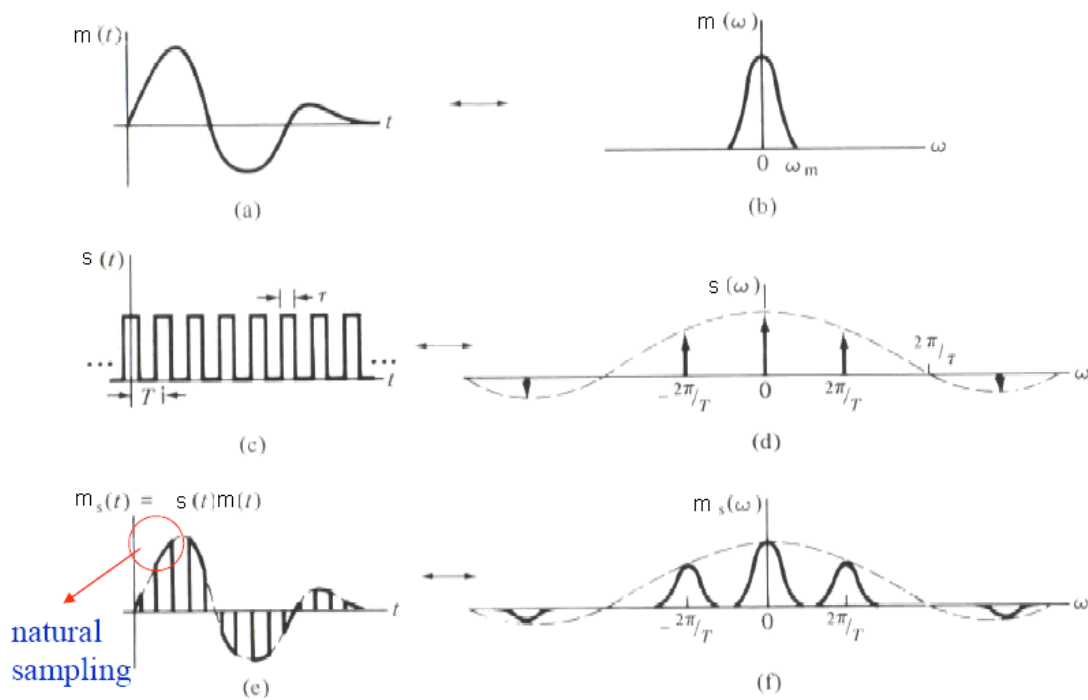


FIG 1A.BLOCK DIAGRAM FOR NATURAL SAMPLING



EQUIPMENTS

DCS-B kit
 Connecting Chords
 Power supply
 20 MHz Dual Trace Oscilloscope

NOTE: Keep All The Switch Faults In Off Position.

A. NATURAL SAMPLING AND ITS RECONSTRUCTION

PROCEDURE

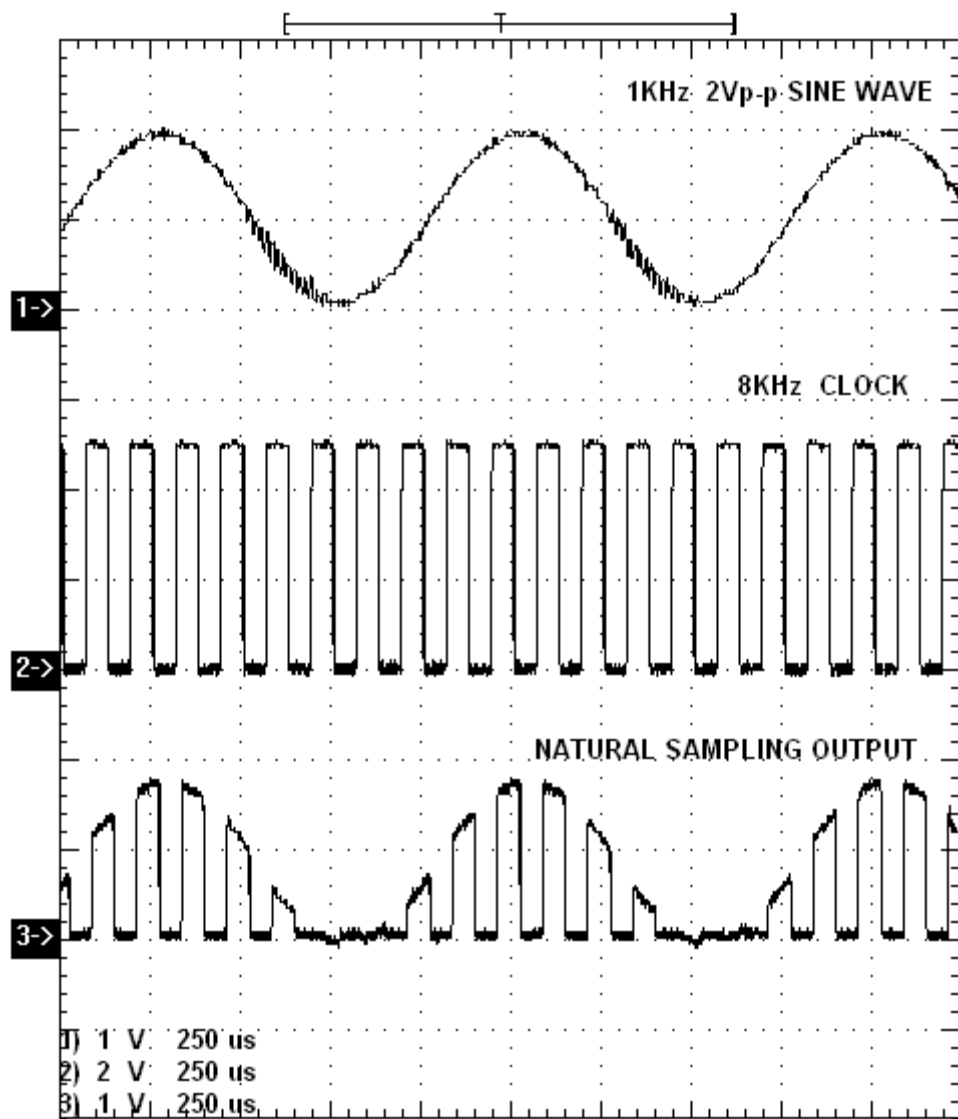
- Ensure that group 1 (**GP1**) clock is selected in the Clock generation section. Selection is done with the help of switch **S1**. Observe the corresponding LED indication.
- Select the sampling clock of frequency 8 KHz using the switch **S2** and observe the corresponding LED indication.
- Adjust the duty cycle of the sampling clock to 50% using Dipswitch **S10** as shown in the block diagram. Sampling clock is available at **TX CLK** post.
- The signal under test, which is in our case, is a sine wave of 1 KHz or 2 KHz, which is available at posts 1 KHz, and **2 KHz** in the Function Generator section.
- Adjust the amplitude of the 1 KHz and 2 KHz sine wave to 2Vpp using pot **P5** and **P6** respectively.
- Connect **TX CLK** post to **CLK 1** post and **1 KHz** sine wave to **IN1** post of the Sampling section. To observe the Natural sampling effect, short the jumper **JP1** to **first** position and check the signal at the **OUT1** post of the sampling section.

OBSERVATIONS

- Pressing Switch S1 to select Group1 (GP1) is indicated by corresponding LED.
- Observed that the natural sampled signal consists of a sequence of pulses of varying amplitude whose tops are not flat but follow the waveform of the signal to be sampled. With natural sampling, a signal sampled at the Nyquist rate may be reconstructed by passing the samples through an ideal Low-pass filter.
- Pressing Switch S2 to select sampling frequency of 8 KHz is indicated by corresponding LED.
- Observe the Sampling clock at **TX CLK** post test point.
- Observe the signal under test, i.e. a sine wave, at posts 1 KHz and 2 KHz in the Function Generator section.
- Observe the adjustment in the Amplitude of 1 KHz and 2 KHz sine wave to 2Vpp using pot **P5** and **P6** respectively.
- Observe the Natural Sampling effect at the **OUT1** post of the sampling section.

CONCLUSION

Observed that the natural sampled signal consists of a sequence of pulses of varying amplitude whose tops are not flat but follow the waveform of the signal to be sampled. With natural sampling, a signal sampled at the Nyquist rate may be reconstructed by passing the samples through an ideal Low-pass filter.



NATURAL SAMPLING

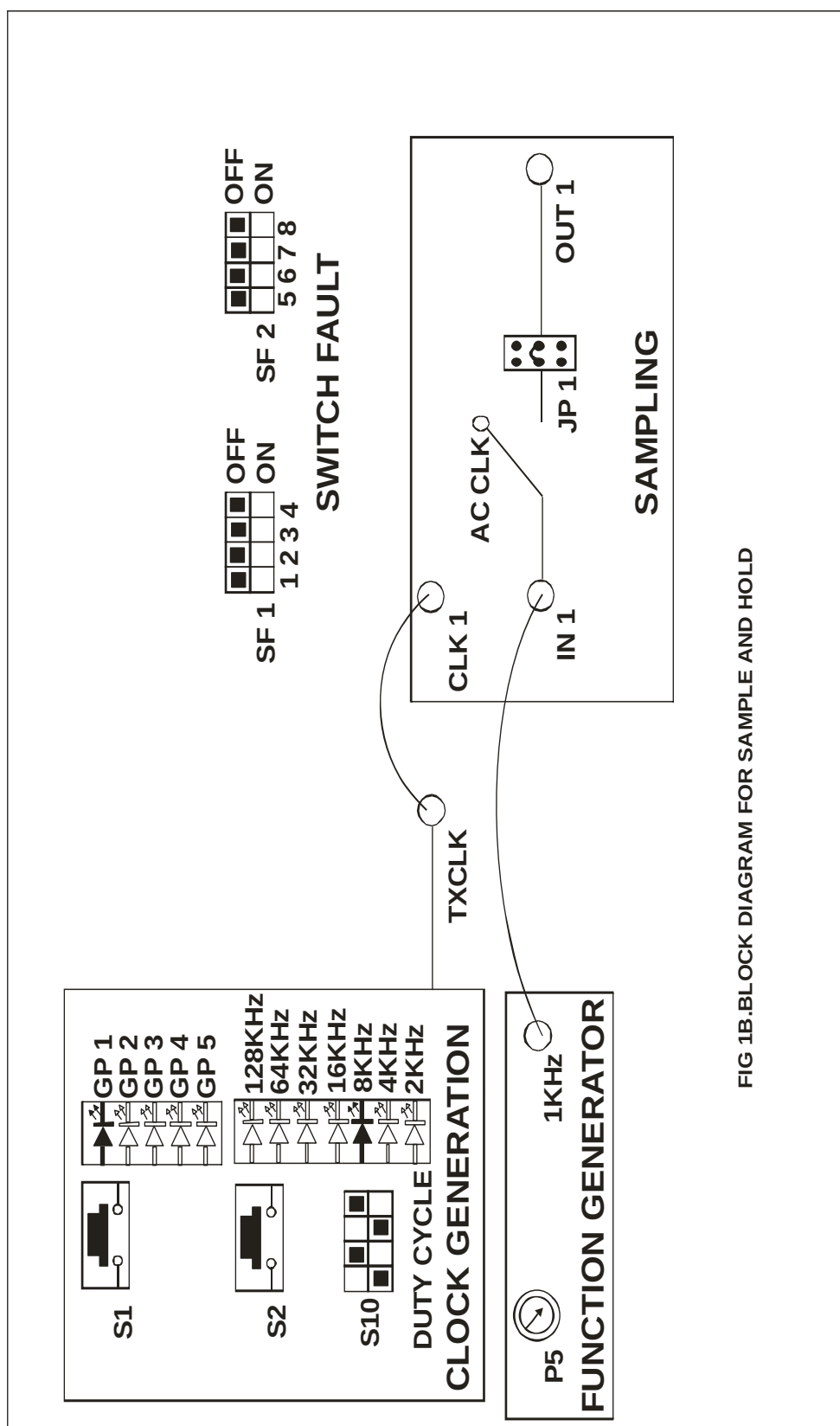
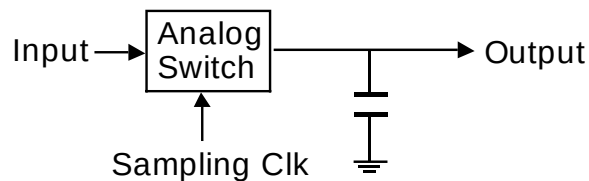


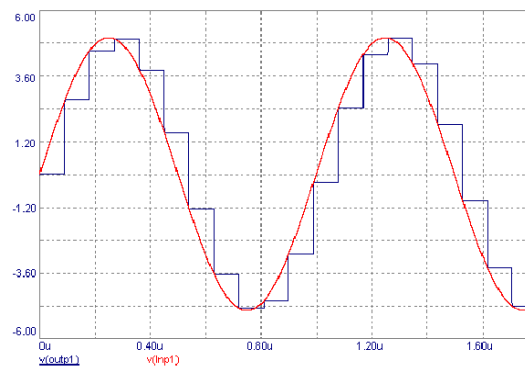
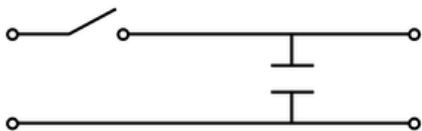
FIG 1B.BLOCK DIAGRAM FOR SAMPLE AND HOLD

B. SAMPLE AND HOLD AND ITS RECONSTRUCTION



In electronics, a sample and hold circuit is used to interface real world, changing analogue signals to a subsequent system such as an analog-to-digital converter. The purpose of this circuit is to hold the analogue value steady for a short time while the converter or other following system performs some operation that takes a little time.

In most circuits, a capacitor is used to store the analogue voltage and an electronic switch or gate is used to alternately connect and disconnect the capacitor from the analog input. The rate at which this switch is operated is the sampling rate of the system.



PROCEDURE

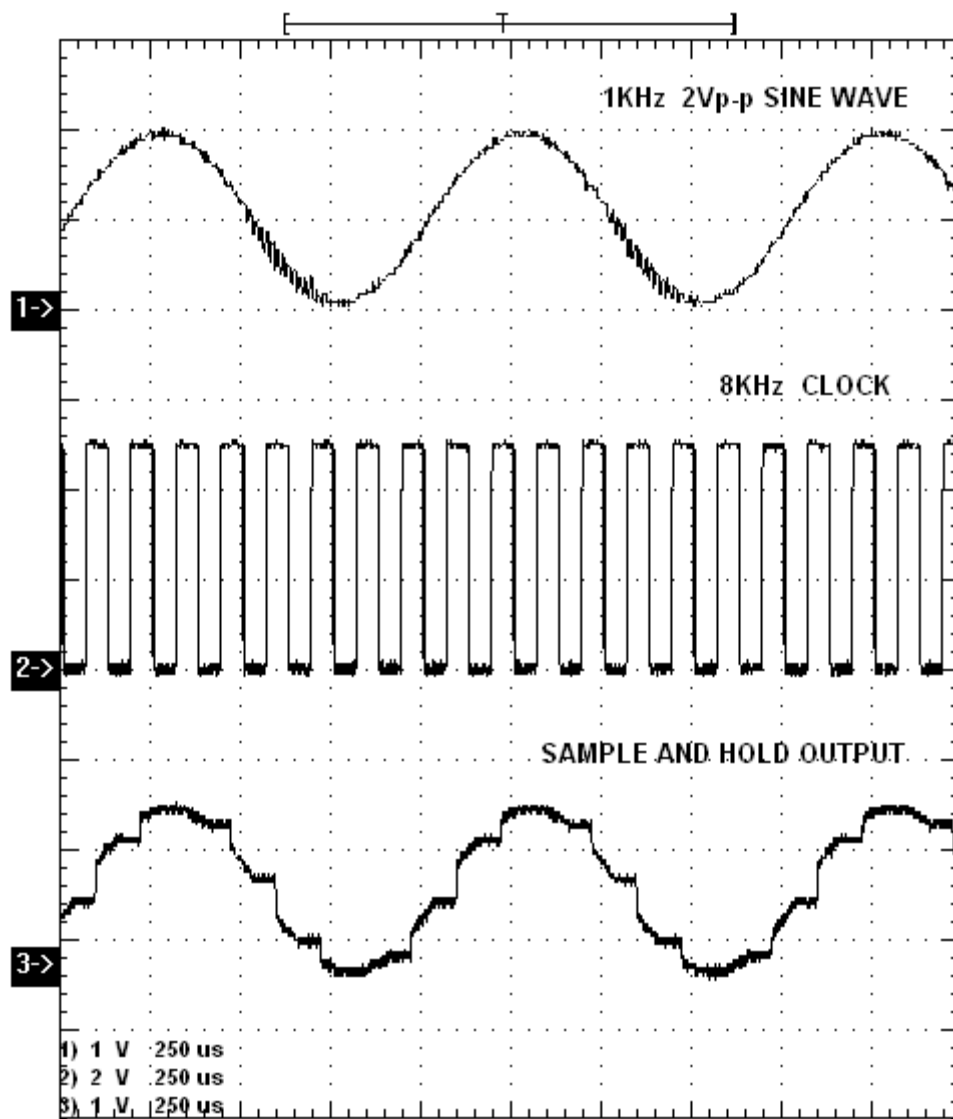
- Ensure that group 1 (**GP1**) clock is selected in the Clock generation section. Selection is done with the help of switch **S1**, observe the corresponding LED indication.
- Select the sampling clock of frequency 8 KHz using the switch **S2** and observe the corresponding LED indication.
- Adjust the duty cycle of the sampling clock to 50% using DIP switch **S10** as shown in the block diagram. Sampling clock is available at **TX CLK** post.
- The signal under test, which in our case is a sine wave of 1 KHz or 2 KHz is available at posts **1 KHz** and **2 KHz** in the Function Generator section.
- Adjust the amplitude of the 1 KHz and 2 KHz sine wave to 2Vpp using pot **P5** and **P6** respectively.
- Connect **TX CLK** post to **CLK 1** post and **1 KHz** sine wave to **IN1** post of the Sampling section.
- To observe the Sample and Hold effect, short the jumper **JP1** to **Second** position and check the signal at the **OUT1** post of the sampling section.

OBSERVATIONS

- Pressing Switch S1 to select Group1 (GP1) is indicated by corresponding LED.
- Pressing Switch S2 to select sampling frequency of 8 KHz is indicated by corresponding LED.
- Observe the Sampling clock at **TX CLK** post test point.
- Observe the signal under test, i.e. a sine wave, at posts 1 KHz and 2 KHz in the Function Generator section.
- Observe the adjustment in the Amplitude of 1 KHz and 2 KHz sine wave to 2Vpp using pot **P5** and **P6** respectively.
- Observe the Sample & Hold effect at the **OUT1** post of the sampling section.

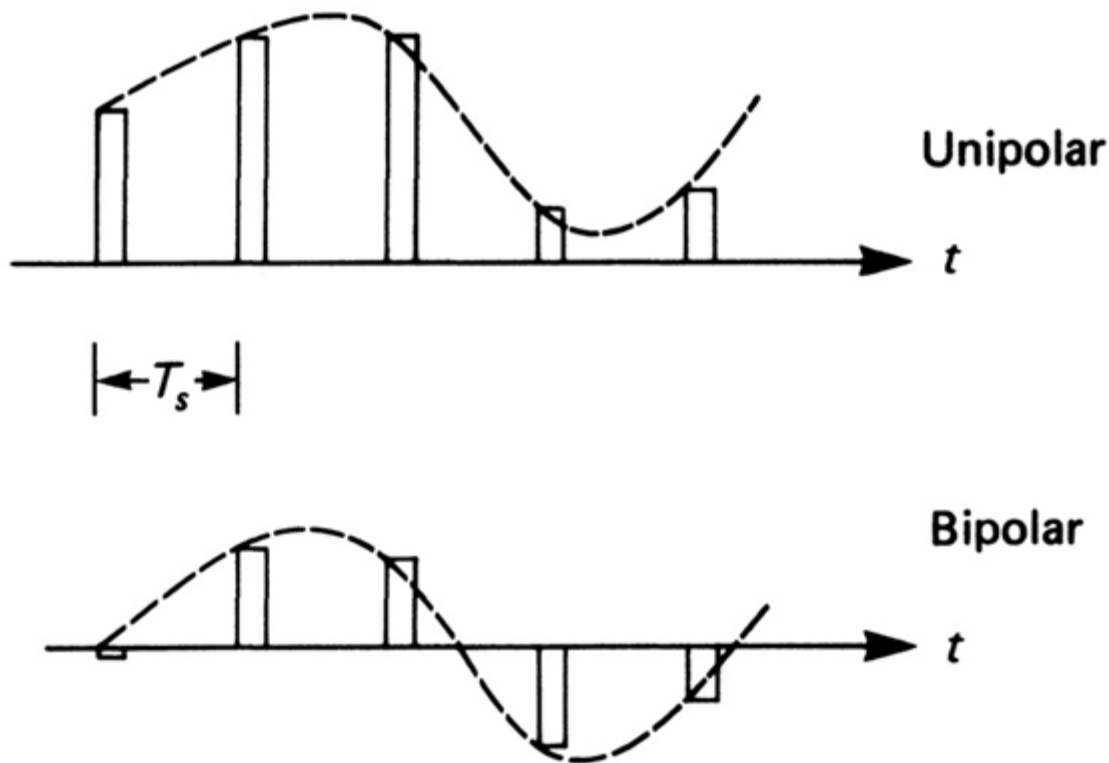
CONCLUSION

In Sample and Hold, we can observe that during on time of Clock capacitor C charges abruptly to a voltage proportional to the sample value and during OFF time of the clock capacitor holds this voltage until the operation is repeated for next sample.



SAMPLE AND HOLD

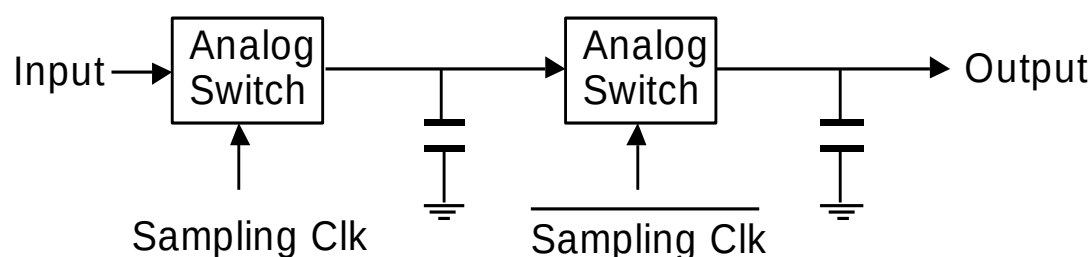
To illustrate the effect of flat-top sampling, let us assume that the signal $m(t)$ has a flat spectral density equal to M_0 over its entire range from 0 to f_M , as shown



in the figure. The form of the transform of the instantaneously sampled signal is shown in the figure.

The sampling frequency $f_s = 1/T_s$ is assumed large enough to allow for a guard band between the spectrum of the base band signal and the DSB-SC signal with carrier f_s . The spectrum of the flat-top sampled signal is shown in the figure. A low-pass filter reconstructs the original signal.

C. FLAT TOP SAMPLING AND ITS RECONSTRUCTION



A flat-top pulse has constant amplitude established by the sample value of the signal at some point within the pulse interval. Flat-top sampling simplifies the design of the electronic circuitry used to perform the sampling operation.

To show the extent of the distortion, consider the signal $m(t)$ having a Fourier transform $M(j\omega)$. The transform of the sampled signal for flat-top sampling is determined by considering that the flat-top pulse can be generated by passing the instantaneously sampled signal

through a network which broadens a pulse of duration τ (an impulse) into a pulse of duration τ . The transform of a pulse of unit amplitude and width τ is

$$F[\text{impulse of strength } \tau \text{ at } t = 0] = \tau \quad \dots\dots\dots (i)$$

The transform of a pulse and unit amplitude is

$$\text{Sin } (\omega\tau/2) [\text{pulse, amplitude} = 1, \text{extending from } t = -\tau/2 \text{ to } t = \tau/2] = \tau \dots\dots\dots (ii) \dots (\omega\tau/2)$$

Hence the transfer function of the network is

$$H(j\omega) = \frac{\tau \sin(\omega\tau/2)}{\omega\tau/2} \dots\dots\dots (iii)$$

Let the signal $m(t)$, with the transform $M(j\omega)$, be band limited to f_M and be sampled at the Nyquist rate or faster. Then in the range of 0 to f_M the transform of the flat-topped sampled signal is given by the product $H(j\omega) M(j\omega)$ or from Eq. (i), (ii) and (iii)

PROCEDURE

- Ensure that group 1 (**GP1**) clock is selected in the Clock generation section. Selection is done with the help of switch **S1**. Observe the corresponding LED indication.
- Select the sampling clock of frequency 8 KHz using the switch **S2** and observe the corresponding LED indication.
- Adjust the duty cycle of the sampling clock to 50% using DIP switch **S10** as shown in the block diagram. Sampling clock is available at **TX CLK** post.
- The signal under test, which in our case is a sine wave of 1 KHz or 2 KHz is available at posts **1 KHz** and **2 KHz** in the Function Generator section.
- Adjust the amplitude of the 1 KHz and 2 KHz sine wave to 2Vpp using pot **P5** and **P6** respectively.
- Connect **TX CLK** post to **CLK 1** post and **1 KHz** sine wave to **IN1** post of the Sampling section.
- To observe the Flat-top sampling effect, short the jumper **JP1** to **FT** position and check the signal at the **OUT1** post of the sampling section.
- Repeat the above procedure from steps 5 to 7 using 2 KHz as an input signal.

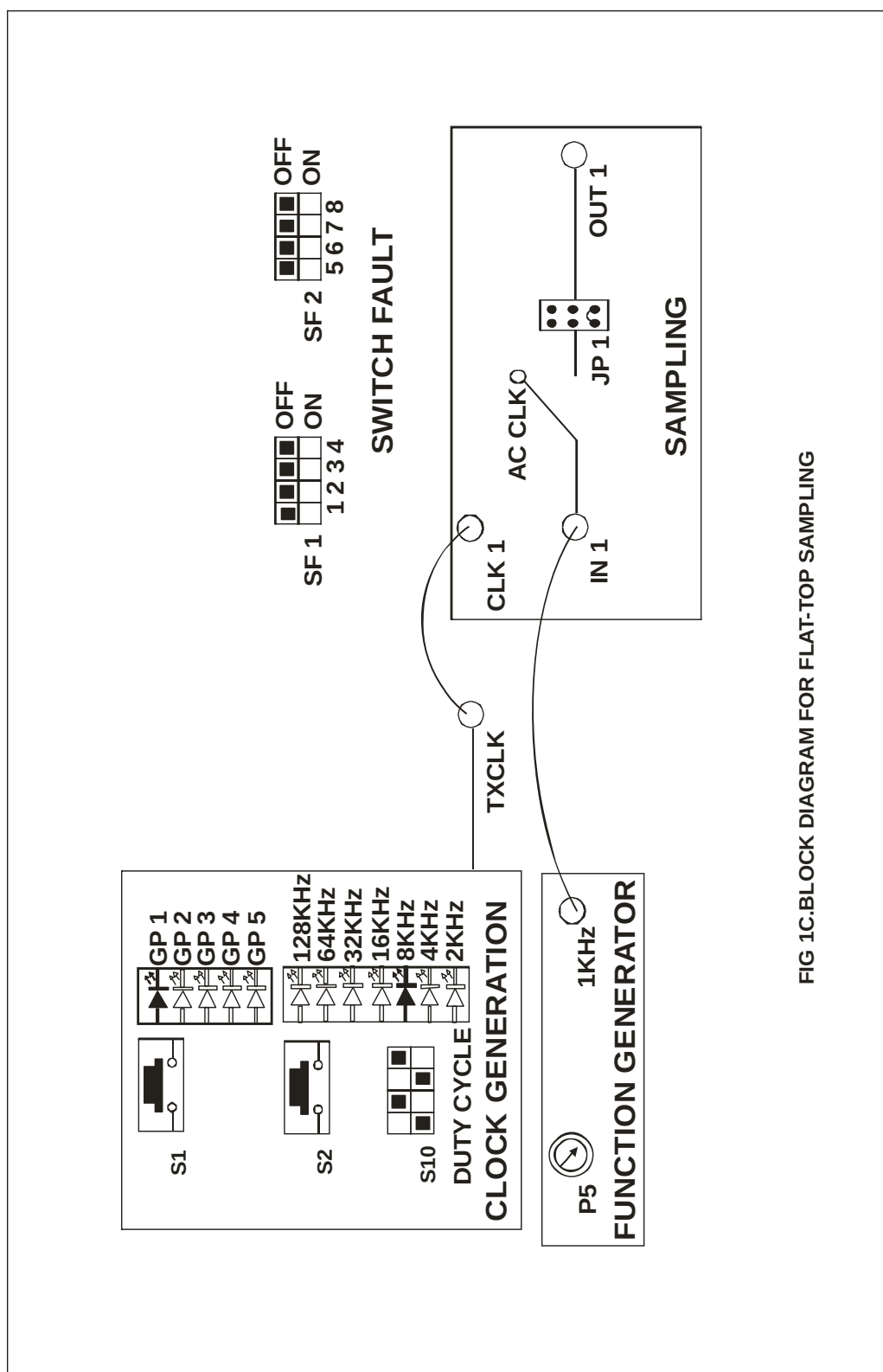


FIG 1C:BLOCK DIAGRAM FOR FLAT-TOP SAMPLING

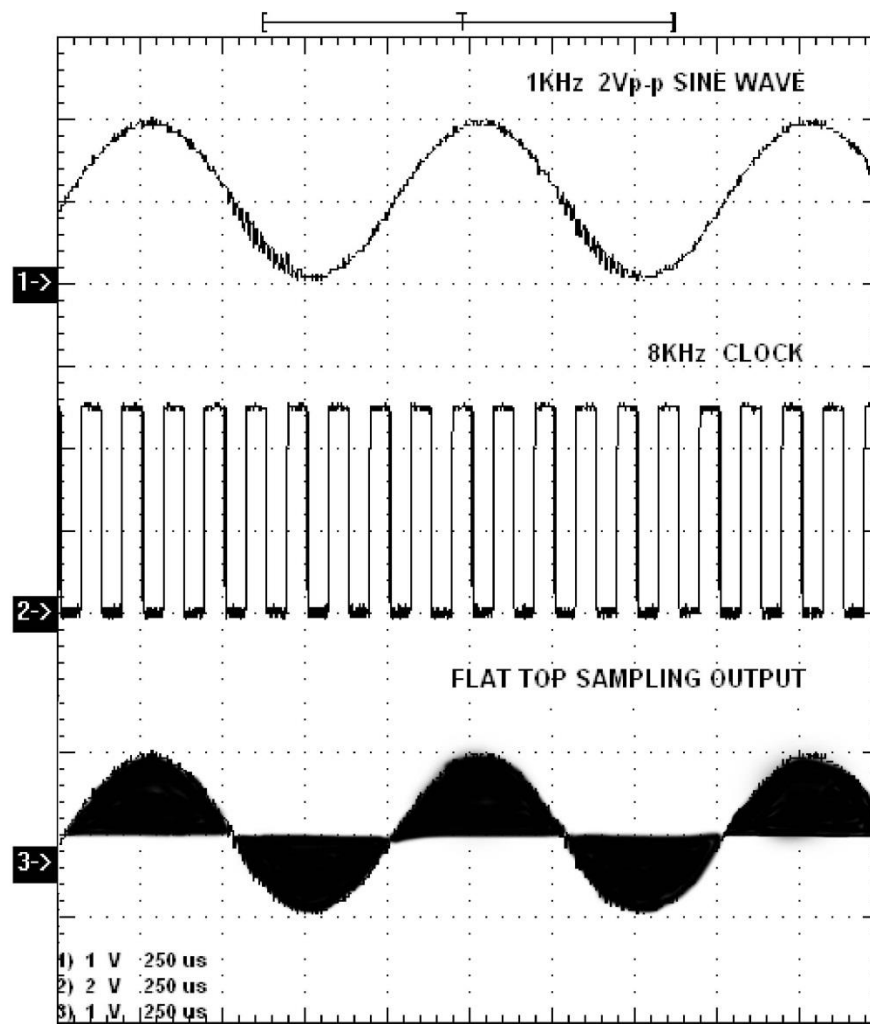
OBSERVATIONS

- Pressing Switch S1 to select Group1 (GP1) is indicated by corresponding LED.
- Pressing Switch S2 to select sampling frequency of 8 KHz is indicated by corresponding LED.
- Observe the Sampling clock at **TX CLK** post test point.
- Observe the signal under test, i.e. a sine wave, at posts 1 KHz and 2 KHz in the Function Generator section.
- Observe the adjustment in the Amplitude of 1 KHz and 2KHz sine wave to 2Vpp using pot **P5** and **P6** respectively.
- Observe the Flat-Top Sampling effect at the **OUT1** post of the sampling section.

CONCLUSION

In Flat top sampling, we arbitrarily sampled the signal at the beginning of the pulse. In sampling of this type the original analog signal cannot be recovered exactly by simply passing the samples through an ideal Low pass filter. However, the distortion need not be large. Flat Top sampling has the merit that it simplifies the design of the electronic circuitry used to perform the sampling operation.

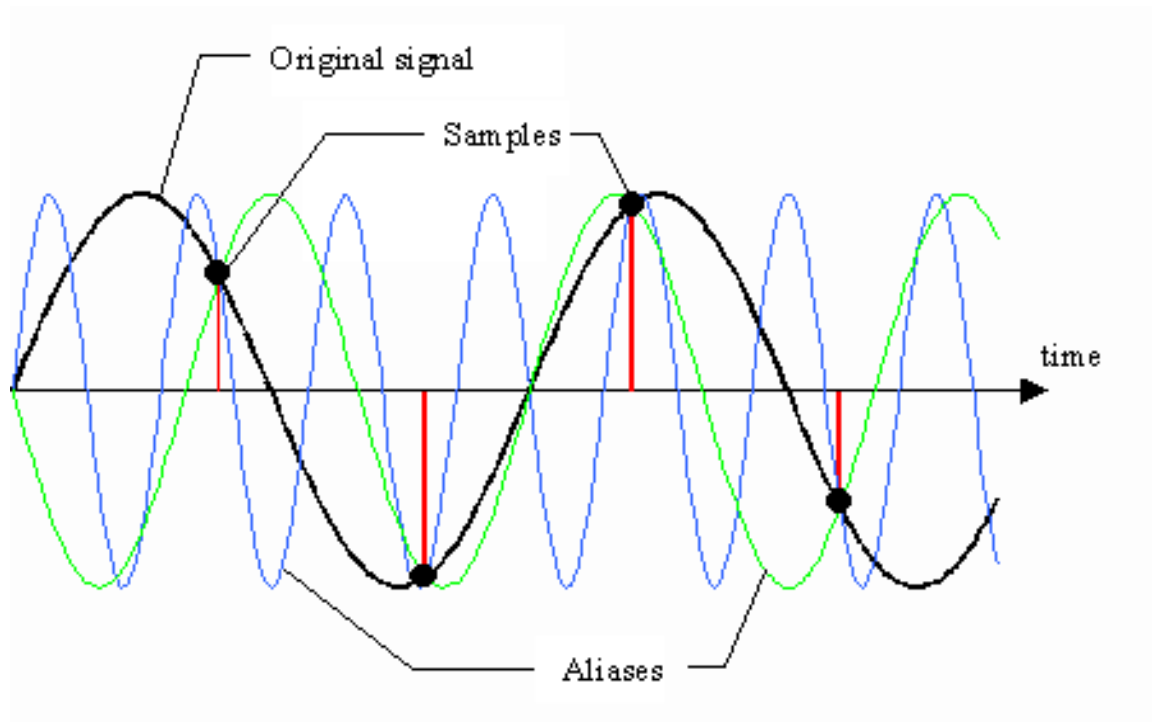
Comparing the reconstructed output of 2nd order Low Pass Butterworth Filter for all the three types of sampling, it is observed that the output of the sample and hold is the best as compared to the output of natural sampling and the output of the flat top sampling.



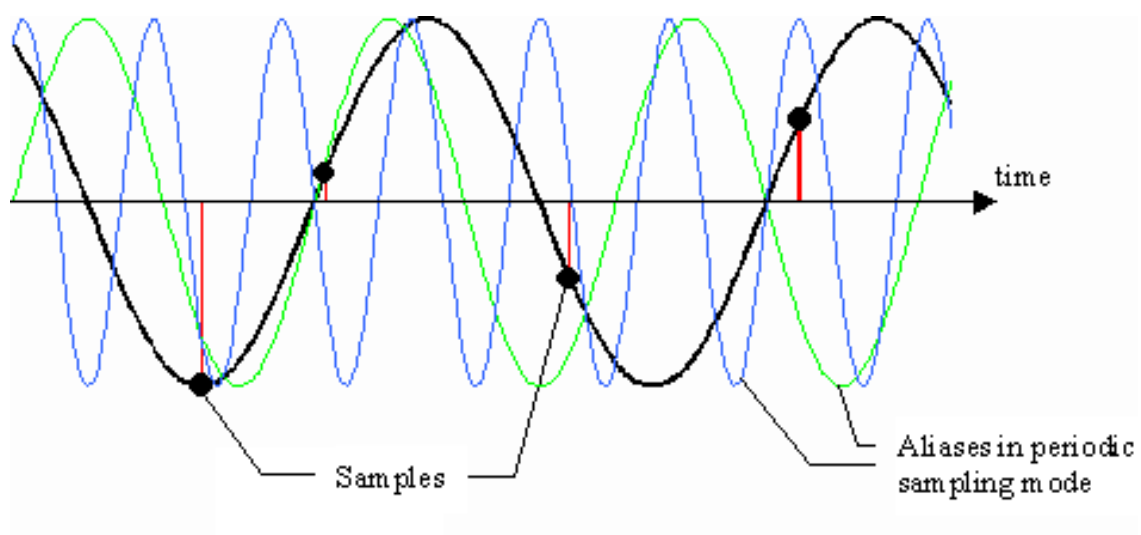
Flat Top Sampling

**EXPERIMENT
NO. 2**

Illustration of the aliasing effect taking place in the case of periodic sampling



Only one sine function goes exactly through the no uniformly spaced sample values:



EXPERIMENT NO.2

NAME

Effect of Various Sampling Frequencies & Duty Cycles

OBJECTIVES

To study the effect of different sampling frequencies on the sampling and reconstructed signal:

- A. Study the Effect of Different Sampling Frequencies
- B. Study the Effect of Duty Cycles

THEORY

When the continuous analog signal is sampled at a frequency F , the resulting discrete signal has more frequency components than the analog signal. To be precise, the frequency components of the analog signal are repeated at the sample rate. That is, in the discrete frequency response they are seen at their original position and are also seen centered around $\pm F$ and around $\pm 2F$, etc. How many samples are necessary to ensure that we are preserving the information contained in the signal? If the signal contains high frequency components, we will need to sample at a higher rate to avoid losing information that is in the signal. In general, to preserve the full information in the signal, it is necessary to sample at twice the maximum frequency of the signal. This is known as the Nyquist rate. The Sampling Theorem states that a signal can be exactly reproduced if it is sampled at a frequency F , where F is greater than twice the maximum frequency in the signal.

What happens if we sample the signal at a frequency that is lower than the Nyquist rate? When the signal is converted back into a continuous time signal, it will exhibit a phenomenon called aliasing. Aliasing is the presence of unwanted components in the reconstructed signal. These components were not present when the original signal was sampled. In addition, some of the frequencies in the original signal may be lost in the reconstructed signal. Aliasing occurs because signal frequencies can overlap if the sampling frequency is too low. Frequencies "fold" around half the sampling frequency - which is why this frequency is often referred to as the folding frequency.

Sometimes, the highest frequency components of a signal are simply noise, or do not contain useful information. To prevent aliasing of this frequency, we can filter out these components before sampling the signal. Because we are filtering out high frequency components and allowing lower frequency components through, this is known as low-pass filtering.

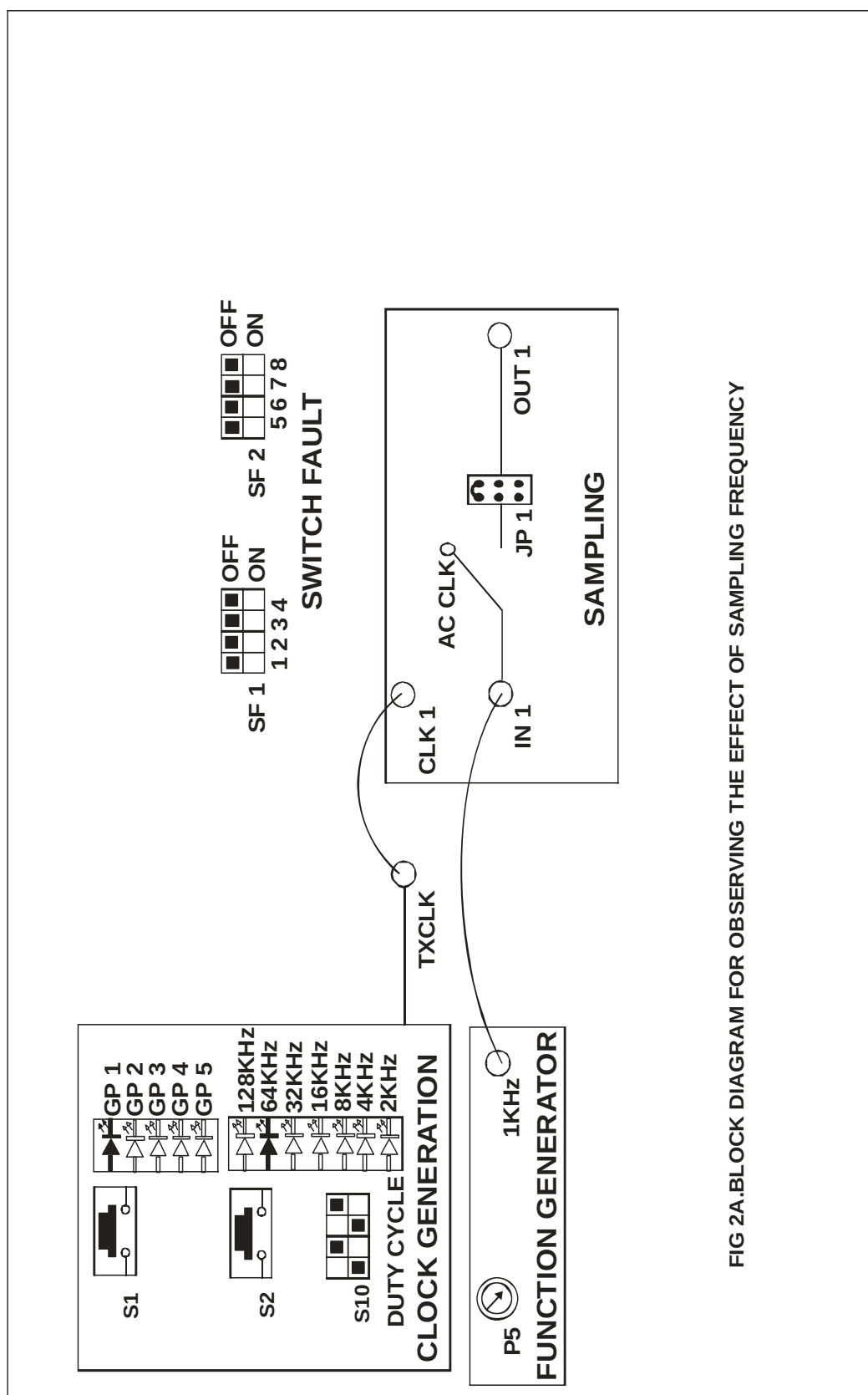


FIG 2A.BLOCK DIAGRAM FOR OBSERVING THE EFFECT OF SAMPLING FREQUENCY

EQUIPMENTS

DCS-B kit
Connecting Chords
Power supply
20 MHz Dual Trace Oscilloscope

NOTE: Keep All The Switch Faults In Off Position.

2A. EFFECT OF SAMPLING FREQUENCY

PROCEDURE

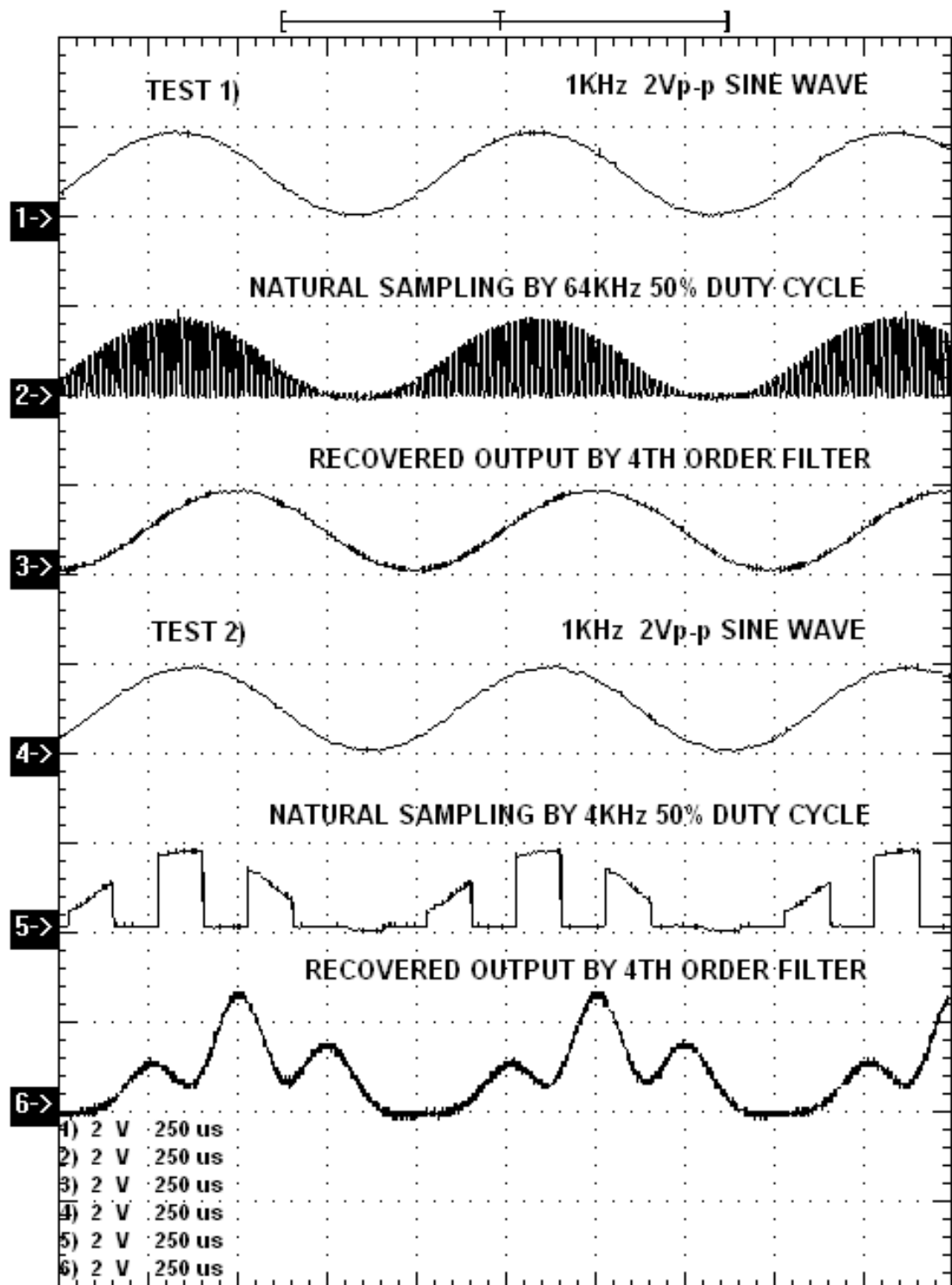
- Ensure that group 1 (**GP1**) clock is selected in the Clock generation section. Selection is done with the help of switch **S1** observe the corresponding LED indication.
- Select the sampling clock of frequency 8 KHz using the switch **S2** and observe the corresponding LED indication.
- Adjust the duty cycle of the sampling clock to 50% using DIP switch **S10** as shown in the block diagram.
- Adjust the amplitude of the 1 KHz and 2 KHz sine wave to 2Vpp using pot **P5** and **P6** respectively.
- Connect **TX CLK** post to **CLK 1** post and **1 KHz** sine wave to **IN1** post of the Sampling section.
- Short the jumper **JP1** to **first** position and check the signal at the **OUT1** post of the sampling section.
- Now select the sampling clock of frequency 64 KHz using the switch **S2**, observe the corresponding LED indication.
- Observe its effect at **OUT1** post of the sampling section.
- Similarly, observe the variation in the sampling output by varying the sampling frequencies from 64 KHz to 2 KHz.
- Repeat the above procedure for different sampling techniques.

OBSERVATION

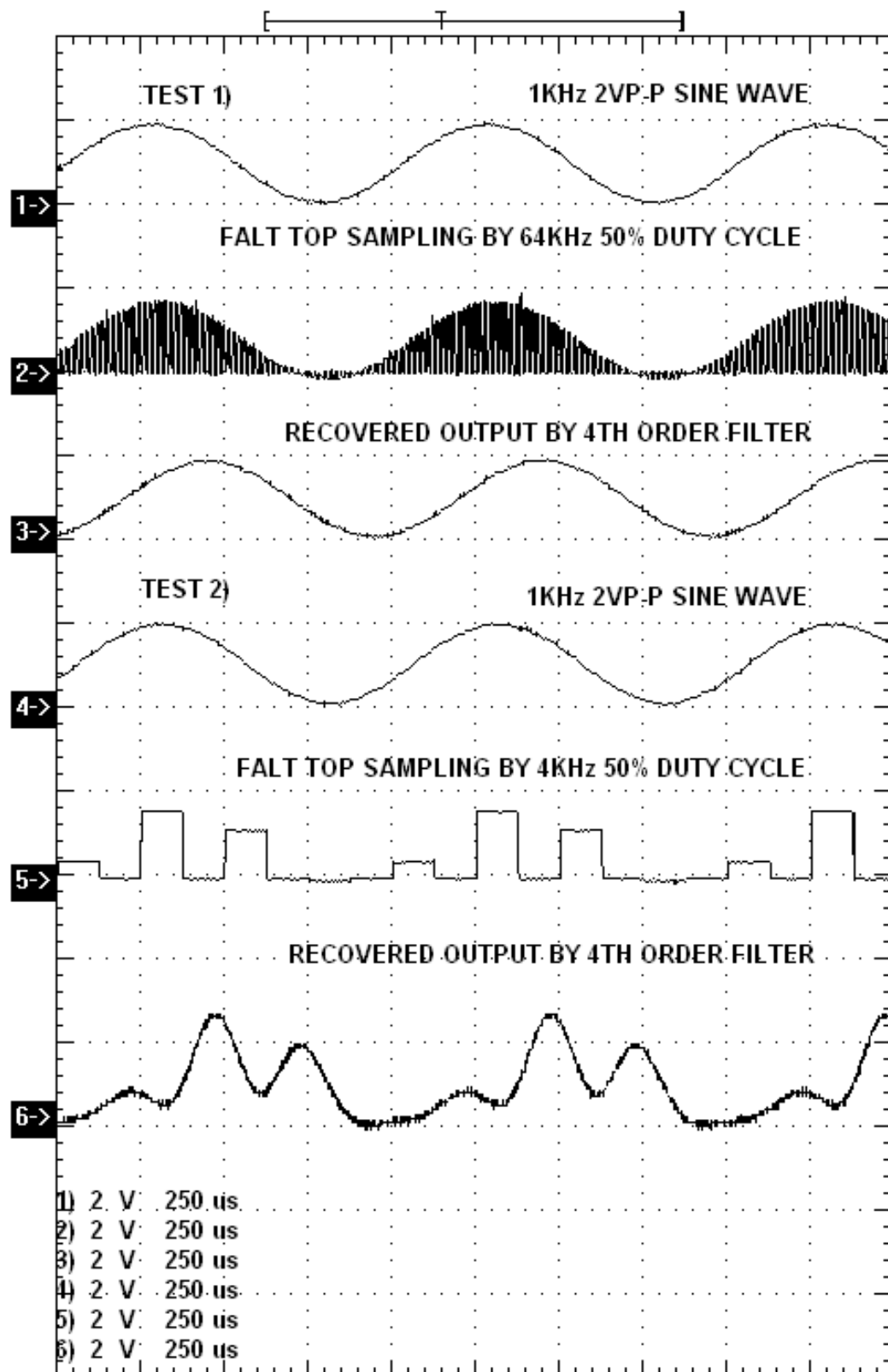
- Pressing Switch S1 to select Group1 (GP1) is indicated by corresponding LED.
- Pressing Switch S2 to select sampling frequency of 8 KHz is indicated by corresponding LED.
- Observe the Sampling clock at **TX CLK** post test point.
- Observe the signal under test, i.e. a sine wave, at posts 1 KHz and 2 KHz in the Function Generator section.
- Observe the adjustment in the Amplitude of 1 KHz and 2 KHz sine wave to 2Vpp using pot **P5** and **P6** respectively.
- Observe the Natural Sampling effect at the **OUT1** post of the sampling section.
- Pressing Switch S2 to select sampling frequency of 64 KHz is indicated by corresponding LED.
- Observe its effect at **OUT1** post of the sampling section.
- Similarly, observe the variation in the sampling output by varying the sampling frequencies from 64 KHz to 2 KHz.

CONCLUSION

From the above observations we conclude that when the sampling frequency is increased the reconstructed output is less distorted and almost original signal is reconstructed. For a sampling frequency of 2 KHz, only 2 samples of the 1 KHz signal are taken, whereas that for a sampling frequency of 8 KHz, 8 samples of 1 KHz signal is taken. Hence, as the number of samples taken of the signal increases, the distortion of the Reconstructed signal decreases. As per the Nyquist Criterion, at least two samples are required for the reconstruction of the signal. If the Nyquist Criterion is not satisfied, or if the signal is not band limited, then spectral overlap, called "aliasing" occurs, causing higher frequencies to show up at lower frequencies in the recovered message.



EFFECT OF SAMPLING FREQUENCY ON NATURAL SAMPLING



EFFECT OF SAMPLING FREQUENCY ON FLAT TOP SAMPLING

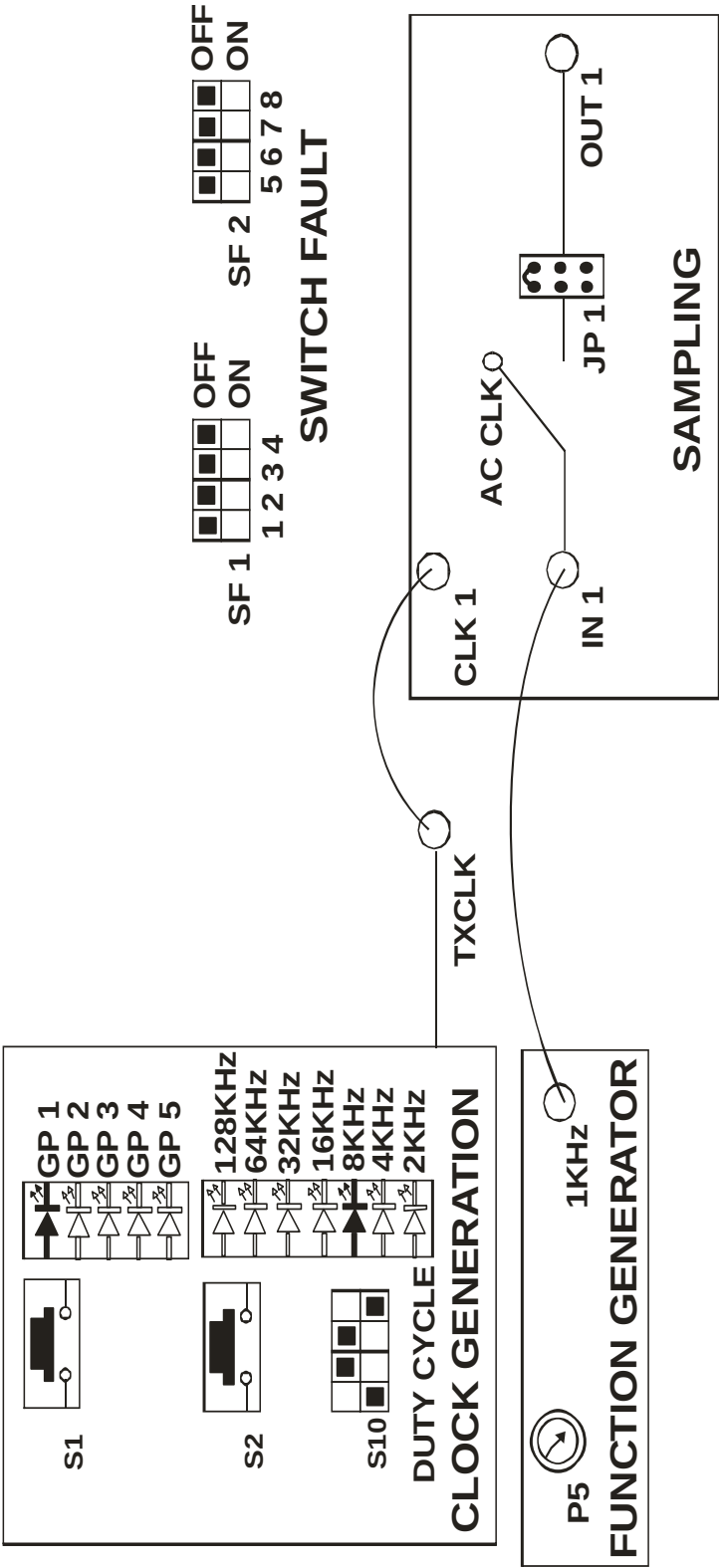


FIG 2B.BLOCK DIAGRAM FOR OBSERVING THE EFFECT OF DUTY CYCLE

2B. EFFECT OF DUTY CYCLE

PROCEDURE

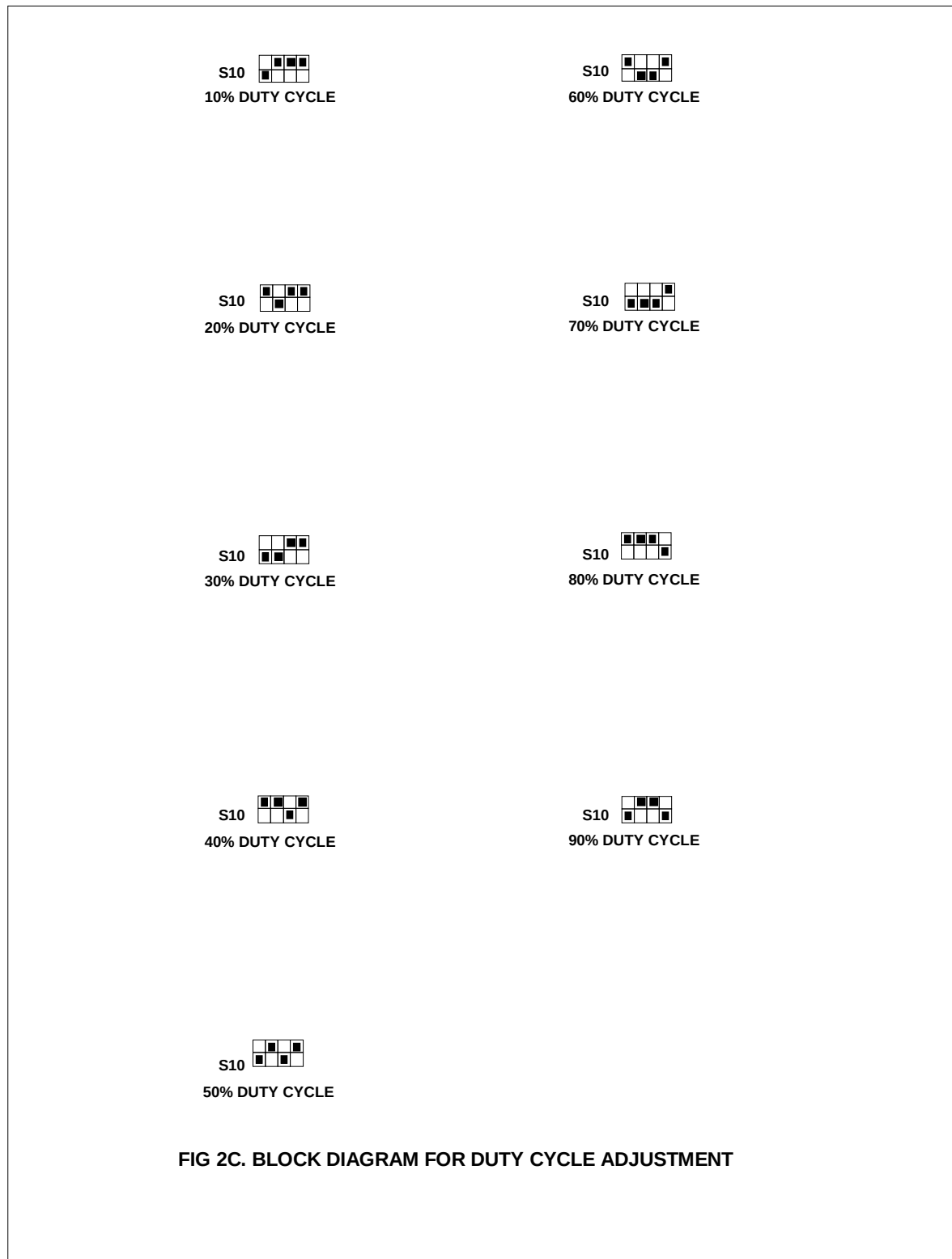
- Ensure that group 1 (**GP1**) clock is selected in the Clock generation section. Selection is done with the help of switch **S1**, observe the corresponding LED indication.
- Select the sampling clock of frequency 8 KHz using the switch **S2** and observe the corresponding LED indication.
- Adjust the duty cycle of the sampling clock to 50% using DIPswitch **S10** as shown in the block diagram.
- Adjust the amplitude of the 1 KHz and 2 KHz sine wave to 2Vpp using pot **P5** and **P6** respectively.
- Connect **TX CLK** post to **CLK 1** post and **1 KHz** sine wave to **IN1** post of the Sampling section.
- Short the jumper **JP1** to **first** position and check the signal at the **OUT1** post of the sampling section.
- Now, adjust the duty cycle of the sampling clock to 10% using DIPswitch **S3** as shown in the block diagram.
- Observe its effect at **OUT1** post of the sampling section.
- Similarly, observe the variation in the sampling output by varying the duty cycle from 10% to 90% duty cycle.
- Repeat the above procedure for different sampling techniques.

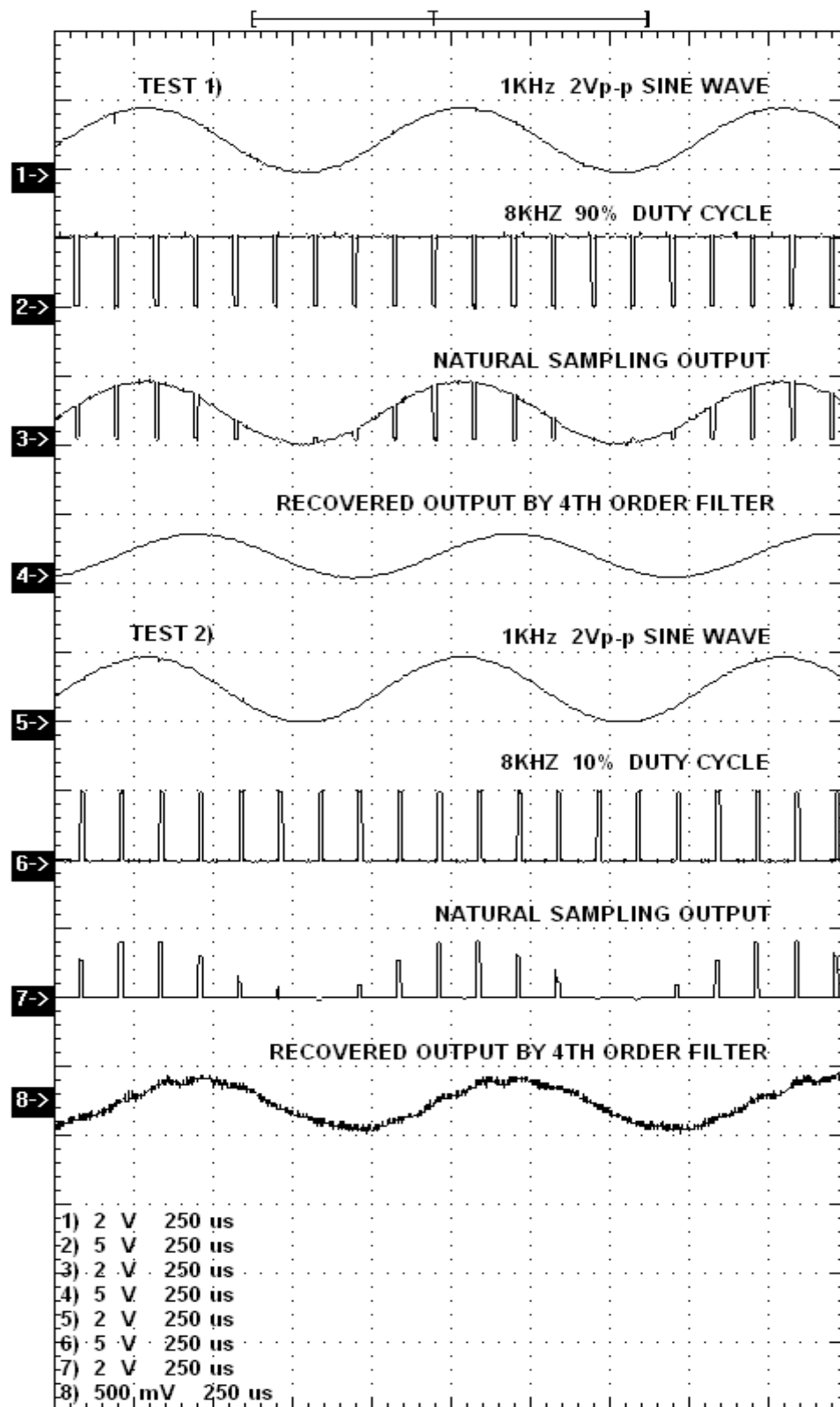
OBSERVATIONS

- Pressing switch S1, the selection of Group 1 (GP1) in the Clock Generation section, is indicated by corresponding LED.
- Pressing switch S2, the selection of clock frequency 8 KHz, is indicated by corresponding LED.
- Adjustment in the amplitude of the 1 KHz and 2 KHz sine wave to 2Vpp can be observed on the corresponding post.
- Check the Sampling effect at the OUT1 post of the sampling section.
- Similarly, observe the variation in the sampling output by varying the duty cycle from 10% to 90% duty cycle.

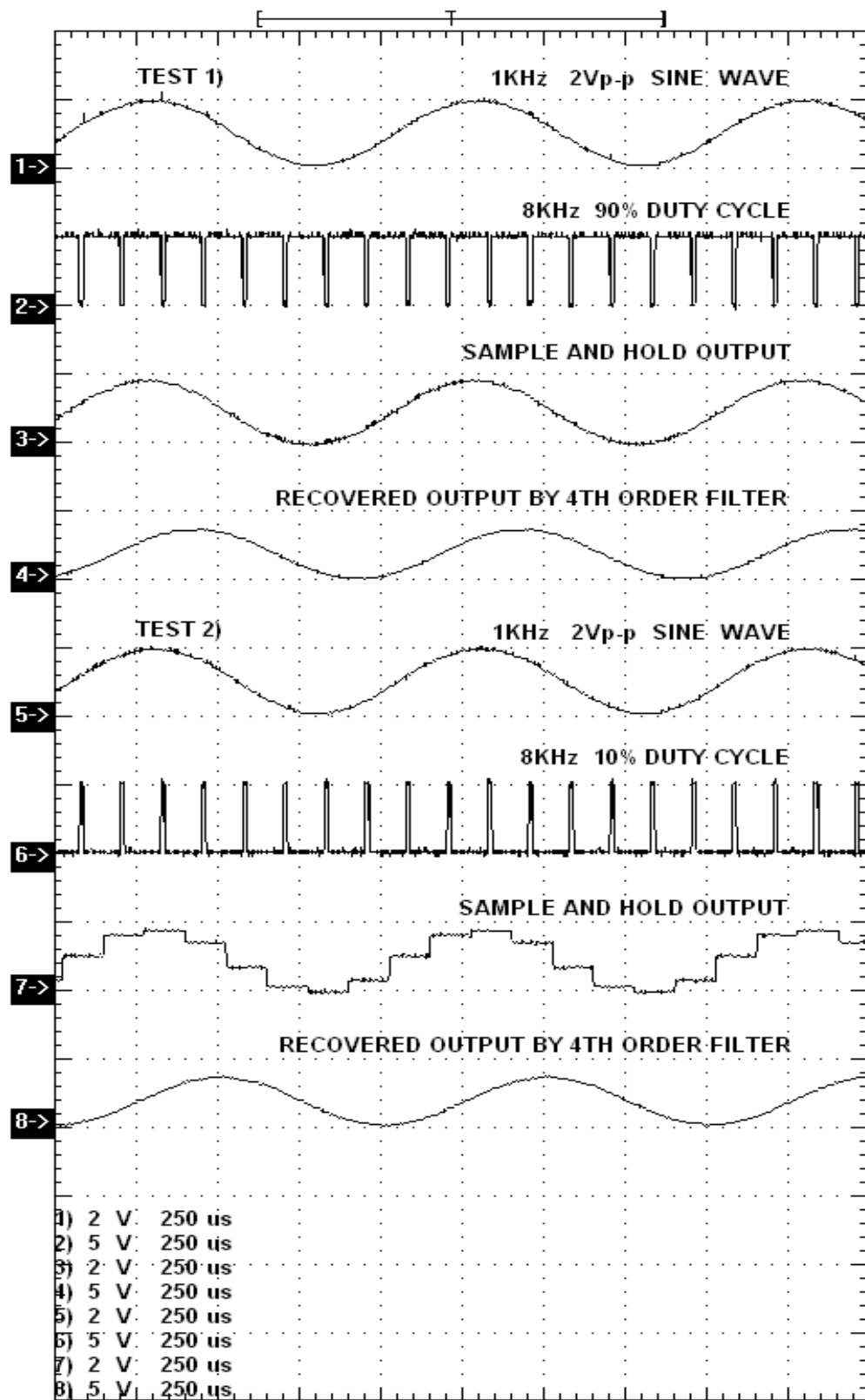
CONCLUSION

From the above observations, we conclude that as the duty cycle increases, the sampling time, i.e. the time period over which the signal information is obtained, is more. Hence, the reconstructed Signal Amplitude approaches that of the original Signal.

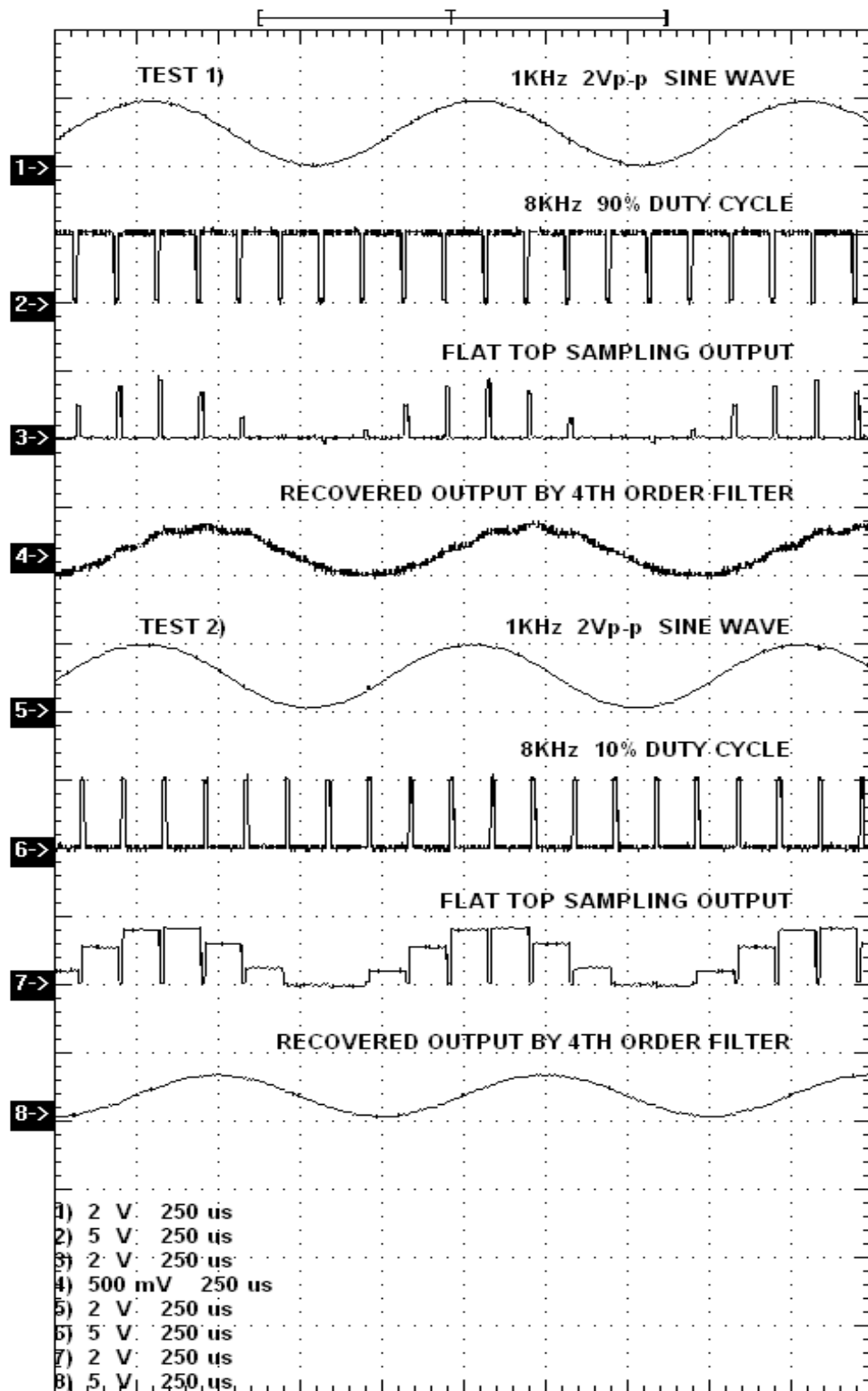




Effect Of Duty Cycle Of Sampling Clock On Natural Sampling

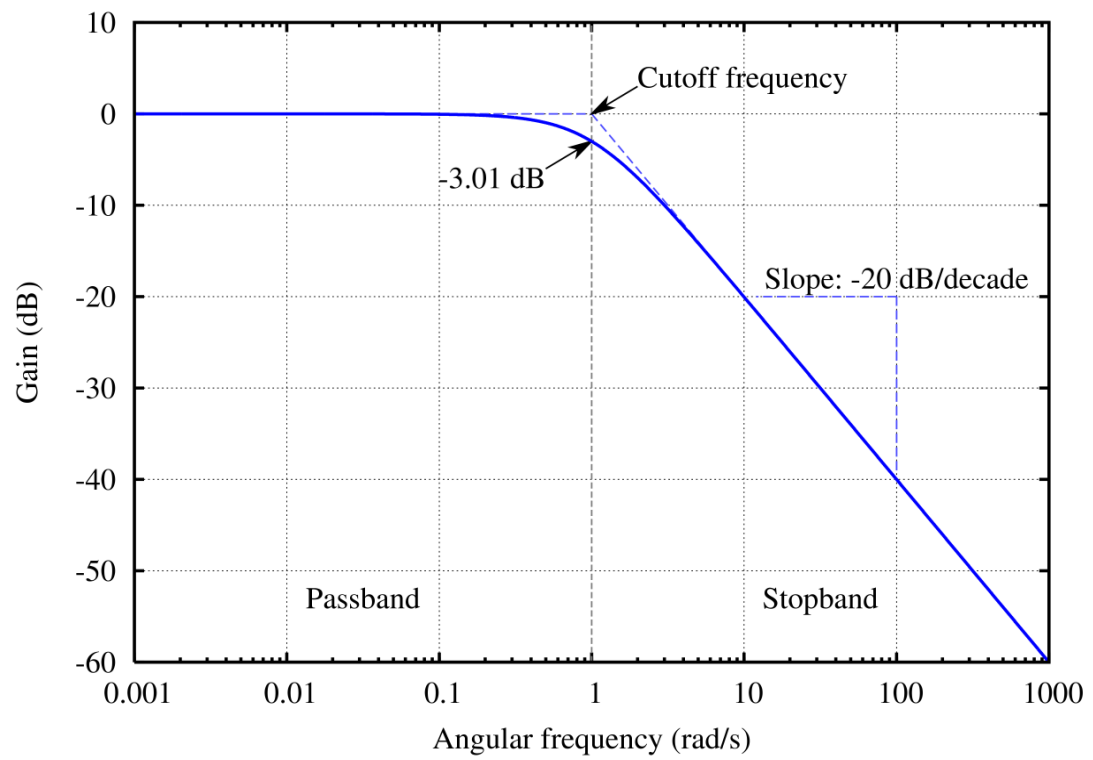


Effect of Duty Cycle Of Sampling Clock On Sample And Hold



Effect of Duty Cycle Of Sampling Clock On Flat Top Sampling

**EXPERIMENT
NO. 3**



EXPERIMENT NO.3

NAME

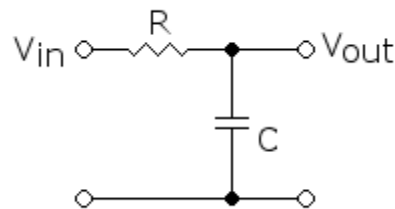
Effect of Order of the Low Pass Filter

OBJECTIVES

To study the effects of the order of the low pass filter:

THEORY

A **low-pass filter** is a filter that passes low frequencies well, but attenuates (or reduces) frequencies higher than the cutoff frequency. The actual amount of attenuation for each frequency varies from filter to filter



There are a great many different types of filter circuits, with different responses to changing frequency. The frequency response of a filter is generally represented using a Bode plot. A **first-order filter**, for example, will reduce the signal strength by half (about -6 dB) every time the frequency doubles (goes up by one octave). The magnitude Bode plot for a first-order filter looks like a horizontal line below the cutoff frequency and a diagonal line above the cutoff frequency. There is also a "knee curve" at the boundary between the two, which shows smooth transitions between the two straight line regions.

A **second-order filter** does a better job of attenuating higher frequencies. The Bode plot for this type of filter resembles that of a first-order filter, except that it falls off more quickly. For example, a second-order Butterworth filter (which is a critically damped RLC circuit, with no peaking) will reduce the signal strength to one fourth its original level every time the frequency doubles (-12 dB per octave). Other second order filters may roll off at different rates initially depending on their Q factor, but approach the same final rate of -12 dB per octave.

Third and higher order filters are defined similarly. In general, the final rate of roll off for an n -order filter is $6n$ dB per octave.

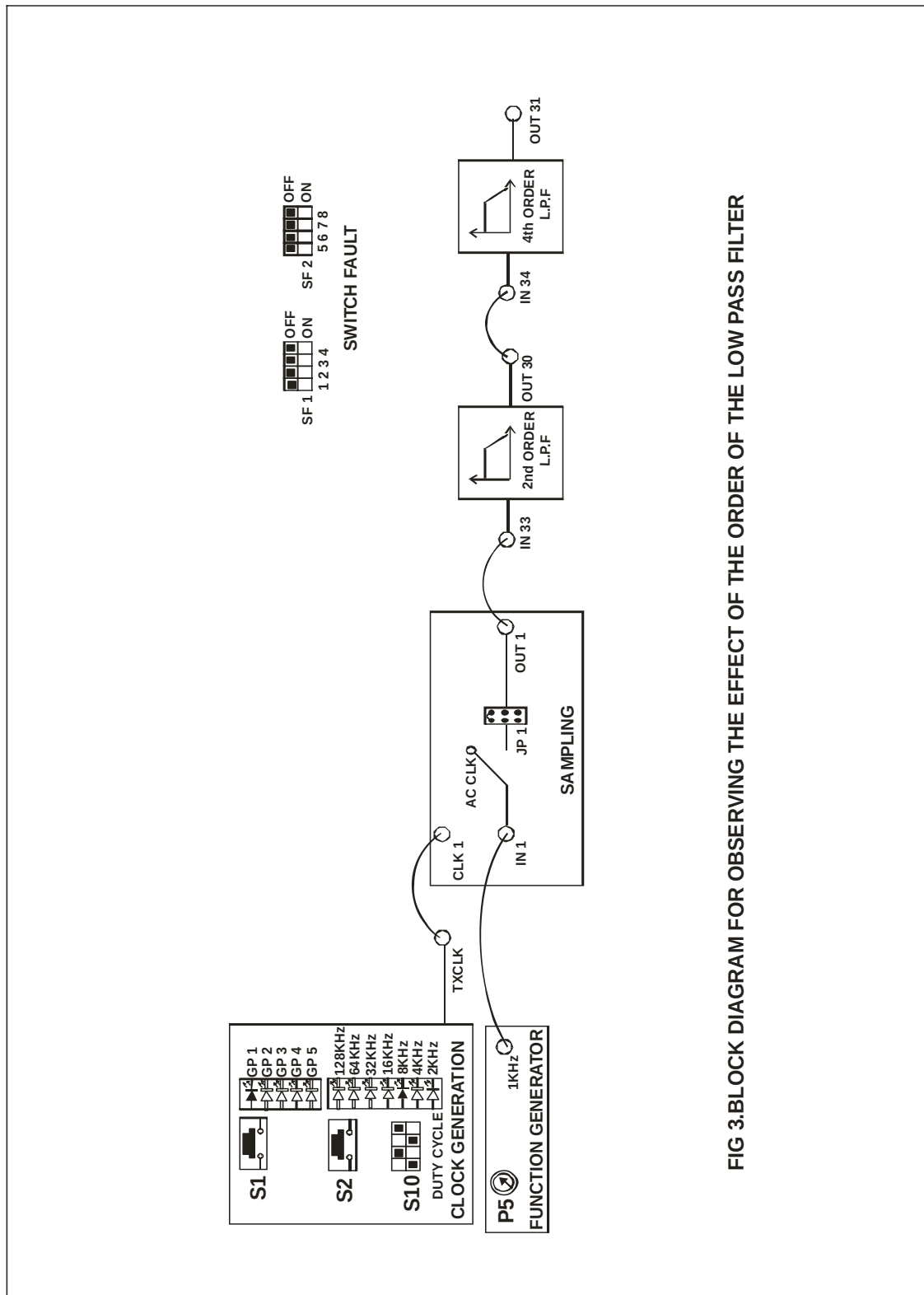


FIG 3.BLOCK DIAGRAM FOR OBSERVING THE EFFECT OF THE ORDER OF THE LOW PASS FILTER

EQUIPMENTS

DCS-B kit
Connecting Chords
Power supply
20 MHz Dual Trace Oscilloscope

NOTE: Keep All The Switch Faults In Off Position.

This experiment is performed to recover the original signal from the modulated one. Here we study the effect of various filters i.e. 2nd order LPF and 4th order LPF.

PROCEDURE

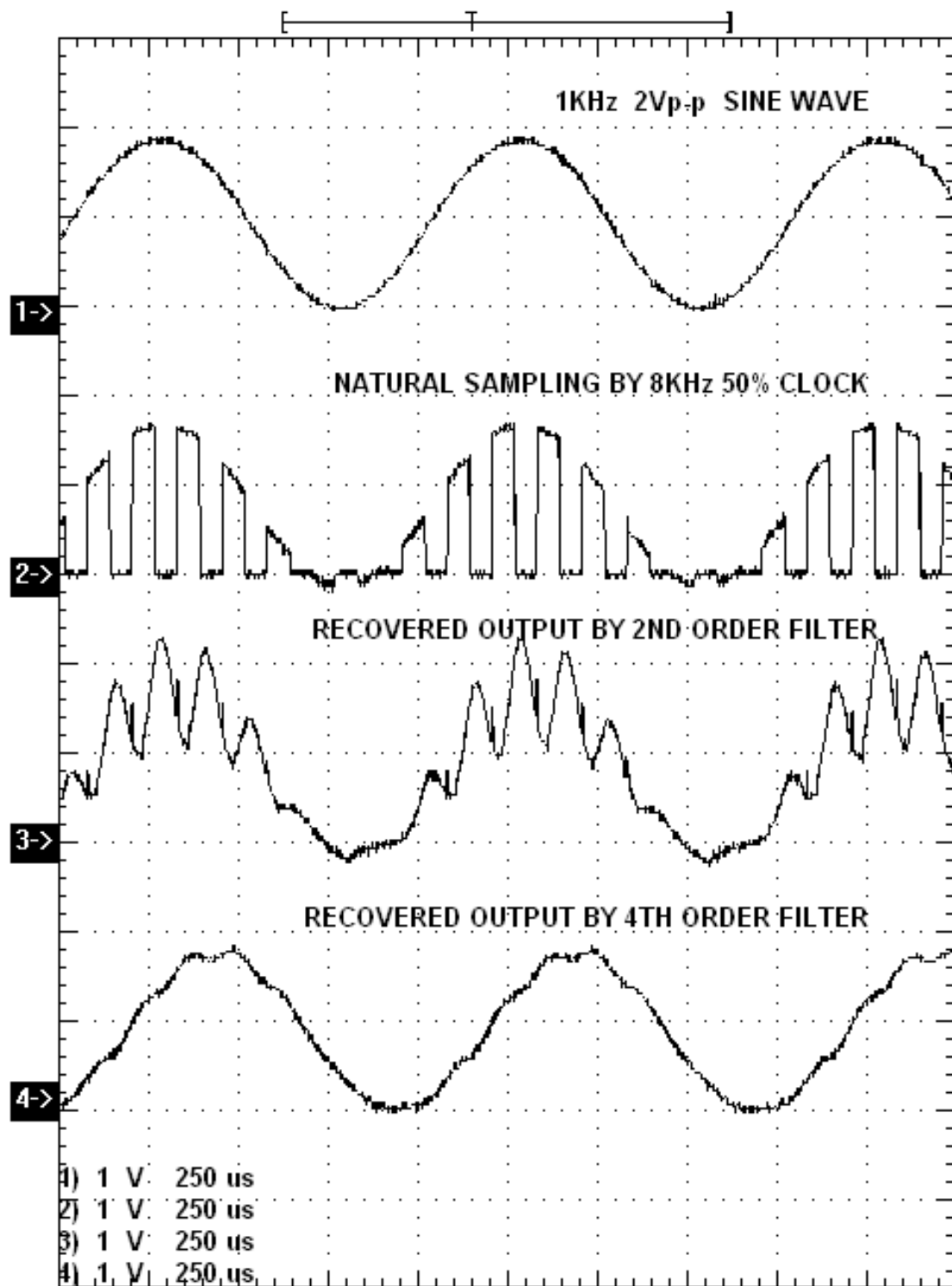
- Ensure that group 1 (**GP1**) clock is selected in the Clock generation section. Selection is done with the help of switch **S1**. Observe the corresponding LED indication.
- Select the sampling clock of frequency 8 KHz using the switch **S2** and observe the corresponding LED indication.
- Adjust the duty cycle of the sampling clock to 50% using DIPswitch **S10** as shown in the block diagram.
- Adjust the amplitude of the 1 KHz and 2 KHz sine wave to 2Vpp using pot **P5** and **P6** respectively.
- Connect **TX CLK** post to **CLK 1** post and **1 KHz** sine wave to **IN1** post of the Sampling section.
- Short the jumper **JP1** to **first** position and check the signal at the **OUT1** post of the sampling section.
- Connect the **OUT1** post of the sampling section to the **IN33** post of the 2nd order LPF and observe the filtered signal at the **OUT30** post.
- Connect the **OUT30** post of the 2nd order LPF to the **IN34** post of the 4th order LPF and observe the filtered signal at the **OUT31** post.
- Compare the two reconstructed signals at **OUT30** post and **OUT31** post which are the outputs of 2nd order LPF and 4th order LPF respectively.
- Repeat the above procedure for different sampling techniques.
- Observe the effect of different sampling frequencies on filter outputs.

OBSERVATIONS

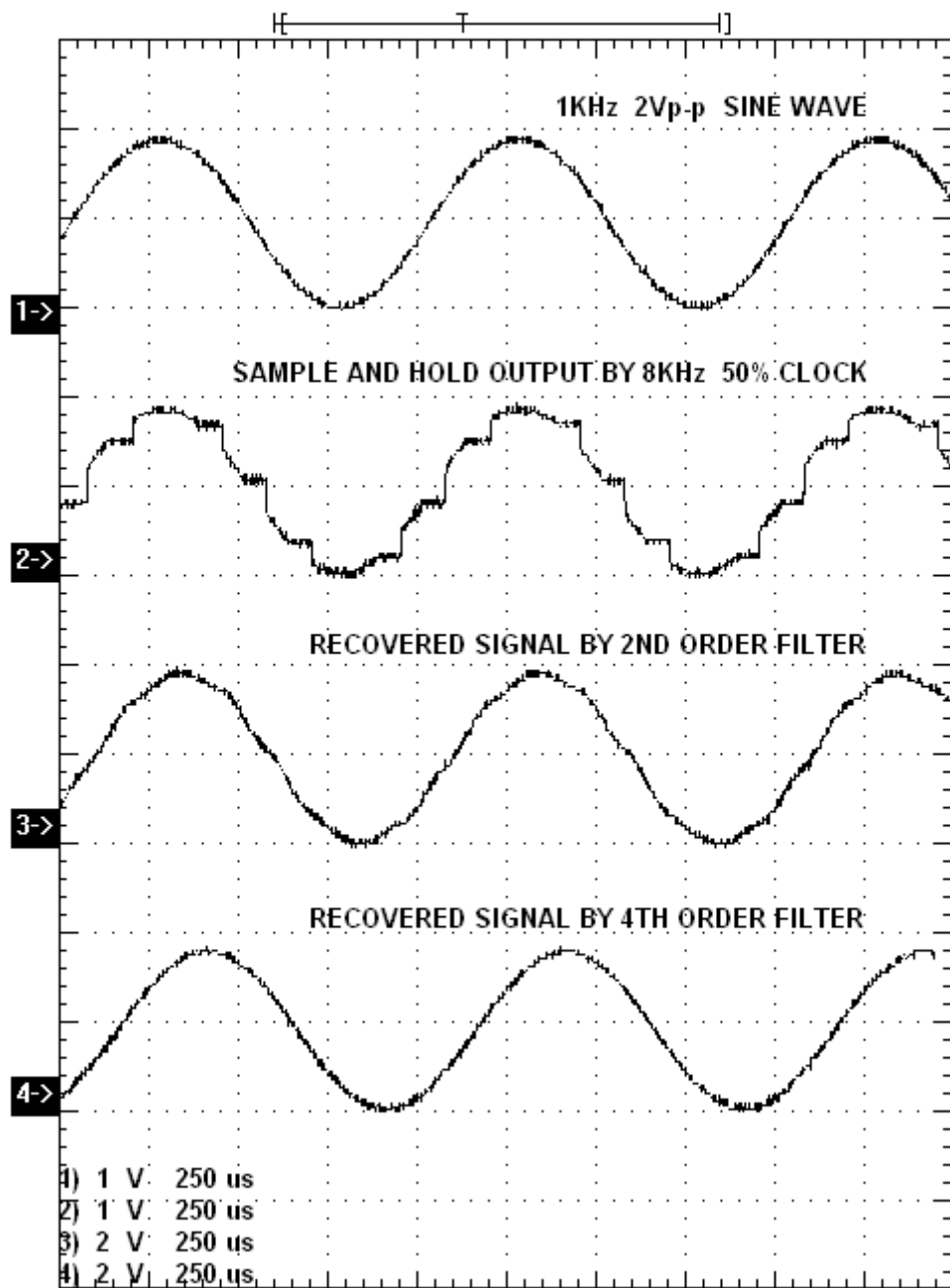
- When switch S1 is pressed, Group 1 (GP1) in the Clock Generation section gets selected, which is indicated by the corresponding LED.
- When switch S2 is pressed, clock frequency 8 KHz gets selected, this is indicated by the corresponding LED.
- Adjustment in the amplitude of the 1 KHz and 2 KHz sine wave to 2Vpp can be observed on the corresponding post.
- Check the Sampling effect at the **OUT1** post of the sampling section.
- Observe the effect of second order-filtered signal at the **OUT30** post.
- Observe the effect of fourth order-filtered signal at the **OUT31** post.
- Observe the effect of different sampling frequencies on filter outputs.

CONCLUSION

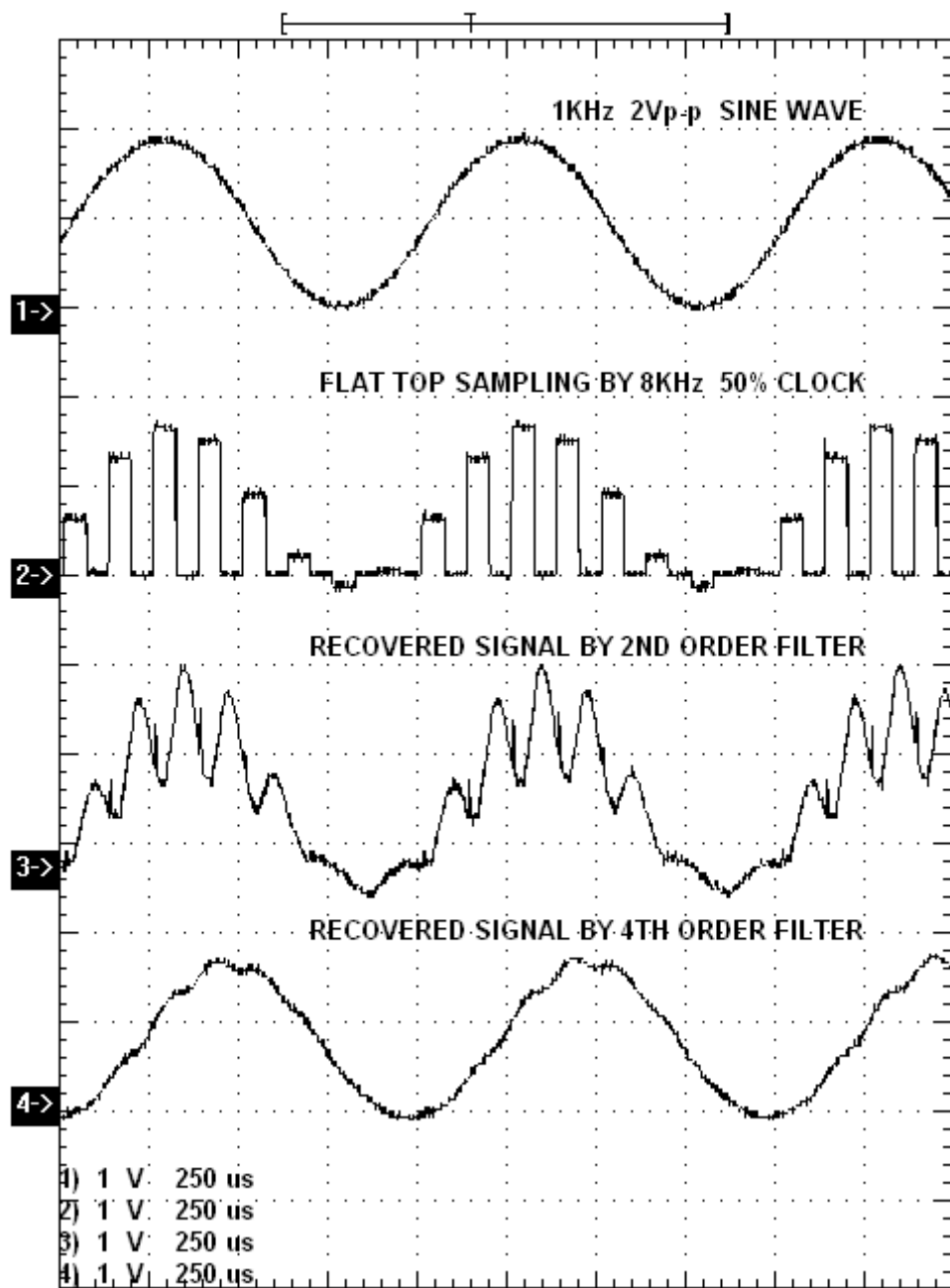
From the above observations, we conclude that as the order of the filter increases, the reconstructed signal characteristics also improve.



Effect of Order of Filter On Natural Sampling



Effect of Order of Filter On Sample And Hold



Effect of Order Of Filter On Flat Top Sampling

**EXPERIMENT
NO. 4**

EXPERIMENT NO.4

NAME

Study of TDM with Different Receiver Synchronization Techniques

OBJECTIVES

To study the Time Division Multiplexing and Demultiplexing of analog signals using various receiver synchronization techniques:

- A. Direct synchronization technique
- B. Clock recovery through Phase Lock Loop
- C. Clock recovery through Threshold Detector

THEORY

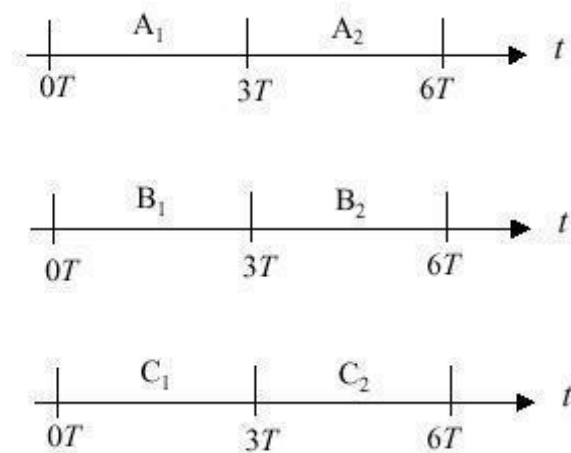
Time-division multiplexing (TDM) is a method of putting multiple data streams in a single signal by separating the signal into many segments, each having a very short duration. Each individual data stream is reassembled at the receiving end based on the timing.

To maintain proper positions of Sample Pulses in the Multiplexer, it is necessary to synchronize the Sampling Process. Because the sampling operations are usually electronic, there is typically a Clock Pulse Train. That serves as a reference for all samples. At the Receiving Station, a similar Clock Synchronization can be derived from the received waveforms by observing the Pulse Sequence over many pulses and averaging the pulses (in a closed loop with the Clock derived on the Voltage Controlled Oscillator).

Clock Synchronization does not guarantee that the proper sequence of samples is synchronized. Proper alignment of the Time Slot Sequence requires Frame Synchronization. Hence one or more Time Slots per Frame may be used to send Synchronization Information. For example, by placing a Special Pulse with larger amplitude than the largest expected Message Amplitude in TIME SLOT-1, the start of a Frame can easily be identified using a suitable Threshold Circuit.

In time-division multiplexing (TDM), the transmission between the multiplexers is provided by a single high-speed digital transmission line. Each connection produces a digital information flow, which is then inserted into the high-speed line. For example in Figure 1(a) each connection generates a signal that produces one unit of information every $3T$ seconds. This unit of information could be a bit, a byte, or a fixed-size block of bits. Typically, the transmission line is organized into frames that in turn are divided into equal-sized slots. For example, in Figure 1(b) the transmission line can send one unit of information every T seconds, and the combined signal has a frame structure that consists of three slots, one for each user. During connection setup, each connection is assigned a slot that can accommodate the information produced by the connection.

(a) Each signal transmits 1 unit every $3T$ seconds



(b) Combined signal transmits 1 unit every T seconds

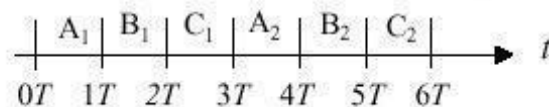


Figure 1 Time Division Multiplexing

TDM was introduced in the telephone network in the early 1960s. The T1 carrier system that carries 24 digital telephone connections is shown in Figure 2

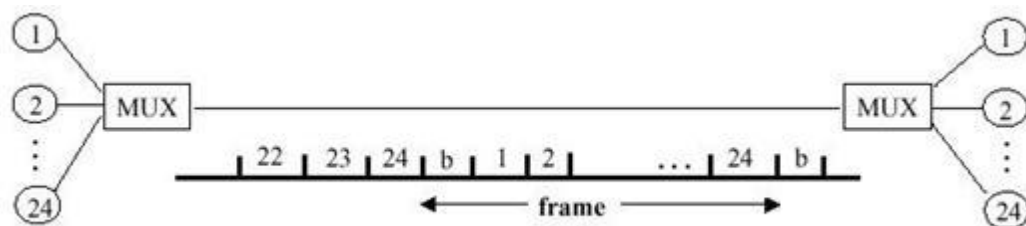


Figure 2 Time Division Multiplexing – T1 Carrier System

EQUIPMENTS

DCS-B kit
Connecting Chords
Power supply
20MHz Dual Trace Oscilloscope

NOTE: Keep All The Switch Faults In Off Position.

PROCEDURE

- Ensure that group 2 (**GP2**) clock is selected in the clock generation section. Selection is done with the help of switch **S1**.
- The selected clock of frequency 32 KHz, 80% duty cycle will be available at the **TXCLK** post.
- The timing signals required for multiplexing four analog channels are derived internally from TX CLK and are available at the test points **TP1, TP2, TP3** and **TP4**.
- Feed four sine waves of frequency **250Hz, 500Hz, 1 KHz** and **2 KHz** having amplitude around 1Vpp from the Function Generator to the four channel inputs **IN7, IN8, IN9** and **IN10** of the analog multiplexer.
- Observe the TDM output at the **OUT5** post of the 4-channel Mux section.
- To recover the individual analog signals from the multiplexed data at the **OUT5** post, connect it to **IN23** post of the 4-channel Demux section.

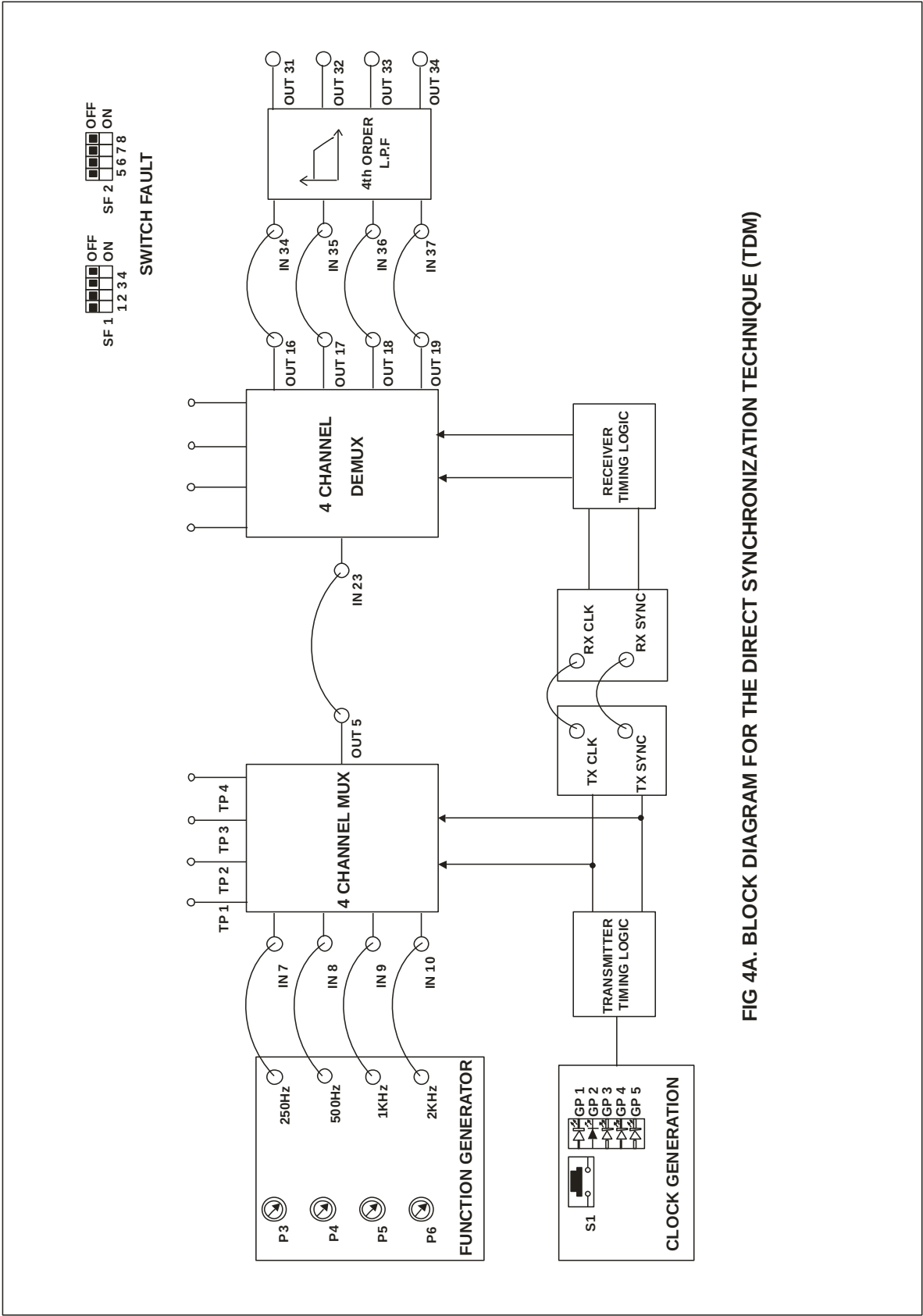


FIG 4A. BLOCK DIAGRAM FOR THE DIRECT SYNCHRONIZATION TECHNIQUE (TDM)

4A. Using the Direct Synchronization Technique

PROCEDURE

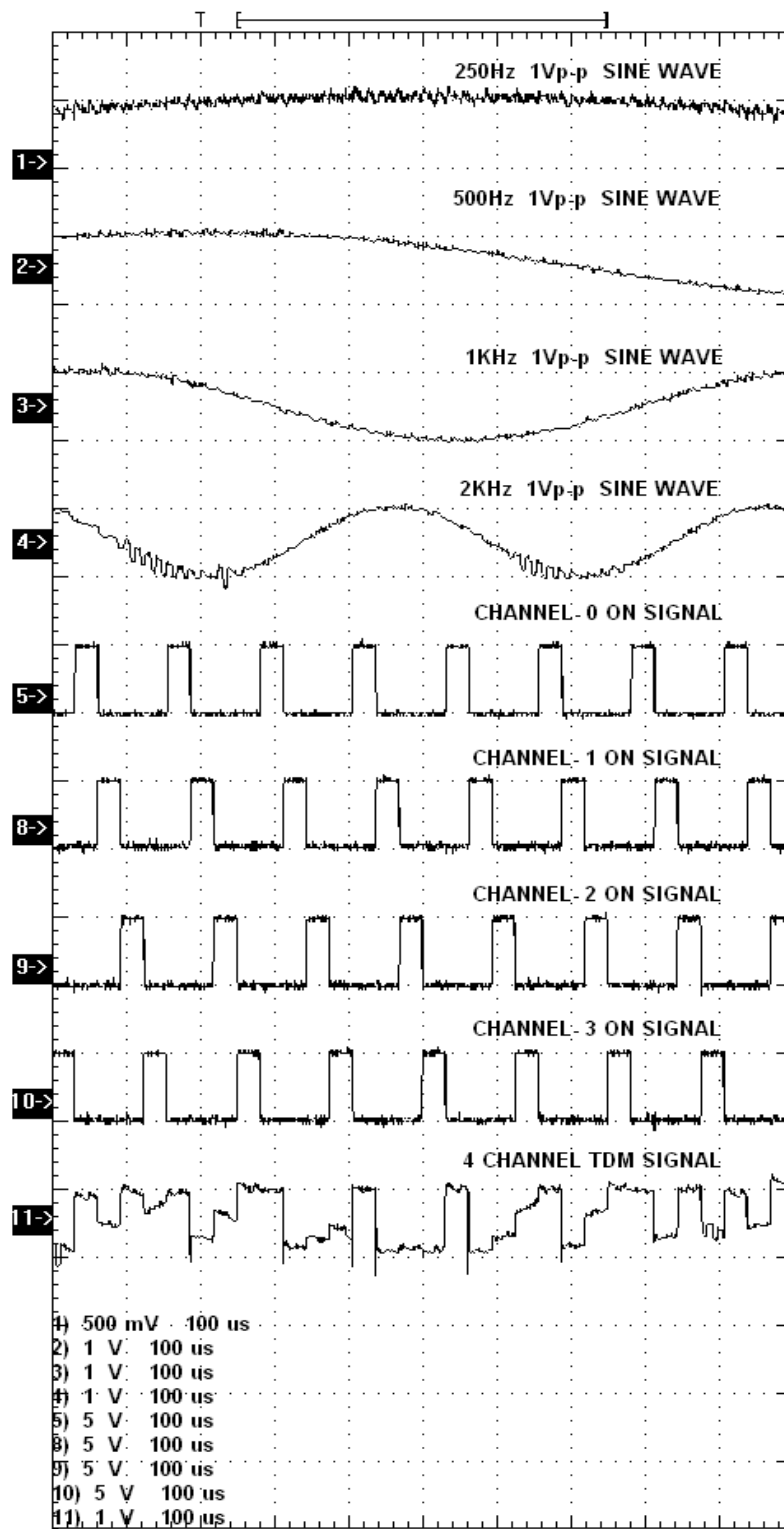
- To provide the timing and synchronization information to the receiver section, connect the **TX CLK** post to the **RX CLK** post and **TX SYNC** post to the **RX SYNC** post. Here, we are connecting the transmitter clock to the receiver clock and transmitter sync to the receiver sync physically.
- The timing signals required for de-multiplexing the four analog channels are derived internally from the **RX CLK** and are available at the test points.
- The four de-multiplexed signals are available at the post's **OUT16, OUT17, OUT18** and **OUT19** of the 4-Channel Demux section.
- The multiplexed sine waves are not pure sine waves. Therefore it needs to be filtered. Connect **OUT16** post to **IN34** post, **OUT17** post to **IN35** post, **OUT18** post to **IN36** post and **OUT19** to **IN37** post of the 4th order LPF.
- Observe the filtered output at posts **OUT31, OUT32, OUT33** and **OUT34** respectively.
- Observe the sequence of the recovered signals, which is in exact compliance with that of the transmitter inputs sequence. Similarly, observe the frequencies of the recovered signals.

OBSERVATIONS

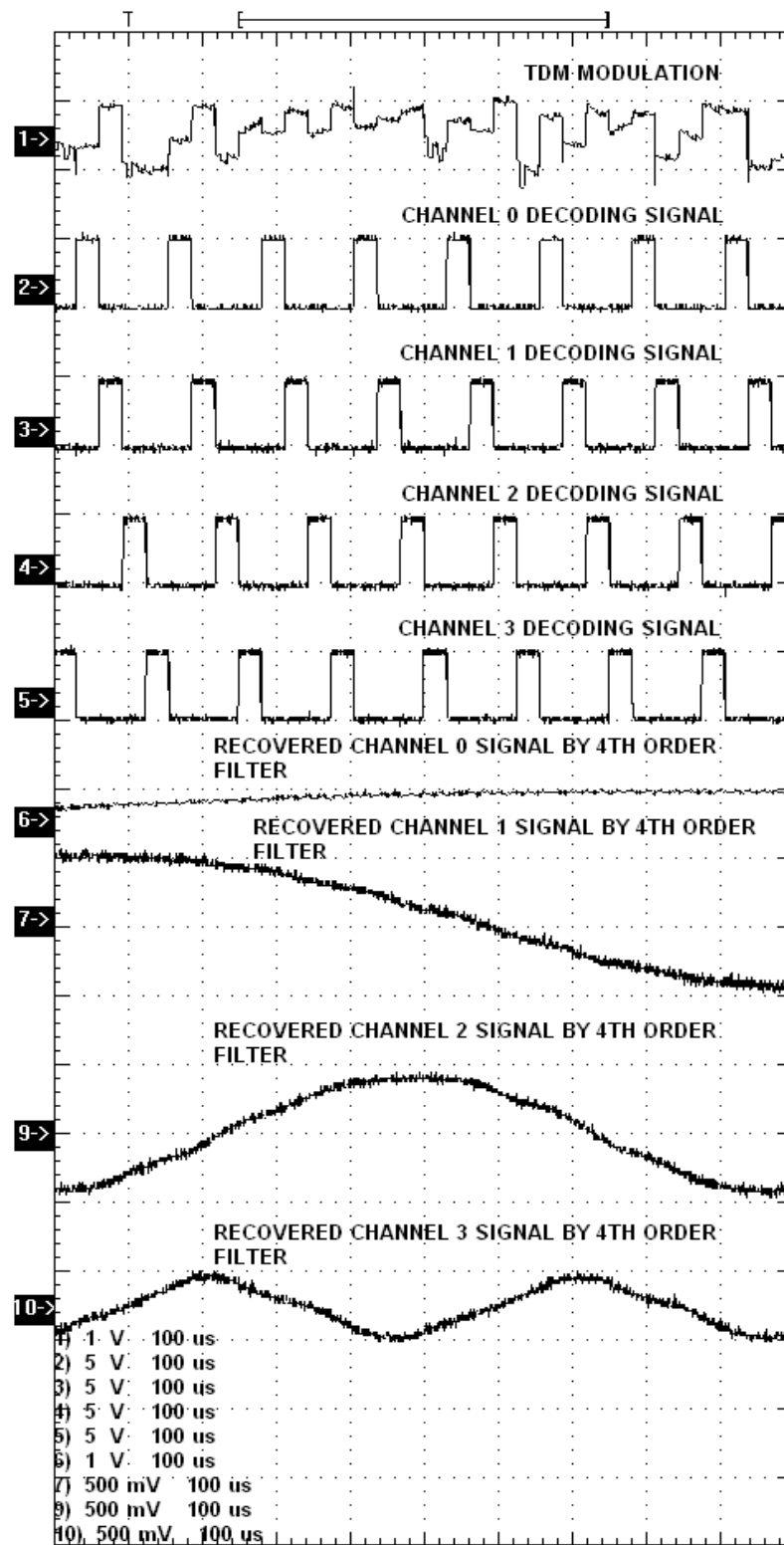
- Input signals **IN7, IN8, IN9, and IN10**.
- Channel Selection Signal at **TP1, TP2, TP3, TP4**.
- **TX CLK** and **RX CLK**.
- Channel Identification Signal **TX SYNC** and **RX SYNC**.
- Multiplexer Output **OUT5**.
- De-multiplexer output **OUT16, OUT17, OUT18, and OUT19**.
- Reconstructed signal **OUT 31, OUT 32, OUT 33, OUT 34**.

CONCLUSION

In this experiment, the transmitter clock and the channel identification clock (Sync) are directly linked to the receiver section. Hence, the transmitter and the receiver are synchronized and proper reconstruction of the signal is achieved.



4 CHANNEL TDM SIGNAL



Demultiplexed 4 Channel TDM Signal

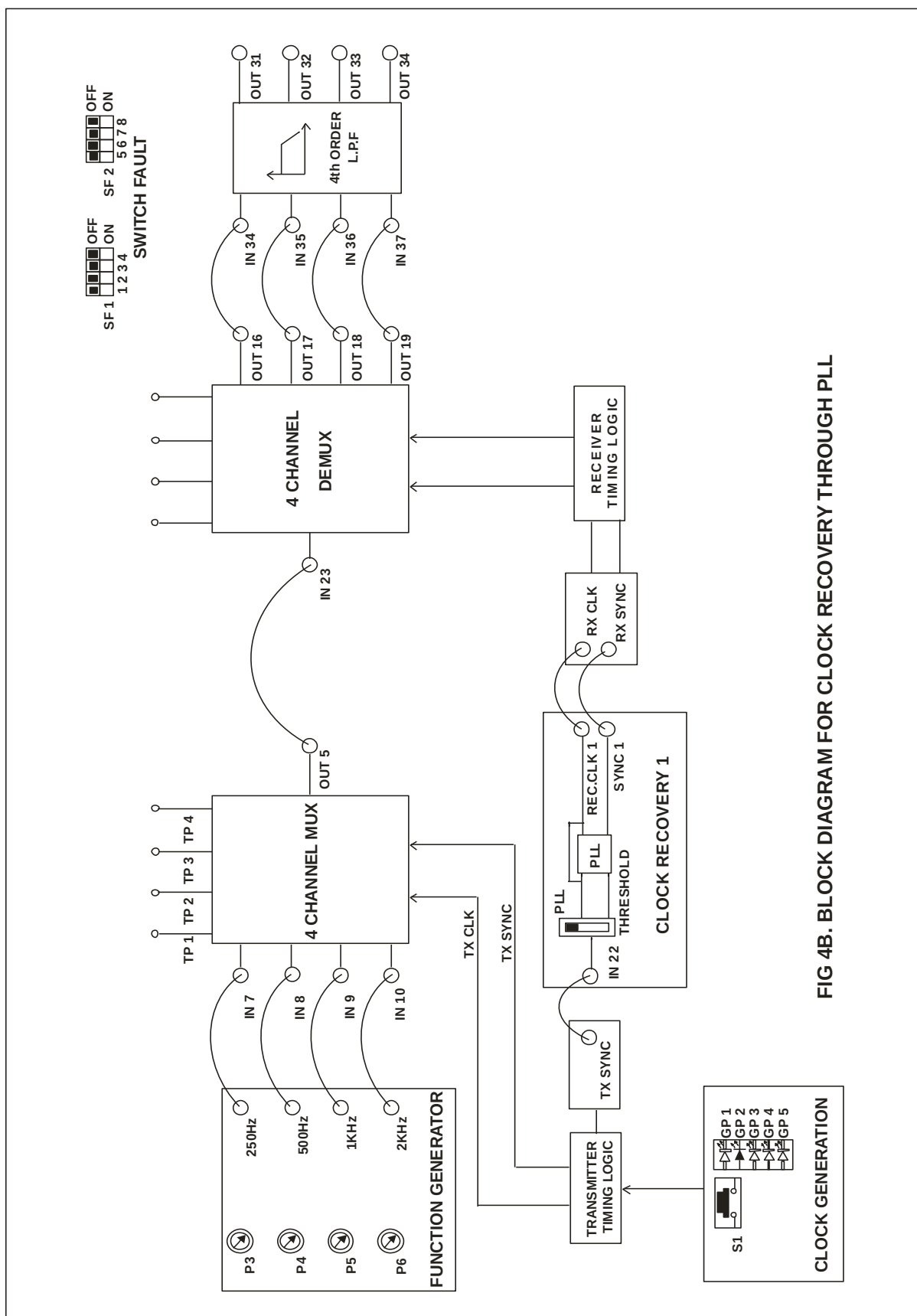


FIG 4B. BLOCK DIAGRAM FOR CLOCK RECOVERY THROUGH PLL

4B. Clock Recovery through PLL

NOTE: In this method we recover the clock and sync from the PLL, instead of transmitting it physically from the transmitter to the receiver.

PROCEDURE

- Ensure that group 2 (**GP2**) clock is selected in the clock generation section. Selection is done with the help of switch **S1**.
- The selected clock of frequency 32 KHz, 80% duty cycle will be available at the **TXCLK** post.
- The timing signals required for multiplexing four analog channels are derived internally from TX CLK and are available at the test points **TP1, TP2, TP3** and **TP4**.
- Feed four sine waves of frequency **250Hz, 500Hz, 1 KHz** and **2 KHz** having amplitude around 1Vpp from the function generator to the four channel inputs **IN7, IN8, IN9** and **IN10** of the analog multiplexer.
- Observe the TDM output at the **OUT5** post of the 4-channel Mux section.
- To recover the individual analog signals from the multiplexed data at the **OUT5** post, connect it to **IN23** post of the 4-channel Demux section.
- Connect the **TX SYNC** post to **IN22** post of the Clock recovery 1 section. Keep the switch **S8** towards the **PLL** position.
- Connect **REC. CLK1** post to **RX CLK** post and **SYNC1** post to **RX SYNC** post.
- Observe the recovered clock and sync at posts **REC. CLK 1** and **SYNC 1** post of the Clock recovery 1 section.
- The demultiplexed sine waves needs to be filtered. Connect **OUT16** post to **IN34** post, **OUT17** post to **IN35** post, **OUT18** post to **IN36** post and **OUT19** to **IN37** post of the 4th order LPF. Observe the filtered output at posts **OUT31, OUT32, OUT33** and **OUT34** respectively.
- Observe the sequence of the recovered signals, which is in exact compliance with that of the transmitter inputs sequence. Similarly, observe the frequencies of the recovered signals.

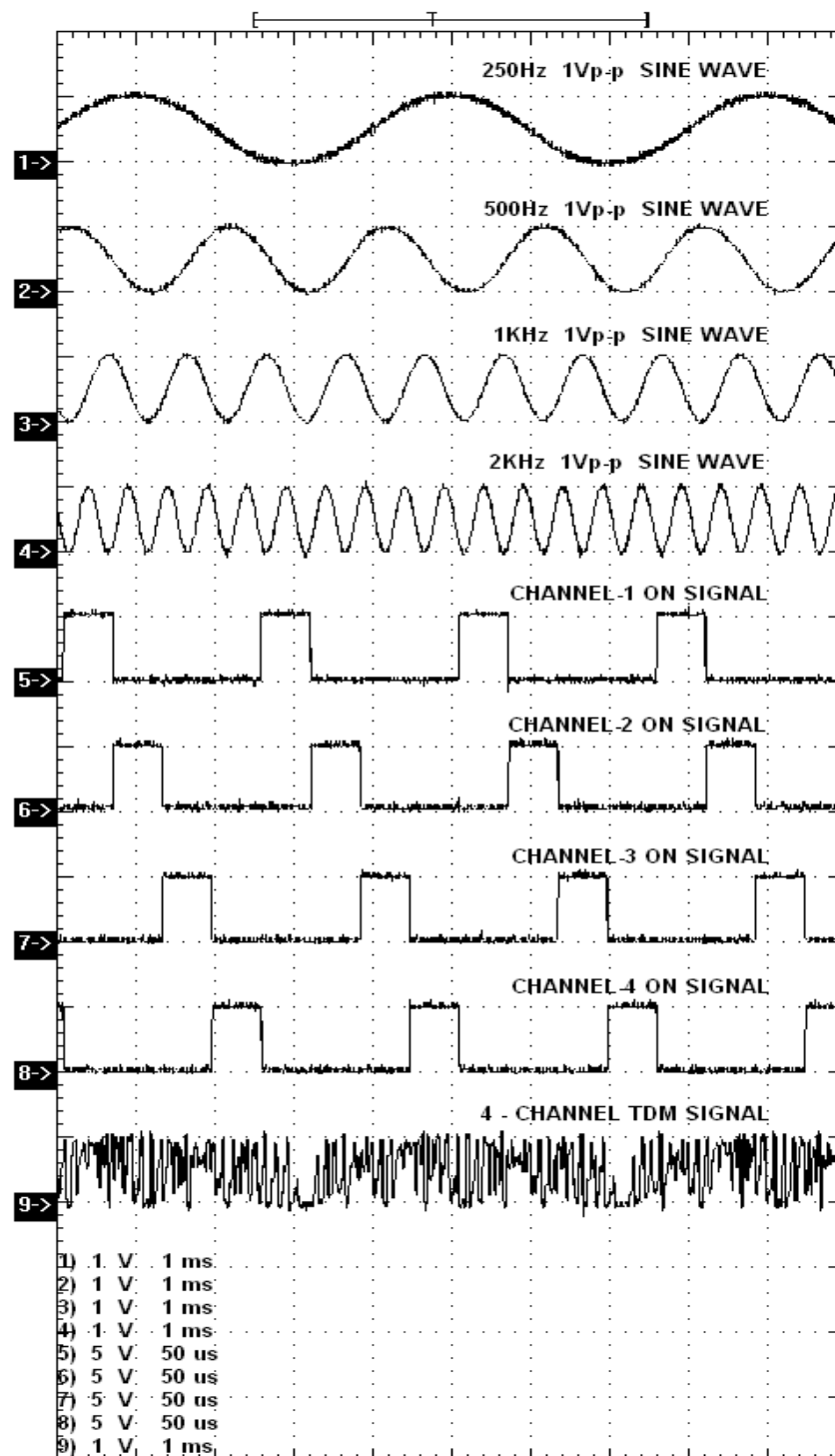
OBSERVATIONS

- Input signals **IN7, IN8, IN9, IN10**.
- Channel Selection Signal at **TP1, TP2, TP3, TP4**.
- **TX CLK** and **RX CLK**.
- Channel Identification Signal **TX SYNC** and **RX SYNC**.
- Multiplexer Output **OUT5**.
- Demultiplexer output **OUT16, OUT17, OUT18, OUT19**.
- Reconstructed signal **OUT 31, OUT 32, OUT 33, OUT 34**.

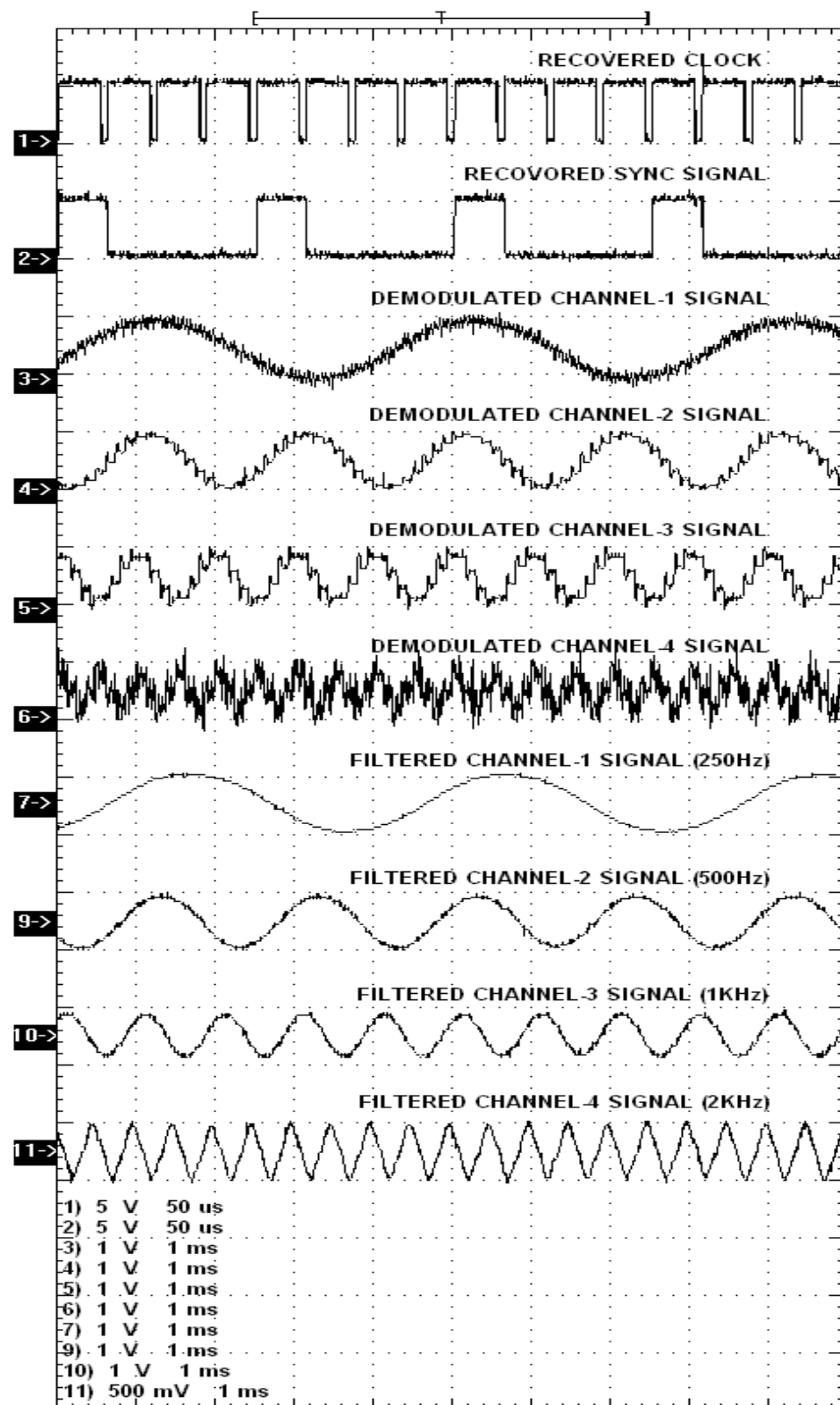
CONCLUSION

In this experiment, the channel identification signal is linked to the PLL input of the receiver section. Here, the PLL locks on the TXSYNC signal Frequency and the PLL, along with its associated circuitry and generates the SYNC and CLOCK signals, which are same in phase and frequency as that of the Transmitter TXSYNC and TXCLK signal and hence

synchronize the Transmitter to the Receiver and proper reconstruction of the signal is achieved.



4 CHANNEL TDM



DEMULTIPLEXED 4 CHANNEL TDM SIGNAL

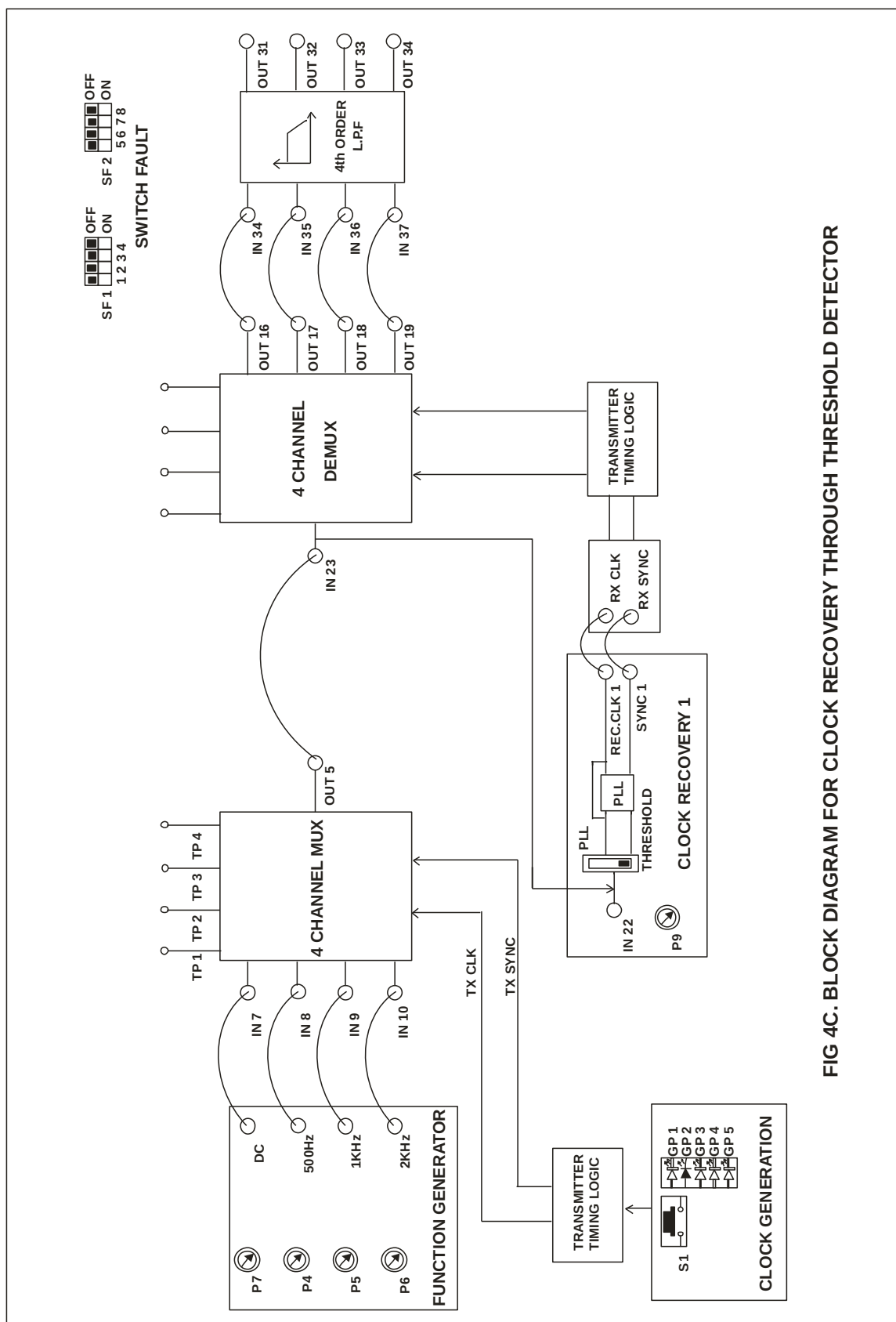


FIG 4C. BLOCK DIAGRAM FOR CLOCK RECOVERY THROUGH THRESHOLD DETECTOR

4 C. Clock Recoveries through Threshold Detector

NOTE: In this method we recover the clock and sync using the Threshold detector, instead of transmitting it physically from the transmitter to the receiver.

PROCEDURE

- Ensure that group 2 (**GP2**) clock is selected in the clock generation section. Selection is done with the help of switch **S1**.
- The selected clock of frequency 32 KHz and 80% duty cycle will be available at the **TXCLK** post.
- The timing signals required for multiplexing four analog channels are derived internally from TX CLK and are available at the test points **TP1, TP2, TP3** and **TP4**.
- For this recovery, we make use of the multiplexed signals.
- Connect the **DC** signal from the DC post of the Function Generator to the **IN7** post of the analog multiplexer.
- Increase the DC voltage up to 5V using pot **P7**.
- Feed three sine waves of frequencies **500Hz, 1 KHz**, and **2 KHz** having amplitude around 1Vpp from the Function Generator to the three channel inputs **IN8, IN9**, and **IN10** respectively.
- Observe the TDM output at the **OUT5** post of the 4-channel MUX.
- Connect the **OUT5** post to **IN23** post of the 4-channel DEMUX.
- Keep the switch **S8** towards **Threshold** position.
- Connect **REC. CLK1** post to **RX CLK** post and **SYNC1** post to **RX SYNC** post.
- Observe the recovered clock and sync at the post's **REC. CLK1** post and **SYNC1** post of the Clock recovery1 section.
- Observe the filtered output at posts **OUT31, OUT32, OUT33** and **OUT34** respectively.
- Observe the sequence of the recovered signals, which is in exact compliance with that of the transmitter inputs sequence. Similarly, observe the frequencies of the recovered signals.

OBSERVATIONS

- Input signals **IN7, IN8, IN9, IN10**.
- Channel Selection Signal at **TP1, TP2, TP3, TP4**.
- **TX CLK** and **RX CLK**.
- Channel Identification Signal **TX SYNC** and **RX SYNC**.
- Multiplexer Output **OUT5**.
- Demultiplexer output **OUT16, OUT17, OUT18, and OUT19**.
- Reconstructed signal **OUT 31, OUT 32, OUT 33, OUT 34**.

CONCLUSION

The PLL will lock into the frequency of generation of the sync pulses and the PLL, along with the associated circuitry and will generate the sync and clock signals, which are same in phase and frequency as that of the transmitter TXSYNC, and TXCLOCK signal, hence synchronizing the transmitter to the receiver.

**EXPERIMENT
NO. 5**

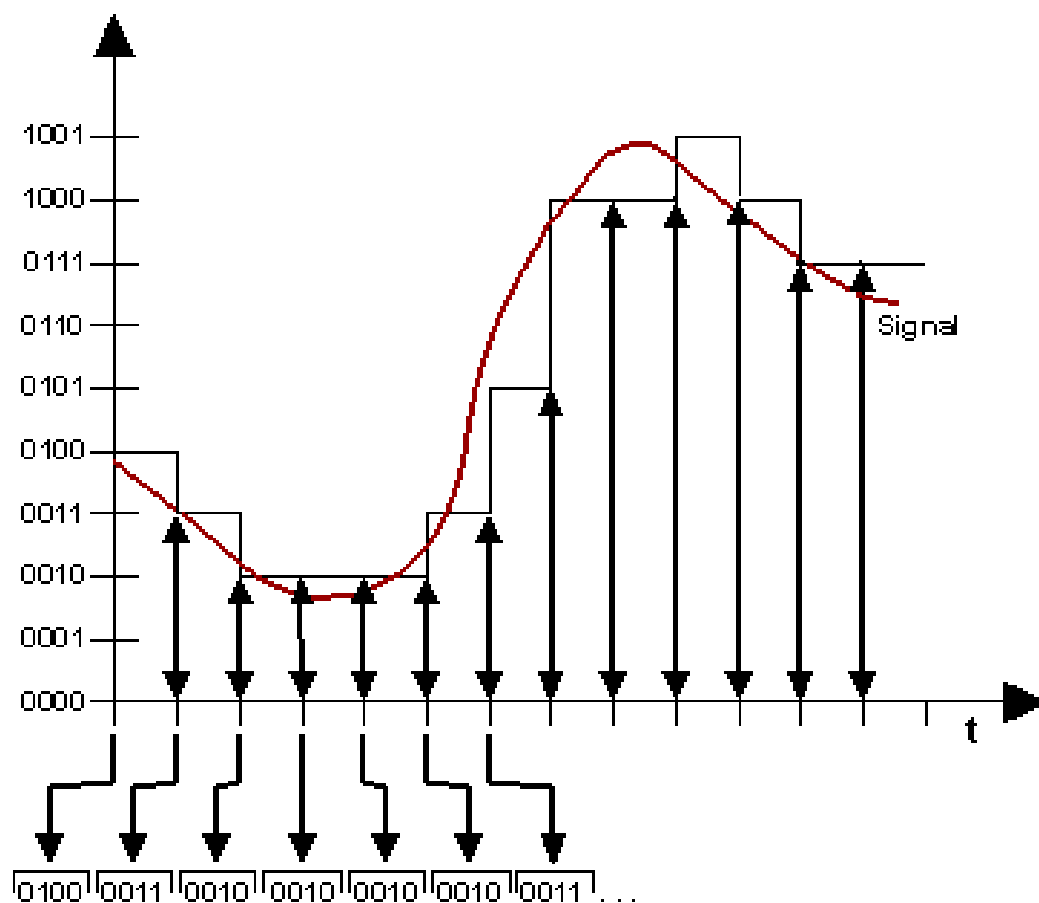
EXPERIMENT NO.5

NAME

OBJECTIVES

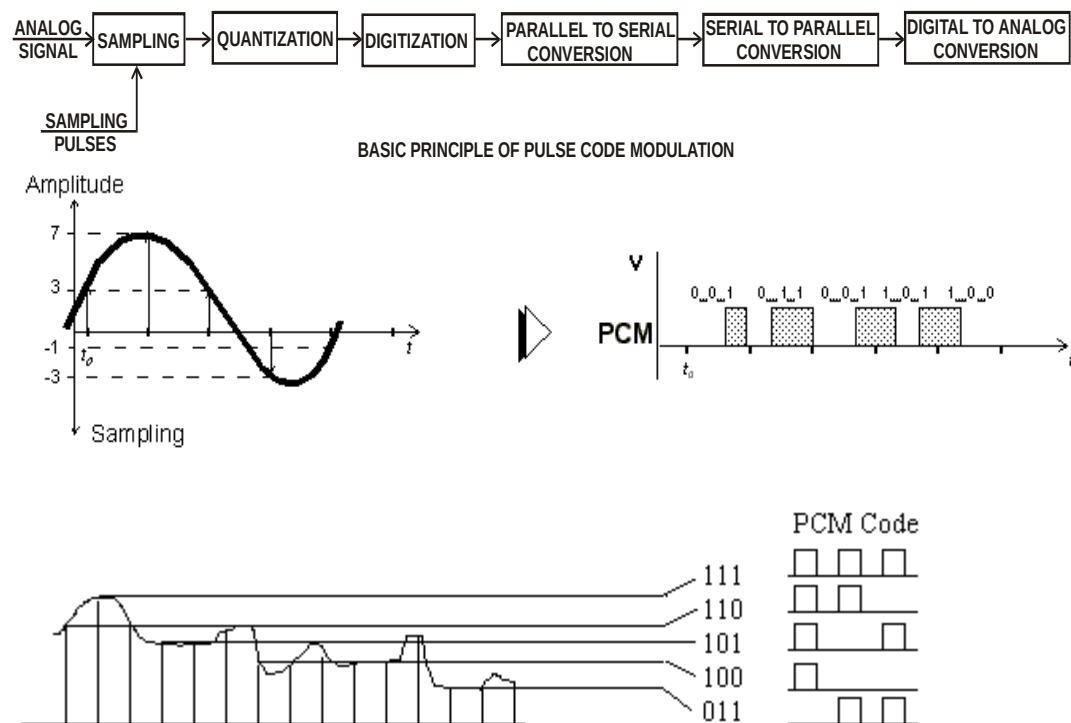
To study the Pulse Code Modulation Technique:

THEORY



Basic Principle of PCM

The analog signal is sampled at a rate more than the Nyquist rate. Through the process of sampling, a PAM signal is generated. The signal is then quantized and digitized. The parallel data word available after the analog to digital conversion is converted to a serial data stream after coding and it is sent through the channel. This coded data stream is said to be PCM coded data and is transmitted serially by the PCM transmitter.



Quantization

Quantization refers to the process of approximating the continuous set of values in the image data with a finite (preferably small) set of values. It refers to the fact that only certain levels of signal are recognized and transmitted. The input to a quantizer is the original data and the output is always one among a finite number of levels. These levels are decided to accommodate the whole range of amplitudes of the input signal. The signal is then digitized by using an A to D converter. In fact, the quantization is done by this converter. The output of A to D converter may be in binary form or in any other digital code such as excess-3, binary, gray, BCD etc. For giving distinct codes to eight levels, 3-bit binary code will be necessary.

In actual practice, there can be more than 8 levels, say 256 levels. In such a case, an 8-bit code will be generated for each sample. These bits are converted to serial bit stream and these bits are transmitted one after the other.

There are two types of quantization - Scalar Quantization and Vector Quantization. In scalar quantization, each input symbol is treated separately in producing the output; while in vector quantization the input symbols are clubbed together in groups called vectors and processed to give the output. This clubbing of data and treating them as a single unit, increases the optimality of the vector quantizer, but at the cost of increased computational complexity. Here, we'll take a look at scalar quantization.

A quantizer can be specified by its input partitions and output levels (also called reproduction points). If the input range is divided into levels of equal spacing, then the quantizer is termed as a Uniform Quantizer, and if not, it is termed as a Non-Uniform Quantizer. A uniform quantizer can be specified easily by its lower bound and the step size. Also, implementing a uniform quantizer is easier than a non-uniform quantizer. Take a look at the uniform quantizer shown below. If the input falls between $n \cdot r$ and $(n+1) \cdot r$, the quantizer outputs the symbol n .

On the receiver side, the serial to parallel converter collects and groups the pulses for one sample. If the transmitter generates an 8-bit code per sample then the serial to parallel

converter collects 8 pulses of the serial bit stream and passes this 8-bit code to the digital to analog converter. This is done in synchronization.

Fast Mode With Sine Wave

EQUIPMENTS

DCS-B kit
Connecting Chords
Power supply
20 MHz Dual Trace Oscilloscope
Power connection cables

NOTE: The PCM experiments have two modes of operation i.e. fast mode and slow mode. In fast mode the transmission clock has a frequency of 250 KHz and in the slow mode the transmission clock has a frequency of 32Hz.

The slow mode is used to observe the various data patterns generated during PCM generation and transmission on the LED indications.

In fast mode, sinusoidal signal as well as DC level signal is used as input for PCM generation and its study. In slow mode, only DC level is used for PCM generation.

PROCEDURE

- Ensure that group 3 (**GP3**) clock is selected in the clock generation section. Selection is done with the help of switch **S1**, observe the corresponding LED indication.
- Keep the jumper **JP3** at the fast mode position.
- The selected clock of frequency 230 KHz will be available at the **TXCLK** post.
- Connect **250Hz** signal having voltage around 4V from the Function generator section to the **ADC IN** post of the A/D converter.
- Keep switch **S12** in the **NONE** parity mode and observe the corresponding LED indication.
- Observe the analog to digitally converted bits on the corresponding LED indication (**B0** to **B6**) in the A/D Converter section.
- Observe the Pseudo random bit sequence at the test point marked **PRBS OUT**.
- **TX DATA** post shows multiplexed data having PRBS and PCM data.
- To recover the analog signal from the PCM data at the **TX DATA** post, connect it to the **RX DATA** post.

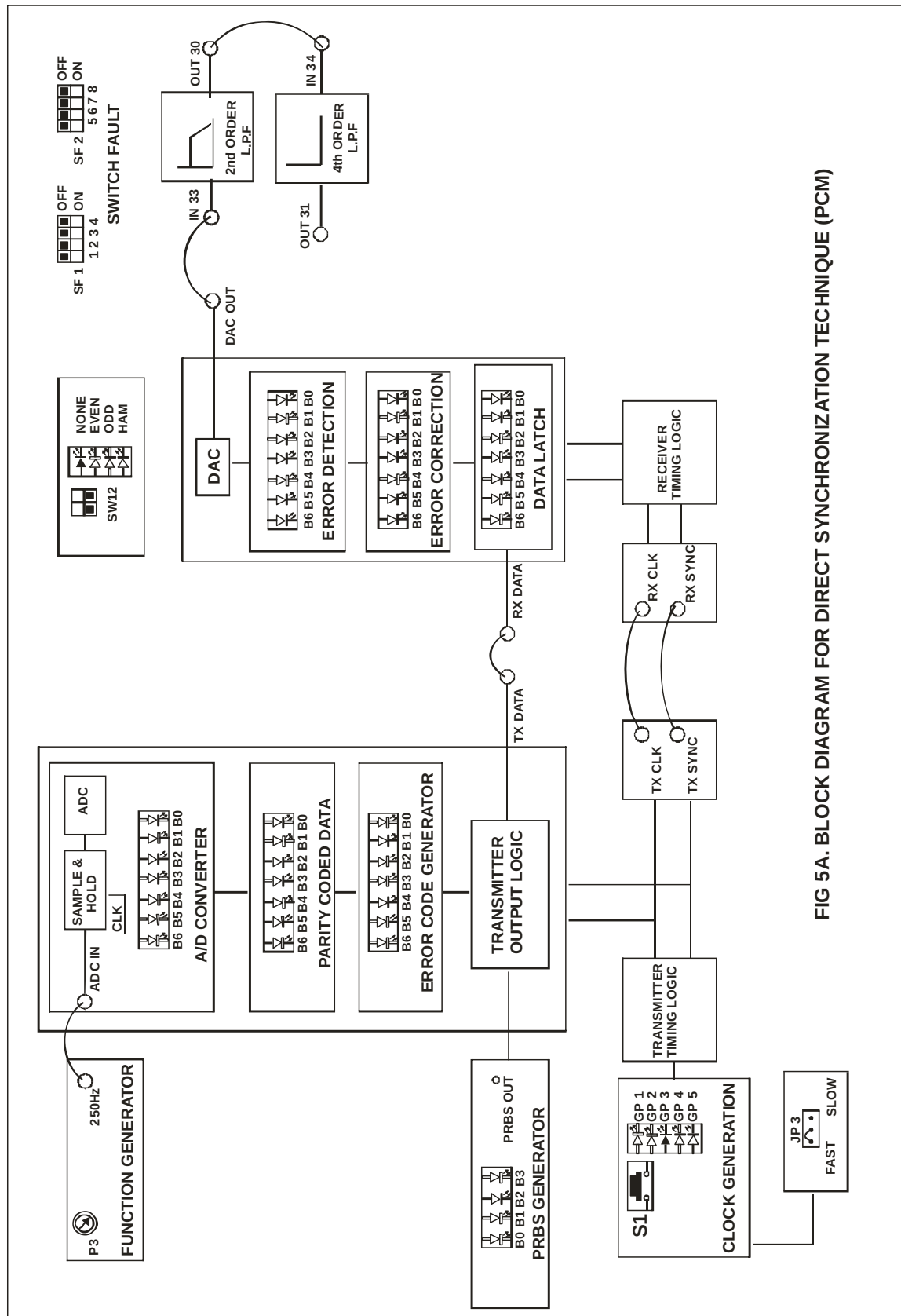


FIG 5A. BLOCK DIAGRAM FOR DIRECT SYNCHRONIZATION TECHNIQUE (PCM)

5A.Using Direct Synchronization Technique (TX CLK &TX SYNC Are Transmitted to the Receiver)

PROCEDURE

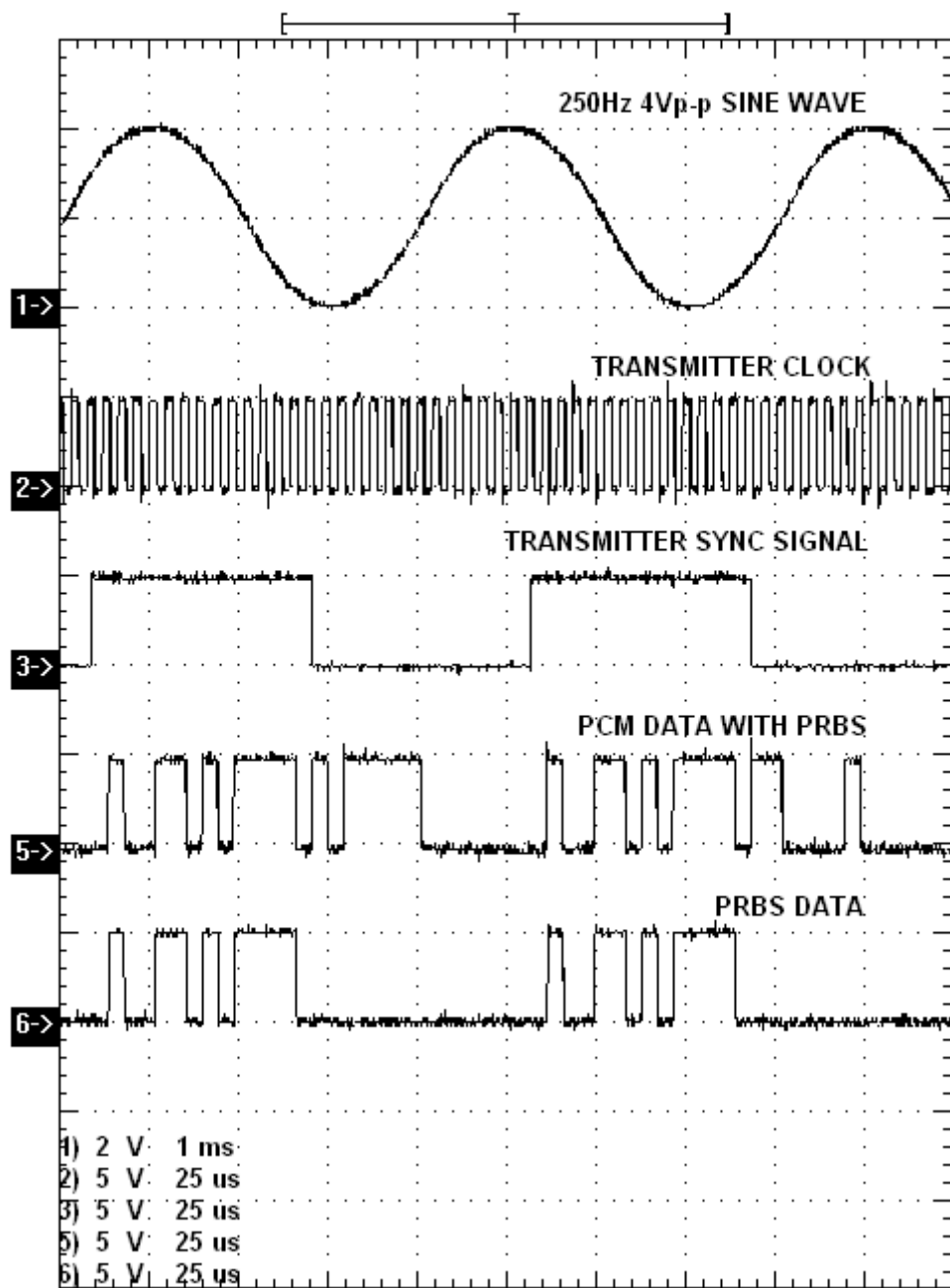
- To provide the timing and synchronization information to the receiver section, connect **TX CLK** post to **RX CLK** post and **TX SYNC** to **RX SYNC** post.
- The serial data is converted to the parallel data & directly observed on the corresponding LED indication at the Data Latch section.
- Observe the D/A converted data at the **DAC OUT** post.
- Connect the **DAC OUT** post to the **IN33** post of the 2nd order LPF. Connect **OUT 30** post of 2nd order LPF to **IN34** post of 4th order LPF.
- Observe the recovered signal at the **OUT31** post of the 4th order LPF.

OBSERVATIONS

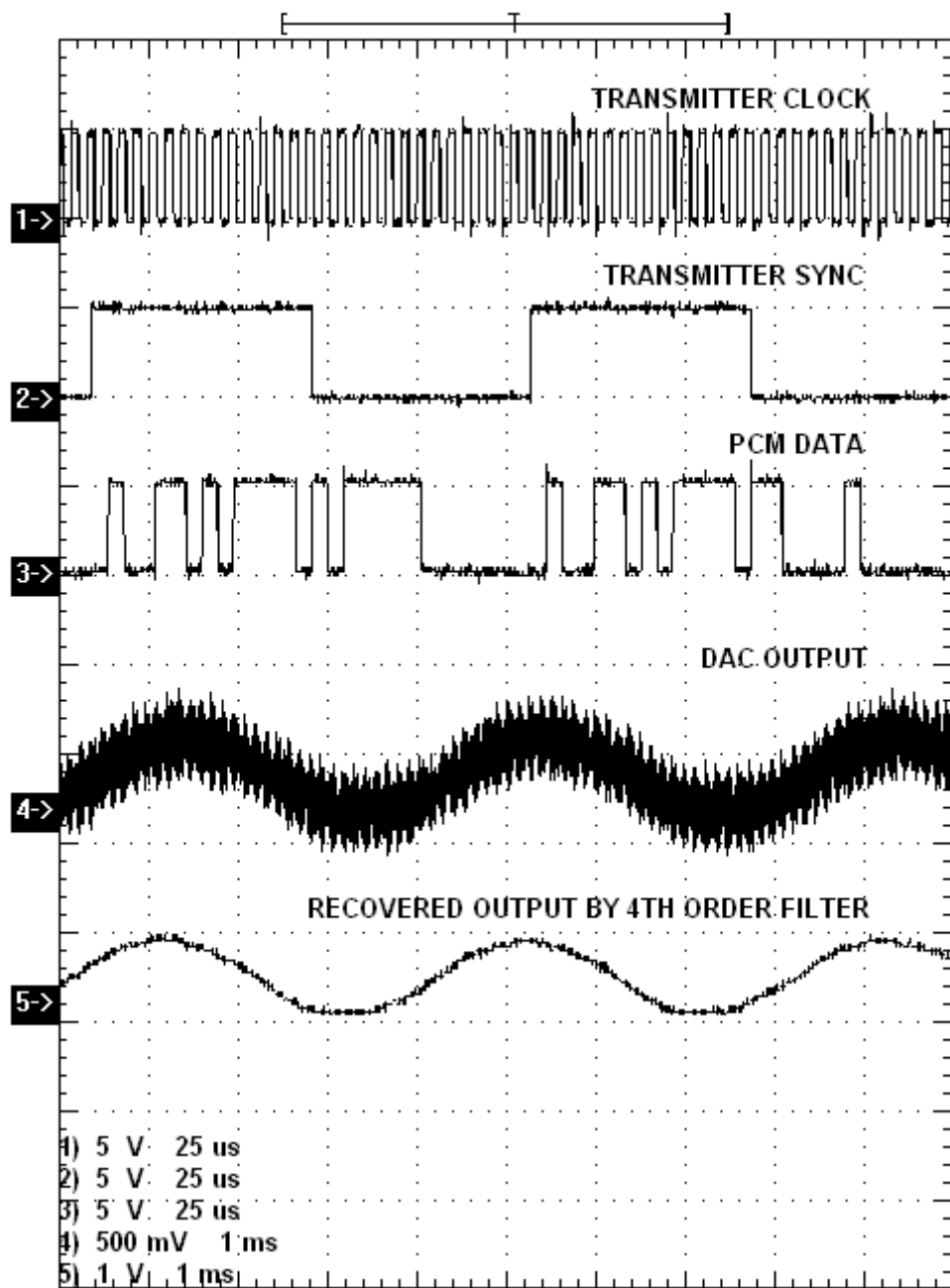
- Input signal
- PCM Data **TX DATA**, **TXCLK**, **TXSYNC**
- PRBS data **PRBS OUT**
- **RXCLK**, **RXSYNC**, **RXDATA**
- **DAC OUT**
- Received signal **OUT 31**

CONCLUSION

At the receiver side, the 4th order low pass butterworth filter is used as a reconstruction unit, which reproduces the signals (sine wave and DC signal levels) same as that of the transmitter side. In this case, it is observed that the reconstructed sine wave has a good linearity.



PCM MODULATION



PCM DEMODULATION USING DIRECT SYNCHRONIZATION TECHNIQUE

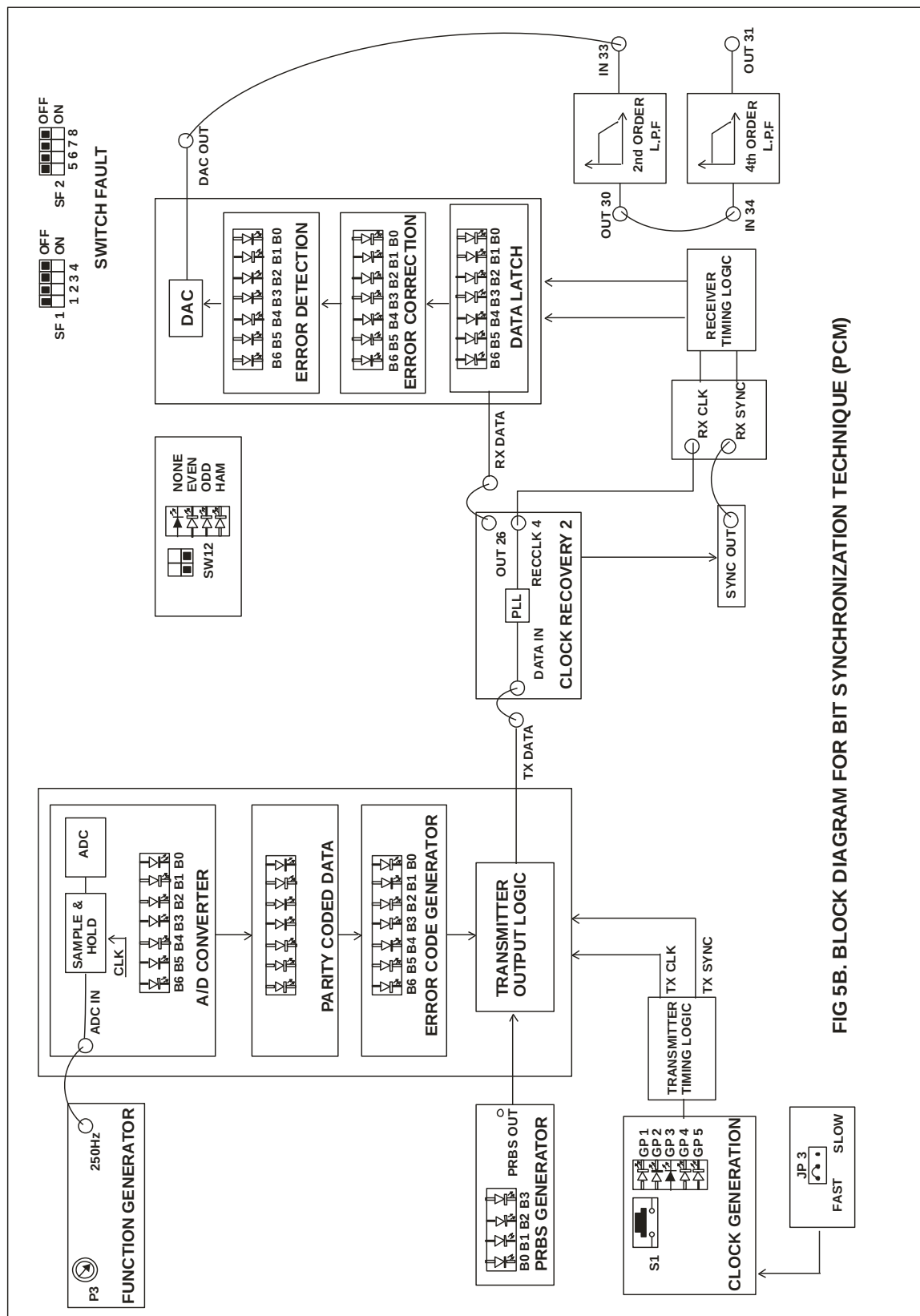


FIG 5B. BLOCK DIAGRAM FOR BIT SYNCHRONIZATION TECHNIQUE (PCM)

5B. Using Bit Synchronization Technique (Clock & sync information is recovered at the receiver from TX DATA)

THEORY

To recover the information from the serial bit stream, the receiver has to know, if the Received data bit at a given instant of time, is a ONE or a ZERO. For determining the Data bit, the clocks of transmitter and receiver are to be perfectly synchronized. Once This Synchronization is achieved, the receiver recovers the data, with reference to the Clock. For achieving this Synchronization, normally known as bit Synchronization, the clock of the transmitter (TXCLK) can be sent along with the data. This calls for an Additional channel for clock transmission. So normally a phase locked loop (PLL) is used at the Receiver for recovering the clock. The phase locked loop is made to lock with the incoming data whose VCO reproduces the clock at the Receiver.

PROCEDURE

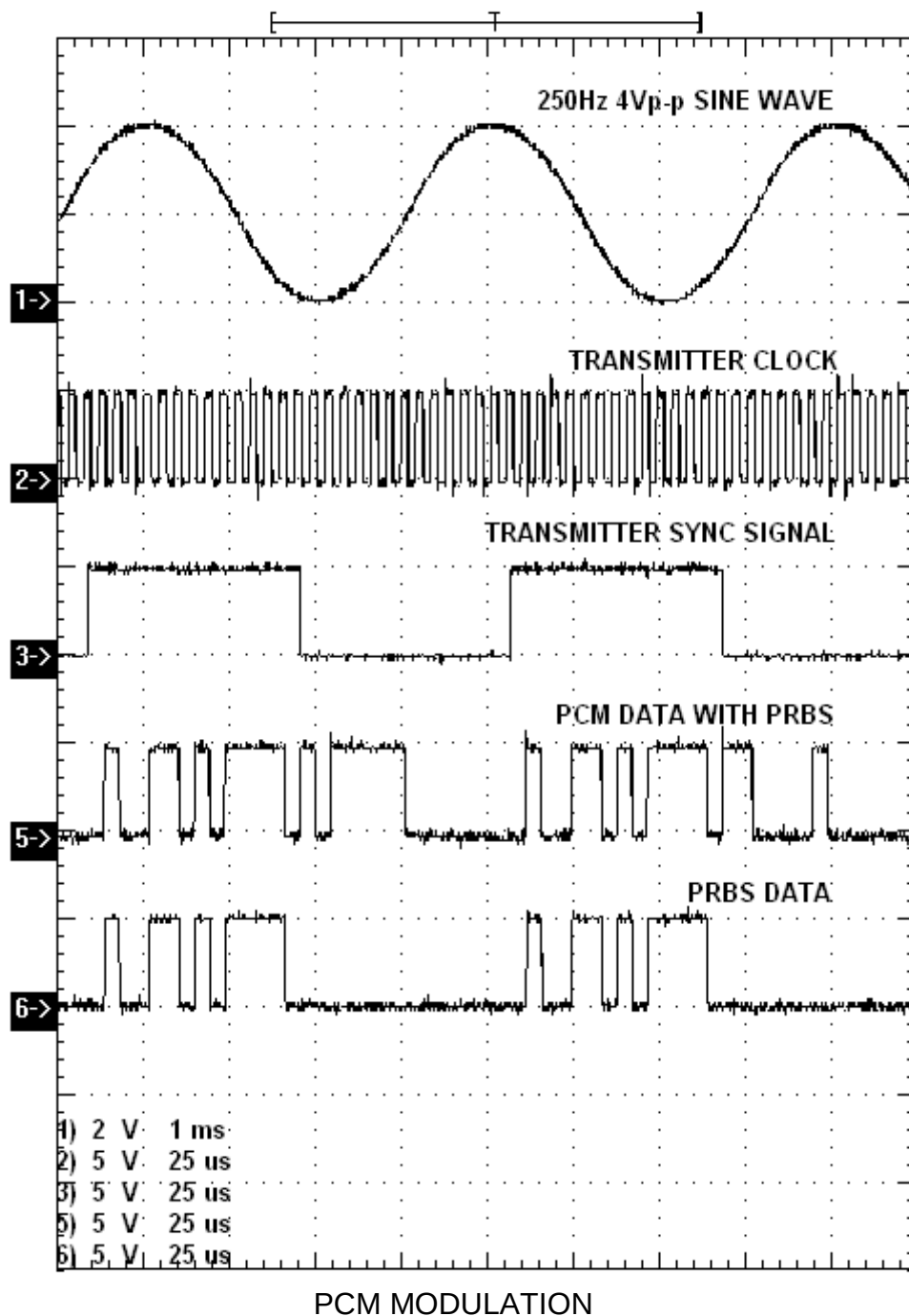
- Ensure that group 3 (**GP3**) clock is selected in the clock generation section. Selection is done with the help of switch **S1**. Observe the corresponding LED indication.
- Keep the jumper **JP3** at the fast mode position.
- The selected clock of frequency 230 KHz will be available at the **TXCLK** post.
- Connect **250Hz** signal having voltage around 4V from the Function generator section to the **ADC IN** post of the A/D converter.
- Keep switch **S12** in the **NONE** parity mode and observe the corresponding LED indication.
- Observe the analog to digitally converted bits on the corresponding LED indication (**B0** to **B6**) in the A/D Converter section.
- Observe the Pseudo random bit sequence at the test point, marked as **PRBS OUT**.
- **TX DATA** post shows multiplexed data having PRBS and PCM data.
- To recover the analog signal from the PCM data at the **TX DATA** post, connect it to the **DATA IN** post in the Clock Recovery 2 section. In this method, we make use of the PLL to recover the clock & sync.
- Observe the recovered clock at the **REC.CLK 4** post at the Clock recovery 2 section and recovered sync at the **SYNC OUT** post.
- Connect **OUT 26** post to the **RX DATA** post, **REC.CLK 3** post to **RX CLK** post and **SYNC OUT** post to **RX SYNC** post.
- Observe the serial to parallel data on the corresponding LED indication at the Data Latch section.
- Observe the D/A converted data at the **DAC OUT** post.
- Connect the **DAC OUT** post to the **IN33** post of the 2nd order LPF. Connect **OUT 30** post of 2nd order LPF to **IN34** post of 4th order LPF.
- Observe the recovered signal at the **OUT31** post of the 4th order LPF.

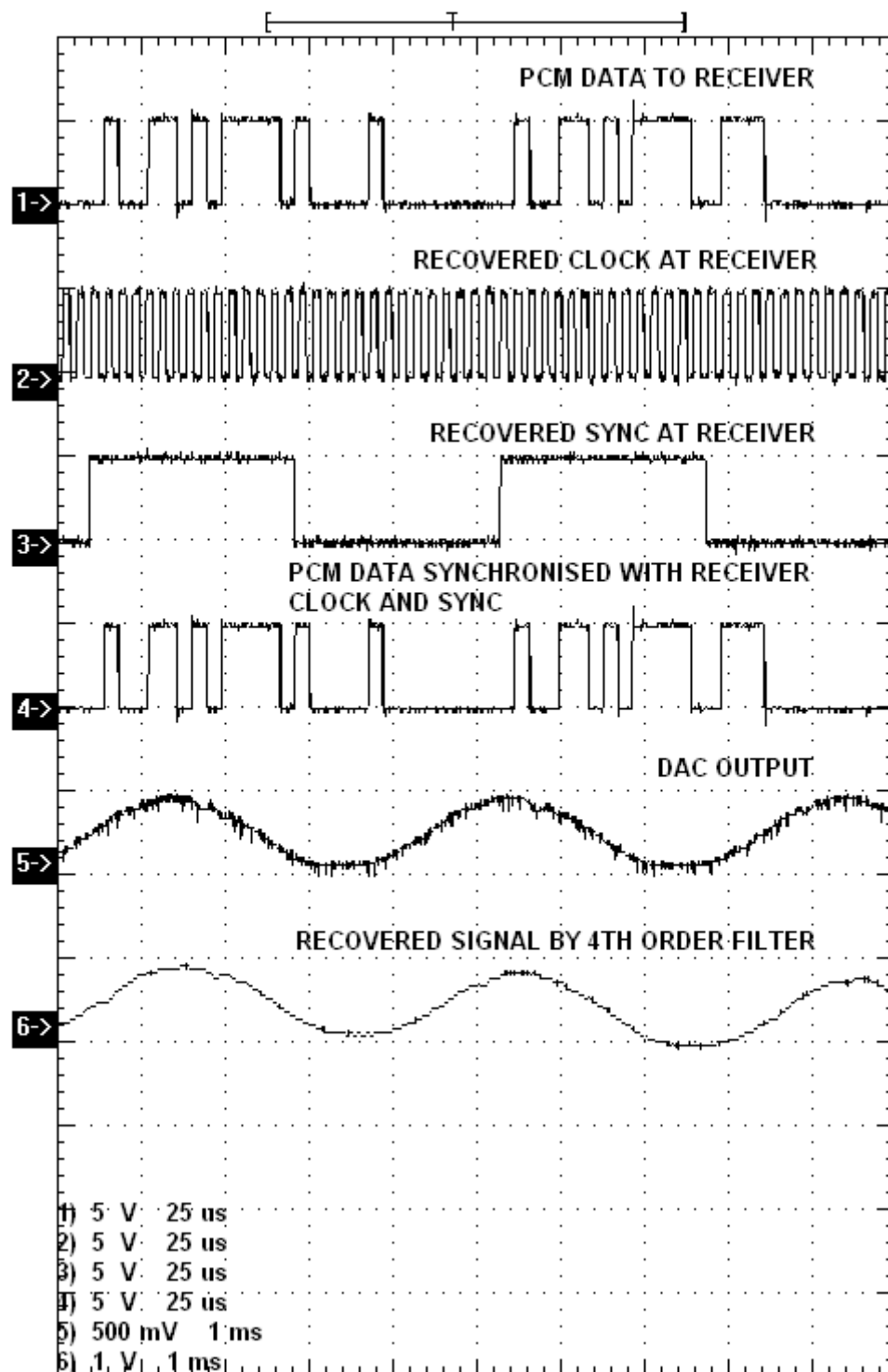
OBSERVATIONS

- Input signal
- PCM Data **TX DATA**, **TXCLK**, **TXSYNC**
- **RXCLK**, **RXSYNC**, **RXDATA**
- **RECCLK4**, **SYNCOUT**
- Received signal **OUT 31**

CONCLUSION

To recover the PCM data in Bit synchronization technique only Txdata is transmitted from Transmitter side, clock and Sync information is recovered at receiver side using PLL.





PCM DEMODULATION USING BIT SYNCHRONIZATION TECHNIQUE

5C. Using Frame Synchronization Technique (RX Sync Signal Is Generated From The PRBS Data)

THEORY

To recover the information transmitted from the data stream, the data is first Converted to parallel code words. Then code words are decoded using the D/A Converter. Demultiplexing logic assigns the samples to the corresponding channel Reconstruction units. For ensuring proper recovery of code words and proper assignment of channel information of the demultiplexing unit, the receiver has to know where the data begins exactly. For this, at the start of every frame, a 14-bit Pseudo random sequence (PRBS) is sent, which forms the header of the data Pattern. Subsequently, the data corresponding to the number of channels multiplexed follows.

The frame sync pulse can be sent by the Transmitter along with the data and clock to Achieve frame Synchronization. But normally it is not sent, since it calls for an Additional channel. To recover the frame sync at the receiver from the data stream, a sequence similar to the one transmitted, is generated at the Receiver. The PRBS detector compares the generated sequence with the incoming data and locks whenever the incoming data matches with the generated sequence. The PRBS sequence is chosen in such a way that it has minimum correlation with the data, to avoid false locks of the detector with the data bits. Once the PRBS detector is locked, it generates a frame sync Pulse, whose rising edge corresponds to the start of the frame and the time to the slot when the pseudo random sequence occurs in the data stream. This frame sync pulse is used to code word and for demultiplexing purposes

PROCEDURE

- Ensure that group 3 (**GP3**) clock is selected in the clock generation section. Selection is done with the help of switch **S1**. Observe the corresponding LED indication.
- Keep the jumper **JP3** at the fast mode position.
- The selected clock of frequency 230 KHz will be available at the **TXCLK** post.
- Connect **250Hz** signal having voltage around 4V from the Function generator section to the **ADC IN** post of the A/D converter.
- Keep switch **S12** in the **NONE** parity mode and observe the corresponding LED indication.
- Observe the analog to digitally converted bits on the corresponding LED indication (**B0** to **B6**) in the A/D Converter section.
- Observe the Pseudo random bit sequence at the test point marked as **PRBS OUT**.
- **TX DATA** post shows multiplexed data having PRBS and PCM data.
- To provide the timing and synchronization information to the receiver section, connect **TX CLK** post to **RX CLK** and **TXDATA** to **RX DATA** post.
- Observe the recovered sync at the **SYNC OUT** post.
- Connect the **SYNC OUT** post to **RX SYNC** post.
- Observe the serial to the parallel converted data, on the corresponding LED indication at the Data Latch section.
- Observe the D/A converted data at the **DAC OUT** post.
- Connect the **DAC OUT** post to the **IN33** post of the 2nd order LPF. Connect **OUT 30** post of 2nd order LPF to **IN34** post of 4th order LPF.
- Observe the recovered signal at the **OUT31** post of the 4th order LPF.

OBSERVATIONS

- Input signal
- PCM Data **TX DATA, TXCLK, TXSYNC**
- **RXCLK, RXSYNC, RXDATA**
- Recovered sync **SYNCOUT**
- Received signal **OUT 31**

CONCLUSION

The PLL will lock onto the frequency of the generated sync pulse and the PLL, along with the associated circuitry and will generate the sync and clock signals, which are same in phase and frequency as that of the TXCLK and TXSYNC, which enables the receiver to get the output.

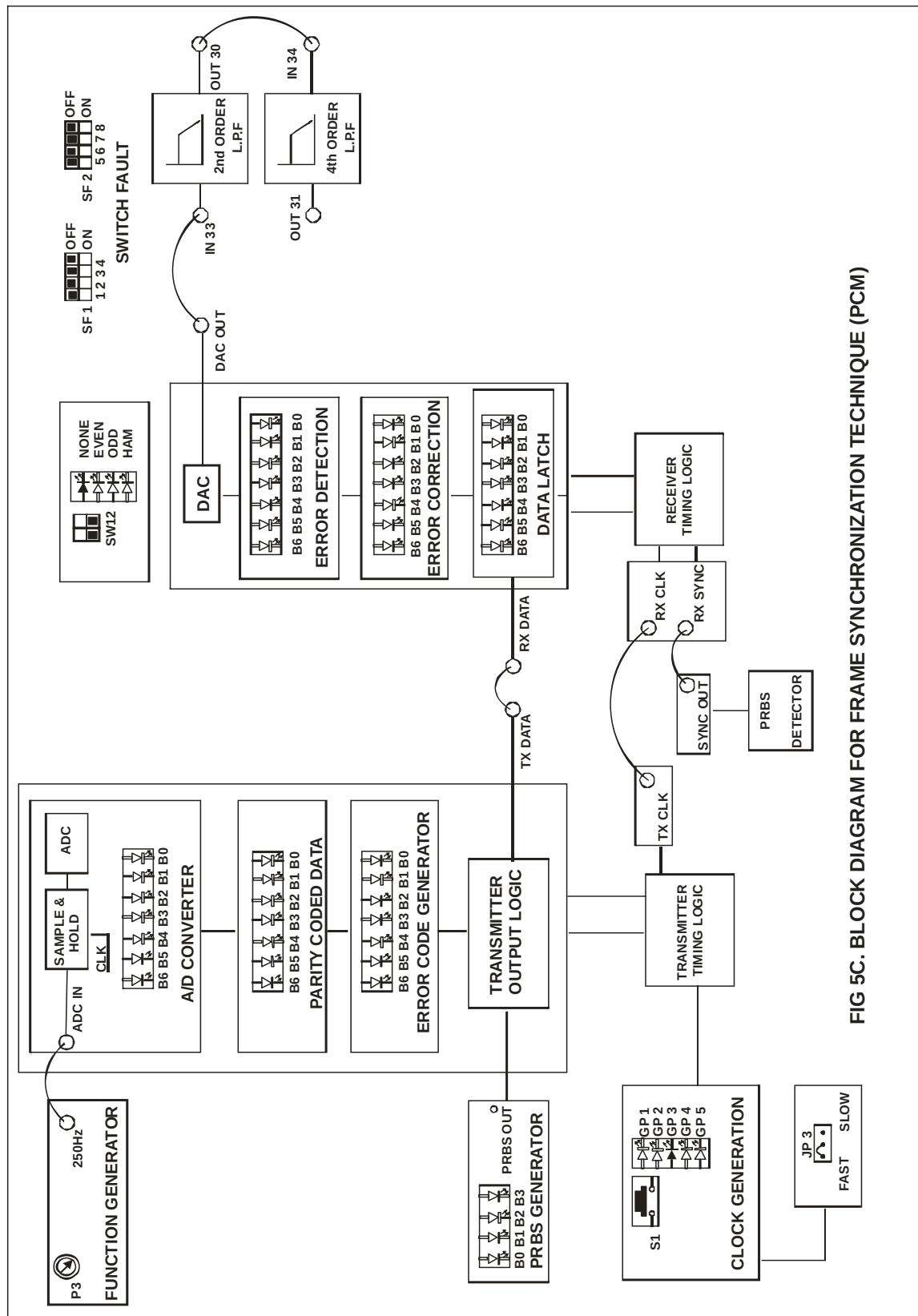
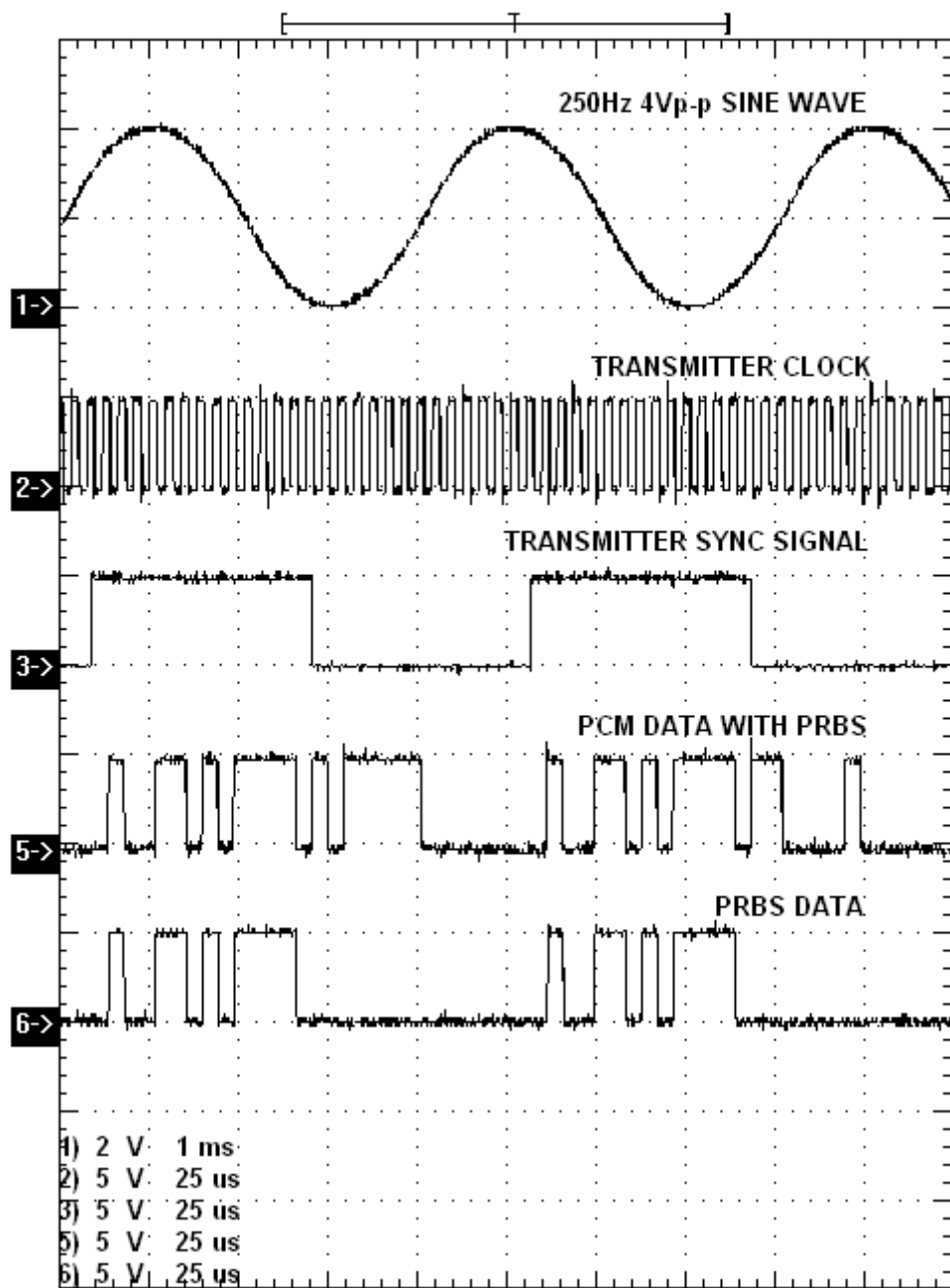
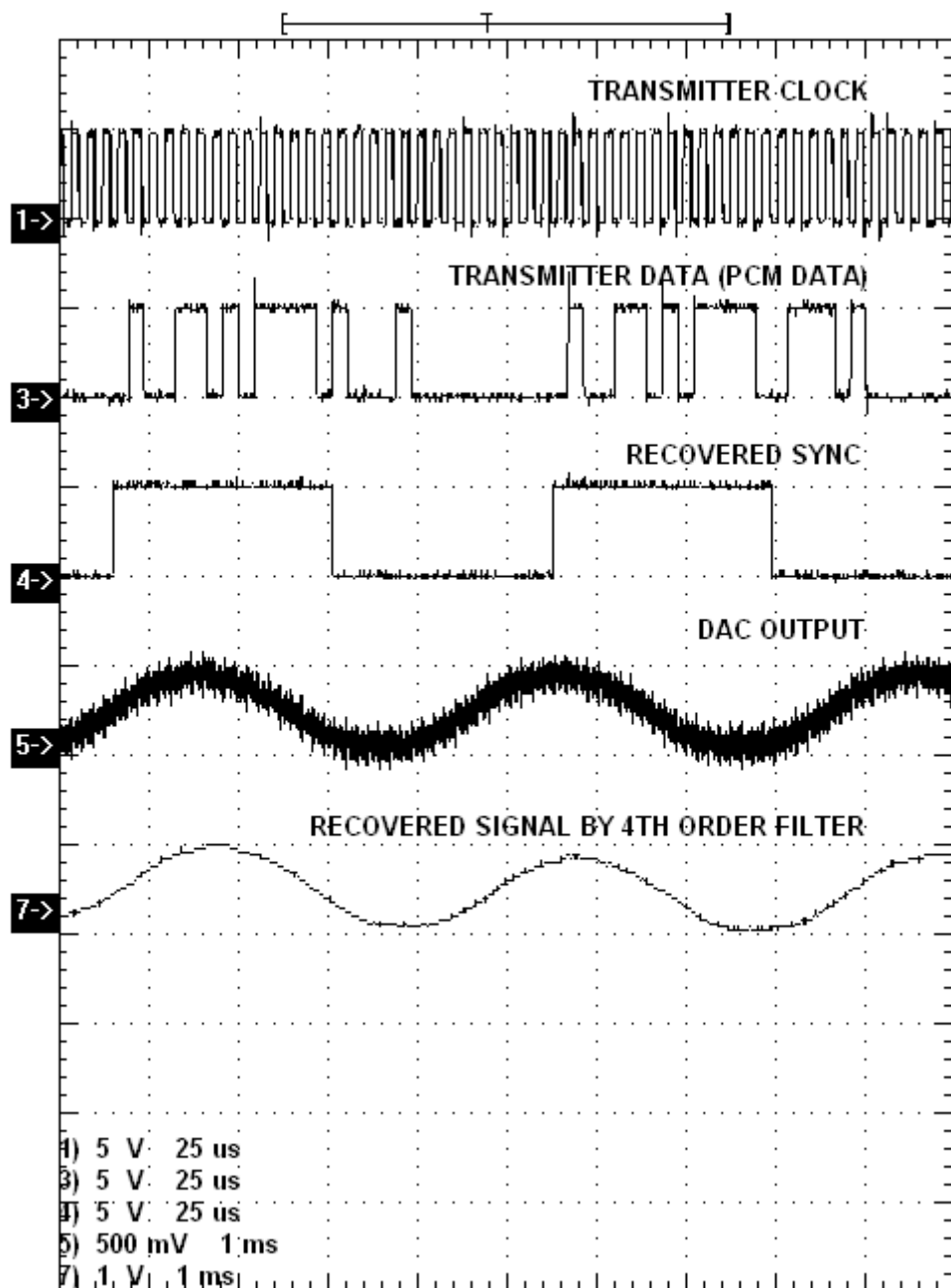


FIG 5C. BLOCK DIAGRAM FOR FRAME SYNCHRONIZATION TECHNIQUE (PCM)



PCM MODULATION



PCM demodulation by frame synchronization technique

**EXPERIMENT
NO. 6**

EXPERIMENT NO.6

NAME

Effects of Parities & Hamming Code on PCM Data

OBJECTIVES

In this experiment we will be studying the effect of Even, Odd and Hamming parity on PCM data during transmission and reception

THEORY

NONE PARITY MODE: Operations without any error check is also available, where the transmitted data consists of 7 bits of data only. The coded sequence will be:

B0, B1, B2, B3, B4, B5, B6 (MSB)

Anyone of the four modes of operations mentioned above can be selected, by using the switches provided onboard.

OBSERVATION TABLE

NONE PARITY MODE

A/D CONVERTO R	ERROR GENERATOR	CODE	DATA LATCH	ERROR CORRECTIO N	PARITY ERROR
0 0 0 0 0 0 0	0 0 0 0 0 0 0		0 0 0 0 0 0 0	0 0 0 0 0 0 0	OFF
0 0 0 0 0 0 0	0 0 0 0 0 0 0		0 0 0 1 0 0 0	0 0 0 1 0 0 0	OFF
1 1 1 1 1 1 1	1 1 1 1 1 1 1		1 1 1 1 1 1 1	1 1 1 1 1 1 1	OFF

It has been observed that on creating a fault in the data stream there is no change in the status of Parity Check LED.

EQUIPMENTS

DCS-B kit
Connecting Chords
Power supply
20 MHz Dual Trace Oscilloscope
Power connection cables

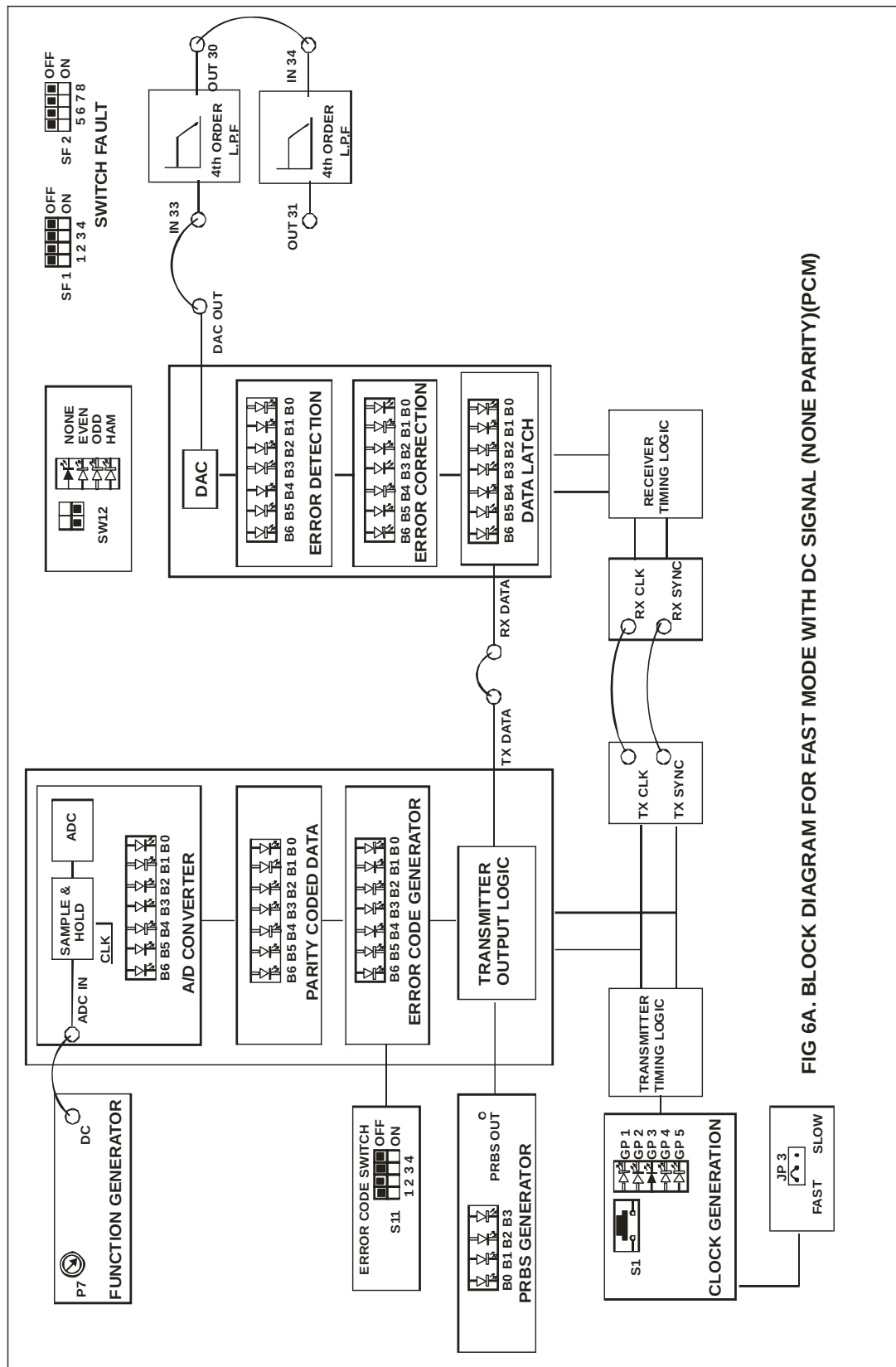


FIG 6A. BLOCK DIAGRAM FOR FAST MODE WITH DC SIGNAL (NONE PARITY)(PCM)

6A. Fast Mode with DC Signal (None Parity)

PROCEDURE

- Ensure that the group 3 (**GP3**) clock is selected in the clock generation section. Selection is done with the help of switch **S1**. Observe the corresponding LED indication.
- Keep the jumper **JP3** at the fast mode position.
- Connect **DC** signal having voltage around 5V from the Function generator section to the **ADC IN** post of the A/D converter.
- Keep switch **S12** in the **NONE** parity mode and observe the corresponding LED indication.
- Observe the analog to digitally converted bits on the corresponding LED indication (**B0** to **B6**) in the A/D converter section by varying the **DC** voltage from 0 to 5V using pot **P7**.
- Observe the Parity code converted data on the corresponding LED indication in the Parity coded data section.
- Observe the None parity-coded data on the corresponding LED indication in the Error code generator section.
- Observe the PCM data at the **TX DATA** post.
- Connect the **TX CLK** post to the **RX CLK** post, **TXDATA** to **RX DATA** post and **TXSYNC** to **RX SYNC** post.
- Observe the serial to parallel converted data on the corresponding LED indication in the Data Latch section, as well as in the Error correction section. We observed that there is no error.
- Observe the D/A converted data at the **DAC OUT** post.
- Connect the **DAC OUT** post to the **IN 34** post of the 4th order LPF post.
- Observe the recovered signal at the **OUT31** post of the 4th order LPF post.

OBSERVATIONS

- A/D Converter
- Parity Coded Data
- Error Code Generator
- Data Latch
- Parity Error Bit

CONCLUSION

In None Parity mode all bits (**B0** to **B6**) are Data bits, no parity is added, hence in None parity mode there is no error correction or detection is possible.

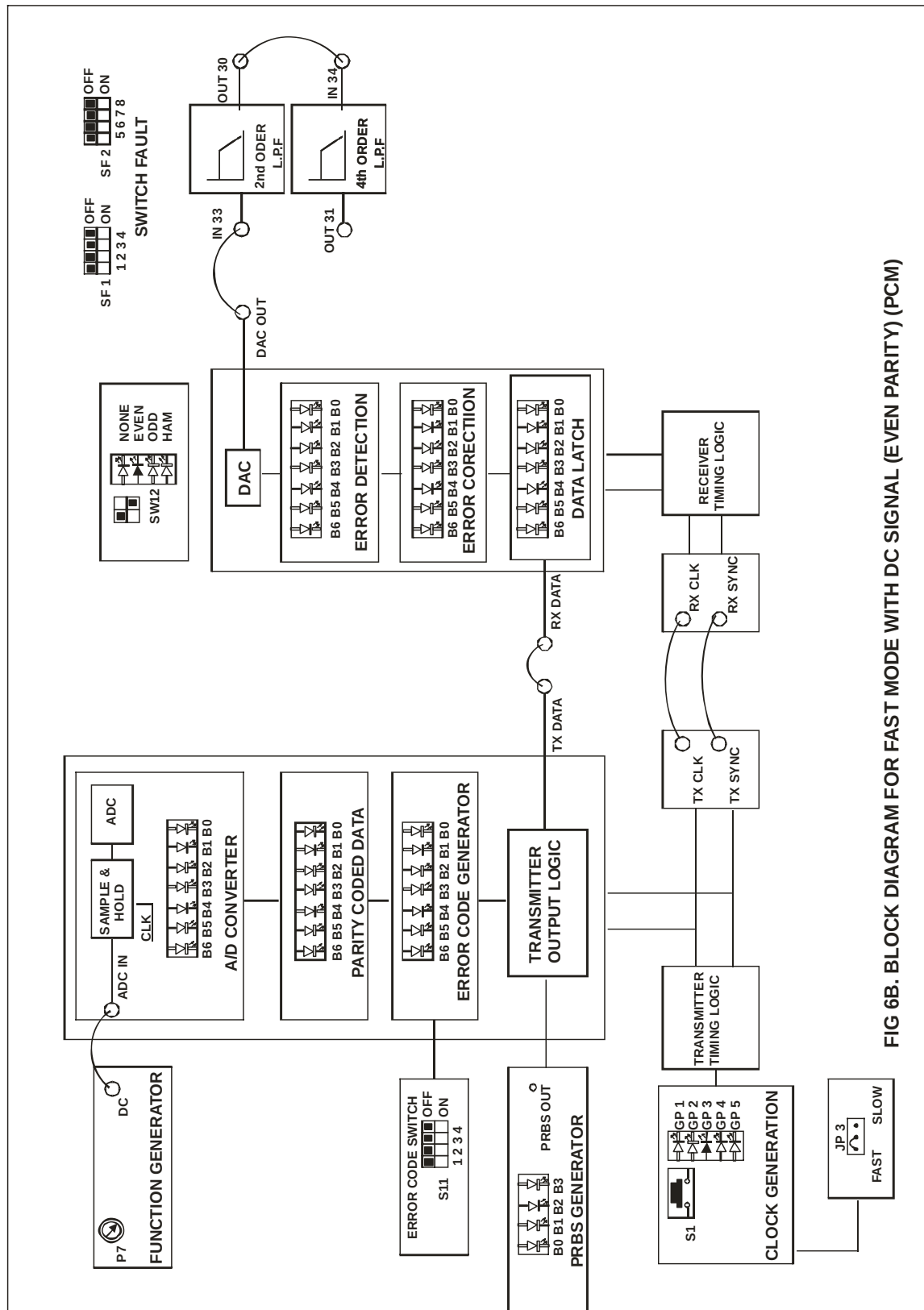


FIG 6B. BLOCK DIAGRAM FOR FAST MODE WITH DC SIGNAL (EVEN PARITY) (PCM)

6B. Fast Mode with DC Signal (Even Parity)

THEORY

EVEN Parity Generator generates the coded sequence, which consists of 6 bits of data pulses and one EVEN Parity Bit for transmission. The least Significant bit (B0) of the data word from the ADC is replaced with the parity bit. The coded sequence will be:

Even Parity bit, B1, B2, B3, B4, B5, B6 (MSB)

Even Parity Checker checks the number of Ones in the data stream and determines if a single bit error had occurred during transmission. PARITY ERROR LED gives the indication.

OBSERVATION TABLE

EVEN PARITY MODE

A/D CONVERTER	ERROR CODE GENERATOR	DATA LATCH	ERROR CORRECTION	PARITY ERROR
0 0 0 0 0 0 0	0 0 0 0 0 0 0	0 0 0 0 0 0 0	0 0 0 0 0 0 0	OFF
0 0 0 0 0 0 0	0 0 0 0 0 0 0	0 0 0 1 0 0 0	0 0 0 1 0 0 0	ON
1 1 1 1 1 1 1	0 1 1 1 1 1 1	0 1 1 1 1 1 1	0 1 1 1 1 1 1	OFF

We observe that the LSB of the A/D converter output is neglected in this mode of operations and the odd parity occupies the positions of LSB in transmission.

Also the LSB of the D/A Converter is always zeroing as in that position odd parity bit was transmitted.

Whenever the transmission of data is error free, the error check LED remains OFF.

Whenever a single bit error occurs the parity check LED remains ON indicating that a single bit error had occurred in transmission. Thus the Even Parity check is able to detect the errors but unable to locate or correct the errors.

PROCEDURE

- Ensure that group 3 (**GP3**) clock is selected in the clock generation section. Selection is done with the help of switch **S1**. Observe the corresponding LED indication.
- Keep the jumper **JP3** at the fast mode position.
- Connect **DC** signal having voltage around 5V from the Function generator section to the **ADC IN** post of the A/D converter.
- Keep switch **S12** in the **EVEN** parity mode and observe the parity indication.
- Observe the EVEN parity coded data on the corresponding LED indication in the Parity coded data section. The LSB i.e. **B0** of the data word from the ADC is replaced with the parity bit.
- Observe the error-coded data on the corresponding LED indication in the Error Code Generator section. There is no error introduced.
- After reception, observe the serial to parallel converted data on the corresponding LED indication in the Data Latch section. This is same as the transmitter data.

- Received parallel converted latched data is presented at input of D/A Converter & the same can be observed on the corresponding LED indication in the Error Correction section.
- Create a single bit fault in any one of the four MSB data bit i.e. **B3** to **B6** by switching on any single bit of switch **S11**.
- Observe the error-coded data on the corresponding LED indication in the Error Code Generator section.
- Observe the serial to parallel converted data on the corresponding LED indication in the Data Latch section.
- Observe the data bits on the corresponding LED indication in the Error Correction section.
- Observe the status of parity error LED in the Error detection /correction section.

OBSERVATIONS

- A/D Converter
- Parity Coded Data
- Error Code Generator
- Data Latch
- Error Correction
- Parity Error Bit

CONCLUSION

EVEN parity checks the even no. of ones in the data string and determines if a single bit error had occurred during transmission. Parity error LED gives the indication.

6 C. Fast Modes with Dc Signal (Odd Parity)

THEORY

ODD Parity Generator generates the coded sequence, which consists of 6 bits of data, plus one ODD parity Bit for transmission. The least significant bit (B0) of the data word from the ADC is replaced with the parity bit. The coded sequence will be: Odd Parity bit, B1, B2, B3, B4, B5, B6 (MSB)

Odd Parity Checker checks the number of Ones in the data stream and determines if a single bit error had occurred during transmission. PARITY ERROR LED gives the indication.

OBSERVATION TABLE

ODD PARITY MODE

A/D Convertor	Error Code Generator	Error Coded Data	Data Latch	D/A Convertor	Parity Error
0 0 0 0 0 0 0	1 0 0 0 0 0 0	1 0 0 0 0 0 0	1 0 0 0 0 0 0	0 0 0 0 0 0 0	OFF
0 0 0 0 0 0 0	1 0 0 0 0 0 0	1 0 0 1 0 0 0	1 0 0 1 0 0 0	0 0 0 1 0 0 0	ON
1 1 1 1 1 1 1	1 1 1 1 1 1 1	1 1 1 1 1 1 1	1 1 1 1 1 1 1	0 1 1 1 1 1 1	OFF

We observe that the LSB of the A/D converter output is neglected in this mode of operation and the odd parity occupies the positions of LSB in transmission.

Also, the LSB of the D/A Converter is always zeroing as noticed in that position, an odd parity bit was transmitted.

Whenever the transmission of data is error free, the error checks LED remains OFF.

Whenever a single bit error occurs, the parity check LED remains ON, indicating that a single bit error had occurred in transmission. Thus, the Odd Parity check is able to detect the errors but unable to locate or correct the errors.

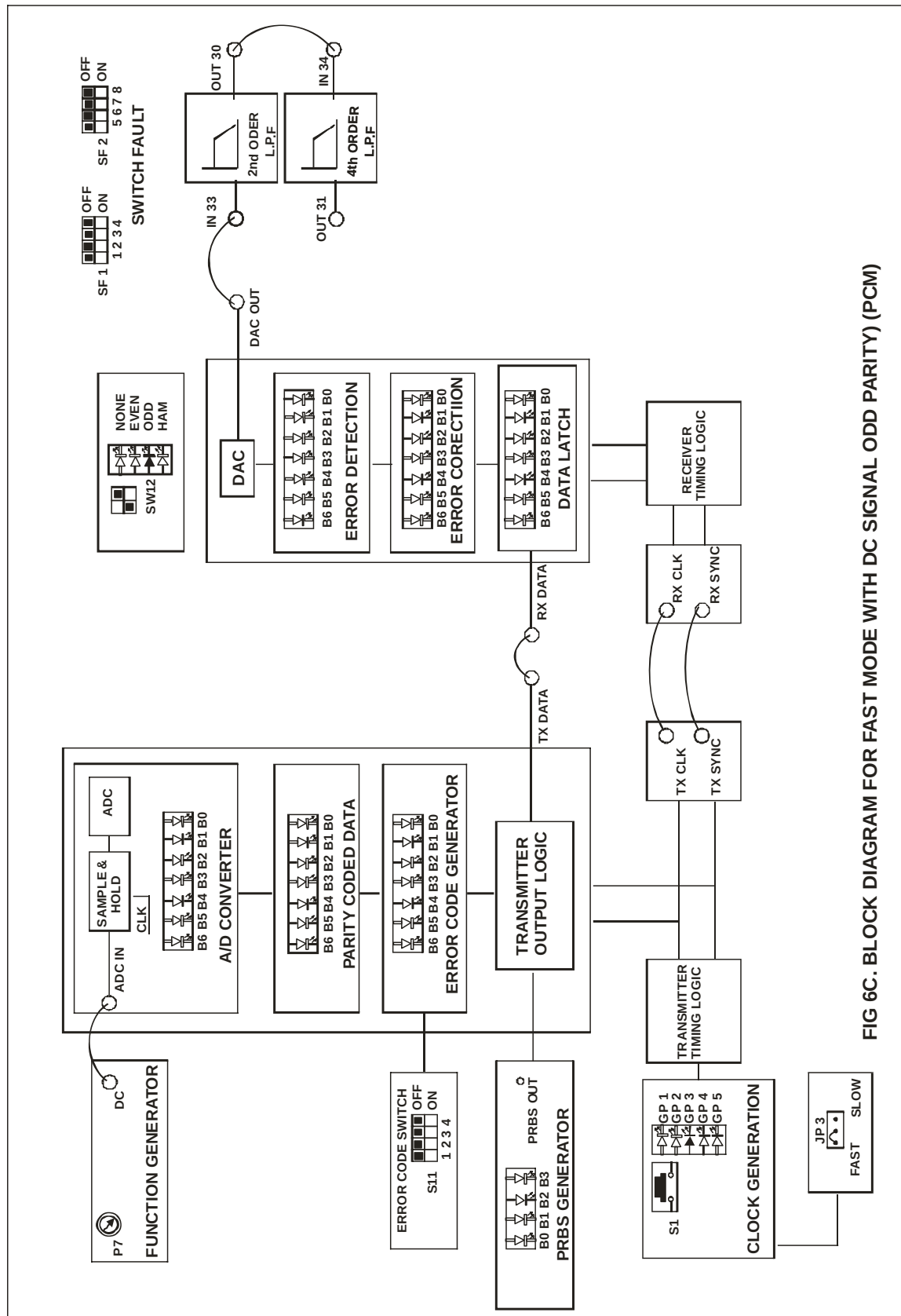


FIG 6C. BLOCK DIAGRAM FOR FAST MODE WITH DC SIGNAL ODD PARITY (PCM)

PROCEDURE

- Ensure that the group 3 (**GP3**) clock is selected in the clock generation section. Selection is done with the help of switch **S1**. Observe the corresponding LED indication.
- Keep the jumper **JP3** at the fast mode position.
- Connect **DC** signal having voltage around 5V from the Function generator section to the **ADC IN** post of the A/D converter.
- Keep switch **S12** in the **ODD** parity mode and observe the parity indication.
- Observe the ODD parity coded data on the corresponding LED indication in the Parity coded data section. The LSB i.e. **B0** of the data word from the ADC is replaced with the parity bit.
- Observe the error-coded data on the corresponding LED indication in the error code generator section. There is no error introduced.
- After reception, observe the serial to parallel converted data on the corresponding LED indication in the Data Latch section. This is same as the transmitter data.
- Received parallel converted latched data is presented at input of D/A Converter & the same can be observed on the corresponding LED indication in the Error Correction section.
- Create a single bit fault in any one of the four MSB data bit i.e. **B3** to **B6** by switching on any single bit of switch **S11**.
- Observe the error-coded data on the corresponding LED indication in the Error Code Generator section.
- Observe the serial to parallel converted data on the corresponding LED indication in the Data Latch section.
- Observe the data bits on the corresponding LED indication in the Error Correction section.
- Observe the status of parity error LED in the Error detection /correction section.

OBSERVATIONS

- A/D Converter
- Parity Coded Data
- Error Code Generator
- Data Latch
- Error Correction
- Parity Error Bit

CONCLUSION

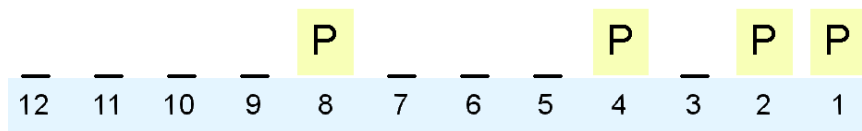
Odd parity checks the odd number of ones in the data string and determines if a single bit error had occurred during transmission. Parity error LED gives the indication.

6D. Fast Mode with DC Signal (Hamming Code)

THEORY

- Hamming codes are code words formed by adding redundant check bits, or parity bits, to a data word.
- The Hamming distance between two code words is the number of bits in which two code words differ.
- This pair of bytes has a Hamming distance of 3: 1 0 0 0 1 0 0 1
1 0 1 1 0 0 0 1
- The minimum Hamming distance for a code is the smallest Hamming distance between all pairs of words in the code.
- The minimum Hamming distance for a code, $D(\min)$ determines its error detecting and error correcting capability.
- For any code word, X , to be interpreted as a different valid code word, Y , at least $D(\min)$ single-bit errors must occur in X .
- Thus, to detect k (or fewer) single-bit errors, the code must have a Hamming distance of $D(\min) = k + 1$.
- Hamming codes can detect $D(\min) - 1$ errors and correct $\lfloor \{D(\min)-1\}/2 \rfloor$ errors.
- Thus, a Hamming distance of $2k + 1$ is required to be able to correct k errors in any data word.
- Hamming distance is provided by adding a suitable number of parity bits to a data word.
- Suppose we have a set of n -bit code words consisting of m data bits and r (redundant) parity bits.
- An error could occur in any of the n bits, so each code word can be associated with n erroneous words at a Hamming distance of 1.
- Therefore, we have $n + 1$ bit patterns for each code word: one valid code word, and n erroneous words.
- With n -bit code words, we have 2^n possible code words consisting of 2^m data bits (where $n = m + r$).
- This gives us the inequality: $(n + 1) \times 2^m \leq 2^n$
- Because $n = m + r$, we can rewrite the inequality as:
- $(m + r + 1) \times 2^m \leq 2^{m+r}$ or $(m + r + 1) \leq 2^r$
 - This inequality gives us a lower limit on the number of check bits that we need in our code words.
- Suppose we have data words of length $m = 4$. Then, $(4 + r + 1) \leq 2^r$
- Implies that r must be greater than or equal to 3.
- This means to build a code with 4-bit data words that will correct single-bit errors; we must add 3 check bits.
- Finding the number of check bits is a tough task. The rest is easy.
- Suppose we have data words of length $m = 8$. Then, $(8 + r + 1) \leq 2^r$
- implies that r must be greater than or equal to 4.
- This means to build a code with 8-bit data words that will correct single-bit errors; we must add 4 check bits, creating code words of length 12.
- So how do we assign values to these check bits?
- With code words of length 12, we observe that each of the digits, 1 to 12, can be expressed in powers of 2. Thus,
 - $1 = 2^0$ $5 = 2^2 + 2^0$ $9 = 2^3 + 2^0$
 - $2 = 2^1$ $6 = 2^2 + 2^1$ $10 = 2^3 + 2^1$

- $3 = 2^1 + 2^0$ $7 = 2^2 + 2^1 + 2^0$ $11 = 2^3 + 2^1 + 2^0$
- $4 = 2^2$ $8 = 2^3$ $12 = 2^3 + 2^2$
- 1 (= 2⁰) contributes to all of the odd-numbered digits.
- 2 (= 2¹) contributes to the digits, 2, 3, 6, 7, 10, and 11.
- And so forth.
- We can use this concept in the creation of our check bits.
- Using our code words of length 12, number each bit position starting with 1 in the low-order bit.
- Each bit position corresponding to an even power of 2 will be occupied by a check bit.
- These check bits contain the parity of each bit position for which it participates in the sum.



A/D CONVERTER	PARITY CODED DATA	ERROR CODED DATA	DATA LATCH	ERROR CORRECTION	ERROR DETECTION
* * * 0 0 0 0	0 0 0 0 0 0 0	0 0 0 0 0 0 0	0 0 0 0 0 0 0	0 0 0 0 0 0 0	0 0 0 0 0 0 0
* * * 0 0 0 0	0 0 0 0 0 0 0	0 0 0 1 0 0 0	0 0 0 1 0 0 0	0 0 0 0 0 0 0	0 0 0 1 0 0 0
* * * 1 0 0 0	1 1 0 1 0 0 0	1 1 0 1 0 0 0	1 1 0 1 0 0 0	0 0 0 1 0 0 0	0 0 0 0 0 0 0
* * * 0 1 0 0	1 0 1 0 1 0 0	1 0 1 0 1 0 0	1 0 1 0 1 0 0	0 0 0 0 1 0 0	0 0 0 0 0 0 0
* * * 1 1 0 0	0 1 1 1 1 0 0	0 1 1 1 1 0 0	0 1 1 1 1 0 0	0 0 0 1 1 0 0	0 0 0 0 0 0 0
* * * 0 0 1 0	0 1 1 0 0 1 0	0 1 1 0 0 1 0	0 1 1 0 0 1 0	0 0 0 0 0 1 0	0 0 0 0 0 0 0
* * * 1 0 1 0	1 0 1 1 0 1 0	1 0 1 1 0 1 0	1 0 1 1 0 1 0	0 0 0 1 0 1 0	0 0 0 0 0 0 0
* * * 0 1 1 0	1 1 0 0 1 1 0	1 1 0 0 1 1 0	1 1 0 0 1 1 0	0 0 0 0 1 1 0	0 0 0 0 0 0 0
* * * 1 1 1 0	0 0 0 1 1 1 0	0 0 0 1 1 1 0	0 0 0 1 1 1 0	0 0 0 1 1 1 0	0 0 0 0 0 0 0
* * * 0 0 0 1	1 1 1 0 0 0 1	1 1 1 0 0 0 1	1 1 1 0 0 0 1	0 0 0 0 0 0 1	0 0 0 0 0 0 0
* * * 1 0 0 1	0 0 1 1 0 0 1	0 0 1 1 0 0 1	0 0 1 1 0 0 1	0 0 0 1 0 0 1	0 0 0 0 0 0 0
* * * 0 1 0 1	0 1 0 0 1 0 1	0 1 0 0 1 0 1	0 1 0 0 1 0 1	0 0 0 0 1 0 1	0 0 0 0 0 0 0
* * * 1 1 0 1	1 0 0 1 1 0 1	1 0 0 1 1 0 1	1 0 0 1 1 0 1	0 0 0 1 1 0 1	0 0 0 0 0 0 0
* * * 0 0 1 1	1 0 0 0 0 1 1	1 0 0 0 0 1 1	1 0 0 0 0 1 1	0 0 0 0 0 1 1	0 0 0 0 0 0 0
* * * 1 0 1 1	0 1 0 1 0 1 1	0 1 0 1 0 1 1	0 1 0 1 0 1 1	0 0 0 1 0 1 1	0 0 0 0 0 0 0
* * * 0 1 1 1	0 0 1 0 1 1 1	0 0 1 0 1 1 1	0 0 1 0 1 1 1	0 0 0 0 1 1 1	0 0 0 0 0 0 0
* * * 1 1 1 1	1 1 1 1 1 1 1	1 1 1 1 1 1 1	1 1 1 1 1 1 1	0 0 0 1 1 1 1	0 0 0 0 0 0 0

We observe that the three LSB data bit of the A/D converter output is neglected in this mode of operations and three Hamming parity bit occupies the positions in transmission.

Also, four MSB bit of the D/A converter forms the data and three LSB bit are always zero as in that position Hamming parity bit were transmitted.

Whenever the transmission of data is error free, all the LED of Error Detection/ Correction logic remains OFF.

Whenever a single bit error occurs, the corresponding bit position is indicated by LED of Error Detection/ Correction Logic and corrected data bit are observed at the input of D/A Converter which are same as the A/D Converter output (4 MSB). Thus, in Hamming Parity mode, single bit error is detected as well as corrected.

PROCEDURE

- Ensure that the group 3 (**GP3**) clock is selected in the clock generation section. Selection is done with the help of switch **S1**. Observe the corresponding LED indication.
- Keep the jumper **JP3** at the fast mode position.
- Connect **DC** signal having voltage around 5V from the Function generator section to the **ADC IN** post of the A/D converter.
- Keep switch **S12** in the **HAMMING** code mode and observe the corresponding LED indication.
- Observe the HAMMING coded data on the corresponding LED indication in the Parity coded data section. Hamming code generator generates the coded sequence, which consists of four bits of data and 3 bits of error check bits. The three LSB i.e. **B0**, **B1**, **B2** of the data word from the ADC are neglected and error check bits are placed in that position.
- Observe the error-coded data on the corresponding LED indication in the Error Code Generator section. There is no error introduced.
- Observe the serial to parallel converted data on the corresponding LED indication in the Data Latch section.
- Observe the data bits on the corresponding LED indication in the Error Correction section.
- Create a single bit fault in any one of the four MSB data bit i.e. **B3** to **B6** by switching on the any single bit of switch **S11**.
- Observe the error-coded data on the corresponding LED indication in the Error Code Generator section. The same is reflected in the Data Latch section.
- Observe the corrected data in the Error Correction section. The error bit is indicated in the Error Detection section.

OBSERVATIONS

- A/D Converter
- Parity Coded Data
- Error Code Generator
- Data Latch
- Error Correction
- Error Detection

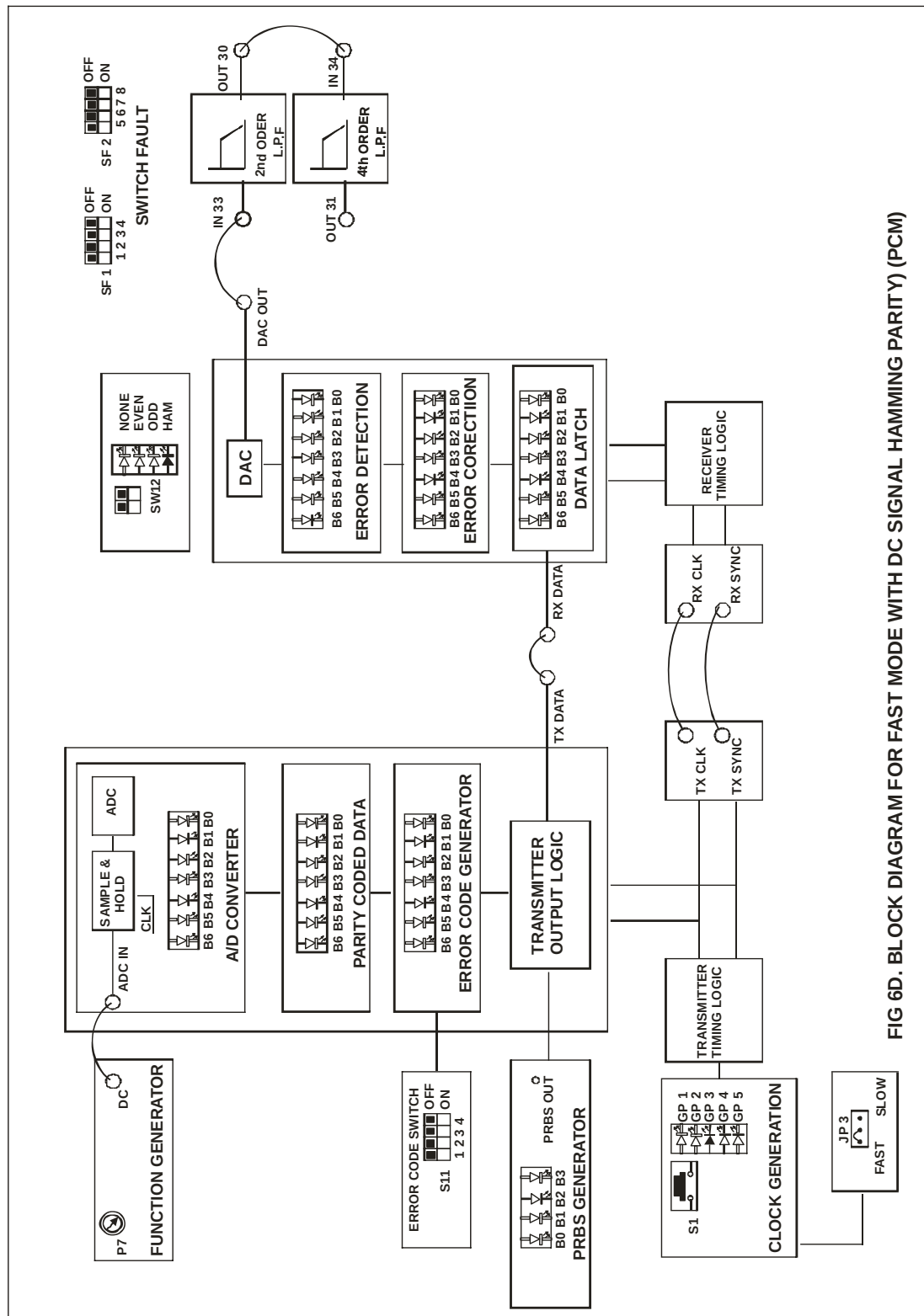


FIG 6D. BLOCK DIAGRAM FOR FAST MODE WITH DC SIGNAL HAMMING PARITY (PCM)

CONCLUSION

Hamming error detection/correction logic computes parity check bits at the receiver and locates the error bit position. Once the error bit position is located, the error bit is corrected by bit reversal.

**EXPERIMENT
NO. 7**

EXPERIMENT NO.7

NAME

Study of PRBS

OBJECTIVES

SLOW MODE WITH DC SIGNAL (NONE PARITY)

It is used to observe the PRBS sequence generation.

THEORY

Pseudo Random Bit Sequences are widely used for Frame Synchronization purpose. They form the header of the transmitted sequence. At the Receiver, these sequences are detected by a detector logic, which recovers the Synchronization signal at the receiver.

Pseudo Random Bit sequence, popularly known as Pseudo Random Code Signals are actually deterministic signals. But, they are called Pseudo Random Signals, because they have the statistical properties of Random Signals.

Example of Random Signal: Sample signal of white noise.

The Pseudo Random Signal has got an advantage that it has minimum correlation with any data signal, so that the false locking of the detector with data is avoided.

For a signal to be Pseudo Random in nature, it should satisfy the following properties of Randomness:

01) Balance Properties:

The given signal is said to possess good balance property, if the number of binary Ones differ from the number of binary Zeroes, by almost one digit in one period.

02) Run Property:

The Run is defined as a sequence of single type of binary digits (either binary Ones or binary Zeroes). The given sequence is said to satisfy the RUN property, if it satisfies the condition mentioned below:

Among the Runs of Ones and Zeroes in each period, it is desirable that about one half the Runs of each type are of length 1, about one fourth are of length 2 and one eighth are of length 3, and so on.

03) Correlation property:

If period of the sequence is compared term by term with any cyclic shift of itself, it is best if the number of agreements differs from the number of disagreements by not more than 1.

PRBS sequence is generated by using a 4-bit right shift register whose feedback is completed by the EX-OR gate.

Initially, let 0001 be the stored nibble on the 4 bit shift registers:

Clock States	B3	B2	B1	B0	
			A	B	C
0	0	0	0	1	1
1	1	0	0	0	0
2	0	1	0	0	0
3	0	0	1	0	1
4	1	0	0	1	1
5	1	1	0	0	0
6	0	1	1	0	1
7	1	0	1	1	0
8	0	1	0	1	1
9	1	0	1	0	1
10	1	1	0	1	1
11	1	1	1	0	1
12	1	1	1	1	0
13	0	1	1	1	0
14	0	0	1	1	0
15	0	0	0	1	1
16	1	0	0	0	0

Thus, the sequence repeats constantly with a period corresponding to 15 clock states.
Length of sequence = $2^4 - 1 = 15$

Now the pattern is 100010011010111

Check for Balance Property:

Number of ones in the sequence = 8

Number of zeros in the sequence = 7

Difference = 1

Hence, the sequence is verified to have good balance property.

Check for Run Property:

Number of zero runs = 4

Number of runs of length 1 = 2 (half of 4)

Number of runs of length 2 = 1 (one fourth of 4)

The sequence is too short to go for further verifications like one eighth.

Thus, the sequence is verified for run property with all ZERO runs. Similar verification can be done for the ONE run.

Check for Correlation Property:

Pseudo Random Sequence = 1 0 0 0 1 0 0 1 1 0 1 0 1 1 1

RRC Shifted Sequence = 1 1 0 0 0 1 0 0 1 1 0 1 0 1 1

a d a a d d a d a d d d d a a

d = represents disagreements

a = represents agreements

RRC = Rotate Right with Carry

We observe the following:

Number of agreements = 7

Number of disagreements = 8

Difference = 1

Therefore, the sequence satisfies correlation property also.

Thus, the Pseudo Random Properties are verified for the Pseudo Random Sequence.

Note: Since the transmitted data contains only 14 bits of data, the generated PRBS is also restricted to have a 14-bit length. The last bit is truncated.

EQUIPMENTS

DCS-B kit
Connecting Chords
Power supply
20 MHz Dual Trace Oscilloscope
Power connection cables

SLOW MODE WITH DC SIGNAL (NONE PARITY)

PROCEDURE

- Ensure that the group 3 (**GP3**) clock is selected in the clock generation section. Selection is done with the help of switch **S1**. Observe the corresponding LED indication.
- Keep the jumper **JP3** at the slow mode position.
- Ensure that the group 3 (**GP3**) clock is selected in the clock generation section. Selection is done with the help of switch **S1**, observe the corresponding LED indication.
- Connect **DC** signal having voltage around 5V from the Function generator section to the **ADC IN** post of the A/D converter.
- Keep switch **S12** in the **NONE** parity mode and observe the corresponding LED indication.
- Observe the none-parity coded data on the corresponding LED indication in the Parity coded data section.
- Observe the flow of the pseudorandom sequence on the corresponding LED's in the PRBS section.

OBSERVATIONS

Observe the flow of the Pseudo Random Sequence in LEDs **B0**, **B1**, **B2**, **B3**. The flow in LEDs is observed to follow the sequence. Here also, the student is advised to verify the truth table. The Pseudo Random Sequence is observed for 14 clock states and the LED remains OFF for another 14 clock states. Whenever the sync remains high (LED off), we can observe the shifting operations visually in LEDs.

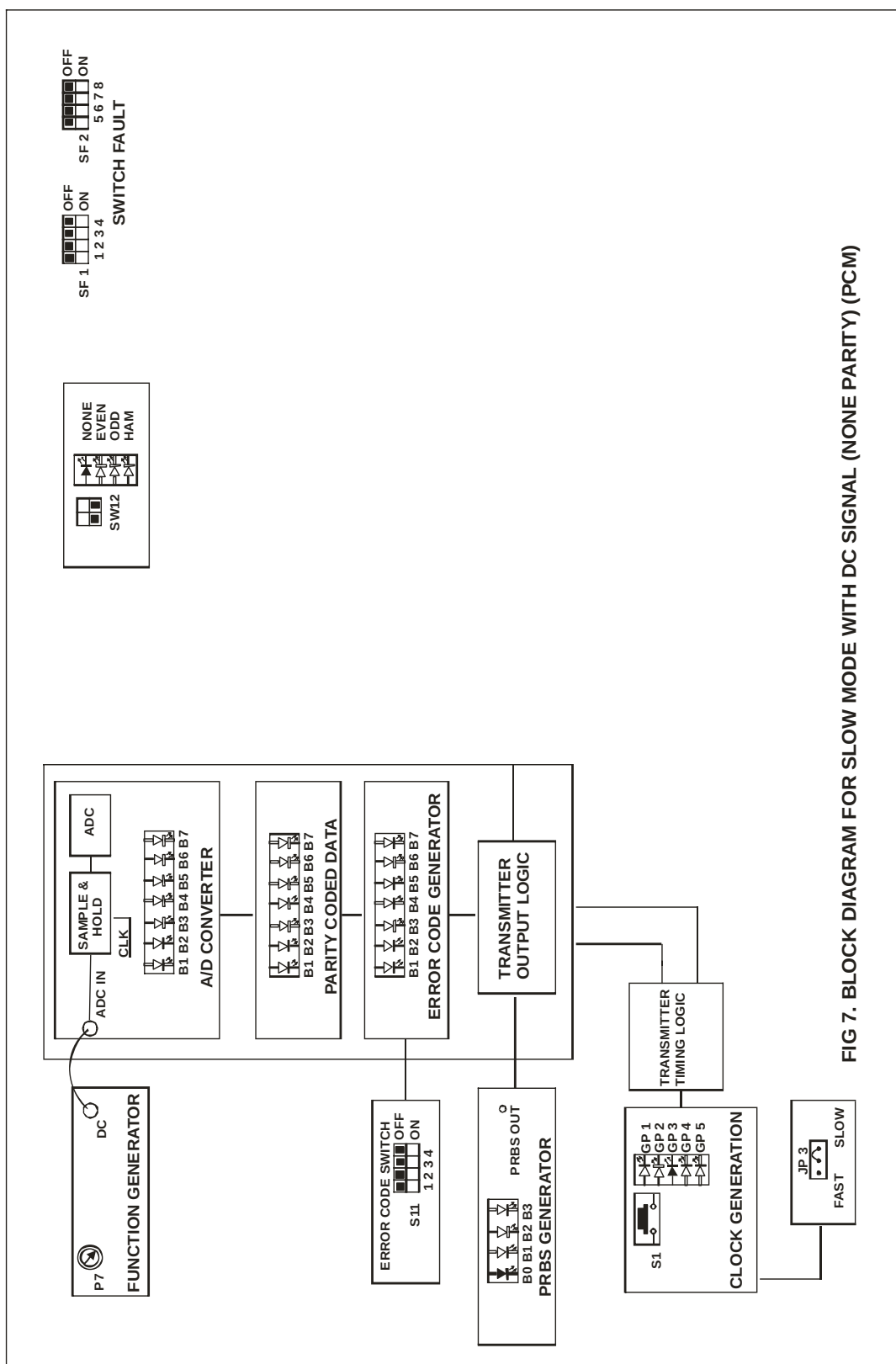


FIG 7. BLOCK DIAGRAM FOR SLOW MODE WITH DC SIGNAL (NONE PARITY) (PCM)

CONCLUSION

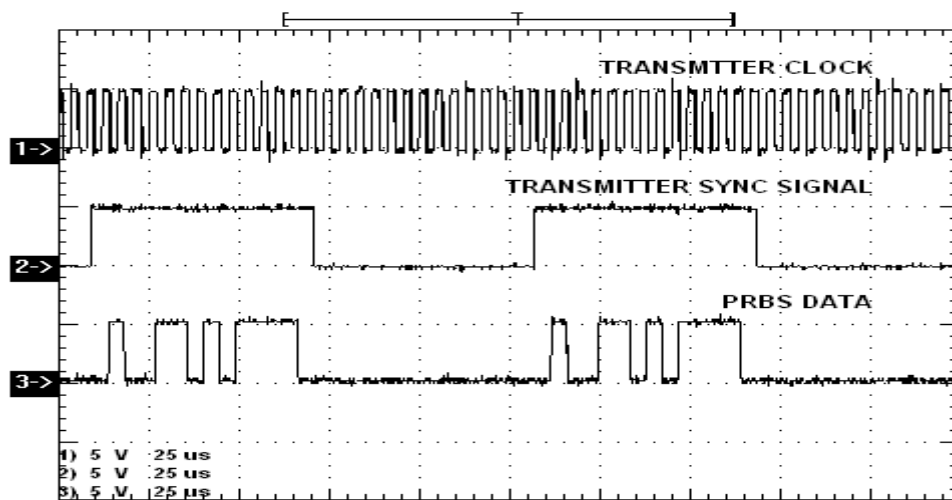
The flow in LEDs is observed to follow the sequence. Here also the student is advised to verify the truth table. The Pseudo Random Sequence is observed for 14 clock states and the LED remains OFF for another 14 clock states. Whenever the sync remains high (LED off), we can observe the shifting operations visually in LEDs

TRUTH TABLE

THEORETICAL AND PRACTICLE

Clock	B3	B2	B1	B0
0	0	0	0	1
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0
4	1	0	0	1
5	1	1	0	0
6	0	1	1	0
7	1	0	1	1
8	0	1	0	1
9	1	0	1	0
10	1	1	0	1
11	1	1	1	0
12	1	1	1	1
13	0	1	1	1
14	0	0	1	1
15	0	0	0	1

WAVEFORM



**EXPERIMENT
NO. 8**

EXPERIMENT NO.8

NAME

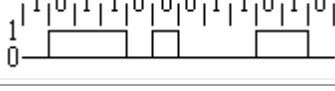
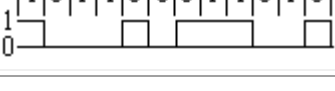
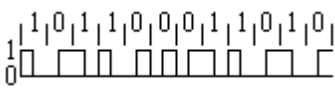
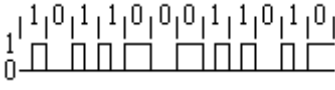
Study of Various Data Encoding and Decoding Techniques

OBJECTIVES

To study the NRZ-L, NRZ-M, NRZ-S, URZ, BIO-L, BIO-M, BIO-S DATA coding & decoding.

THEORY

ENCODING SCHEMES:

Code Name	Binary Code	Code Definition
NRZ-L		Non-Return-to-Zero Level "One" is represented by one level "Zero" is represented by another level lower the one but not zero.
NRZ-M		Non-Return-to-Zero Mark "One" is represented by a change in level "Zero" is represented by no change in level.
NRZ-S		Non-Return-to-Zero Space "One" is represented by no change in level "Zero" is represented by change in level.
NRZ-I		Non-Return-to-Zero Inverse "One" is represented by no change in level "Zero" is represented by change in level.
Bi-Phase-L		Bi-Phase Level (Split Phase) Level change occurs at the beginning of every bit period "One" is represented by a "One" level with transition to the "Zero" level "Zero" is represented by a "Zero" level with transition to the "One" level.
Bi-Phase-M		Bi-Phase Mark Level change occurs at the beginning of every bit period "One" is represented by a midbit level change "Zero" is represented by no midbit level change.
Bi-Phase-S		Bi-Phase Space Level change occurs at the beginning of every bit period "One" is represented by no midbit level change "Zero" is represented by a midbit level change.

Once the carrier is recovered, the data can be detected by comparing the phase of the received modulated carrier with the phase of the reference carrier.

NON - RETURN TO ZERO signal are the easiest formats that can be generated. These signals do not return to zero with the clock. The frequency component associated with these signals are half that of the clock frequency. The following data formats come under this category. Non-return to zero encoding is commonly used in slow speed communications

interfaces for both synchronous and asynchronous transmission. Using NRZ, logic 1 bit is sent as a high value and logic 0 bit is sent as a low value.

- A) Non-Return to zero - LEVEL - NRZ-L
- B) Non-Return to zero - MARK - NRZ-M
- C) Non-Return to zero - SPACE - NRZ-S

A) NON-RETURN TO ZERO - LEVEL (NRZ-L):

This is the most extensively used waveform in digital logics. The data format is very simple where all 'ones' are represented by 'high' and all 'zeros' by 'low'. The data format is directly available at the output of all digital data generation logics and hence very easy to generate. Here, all the transitions take place at the rising edge of the clock.

B) NON-RETURN TO ZERO - MARK (NRZ-M):

These waveforms are extensively used in magnetic tape recording. In this data format, all 'ones' are marked by change in levels and all 'zeros' by no transitions and the transitions take place at the rising edge of the clock.

C) NON-RETURN TO ZERO - SPACE (NRZ-S),

This type of waveform is marked by change in levels for 'zeros' and no transition for 'ones' and the transitions take place at the rising edge of the clock. This format is also used in magnetic tape recording.

D) UNIPOLAR AND BIPOLAR:

Unipolar signals are those signals, which have transition between 0 to +VCC. Bipolar signals are those signals, which have transition between +VCC to -VCC.

THE CODING OF BASIC DATA NRZ-L INTO NRZ-M AND NRZ-S FORMAT CAN BE UNDERSTOOD BY REFERING TO THE CIRCUIT DESCRIPTION MANUAL AND THE WAVEFORMS.

For the decoding the NRZ coded data, firstly the clock is recovered from the incoming coded data by using Phase Locked Loop techniques on kit DCL-06. Waveforms show the data stream and clock using a typical NRZ-L protocol. In this data format, 'ones' are represented by highs and 'zeros' are represented by lows. In the data stream, it should be guaranteed that continuously more than 5 zeros or 5 ones should not occur in succession while transmitting data.

This ensures that the slowest rate of changes will then be a frequency corresponding to one-twelfth clock. The fastest rate of change of frequency will correspond to one-half of the clock frequency with alternate one and zeroes occurring in the data stream.

This phase -encoded -group consists of:

- a) Biphasic - Level (Popularly known as Manchester Coding)
- b) Biphasic - Mark
- c) Biphasic - Space

These schemes are used in magnetic recording, optical communications and in satellite links. These phase encoded signals are special because they are composed of both the in phase and out of phase components of the clock.

A) BIPHASE-LINE CODING (BIPHASE-L) - (MANCHESTER CODING):

With the Biphase-L, 'one' is represented by a half bit wide pulse, positioned during the first half of the bit interval and a 'zero' is represented by a half bit wide pulse positioned during the second half of the bit interval.

B) BIPHASE MARK CODING (BIPHASE-M):

With the Biphase-M, a transition occurs at the beginning of every bit interval. A 'one' is represented by a second transition, half bit later, whereas a zero has no second transition.

C) BIPHASE SPACE CODING (BIPHASE-S):

With a Biphase-S, a transition occurs at the beginning of every bit interval. A 'zero' is marked by a second transition, one half bit later; 'one' has no second transition.

The coding of basic data NRZ-L into Biphase-L, Biphase-M, and Biphase-S format can be understood easily by referring to circuit diagram and Waveforms.

Biphase-Line Coding is a synchronous clock coding technique used by the OSI physical layer (in LAN) to encode the clock and data of a synchronous bit stream. In this technique, the actual binary data to be transmitted over the cable are not sent as a sequence of logic 1's and 0's (known technically as Non Return to Zero (NRZ)). Instead, the bits are translated into a slightly different format that has a number of advantages over the use of straight binary encoding (i.e. NRZ).

Biphase-Line Coding follows the rules shown below:

Original	Data Value Sent
Logic 0	0 to 1 (upward transition at bit centre)
Logic 1	1 to 0 (downward transition at bit centre)

Example of Biphase-Line Coding:

The pattern of bits " 0 1 1 1 0 0 1 " encodes to " 01 10 10 10 10 01 01 10".

Another more curious example is the pattern " 1 0 1 0 1 etc" which encodes to "10 01 10 01 10 " which could also be viewed as "1 00 11 00 11 0 ". Thus, for a 10 Mbps Ethernet LAN, the preamble sequence encodes to a 10 MHz square wave.

The recovery circuit of Biphase coded signal is quite simple as compared to NRZ or URZ signals because the bi-phase signals have one transition per cycle of the clock, irrespective of the transmitted data. So these signals ensure sufficient number of transitions in the data stream. But, one disadvantage in bi-phase signals is that they do not provide phase continuity in the transmitted data stream.

A) RETURN TO ZERO SIGNALS:

These signals are called "Return to Zero signals", since they return to 'zero' with the clock. In this category, there is only one data format, i.e., the unipolar return to zero (URZ).

In the URZ a 'one' is represented by a half bit wide pulse and a 'zero' is represented by the absence of a pulse.

EQUIPMENTS

DCS-B kit
 Connecting Chords
 Power supply
 20 MHz Dual Trace Oscilloscope
 Power connection cables

8A. NRZ-L, NRZ-M, NRZ-S, URZ, BIO-L, BIO-M, BIO-S CODING & DECODING

PROCEDURE

- Ensure that the group 4 (**GP4**) clock is selected in the Clock Generation section. Selection is done with the help of switch **S1**. Observe the corresponding LED indication.
- Observe the transmitter clock of frequency **250 KHz** at **TXCLK** post.
- Set the data pattern using switch **S4** as shown in the block diagram.
- Observe the 8-bit data pattern at **S DATA** post.
- Connect **S DATA** to **IN16** post and **TXCLK** to **CLK2** post of the Encoded Data section.
- Observe the encoded data at the **OUT10** post of the Encoded Data section. Selection of the different encoded data's are done using switch **S3**. The selected encoded data is indicated on the corresponding LED indication in the Encoded Data section.
- Connect **OUT10** to **IN27** post of the Decoded Data section.
- Observe the recovered clock at **REC.CLK2** test point of the Decoded Data section.
- Observe the decoded data at **OUT23** post of the Decoded Data section.
- We can observe the decoded data as per the selected encoded data.

OBSERVATIONS

Transmitter Clock	Txclk
Simulated Data	Sdata
Encoded Data	Out 10
Ami Encoded Data	Out 11
Recovered Clock	Rec.Clk2
Decoded Data	Out 23
Ami Decoded Data	Out 22
Bipolar Data	Out 6
Unipolar Data	Out 7

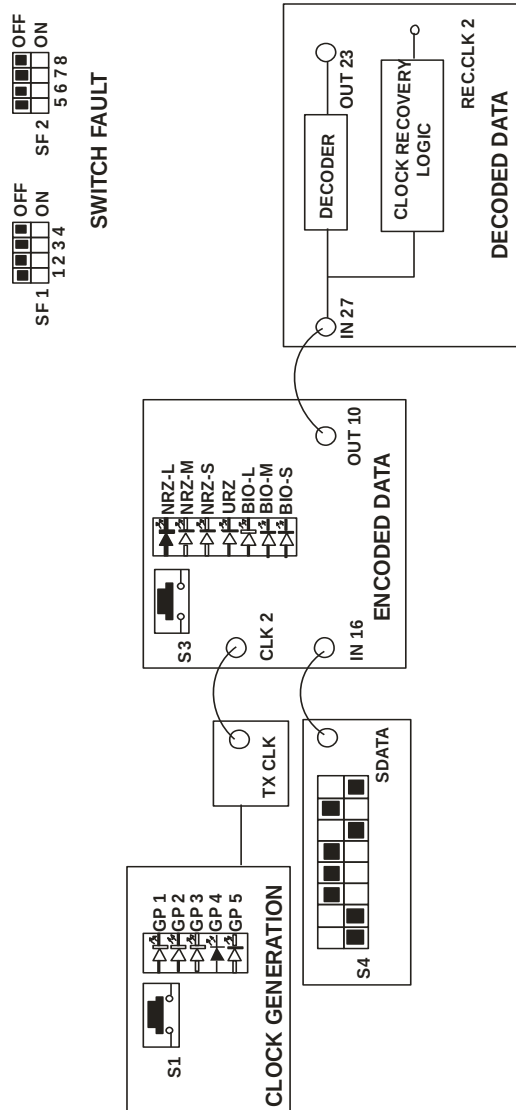
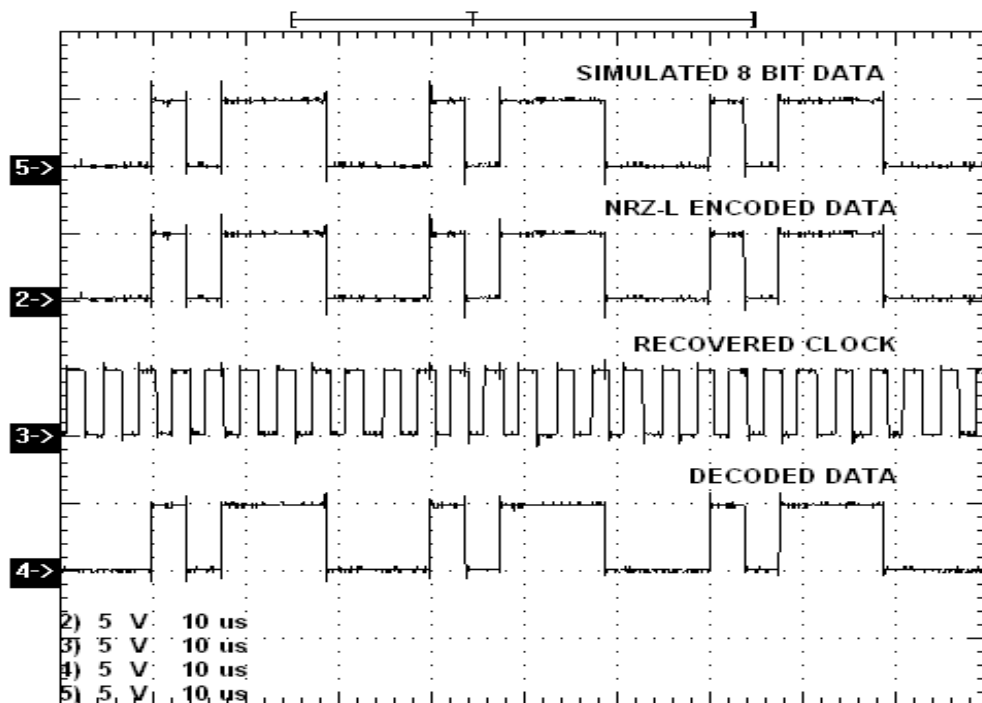
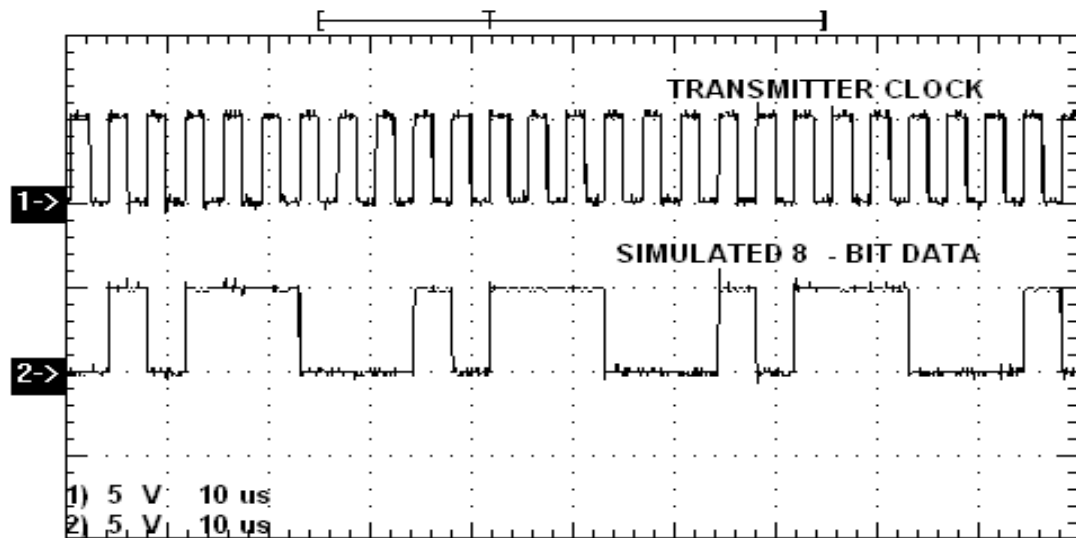


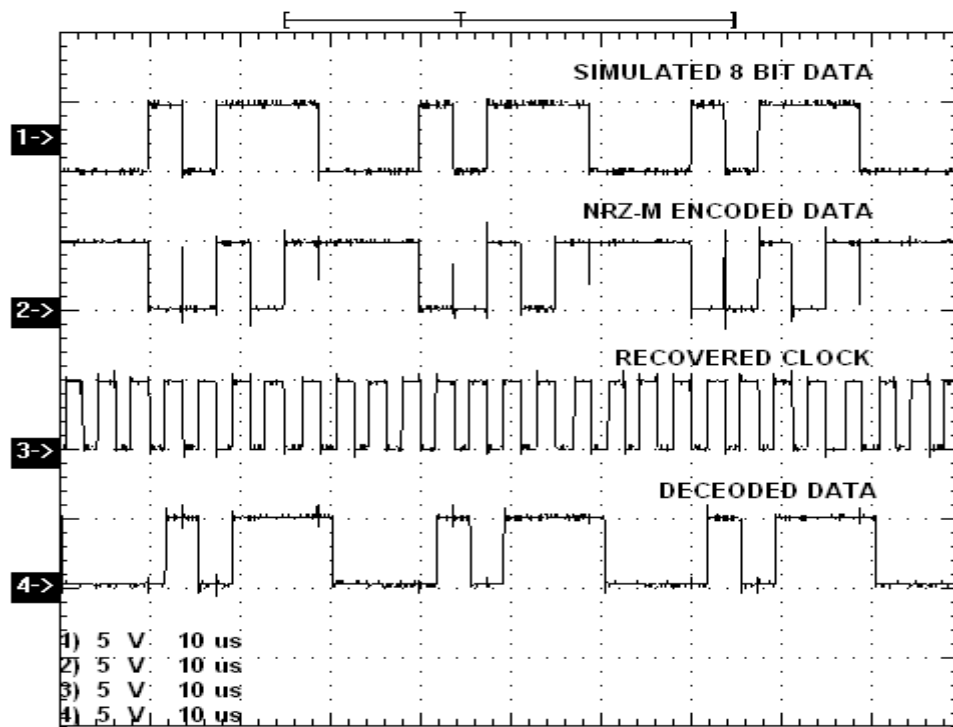
FIG 8A.BLOCK DIAGRAM FOR DATA CODING AND DECODING TECHNIQUES (NRZ-L, NRZ-M, NRZ-S, URZ, BIO-L, BIO-M, BIO-S)

CONCLUSION

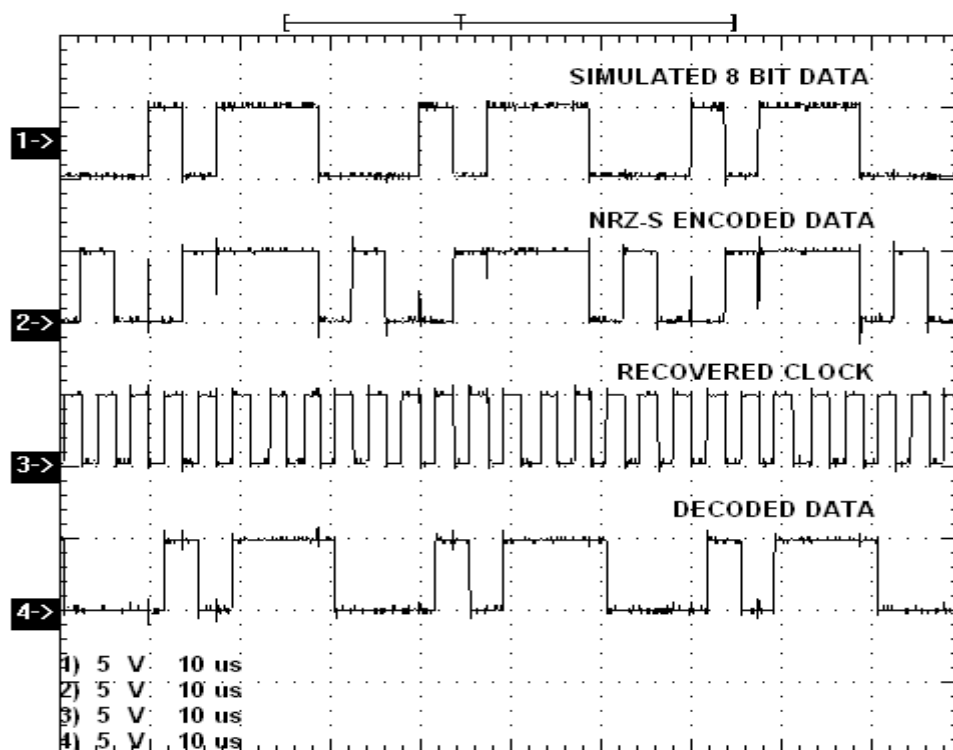
The NRZ-L, NRZ-M, NRZ-S, URZ, BIO-L, BIO-M, BIO-S coded signals have a frequency component, that is equal to half the frequency of the coding (reference) clock. The recovered data and clock is observed to have a very small phase lag with respect to the transmitted data and clock.



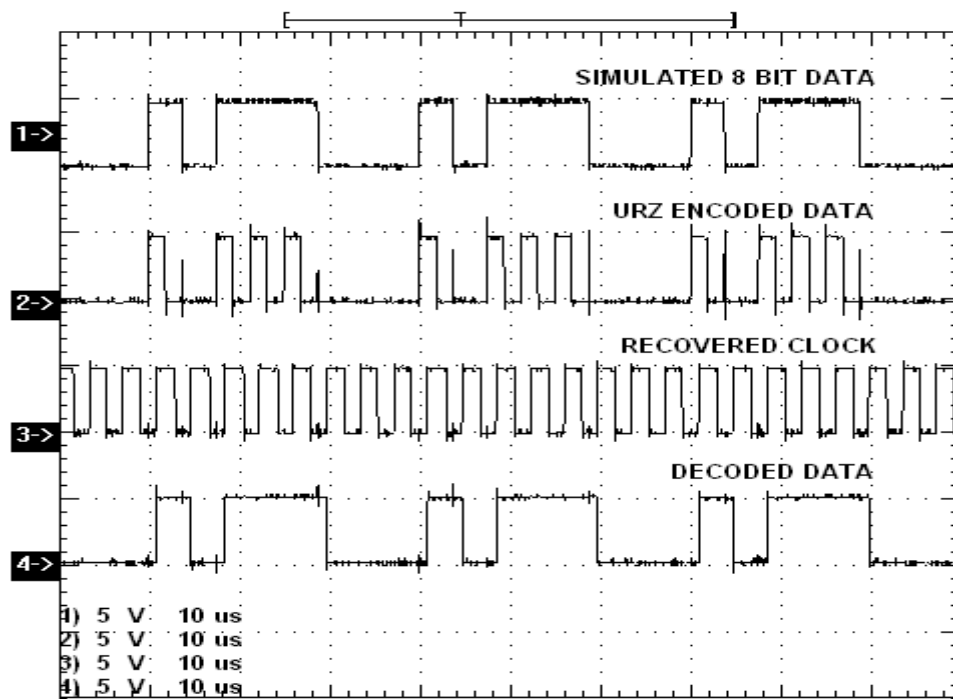
NRZ-L Encoding And Decoding



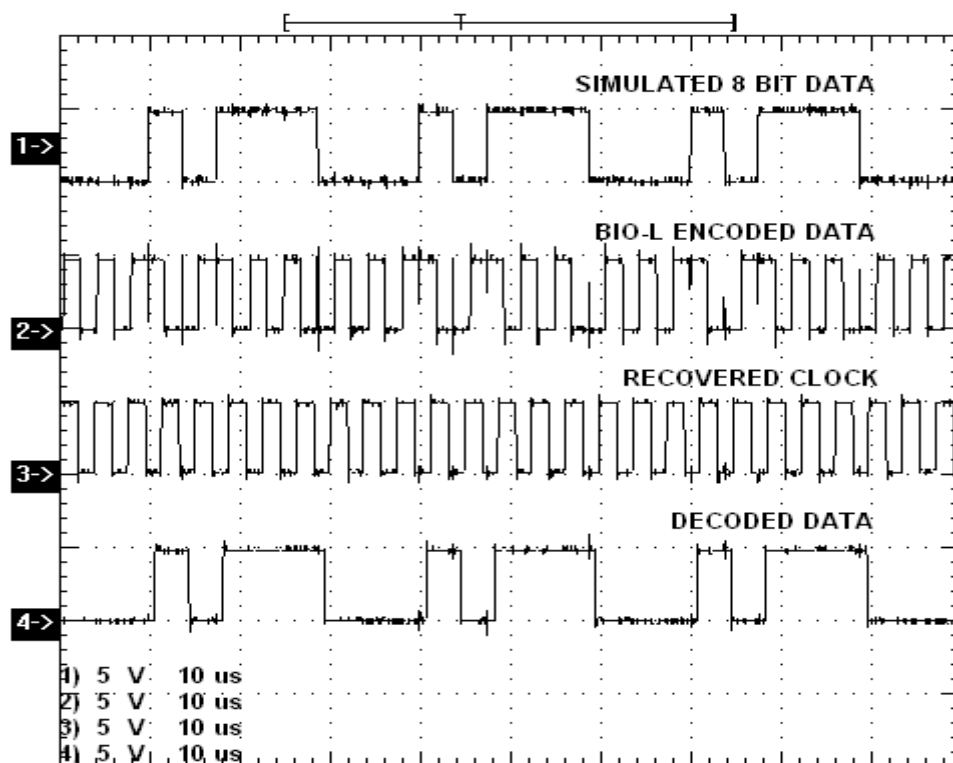
NRZ-M Encoding And Decoding



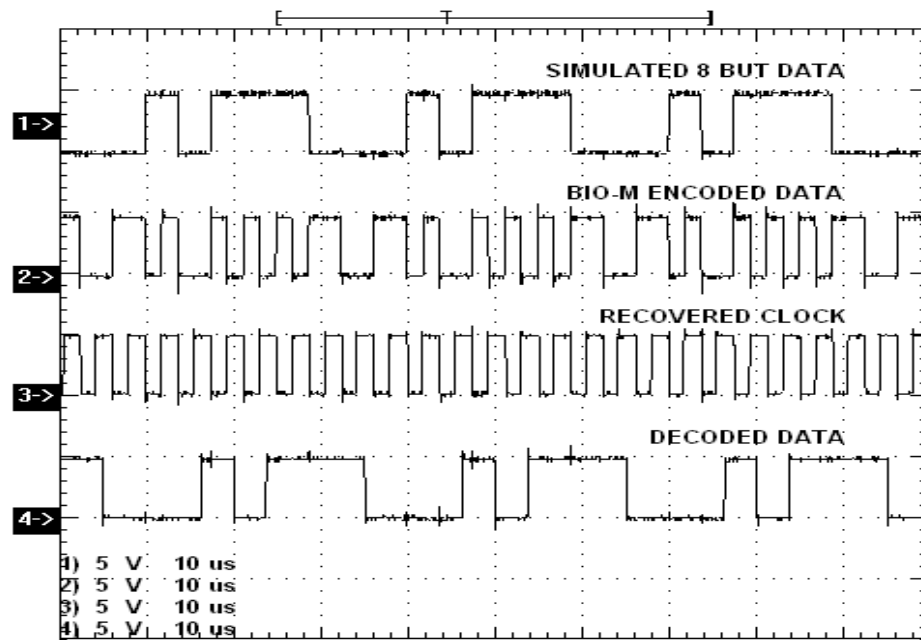
NRZ-S Encoding And Decoding



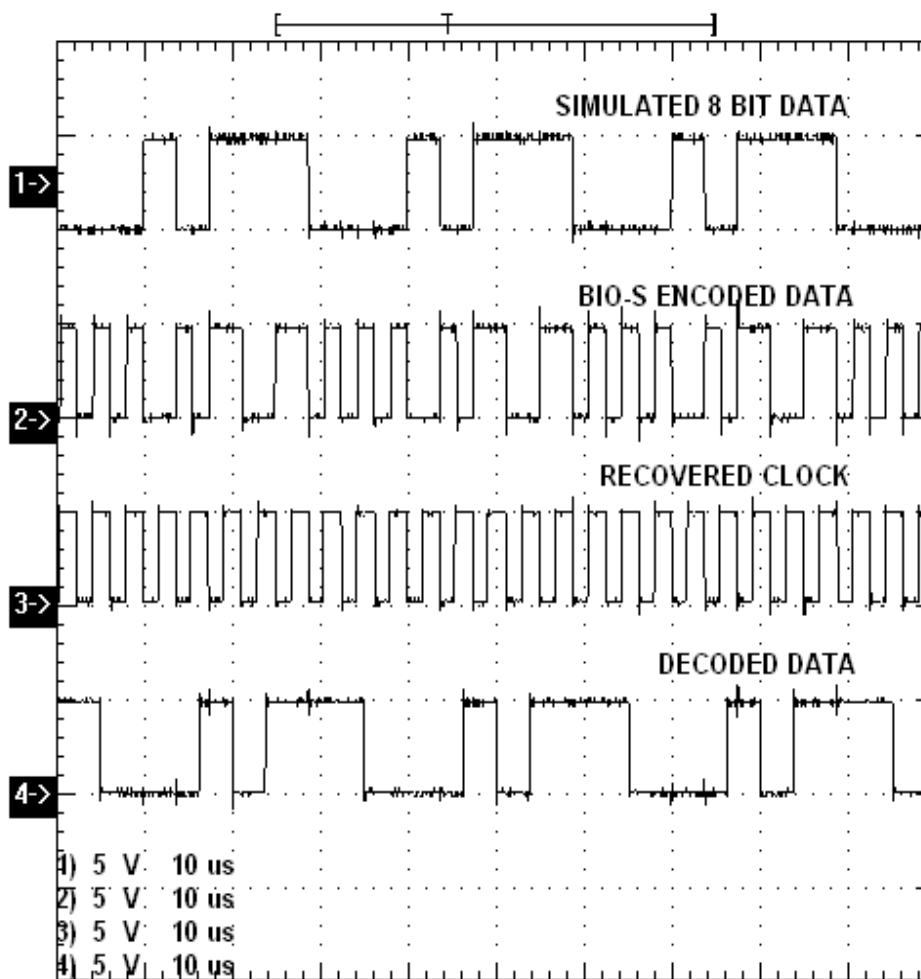
URZ Encoding And Decoding



BIO-L Encoding And Decoding



BIO-M Encoding And Decoding



BIO-S Encoding And Decoding

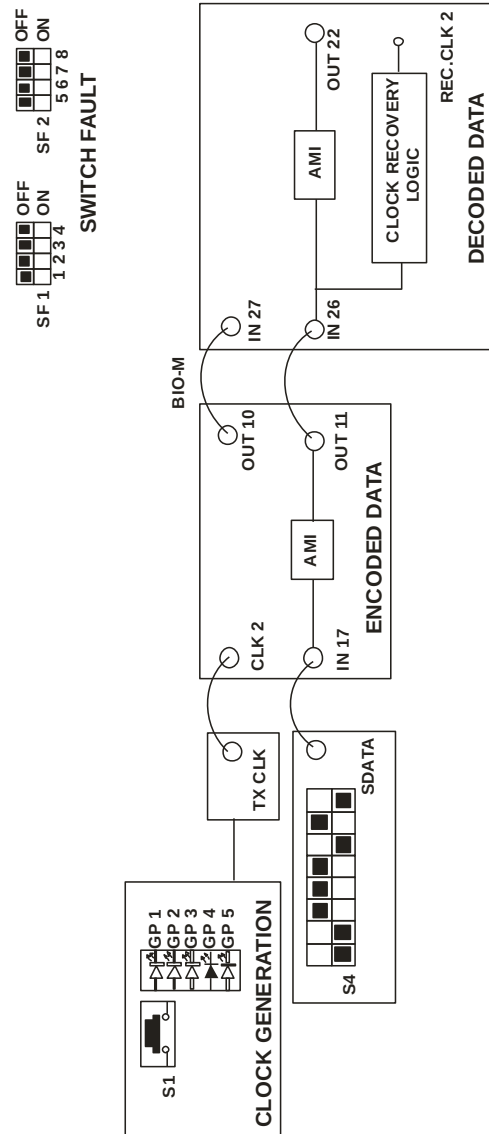


FIG 8B.BLOCK DIAGRAM FOR DATA CODING AND DECODING TECHNIQUES (AMI)

8B. Ami Encoding and Decoding

THEORY

Multilevel Signals

Multilevel signals use three or more levels of voltages to represent the binary digits, 'one' and 'zero' - instead of the normal 'highs' and 'lows'

Return to zero - Alternative Mark Inversion (RZ - AMI) is the most commonly used multilevel signal. This coding scheme is most often used in telemetry systems. In this scheme, 'ones' are represented by equal amplitude of alternating pulses, which alternates between +5V and -5V. These alternating pulses return to 0 volt, after every half bit interval. The 'zeros' are marked by the absence of pulses.

The coding of basic data NRZ-L into URZ and RZ-AMI format can be understood easily by referring to circuit diagram and Waveforms.

For decoding the URZ coded data, first of all, the clock is recovered from the incoming coded data by using phase locked loop techniques. Since the URZ signal has a frequency component equal to the clock, the bi-directional monoshot and / OR the gate should be eliminated in this application.

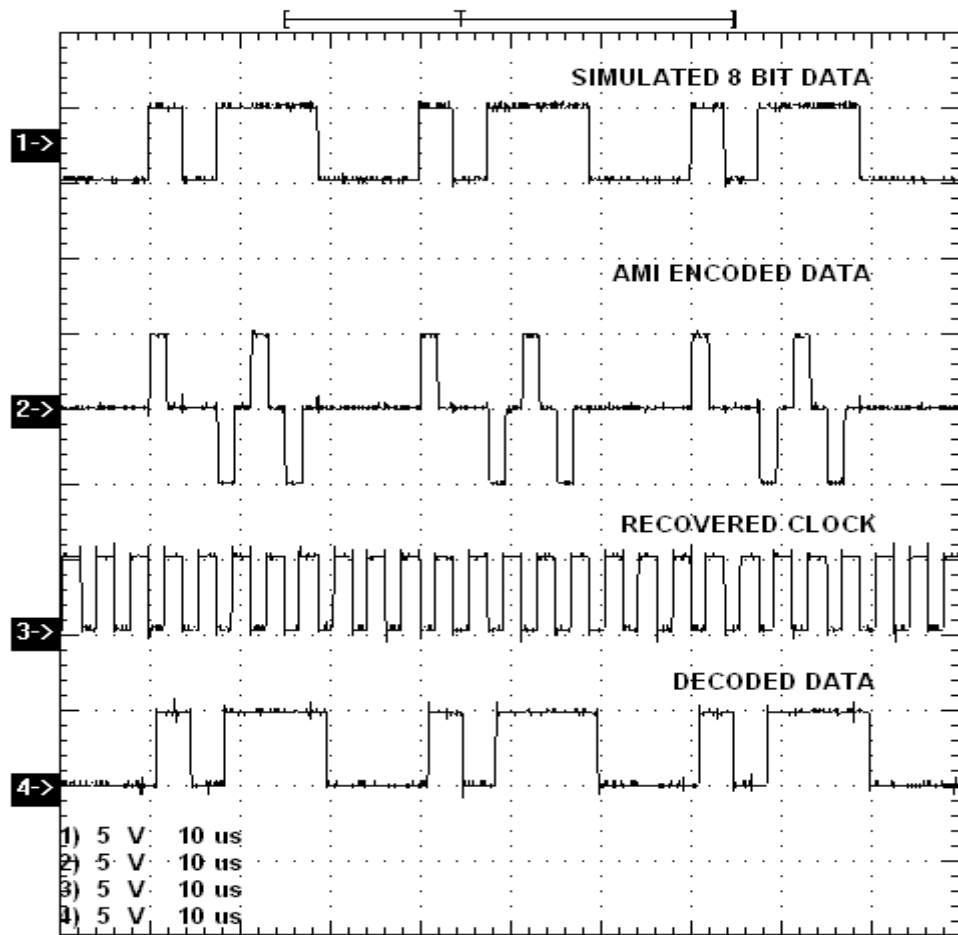
For recovery of clock from RZ-AMI, first the RZ-AMI should be converted to URZ by standard techniques. Then the clock is recovered.

Refer the circuit diagram and waveforms for the decoding techniques.

- Ensure that group 4 (**GP4**) clock is selected in the Clock Generation section. Selection is done with the help of switch **S1**.
- Observe the transmitter clock of frequency **250 KHz** at **TXCLK** post.
- Set the data pattern using switch **S4** as shown in the block diagram.
- Observe the 8-bit data pattern at **S DATA** post.
- Connect **S DATA** to **IN17** post and **TXCLK** to **CLK2** post of the Encoded Data section.
- Observe the AMI encoded data at the **OUT11** post of the Encoded Data section.
- Connect **OUT11** to **IN26** post of the Decoded Data section.
- For clock recovery connect **OUT 10** post of Encoded Data section to **IN 27** post of data decoder section.
- Select **BIO-M** data Using Switch S3 and observe the corresponding LED indication..
- Observe the decoded AMI data at the **OUT22** post of the Decoded Data section.

CONCLUSION

The AMI coded signal have alternate pulse of +5V and -5V for half bit interval when data bit is ONE. In decoding clock is recovered from incoming data bit using PLL technique.



AMI Encoding And Decoding

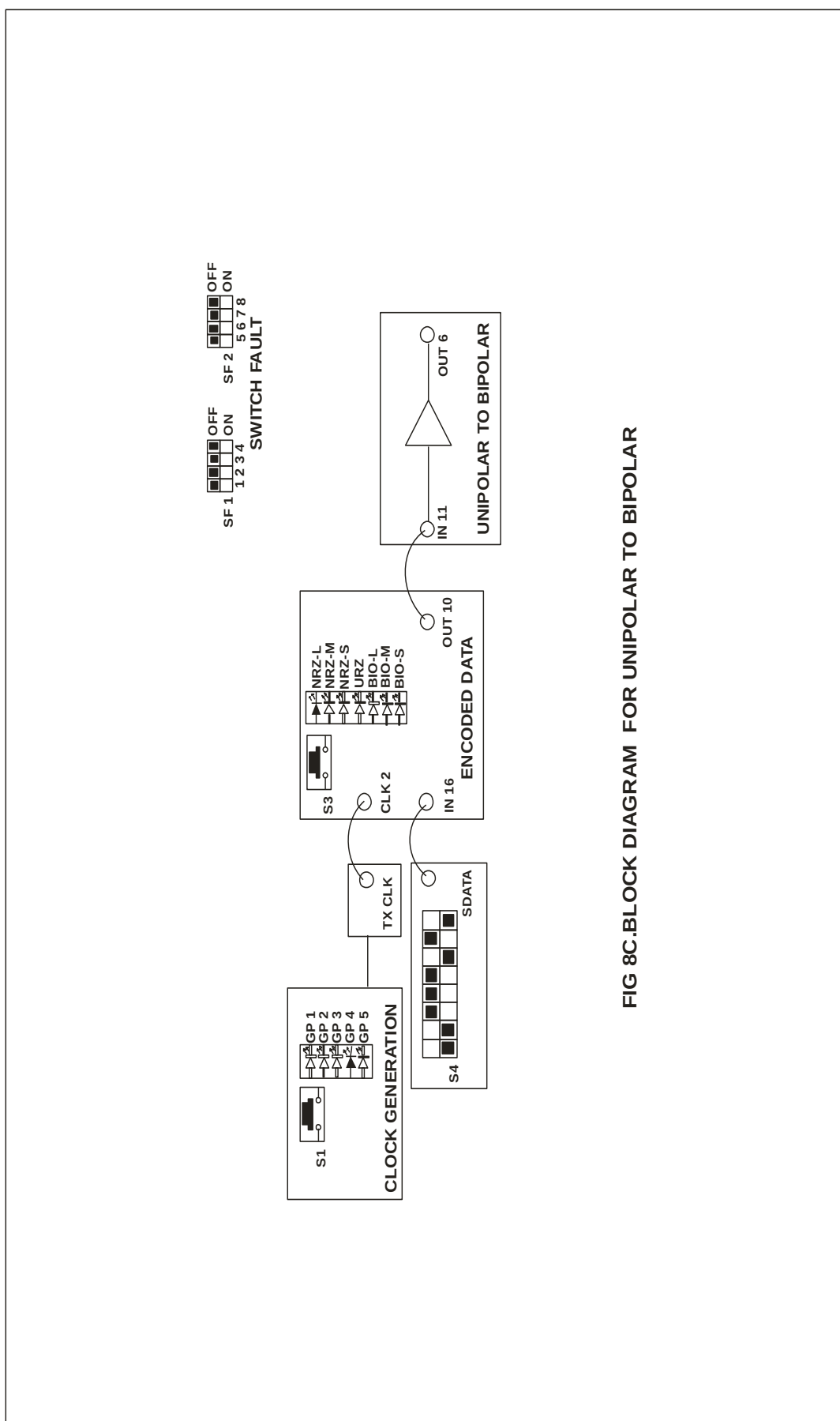


FIG 8C:BLOCK DIAGRAM FOR UNIPOLAR TO BIPOLAR

8C. UNIPOLAR TO BIPOLAR

PROCEDURE

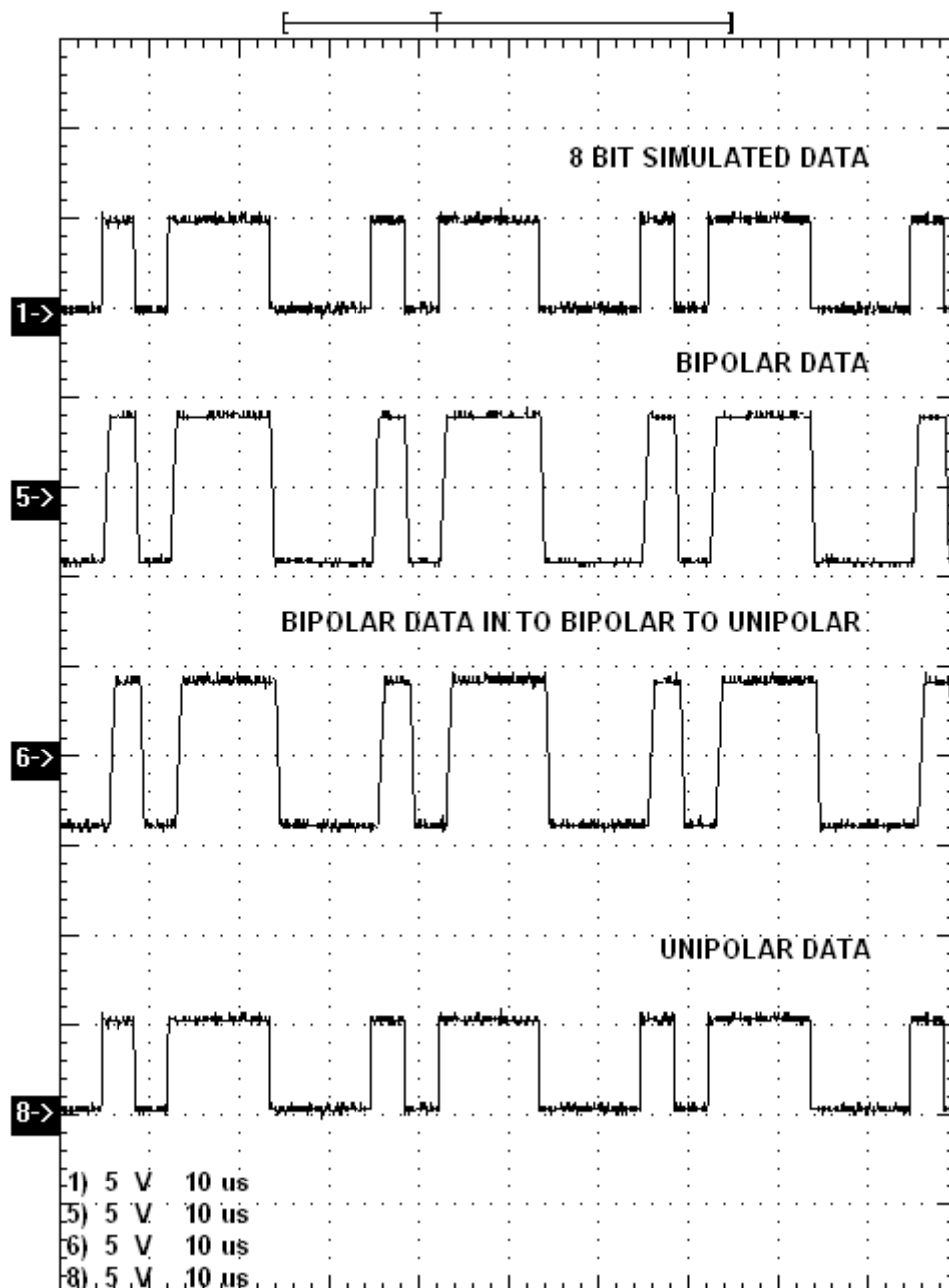
- Ensure that the group 4 (**GP4**) clock is selected in the clock generation section. Selection is done with the help of switch **S1**. Observe the corresponding LED indication.
- Observe the transmitter clock of frequency **250 KHz** at **TXCLK** post.
- The AMI coded signal have alternate pulse of +5V and -5V for half bit interval when data bit is ONE. In decoding clock is recovered from incoming data bit using PLL technique.
- Set the data pattern using switch **S4** as per the given block diagram.
- Observe the 8-bit data pattern at **S DATA** post.
- Connect **SDATA** to **IN16** post and **TXCLK** to **CLK2** post of the AMI coded signal have alternate pulse of +5V and -5V for half bit interval when data bit is ONE. In decoding clock is recovered from incoming data bit using PLL technique.
- Encoded Data section.
- Select NRZ-L data with the help of the switch **S2** and observe the corresponding LED indication in the Encoded Data section.
- Connect **OUT10** post of the Encoded Data section to the **IN11** post of the Unipolar to Bipolar section.
- Observe the Bipolar data at the **OUT6** post of the Unipolar to Bipolar section.
- The AMI coded signal have alternate pulse of +5V and -5V for half bit interval when data bit is ONE. In decoding clock is recovered from incoming data bit using PLL technique.
- The AMI coded signal have alternate pulse of +5V and -5V for half bit interval when data bit is ONE. In decoding clock is recovered from incoming data bit using PLL technique.
- The AMI coded signal have alternate pulse of +5V and -5V for half bit interval when data bit is ONE. In decoding clock is recovered from incoming data bit using PLL technique.

CONCLUSION

In Unipolar to Bipolar technique NRZ-L data which is unipolar (between GND and +5V) is translated into bipolar (+5V and -5V)

AMI, UNIPOLAR TO BIPOLAR AND BIPOLAR TO UNIPOLAR ENCODING

The AMI coded signal have alternate pulse of +5V and -5V for half bit interval when data bit is ONE. In decoding clock is recovered from incoming data bit using PLL technique



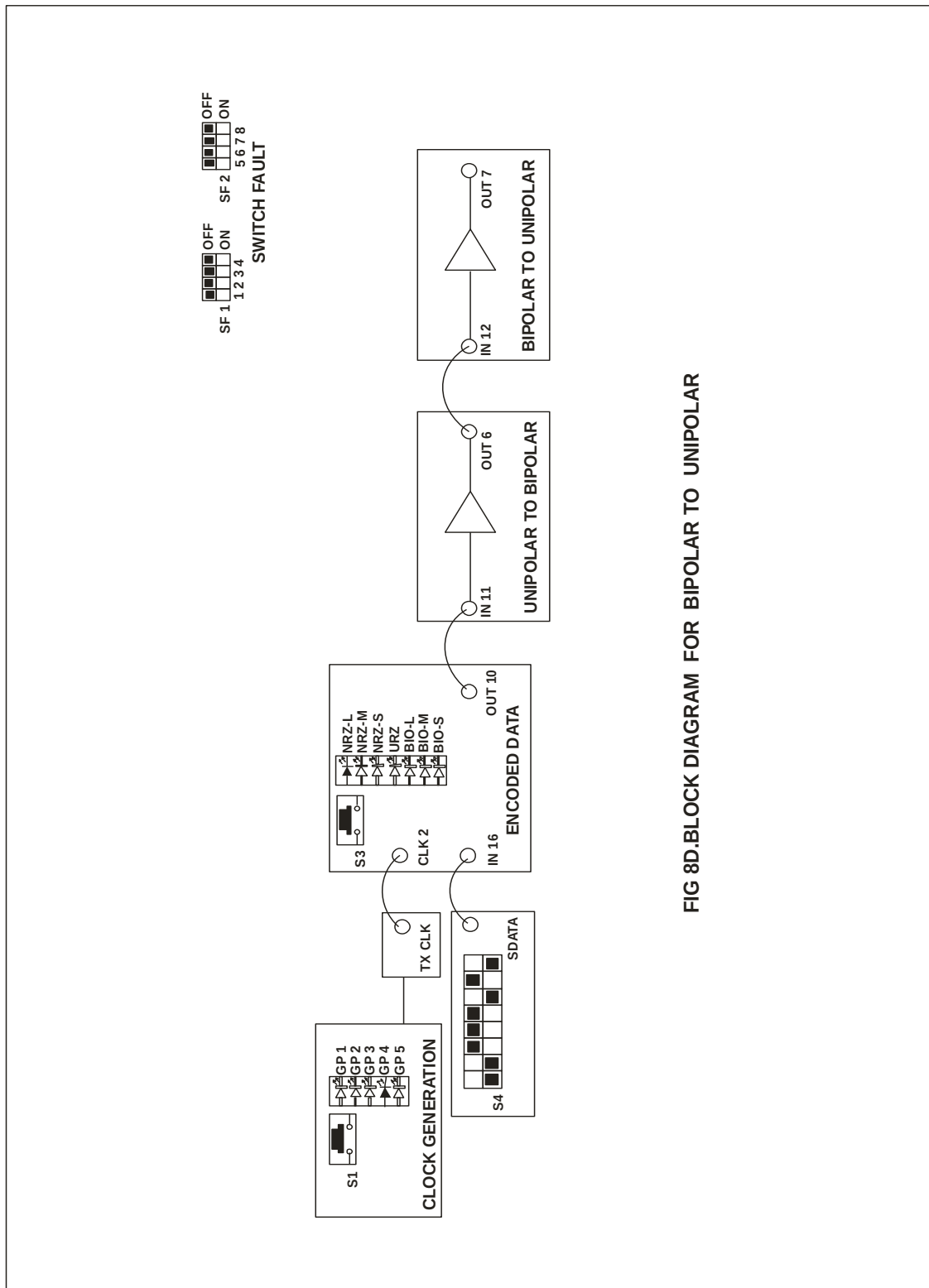


FIG 8D.BLOCK DIAGRAM FOR BIPOLAR TO UNIPOLAR

8D. Bipolar To Unipolar

PROCEDURE

- Ensure that the group 4 (**GP4**) clock is selected in the clock generation section. Selection is done with the help of switch **S1** and observes the corresponding LED indication.
- Observe the transmitter clock of frequency **250 KHz** at **TXCLK** post.
- Set the data pattern using switch **S4** as per the given block diagram.
- Observe the 8-bit data pattern at **S DATA** post.
- Connect **SDATA** to **IN16** post and **TXCLK** to **CLK2** post of the Encoded Data section.
- Select NRZ-L data with the help of the switch **S2** and observe the corresponding LED indication in the Encoded Data section.
- Connect **OUT10** post of the Encoded Data section to **IN12** post of the Bipolar to Unipolar section.
- Observe the Bipolar data at the **OUT7** post of Bipolar to Unipolar section.

CONCLUSION

In Bipolar to Unipolar technique bipolar data (+5V and –5V) is converted back into NRZ-L data, which is Unipolar. (Between GND and +5V)

**EXPERIMENT
NO. 9**

EXPERIMENT NO.9

NAME

Various carrier modulation and demodulation techniques

OBJECTIVES

To study the various carrier modulation and demodulation techniques:

- A. ASK
- B. FSK
- C. PSK

THEORY

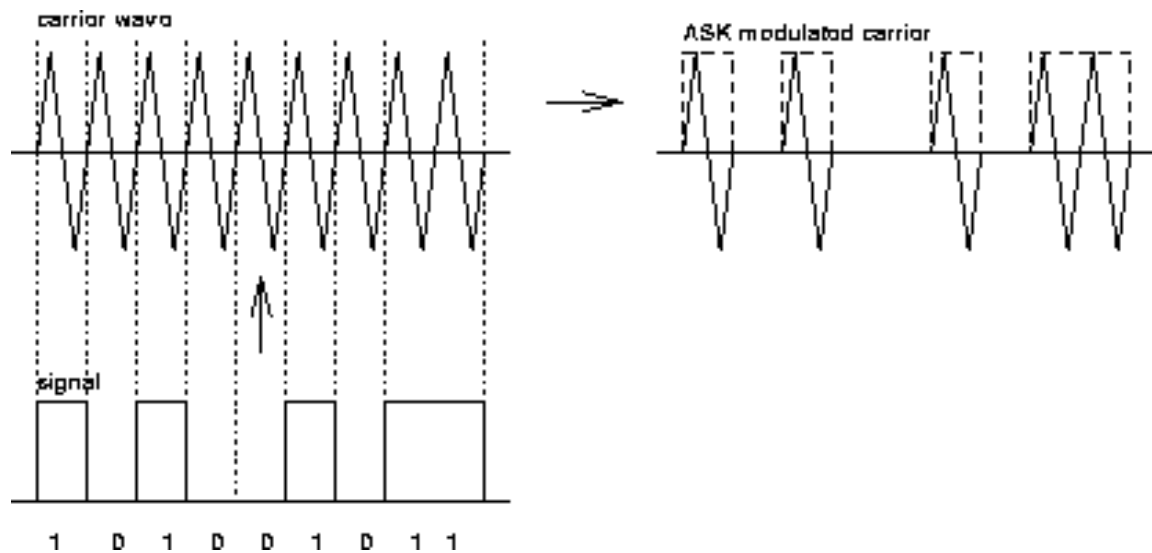
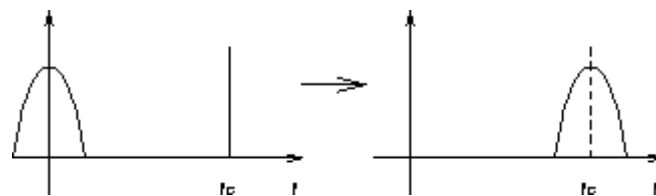
Carrier Modulation Schemes

Digital Communication is the technique by which all the information is represented in terms of binary digits, i.e., 'ones' and 'zeros'. These digits are represented by discrete voltage levels and the clock frequency of a digital communication scheme is generally low. For long distance transmission, the data is made to modulate a continuous wave (sine wave) carrier. These techniques are called Carrier Modulation Techniques. The various types of Carrier Modulation Techniques normally adopted in practice fall under three broad categories.

- a) Amplitude Shift Keying (ASK)
- b) Frequency Shift Keying (FSK)
- c) Phase Shift Keying (PSK)

EQUIPMENTS

DCS-B kit
Connecting Chords
Power supply
20 MHz Dual Trace Oscilloscope
Power connection cables

**FIGURE: AMPLITUDE SHIFT KEYING****Figure: Amplitude shift keying -- frequency domain**

9A. Ask Modulation And Demodulation

THEORY

For all types of Carrier Modulation, the Carrier frequency should be at least 2 times the base band of the modulating signal.

Assuming that the modulated carrier is a sine wave represented by the following equation,

$$C(t) = A(t) \cos \omega t$$

Where $C(t)$ = Carrier sine wave.
 $A(t)$ = Time varying amplitude
 ωt = Time varying angle.

Now in amplitude shift keying, the carrier is being transmitted only when the modulating data is 'one'. When the data is 'zero' the carrier is rejected from transmission. Thus the resulting modulated output of this type of modulation can be represented as follows:

$$M(t) = r(t) * C(t)$$

Where $M(t)$ = Modulated Carrier
 $r(t)$ = Time varying modulating data which is either 'one' or 'zero'

Now $M(t) = A(t) \cos \omega t$ when the modulating data is one.

0 when the modulating data is zero.

An Envelope detector is used to recover the data from the modulated carrier.

ASK describes the technique in which the carrier wave $f_c(t)$ is multiplied by the digital signal $f(t)$. Mathematically, the modulated carrier signal is given by the equation,

$$s(t) = f(t)\sin(2\pi f_c t + \phi) \quad (21)$$

It is a special case of **amplitude modulation (AM)**. Amplitude modulation has the property of translating the spectrum of the modulation $f(t)$ to the carrier frequency. The bandwidth of the signal remains unchanged.

The fact that AM simply shifts the signal spectrum is often used to convert the carrier frequency to a more suitable value without altering the modulation. This process is known variously as **mixing**, **up-conversion** or **down-conversion**. Some form of conversion will always be present when the channel carrier occupies a frequency range outside the modulation frequency range.

PROCEDURE

- Ensure that the group 4 (**GP4**) clock is selected in the Clock Generation section. Selection is done with the help of switch **S1**. Observe the corresponding LED indication.
- Observe the transmitter clock of frequency **250KHz** at **TXCLK** post.
- Set the data pattern-using switch **S4** as per the given block diagram.
- Observe the 8-bit data pattern at **S DATA** post.
- Observe the carrier sine wave of frequency **1MHz** at **SIN3** post in the carrier section.
- Connect **SIN3** post to the **IN2** post and **IN3** post to the Ground in carrier modulation section
- Connect **SDATA** to **IN16** post and **TXCLK** to **CLK2** post of the Encoded Data section.
- Select **NRZ-L** data with the help of the switch **S3** and observe the corresponding LED indication in the Encoded Data section.
- Connect **OUT10** post of the Encoded Data section to **IN4** post as a control input for the carrier modulator section.
- Observe the ASK modulated signal at the **OUT2** post of the Carrier modulator section.
- For the demodulation of the ASK modulated data, connect the **OUT2** post of the carrier modulator to the **IN24** post of the ASK demodulator section.
- Observe the ASK demodulated data at **OUT20** post of the ASK demodulator section.
- Verify the recovered data with the S DATA.

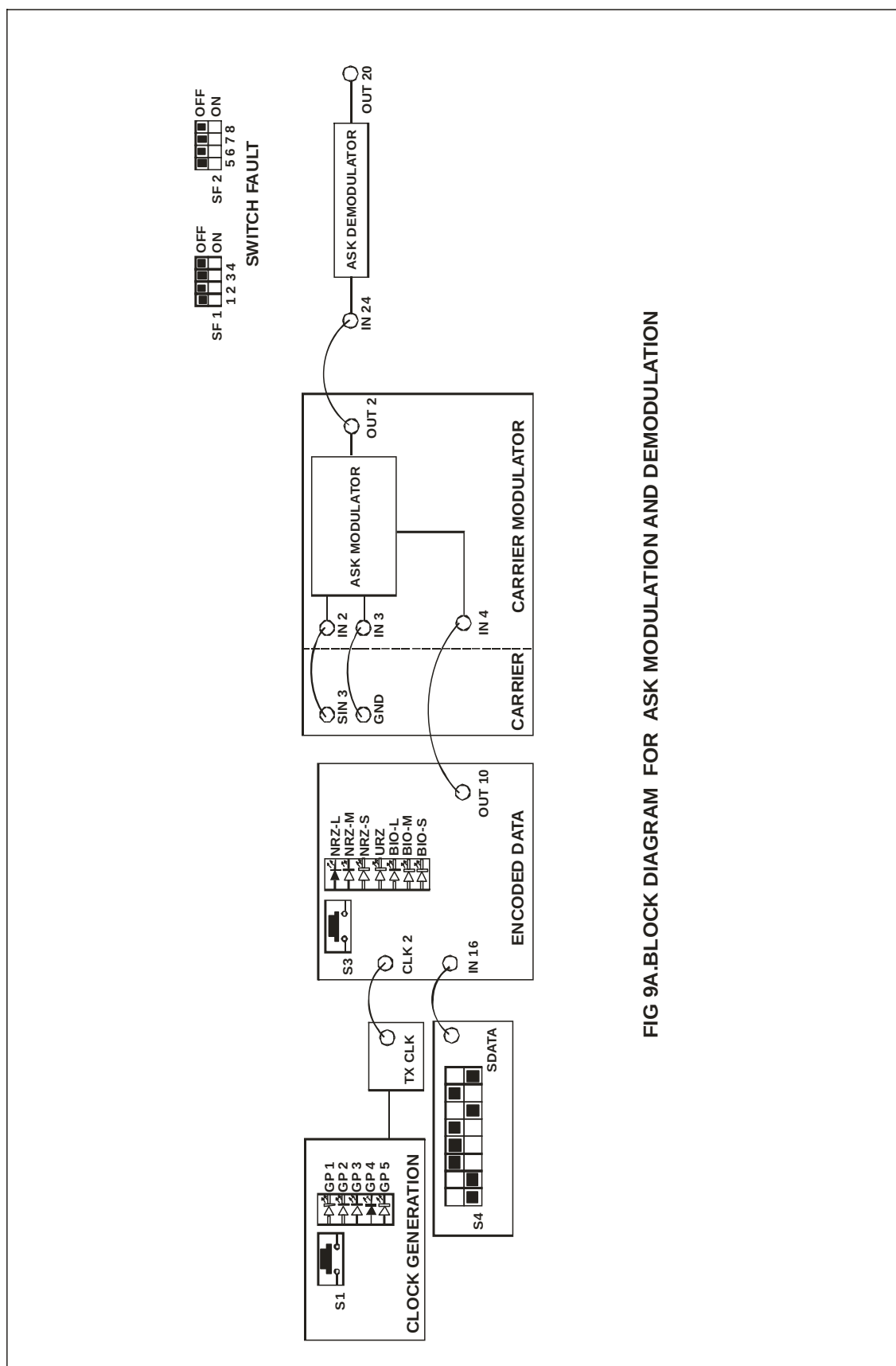


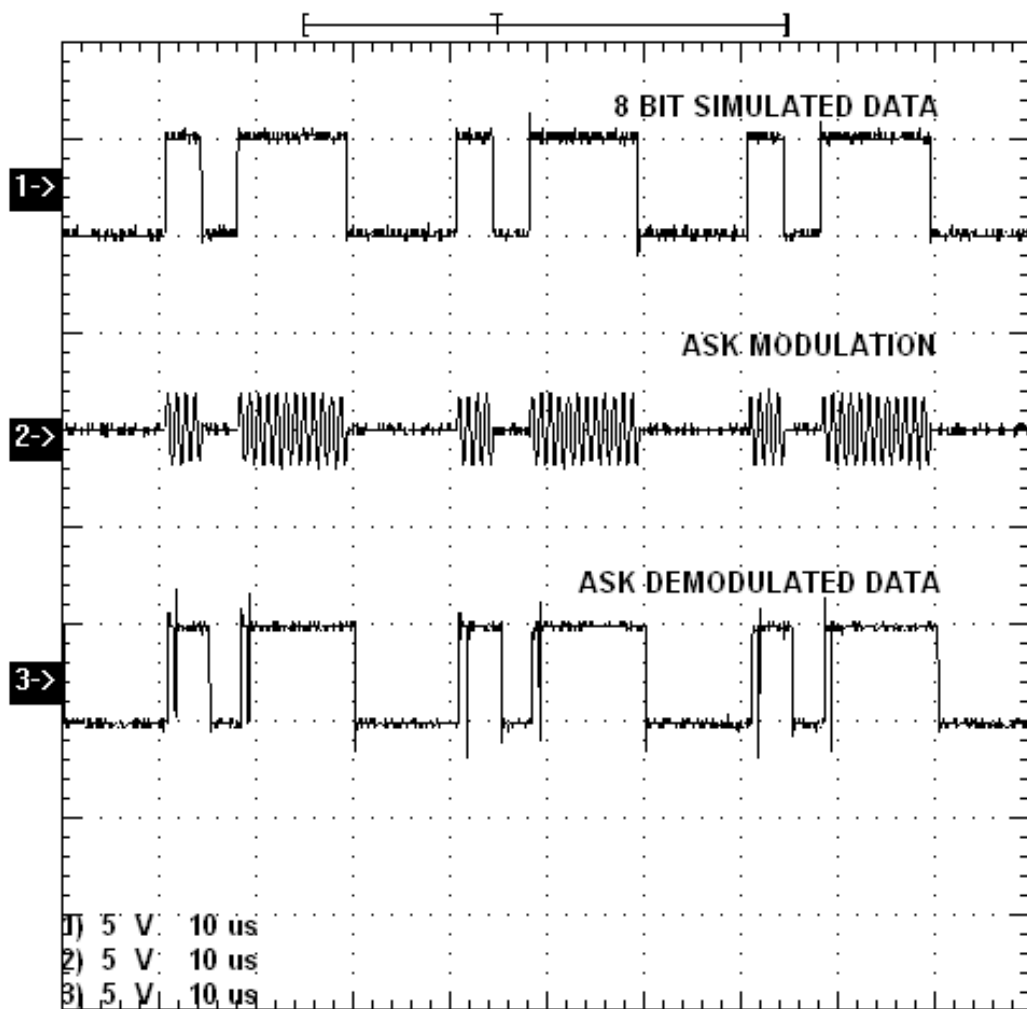
FIG 9A.BLOCK DIAGRAM FOR ASK MODULATION AND DEMODULATION

OBSERVATIONS

- Input **NRZ-L** Data at **IN 4**
- Carrier frequency **SIN 3**
- ASK modulated signal at **OUT 2**
- ASK Demodulated signal at **OUT 20**

CONCLUSION

It has been observed that there is a very small time lag between the modulating data and the recovered data.



ASK MODULATION AND DEMODULATION

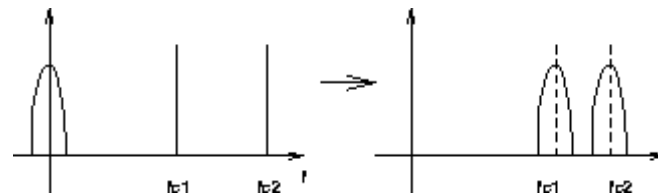


Figure: Frequency shift keying -- frequency domain

FSK is classified as wide-band if the separation between the two carrier frequencies is larger than the bandwidth of the spectrums of $f_1(t)$ and $f_2(t)$. In this case the spectrum of the modulated signal appears as two separate ASK signals. Narrow-band FSK is the term used to describe an FSK signal whose carrier frequencies are separated by less than the width of the spectrum than ASK for the same modulation.

9B. FSK MODULATION AND DEMODULATION

THEORY

In this type of modulation, the modulated output shifts between two frequencies for all the 'one' to 'zero' transitions.

Let the carrier frequencies be represented by w_1 and w_2 . Thus, we have,

$$M(t) = \begin{cases} A(t) \cos w_1 t & \text{if data is 'One'} \\ A(t) \cos w_2 t & \text{if data is 'Zero'} \end{cases}$$

Where $A(t)$ = Time varying amplitude of the sine wave

$M(t)$ = Modulated carrier

FSK Demodulator employs PLL logic for the recovery of data.

FSK describes the modulation of a carrier (or two carriers) by using a different frequency for 1 or 0. The resultant modulated signal may be regarded as the sum of two amplitude modulated signals of different carrier frequency (Figures  and ).

$$s(t) = f_1(t) \sin(2\pi f_{C1}t + \phi) + f_2(t) \sin(2\pi f_{C2}t + \phi) \quad (22)$$

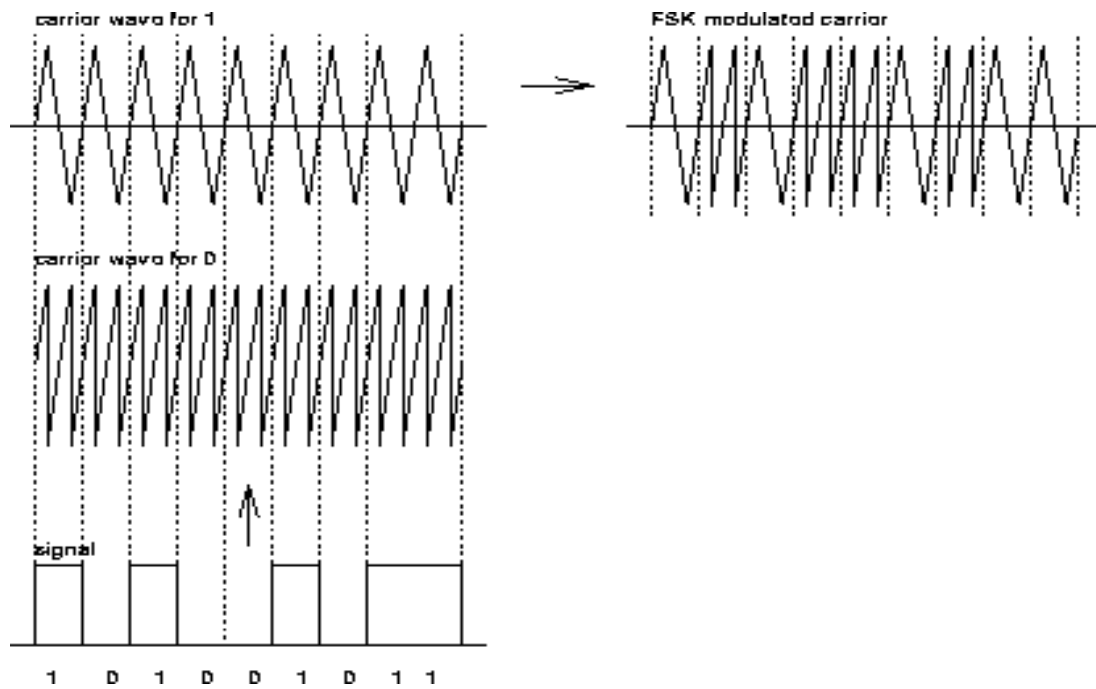


Figure: Frequency shift keying

PROCEDURE

- Ensure that the group 4 (**GP4**) clock is selected in the Clock Generation section. Selection is done with the help of switch **S1** and observes the corresponding LED indication.
- Observe the transmitter clock of frequency **250 KHz** at **TXCLK** post.
- Set the data pattern using switch **S4** as per the given block diagram.
- Observe the 8-bit data pattern at **S DATA** post.
- Observe the carrier sine wave of frequencies **500 KHz** at **SIN1** post and **1MHz** at **SIN3** post in the carrier section.
- Connect the **SIN1** post to the **IN3** post and **SIN3** post to the **IN2** post of the Carrier modulator section.
- Connect **S DATA** to **IN16** post and **TXCLK** to **CLK2** post of the Encoded Data section.
- Select NRZ-L data with the help of the switch **S3** and observe the corresponding LED indication in the Encoded Data section.
- Connect **OUT10** post of the Encoded Data section to **IN4** post as a control input for the carrier modulator section.
- Observe the FSK modulated signal at the **OUT2** post of the Carrier modulator section.
- For the demodulation of the FSK modulated data, connect the **OUT2** post of the carrier modulator to the **IN28** post of the FSK demodulator section.
- Observe the FSK demodulated data at **OUT24** post of the FSK demodulator section.
- Verify the recovered data with the S DATA

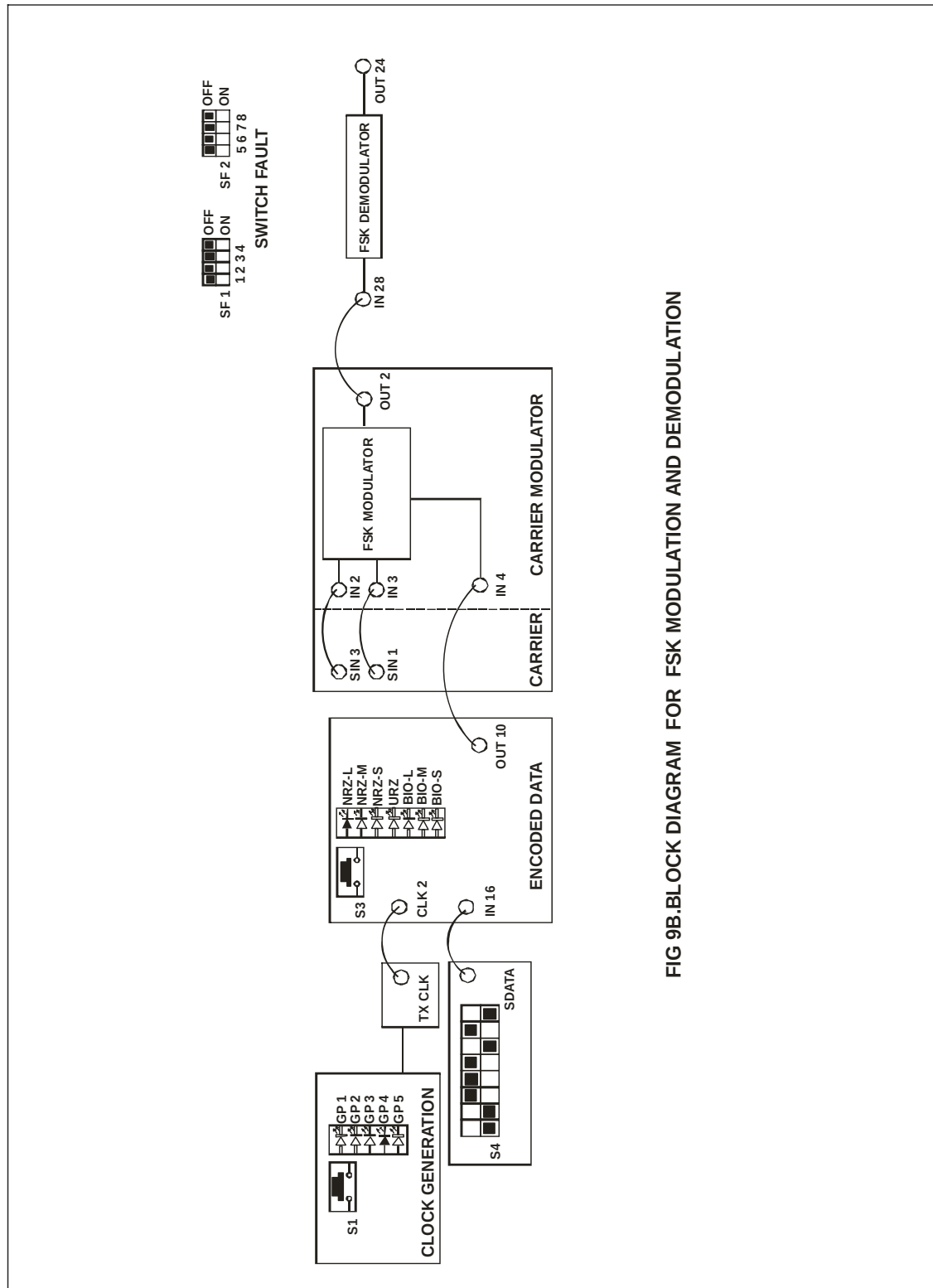


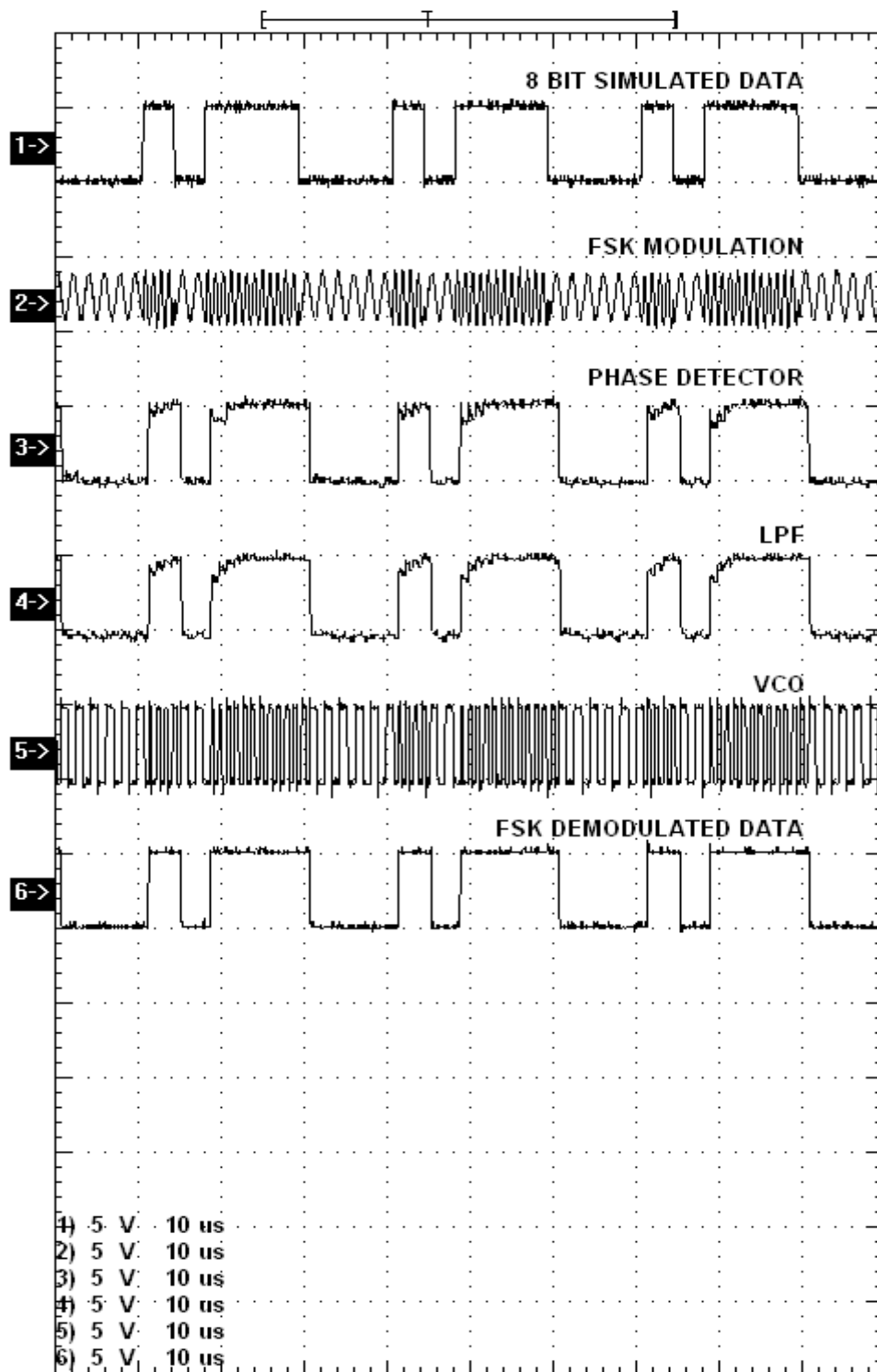
FIG 9B.BLOCK DIAGRAM FOR FSK MODULATION AND DEMODULATION

OBSERVATIONS

- INPUT **NRZ-L** DATA AT IN 4
- Carrier frequency **SIN 1** and **SIN 3**
- FSK modulated signal at **OUT 2**
- FSK DEMODULATED SIGNAL AT **OUT 24**
- Observe output of **PHASE DETECTOR**, **LPF**, **VCO** on test points provided

CONCLUSION

A small phase lag exists between the modulating data and the recovered data because of the limitation of the tracking ability and the time response of PLL.



FSK MODULATION AND DEMODULATION

9 C. PSK Modulations and Demodulation

THEORY

In the PSK modulation or phase shift keying, for all the 'one' to 'zero' transitions of the modulating data, the modulated output switches between the in phase and out of phase components of the modulating frequency. If the modulated carrier is represented by,

$$M(t) = A(t) \cos(\omega t + \text{Phase})$$

Where A(t) = Time varying amplitude
 ωt = Time varying angle
 M(t) = Modulated carrier

PSK describes the modulation technique that alters the phase of the carrier.

Mathematically,

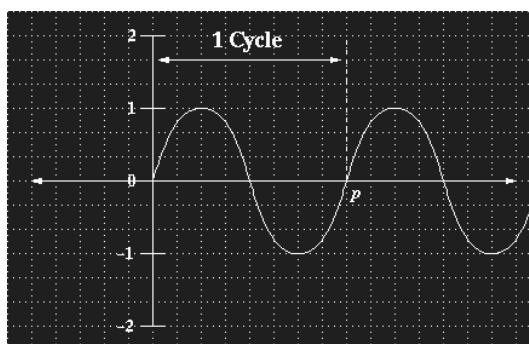
$$s(t) = \sin(2\pi f_C t + \phi(t)) \quad (23)$$

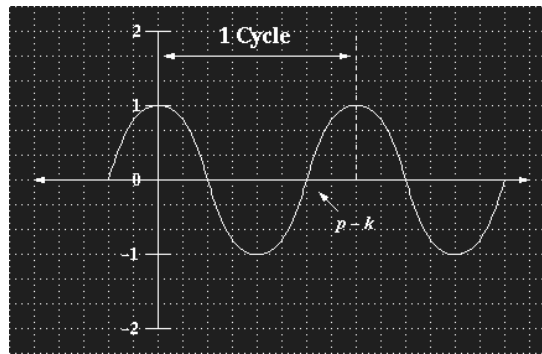
1. In angle modulation, the phase of the carrier is discretely varied in relation either to a reference phase or to the phase of the immediately preceding signal element, in accordance with data being transmitted.

2. In a communication system, the representation of characters, such as bits or quaternary digits, is done by a shift in the phase of an electromagnetic carrier wave with respect to a reference, by an amount corresponding to the symbol being encoded. *Note 1:* For example, While encoding bits, the phase shift could be 0° for encoding a "0," and 180° for encoding a "1," or the phase shift could be -90° for "0" and $+90^\circ$ for a "1," thus making the representations for "0" and "1" a total of 180° apart.

Note 2: PSK systems are designed so that the carrier can assume only two different phase angles, each change of phase carries one bit of information, i.e., the bit rate equals the modulation rate. If the number of recognizable phase angles is increased to 4, then 2 bits of information can be encoded into each signal element; likewise, 8 phase angles can encode 3 bits in each signal element.

Phase shift keying is a technique which shifts the period of a wave.

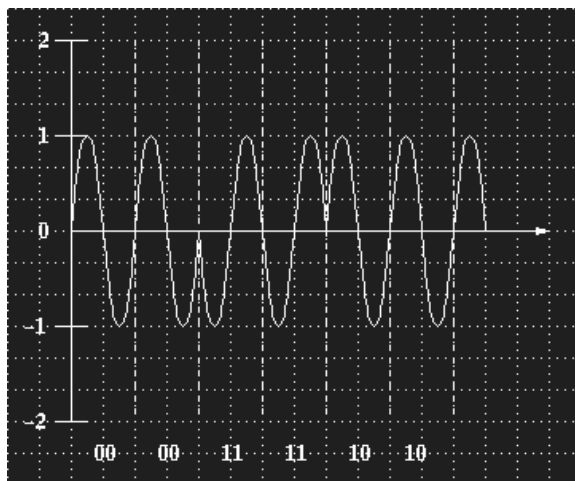




This wave 1 has a period of p , noted above. Also notice that the start of the wave's period is at 0.

Wave 2 is the same wave as the first, but its phase has been **shifted**. Notice that the period starts at the wave's highest point (1). So what's the point? Such a behavior is seen because we have shifted this wave by **one quarter** of the wave's full period. We can shift it to another quarter, if we want to, so the original wave would be shifted by **half** its period. And, we could do it one more time, so that it would be shifted **three quarters** of its original period.

This means that, we have 4 separate waves. So why not let each wave stand for some binary value? Since there are 4 waves, we can let each wave signify 2 bits (00, 01, 10, 11):



Bit value	Amount of shift
00	None
01	1/4
10	1/2
11	3/4

This technique of allowing each shift of a wave represent some bit value is **phase shift keying**. But, the real key is to shift each wave relative to the wave that came before it. An example can be seen in the above diagram.

Please note that when binary values were chosen randomly, for each wave, the values shown are incorrect. Thus, the correct pattern should be: 00 00 10 00 10 00.

PROCEDURE

- Ensure that the group 4 (**GP4**) clock is selected in the Clock Generation section. Selection is done with the help of switch **S1** and observes the corresponding LED indication.
- Observe the transmitter clock of frequency **250 KHz** at **TXCLK** post.
- Set the data pattern-using switch **S4** as per the given block diagram.
- Observe the 8-bit data pattern at **S DATA** post.
- Observe the carrier sine waves of frequencies **1MHz** at **SIN2** post and 1MHz with 180° phase at **SIN3** post in the carrier section.

- Connect the **SIN2** post to the **IN2** post and **SIN3** post to the **IN3** post of the Carrier modulator section.
- Connect **S DATA** to **IN16** post and **TXCLK** to **CLK2** post of the Encoded Data section.
- Select NRZ-L data with the help of the switch **S3** and observe the corresponding LED indication in the Encoded Data section.
- Connect **OUT10** post of the Encoded Data section to **IN4** post as a control input for the carrier modulator section.
- Observe the PSK modulated signal at the **OUT2** post of the Carrier modulator section.
- For the demodulation of the PSK modulated data, connect the **OUT2** post of the carrier modulator to the **IN30** post of the PSK demodulator section.
- Observe the PSK demodulated data at **OUT27** post of the PSK demodulator section.
- Verify the recovered data with the S DATA.

OBSERVATIONS

Input **NRZ-L** Data at **IN 4**

Carrier frequency **SIN 2** and **SIN 3**

PSK modulated signal at **OUT 2**

PSK Demodulated signal at **OUT 27**

Observe the output of the **SINE TO SQUARE CONVERTOR, SQUARING LOOP, DIVIDE BY 2** on the test points provided

CONCLUSION

It is observed that the successful operation of the PSK detector is fully dependent on the phase components of the transmitted modulated carrier. If the phase reversal of the modulated carrier, along with the rising and falling edges of the data are not proper, then the efficient detection of data from PSK modulated carrier becomes impossible.

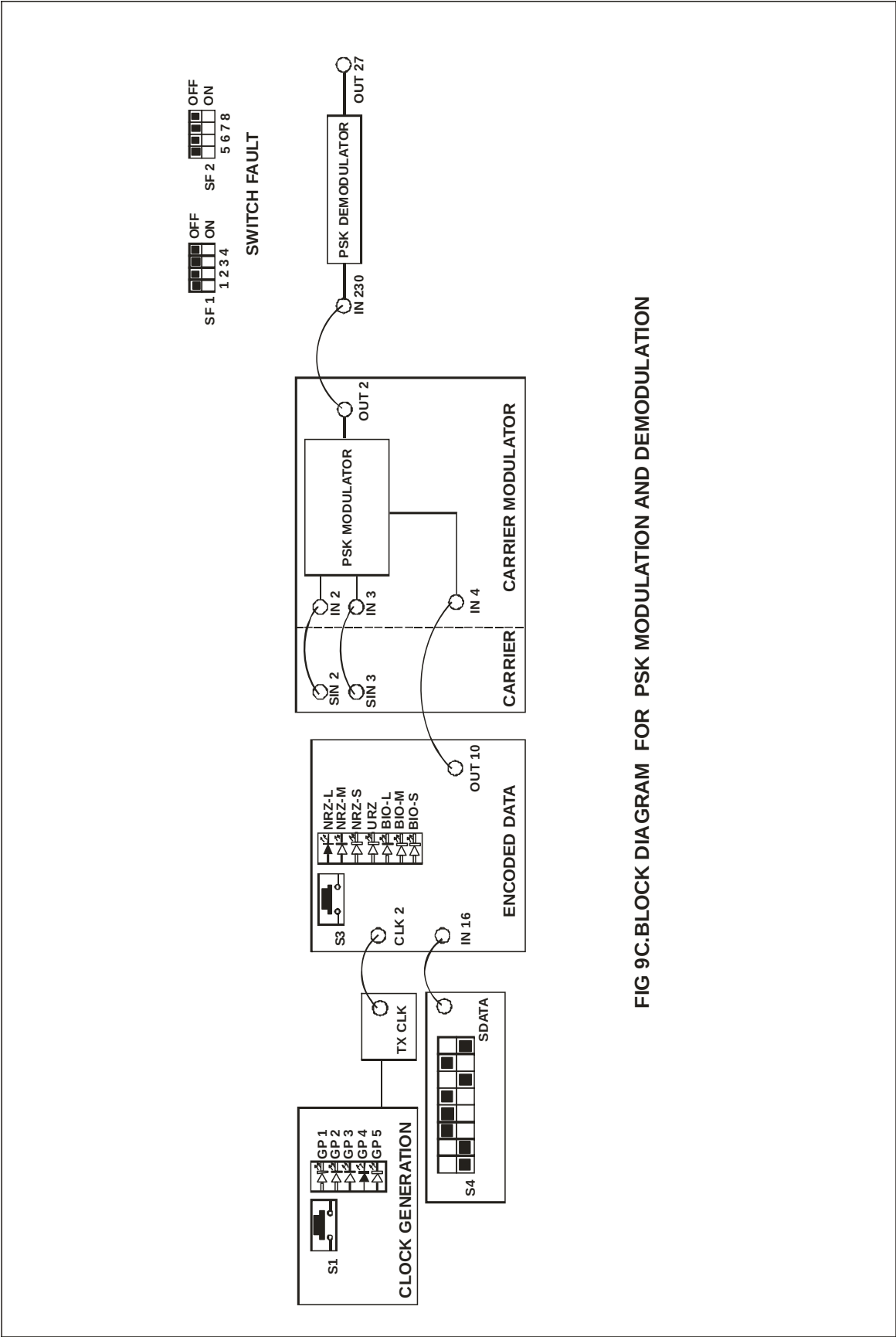
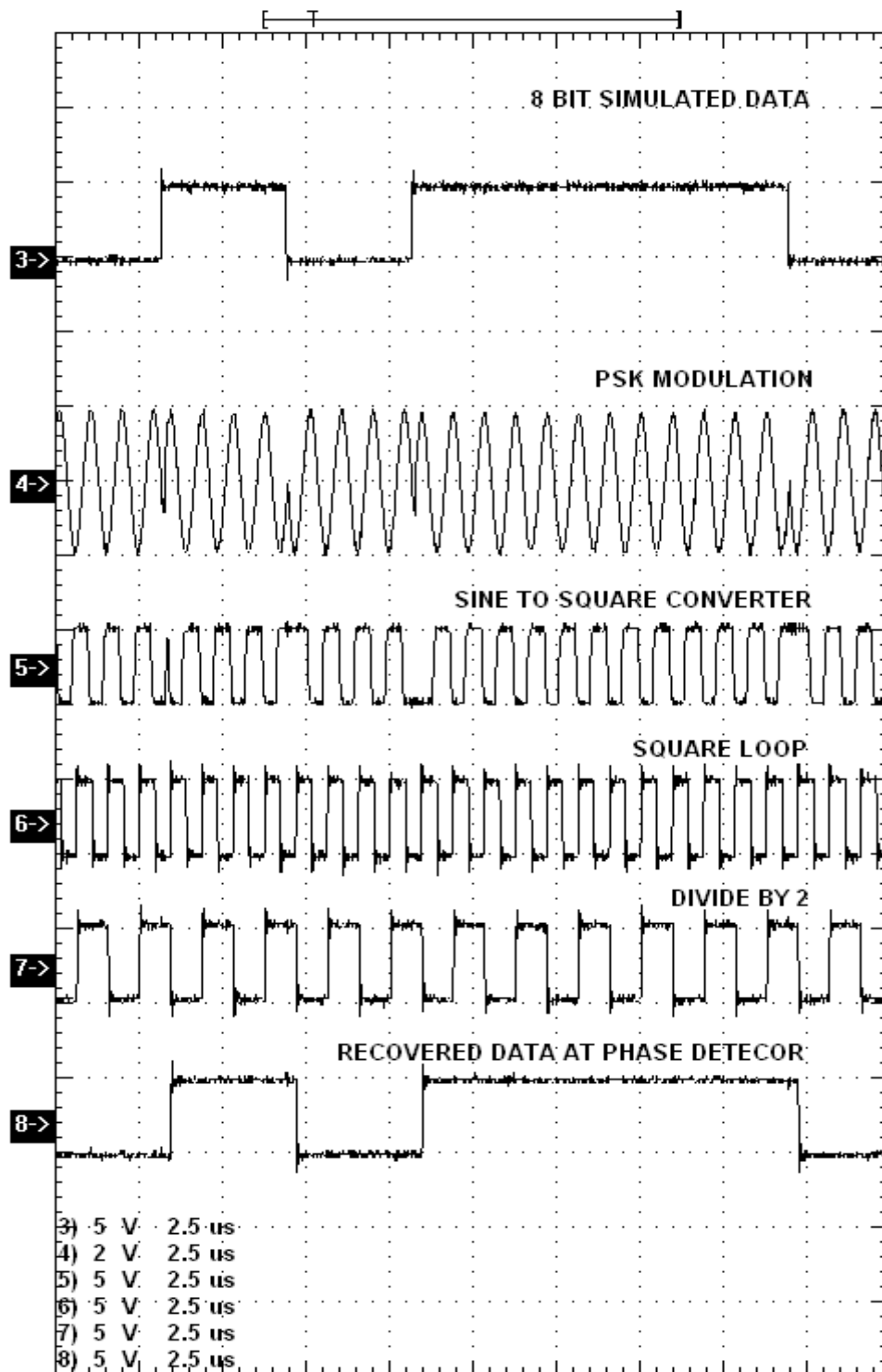


FIG 9C.BLOCK DIAGRAM FOR PSK MODULATION AND DEMODULATION



PSK Modulation and Demodulation

**EXPERIMENT
NO. 10**

EXPERIMENT NO.10

NAME

Study of DM And ADM

OBJECTIVES

To study the signal modulation technique:

- A. Delta modulation & demodulation technique
- B. Slope Overload & Increased integrator Gain
- C. Adaptive Delta Modulation & Demodulation

THEORY

10A. Delta Modulation & Demodulation Technique

Delta modulation is a process of converting Analog signal into one bit code. In Delta modulation only one bit is sent per sample. This bit indicates whether the signal is larger or smaller than the previous samples. Delta Modulation system cannot react to rapid amplitude changes between samples. It is primarily used in voice communication.

The analog signal $m(t)$ is compared with the approximate signal $\bar{m}$. i.e. It is expected to be a reconstructed signal. If $m(t)$ is more than $\bar{m}$ then a positive pulse is issued and if $m(t)$ is lesser than $\bar{m}$, a negative pulse is generated.

The comparator is a high gain differential comparator. If the analog signal $m(t)$ applied to its non-inverting input is more than the reconstructed signal (that the receiver is expected to produce) then its output is highly positive. On contrary, if the reconstructed signal $m(t)$ is more than the analog signal $m(t)$, then its output is highly negative.

The output of the comparator is transmitted to the modulator. The modulator receives a train of pulses at constant rate from the pulse generator. If the comparator output is positive [i.e. if $m(t)$ is more than $\bar{m}(t)$] then the pulses are transmitted to the output of the modulator without any change. If however, the comparator output is negative [i.e. if $m(t)$ is lesser than $\bar{m}(t)$] then the pulses are inverted and then transmitted to the output.

The integrator constructs the signal $\bar{m}(t)$ by increasing its output by a fixed amount Δ on the receipt of a positive pulse and decreasing its output $\bar{m}(t)$ by the same fixed amount Δ on receipt of negative pulse (as is the case with the expected action of the receiver).

The receiver receives highly distorted pulses, which are reshaped by the pulse regenerator, into pulses of constant amplitude and width. The output of integrator is the same as $\bar{m}(t)$ and is a stepped waveform. The stepped waveform is smoothened by the low pass filter to produce smooth received signal.

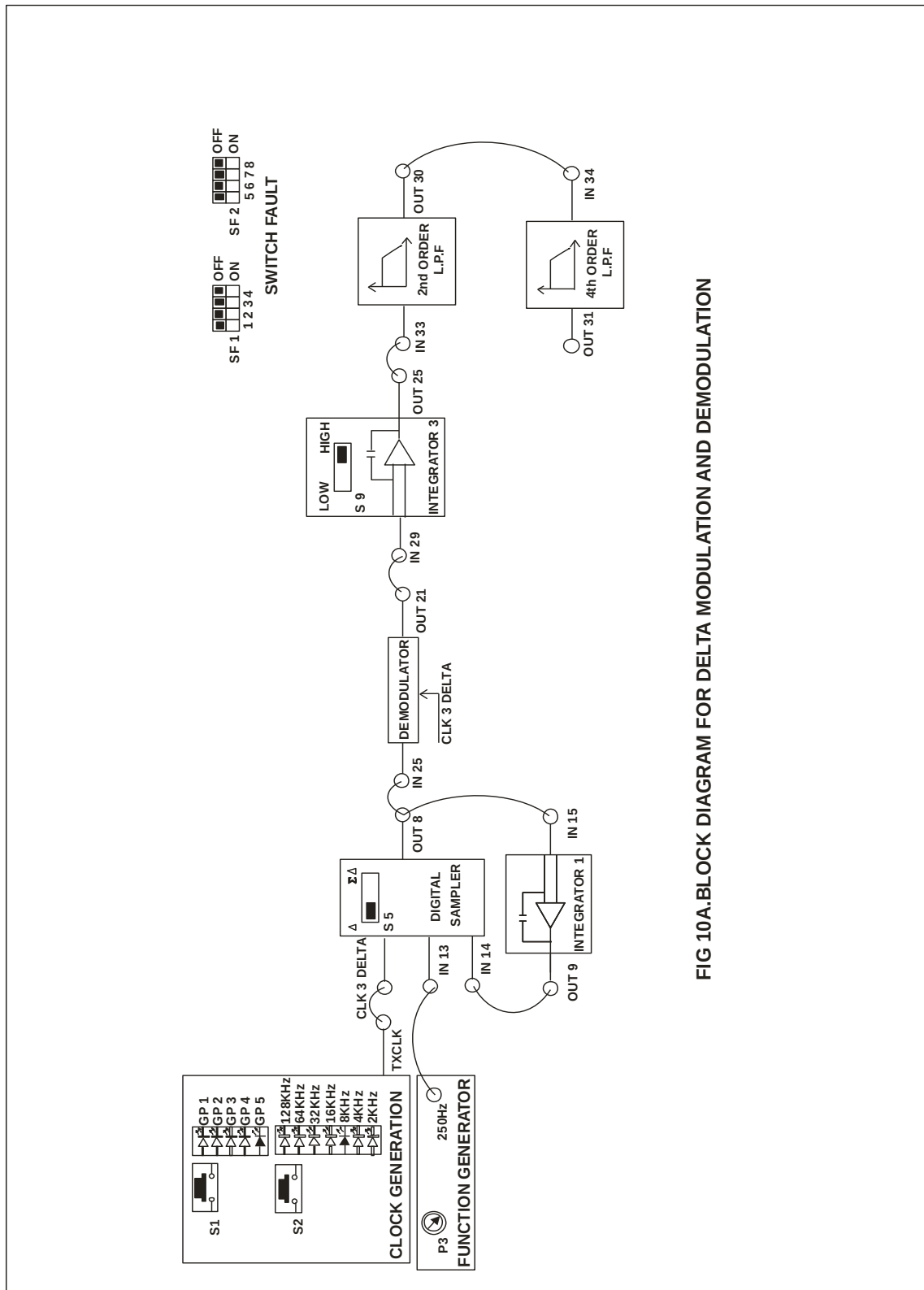


FIG 10A.BLOCK DIAGRAM FOR DELTA MODULATION AND DEMODULATION

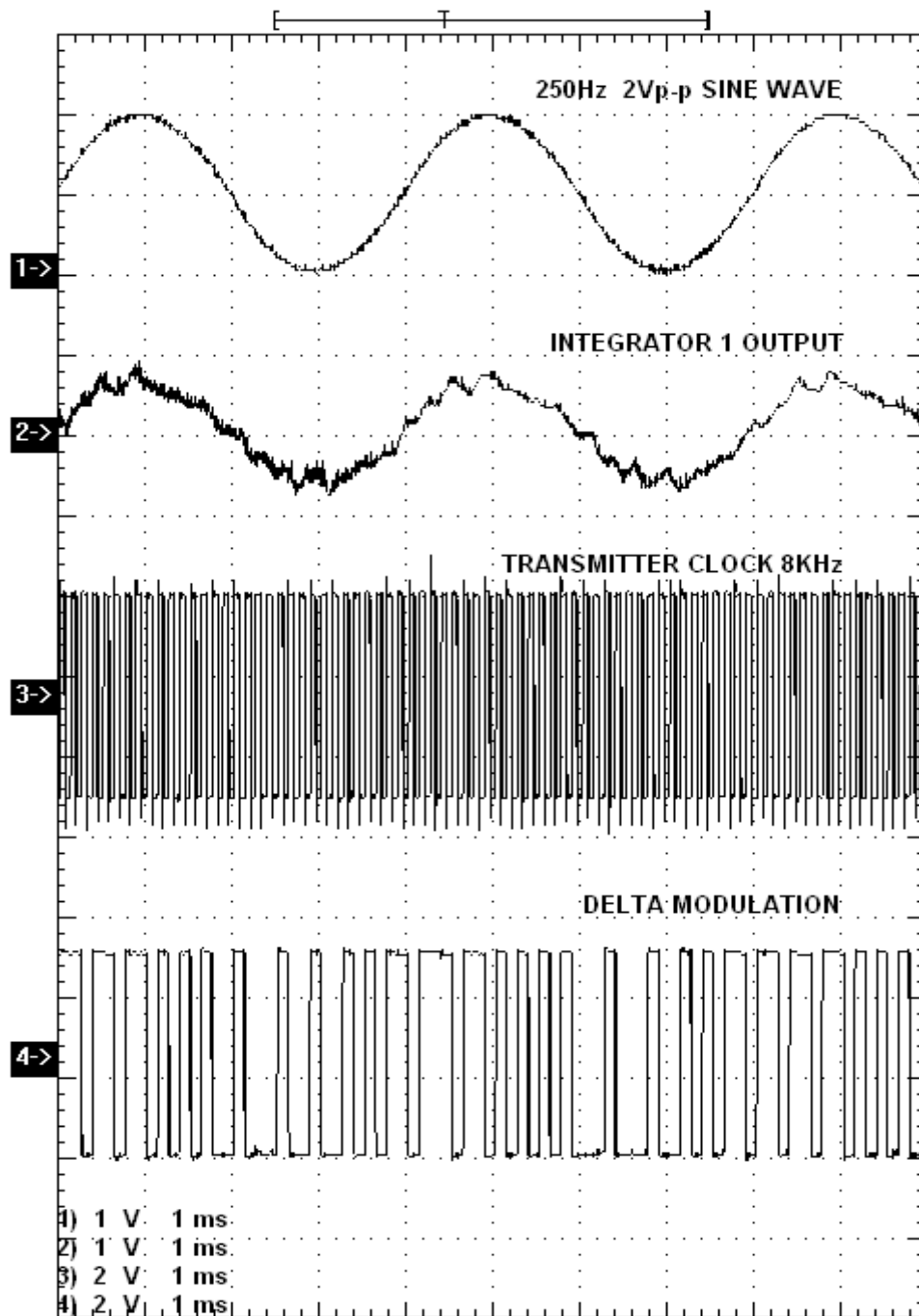
EQUIPMENTS

DCS-B kit
Connecting Chords
Power supply
20 MHz Dual Trace Oscilloscope
Power connection cables

10A. Delta Modulation & Demodulation Technique

PROCEDURE

- Ensure that the group 5 (**GP5**) clock is selected in the Clock Generation section. Selection is done with the help of switch **S1** and observes the corresponding LED indication.
- Select the transmitter clock of frequency 8KHz using switch **S2** and the selected clock is indicated on the corresponding LED indication in the Clock Generation section.
- Connect **250Hz** sine wave having amplitude of 0V using pot **P3** to **IN13** post and **TXCLK** post to the **CLK3 DELTA** post of the Digital Sampler.
- Keep switch **S5** in Delta position.
- Connect **OUT 8** post of the Digital sampler to **IN15** post of Integrator 1.
- Connect **OUT9** post of Integrator 1 to **IN14** post of the Digital sampler section.
- Observe the delta modulated output at **OUT8** post of the Digital Sampler
- Observe the integrated output at the **OUT9** post of the Integrator1 section. We observe that as the clock rate increases the amplitude of the triangular wave decreases.
- Increase the amplitude of the 250Hz sine wave up to 0.5V using pot **P3** in the Function Generator section. Signal approximating 250Hz sine wave is available at **OUT9** post of the Integrator 1 section. This signal is obtained by integrating the digital output resulting from delta modulation.
- Increase the amplitude of the 250Hz sine wave up to 2Vpp using pot **P3** in the Function Generator section, observe that the digital high makes the integrator output to go upward and digital low makes the integrator output to go downwards.
- Increase the amplitude of the 250Hz sine wave further high, using pot **P3** and observe that the integrated output cannot follow the input signal. This is because of the fact that as the frequency of the input analog signal increases there is less conversion of ones and zero's in the Delta modulated output.
- Connect the **OUT8** post of the digital sampler to **IN25** post of the Demodulator section.
- Connect the **OUT21** of the demodulator section to the **IN29** post of the Integrator 3 section.
- Keep switch **S9** in the Integrator 3 section towards High position. Connect **OUT25** post to the **IN33** post of the 2nd order LPF.
- Connect **OUT30** post of 2nd order LPF to **IN34** post of 4th order LPF.
- Observe the reconstructed signal through 2nd order LPF and 4th order LPF.
- Repeat the above procedure for different input signals & clock frequencies



OBSERVATIONS

Sampling clock **TXCLK**

Input Signal

Integrator 1 output at feedback loop for Delta modulator **OUT 9**

Digital sampler Output **OUT 8**

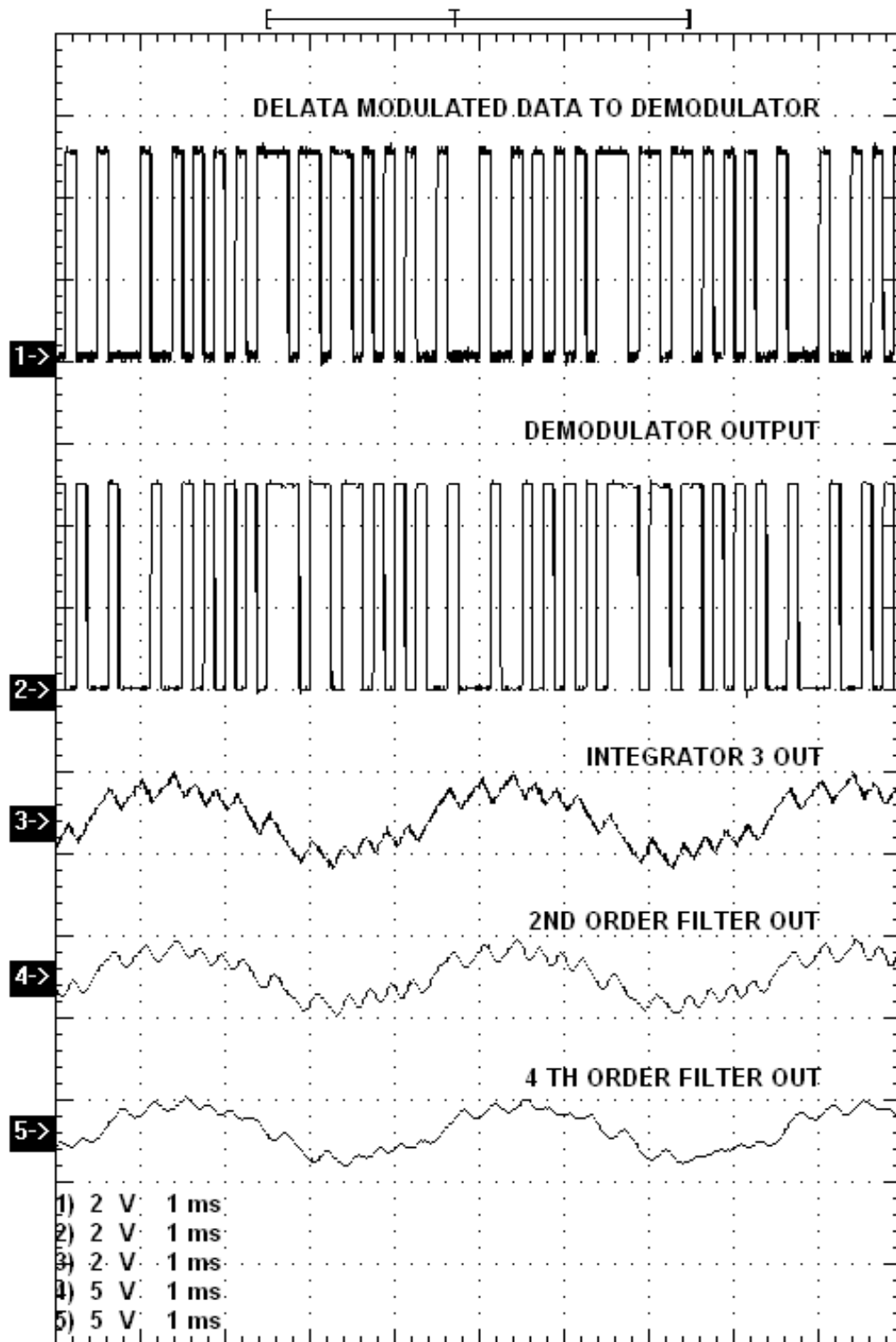
Demodulator Output **OUT 21**

Integrator 3 output **OUT 25**

Filter Output **OUT 31**

CONCLUSION

Delta modulation (DM or Δ -modulation) is an analog-to-digital and digital-to-analog signal conversion technique used for transmission of voice information where quality is not of primary importance. DM is the simplest form of differential pulse-code modulation (DPCM) where the difference between successive samples is encoded into n-bit data streams. In delta modulation, the transmitted data is reduced to a 1-bit data stream.



Delta Modulation & Delta Demodulation

10B. To Study The Slope Over Load And Increased Integrator Gain

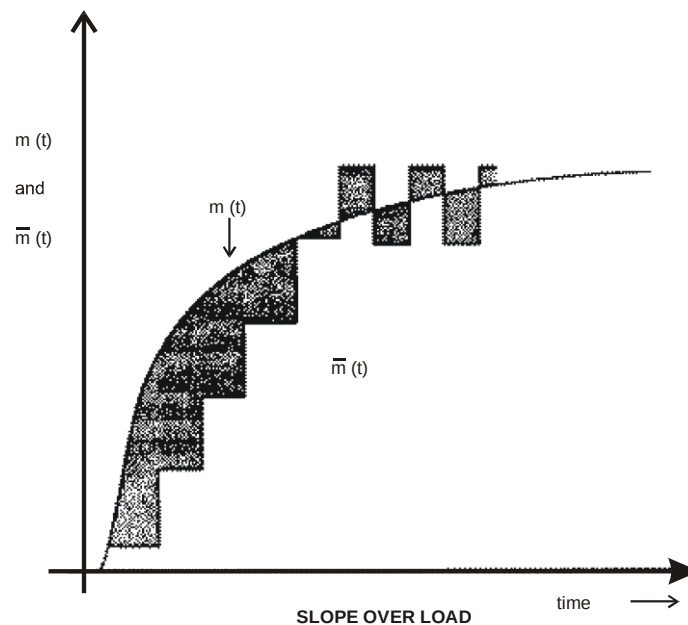
Slope Over Load

A small value of Δ can create a considerable difference when the signal increases at a fast rate. We can see that the reconstructed signal increases by Δ at each pulse instant, for an increasing analog signal. The maximum increase in $\bar{m}(t)$ per second is $\Delta * f$ where, f is the pulse rate. If the analog signal increases at a rate more than $(\Delta * f)$, then the reconstructed signal $\bar{m}(t)$ will be lower than the analog signal. This situation is known as the slope over load i.e. $dm(t)/dt$ is more than that the receiver can create. The receiver output [nearly same at $\bar{m}(t)$] does not truly represent the analog signal $m(t)$.

A simple solution to avoid slope overload could be to increase the step size (Δ). However, this increases the difference between analog signal $m(t)$ and the received signal $\bar{m}(t)$ at the time of hunting.

Another solution can be to increase the pulse repetition rate f . However, the main objective in development of DM was to reduce the bit rate, which increases by increasing f . Hence, a very large increase in f cannot be tolerated.

An intelligent solution offered was to use increasingly step size only when slope overload is likely to occur i.e. when the analog signal $m(t)$ is either increasing or decreasing at a fast rate. In fact, this solution was found to be highly effective and it resulted in a system known as Adaptive Delta Modulation (ADM).



Slope overload distortion: This type of distortion is due to the use of a step size Δ that is too small to follow the portions of the waveform that have a steep slope. It can be reduced by increasing the step size.

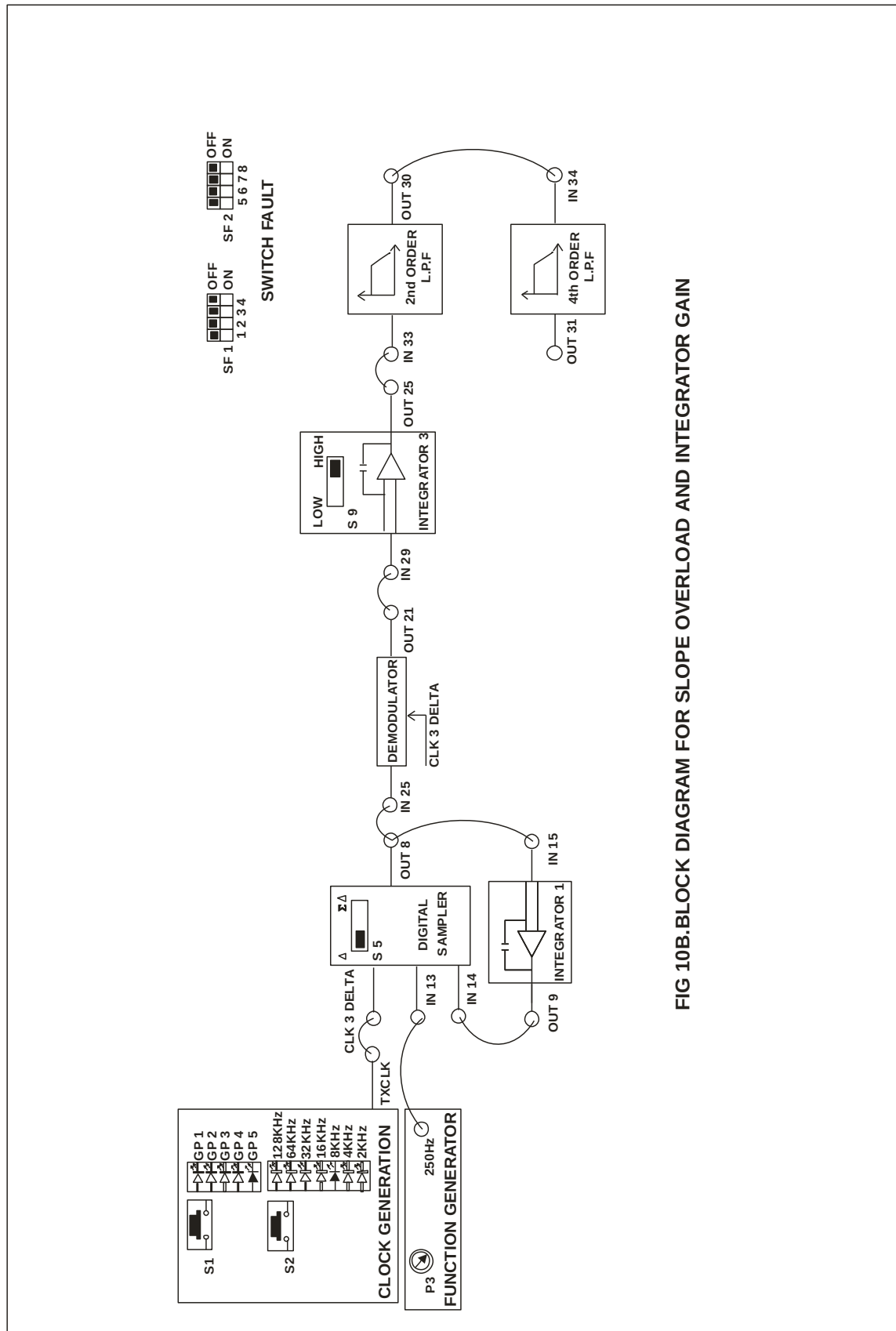


FIG 10B.BLOCK DIAGRAM FOR SLOPE OVERLOAD AND INTEGRATOR GAIN

PROCEDURE

- Ensure that the group 5 (**GP5**) clock is selected in the Clock Generation section. Selection is done with the help of switch **S1**. Observe the corresponding LED indication.
- Select the transmitter clock of frequency 8 KHz using switch **S2**. The selected clock is indicated on the corresponding LED indication in the Clock Generation section.
- Connect **250Hz** sine wave having amplitude of 4Vpp using pot **P3** to **IN13** post and **TXCLK** post to the **CLK3 DELTA** post of the Digital Sampler.
- Keep switch **S5** in Delta position.
- Connect **OUT8** post of the Digital sampler to **IN15** post of Integrator 1.
- Connect **OUT9** post of Integrator 1 to **IN14** post of the Digital Sampler.
- Connect the **OUT8** post of the Digital Sampler to **IN25** post of the Demodulator section.
- Connect the **OUT21** of the demodulator section to the **IN29** post of the Integrator 3 section.
- Keep switch **S9** in the Integrator 3 section towards High position.
- Connect **OUT25** post to the **IN33** post of the 2nd order LPF.
- Connect **OUT30** post of 2nd order LPF to **IN34** post of 4th order LPF.
- Observe the integrated output at the **OUT25** post integrator3 section at different clock frequencies.
- Perform the experiment by changing the sine wave frequencies from 500 Hz to 2 KHz.
- Select 1 KHz sine wave of 4V amplitude as input signal. We observe the slope overload effect.

OBSERVATIONS

Sampling clock **TXCLK**

Input Signal

Integrator 1 output at feedback loop for Delta modulator **OUT 9**

Digital sampler Output **OUT 8**

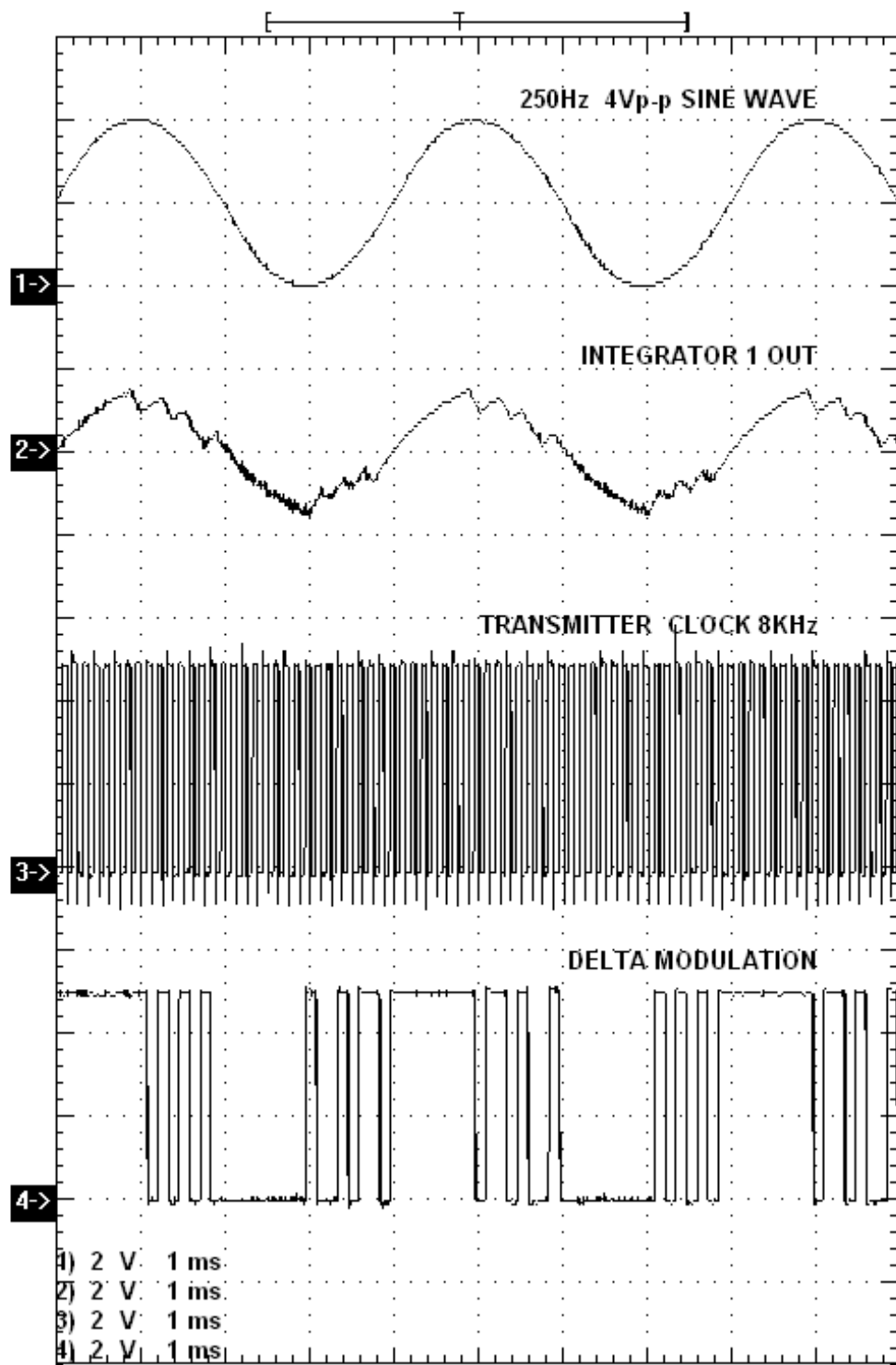
Demodulator Output **OUT 21**

Integrator 3 output **OUT 25**

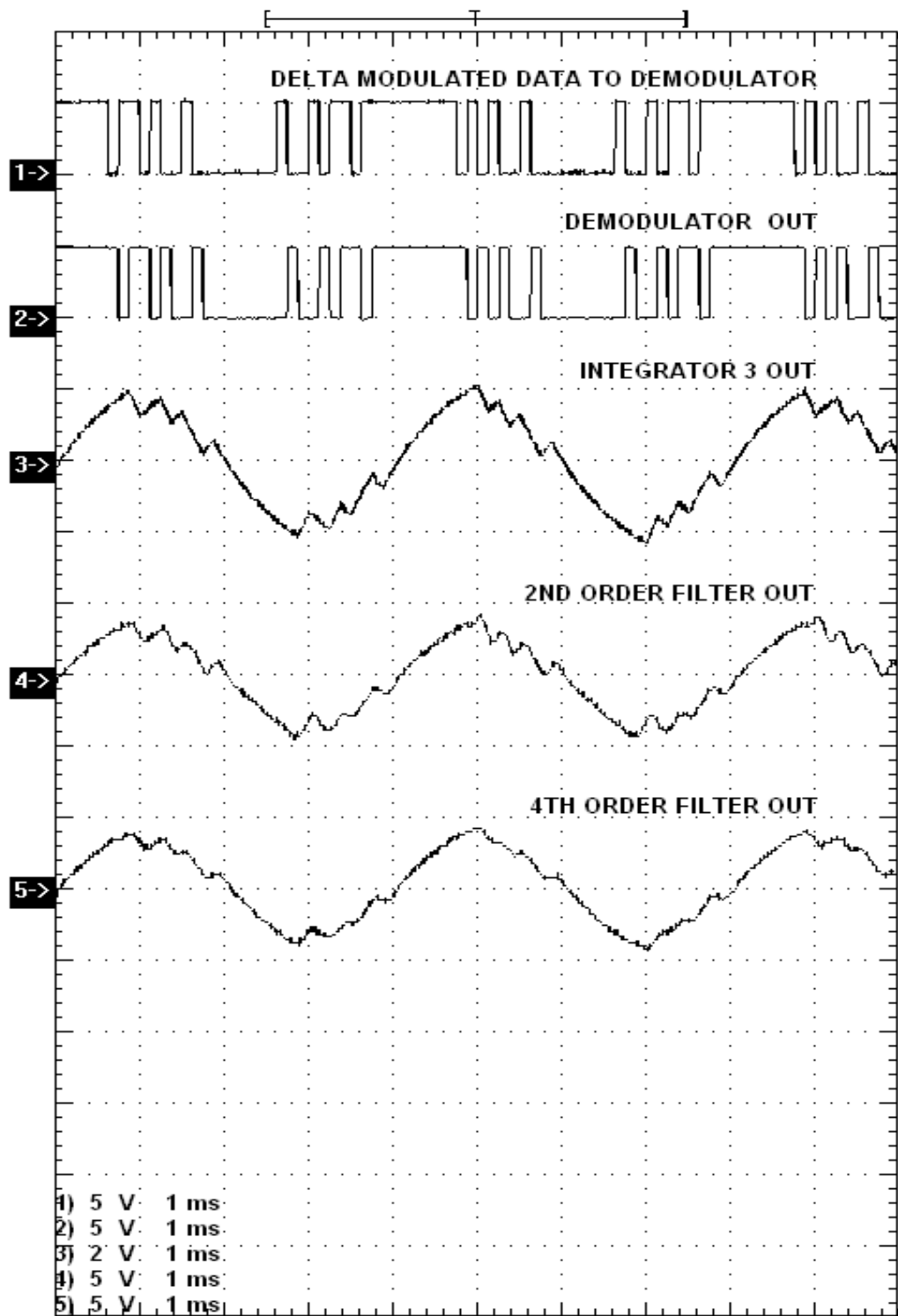
Filter Outputs **OUT 31**

CONCLUSION

Slope overload distortion is due to the use of a step size delta that is too small to follow portions of the waveform that have a steep slope. It can be reduced by increasing the step size.



Study of Slop Overload Effect



Study of Slop Overload Effect

10C. Adaptive Delta Modulation & Demodulation

THEORY

Delta Modulation system is unable to chase the rapidly changing information of the analog signal. That is the reason why more distortions are observed, which leads to poor quality reception. This is known as slope overloading phenomenon. The problem can be overcome by increasing the integrator gain. Adaptive Delta Modulation is a variation of Delta Modulation, which offers relief from disadvantage of DM by adopting the step size to accommodate changing signal conditions. If the input signal is large, step is caused to increase, thereby reducing slope overload effects. The block diagram of ADM is as shown in fig 3.1. It is same as Delta Modulation, except the variable gain circuit and step size controller. The controller keeps sensing the slope condition of the message conveyed. If the slope is large the controller output causes the variable gain circuit to have large gain. If the slope is small, the controller output causes a small gain.

In certain cases Adaptive Delta Modulation does not change step size on a pulse-to-pulse basis, but changes are made much more slowly. Such slow control is referred to as syllabic. The usual implementation involves a continuously variable slope Delta (CVSD). There are varieties of IC's for CVSD encoding and decoding in today's semiconductor market.

The CVSD is the simple alternative to more complex conventional conversion techniques in system requiring digital communication of analog signals. The CVSD A/D is well suited for the requirements of digital communications. A Delta Modulator consists of a comparator in the forward path and an integrator in the feedback path of a simple control loop. The input to the comparator is the simple analog signal and the integrator output. The comparator output is the difference between the input voltage and the integrator output. That sign bit is the digital output and also controls the direction of ramp in the integrator. The output of comparator is fed to the sampler. Then, the sampler output is fed to the slope polarity switch and level detect algorithm. The level detect algorithm is again fed to the slope magnitude control followed by the slope polarity switch. The output slope polarity switch is fed to the integrator in the control loop. With no input at the transmitter, continuous 1 and 0 alternations are transmitted. The outstanding characteristic is its ability to transmit the intelligible voice out at a relatively low data rate. Companded PCM for telephone quality transmission requires about 64Kbits/sec. data rate/channel. CVSD produces equal quality at 32Kbit/sec. In CVSD Decoder CVSD mod output is fed to the input of comparator.

The comparator output is fed to the internal shift register. Then, the output of internal shift register is fed to the digital logic followed by slope polarity switch and integrator. The output of integrator is fed to the low pass filters for the reconstruction of original signal.

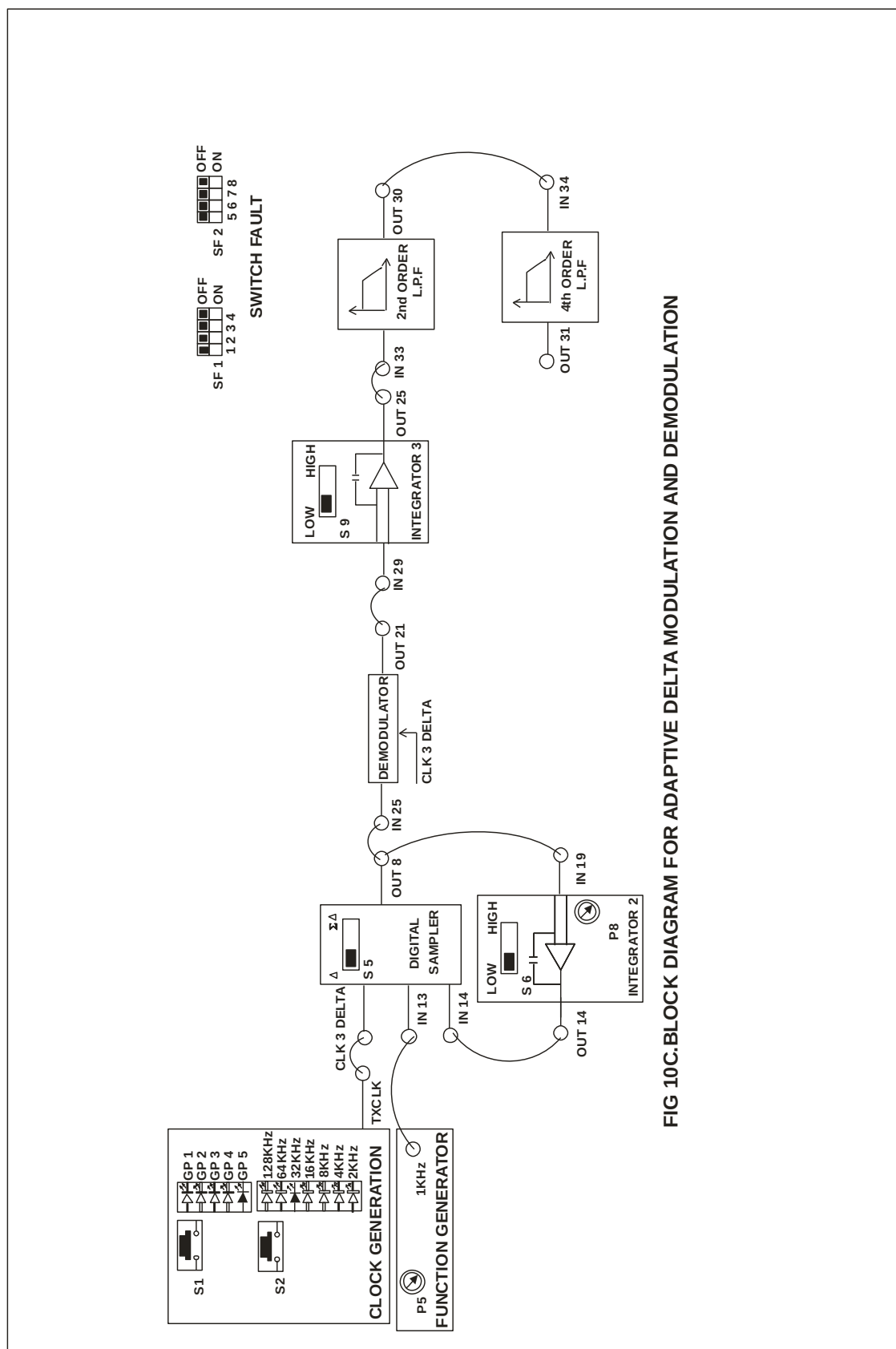


FIG 10C.BLOCK DIAGRAM FOR ADAPTIVE DELTA MODULATION AND DEMODULATION

PROCEDURE

- Ensure that the group 5 (**GP5**) clock is selected in the clock generation section. Selection is done with the help of switch **S1**. Observe the corresponding LED indication.
- Select the transmitter clock of frequency 32 KHz using switch **S2** and the selected clock is indicated on the corresponding LED indication in the Clock Generation section.
- Connect **1 KHz** sine wave having amplitude of 2Vpp, using pot **P5** to **IN13** post and **TXCLK** to **CLK3 DELTA** post of the Digital Sampler.
- Keep switch **S5** in Delta position.
- Connect **OUT 8** post of the Digital sampler to **IN19** post of Integrator 2.
- Keep the switch **S6** of the Integrator 2 to Low position.
- Connect **OUT14** post of Integrator 2 to **IN14** post of the Digital Sampler.
- Observe the modulated output at **OUT8** post of the Digital Sampler.
- Adjust Pot **P8** and observe the integrated output at the **OUT14** post Integrator 2 section.
- Connect the **OUT8** post of the Digital Sampler to **IN25** post of the Demodulator section.
- Connect the **OUT21** of the Demodulator section to the **IN29** post of the Integrator 3 section.
- Keep switch **S9** in the Integrator 3 section towards Low position. Connect **OUT25** post to the **IN33** post of the 2nd order LPF.
- Connect **OUT30** post of 2nd order LPF to **IN34** post of 4th order LPF.
- Observe the reconstructed signal through 2nd order LPF and 4th order LPF.
- Repeat the above procedure for different input signals and clock frequencies.

OBSERVATION

Sampling clock **TXCLK**

Input Signal

Integrator 1 output at feedback loop for Delta modulator **OUT 9**

Digital sampler Output **OUT 8**

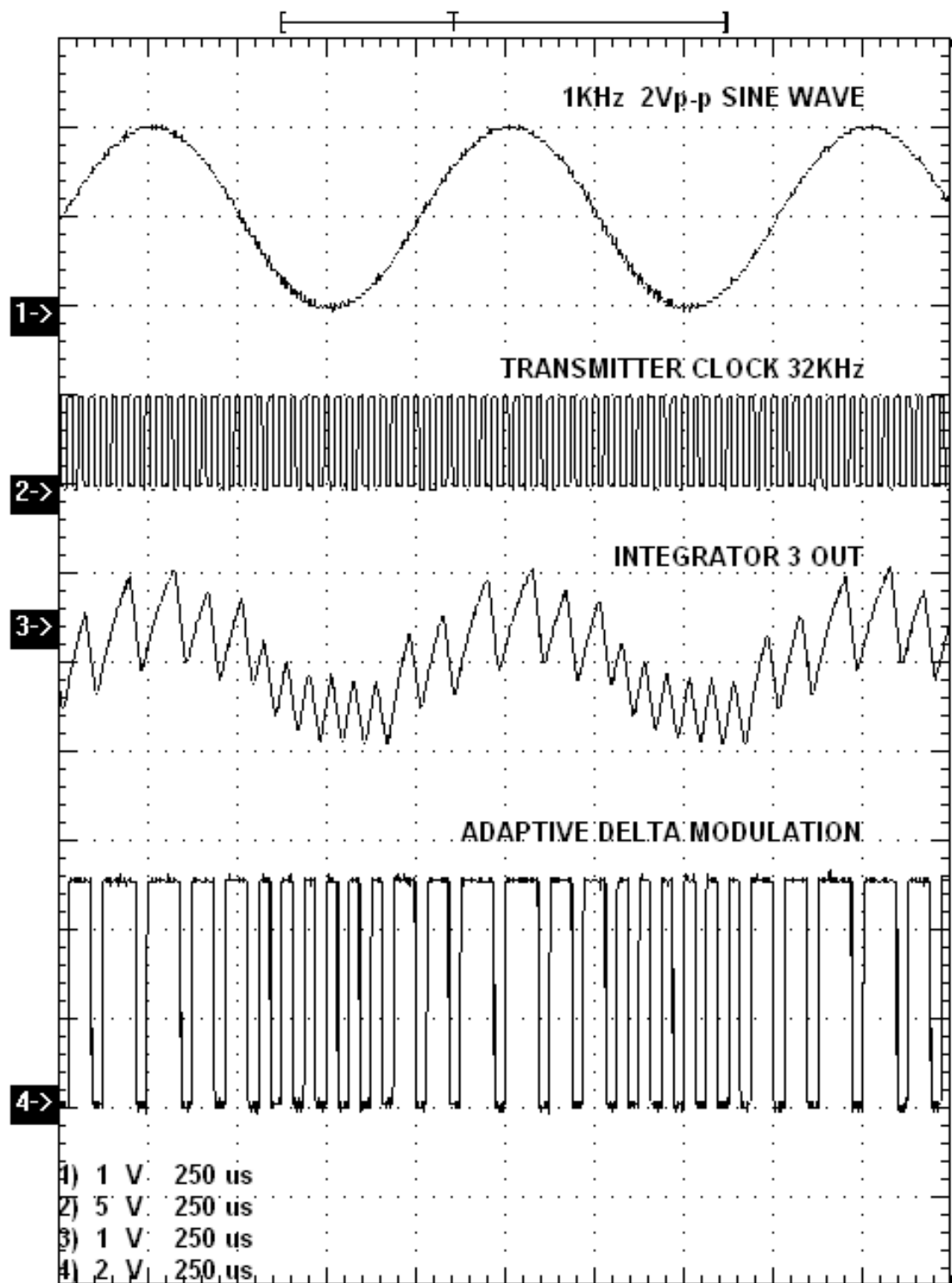
Demodulator Output **OUT 21**

Integrator 3 output **OUT 25**

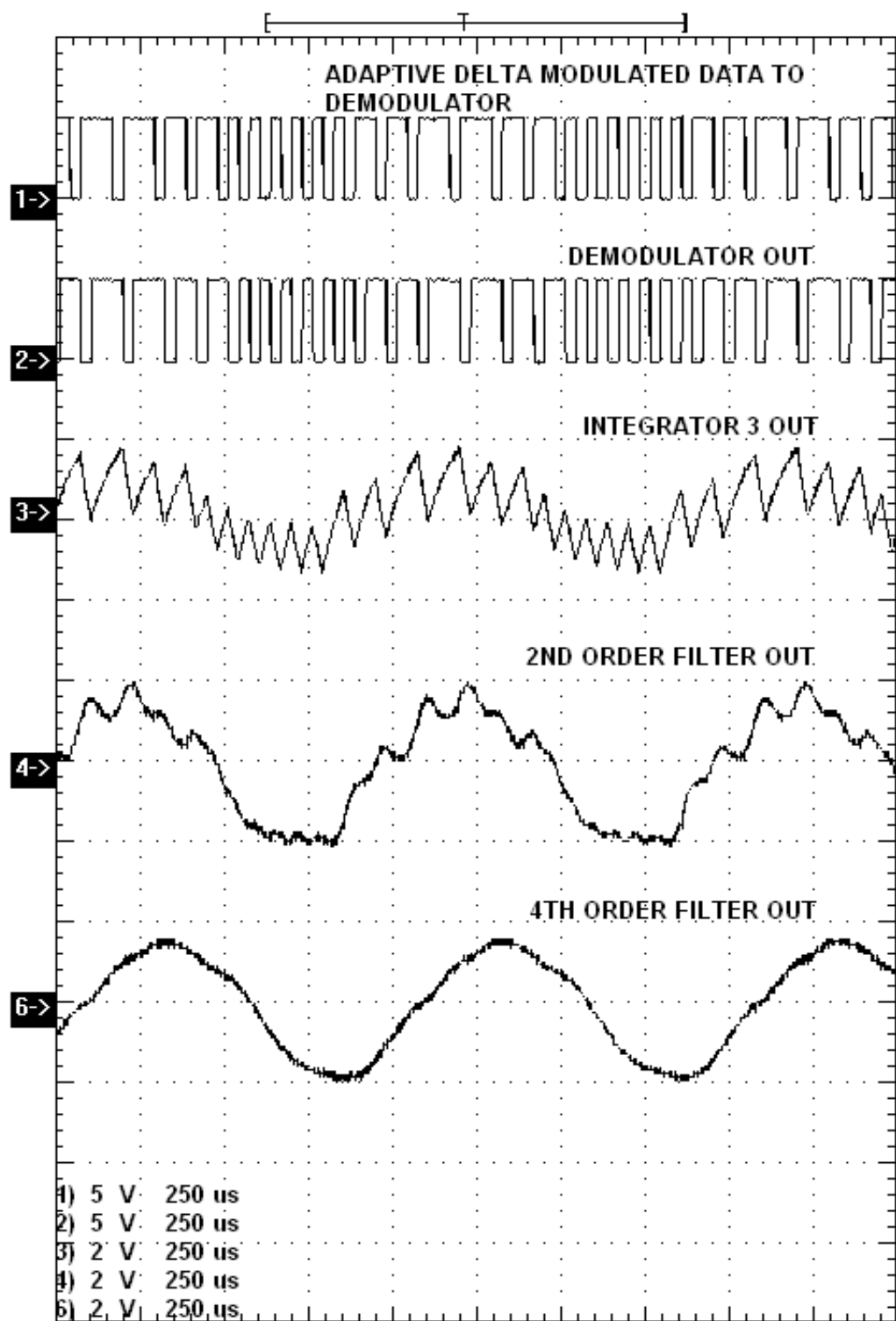
Filter Outputs **OUT 31**

CONCLUSION

ADM is similar to Delta Modulation with the ability to adjust the slope of the tracking signal. ADM attempts to increase the dynamic range and the tracking capabilities of futed step-size Delta Modulation. In feed forward adaptation, the step size of the quantizer is adapted in proportion to the input signal strength. In feedback adaptation, the adaptation is made based on the history of the quantizer's output.



Adaptive Delta Modulation

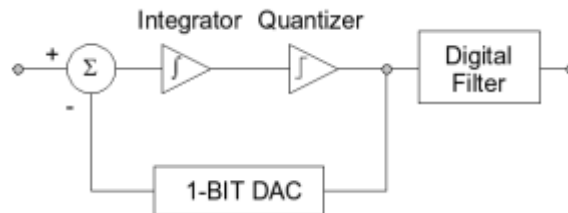


Adaptive Delta Demodulation

10D. To Study The Sigma – Delta Modulation And Demodulation

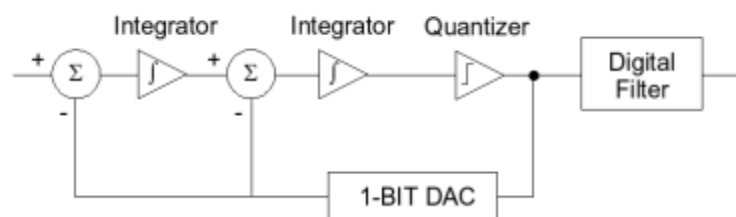
THEORY

The **Delta-Sigma** ($\Delta\Sigma$) modulation is a kind of an analog-to-digital signal or digital-to-analog conversion derived from delta modulation.



The quantizer (feedback) input in Delta Modulation is viewed as an approximation to the derivative of the incoming signal. This behavior leads to a drawback of delta modulation is that transmission disturbances such as noise result in an accumulative error in demodulated signal. This drawback is overcome by integrating the incoming signal prior to delta modulation. In this way, the modulator now responds to the actual voltage levels rather than the change in voltage level. Integrator at input of Delta modulator is called Sigma Delta modulator. Through Sigma delta modulator, we can transmit the DC signal and the Receiver design becomes simple. We require only low pass filter at the receiver side.

Principle



The principle of the sigma-delta architecture is to make rough evaluations of the signal, to measure the error, integrate it and then to compensate for that error. The mean output value is then equal to the mean input value if the integral of the error is finite.

The number of integrators, and consequently, the numbers of feedback loops, indicates the *order* of a $\Delta\Sigma$ -modulator, a 2nd order $\Delta\Sigma$ modulator. First order modulators are stable, but for higher order ones stability must be taken into great account.

Quantization theory formulas

When a signal is quantized, the resulting signal approximately has the second-order statistics of a signal with independent additive white noise. Assuming that the signal value is in the range of one step of the quantized value with an equal distribution, the mean square value of this quantization noise is

$$e_{\text{rms}}^2 = \frac{1}{\Delta} \int_{-\Delta/2}^{+\Delta/2} e^2 de = \frac{\Delta^2}{12}$$

In reality, the quantization noise is of course not independent of the signal; this dependence is the source of idle tones and pattern noise in Sigma-Delta converters.

Oversampling ratio, where f_s is the sampling frequency and $2f_0$ is Nyquist rate

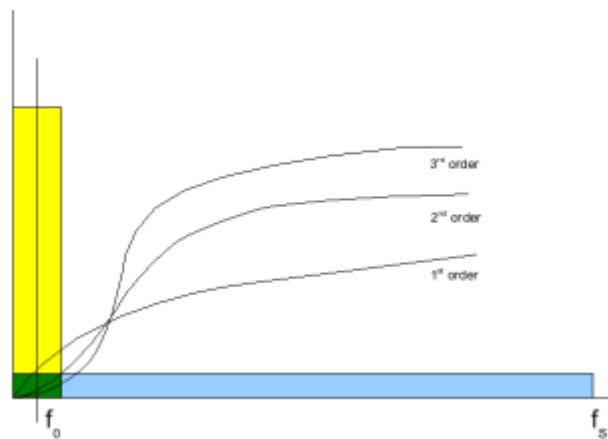
$$\text{OSR} = \frac{f_s}{2f_0} = \frac{1}{2f_0\tau}$$

The noise power within the band of interest can be expressed in term of OSR

$$\eta_0 = \frac{e_{\text{rms}}}{\sqrt{\text{OSR}}}$$

Oversampling:

Let us consider a signal at frequency f_0 and a sampling frequency of f_s much higher than the Nyquist rate.



$\Delta\Sigma$ modulation is based on oversampling technique to reduce the noise in the band of interest (green), which also avoids the use of high-precision analog circuits for the anti-aliasing filter. The quantization noise is the same both in a Nyquist converter (in yellow) and in an oversampling one (in blue), but it is distributed in a larger spectrum, in $\Delta\Sigma$ -converters, noise is further reduced at low frequencies, that is the band of interest where the signal is, and it is increased at the highest one, where it can be filtered. This property is known as noise shaping.

From a mathematical point of view, the previous noise power formula can be re-written for an N-order $\Delta\Sigma$ -modulator.

$$\eta_0 = \frac{e_{\text{rms}} \pi^N}{\sqrt{2N+1}} (2f_0\tau)^{(N+\frac{1}{2})}$$

It means that, higher the oversampling ratio, higher is the Signal-to-noise ratio and the higher is the resolution in bit.

Another key aspect given by oversampling is the exchange speed-resolution. In fact, the decimation filter put after the modulator not only filters the whole sampled signal in the band of interest (cutting the noise at higher frequencies), but also reduces the frequency of the signal increasing its resolution. This is obtained by a sort of averaging of the higher data rate bit stream.

EXAMPLE OF DECIMATION:

Let us have, for instance, an 8:1 decimation filter and a 1-bit stream. If we have an input stream like 10010110, counting the number of ones, the decimation result is $4/8 = 0.5 = 100$ in binary. In other words, we

- reduce by eight the frequency of the stream and
- the serial (1-bit) input bus become a parallel (3-bits) output bus.

Changes from Δ -modulation:

Δ -modulation requires an integrator to reconstruct the analog signal; moving this integrator (Σ) in front of the Δ -modulator simplify the design of the last stage filter. This is due to the different spectrum shaping of the two types of modulation: $\Delta\Sigma$ -modulator shapes the noise, leaving the signal as it is, while Δ -modulator leaves the noise as it is and shapes the spectrum of the signal, which has to be reconstructed by the aforementioned integrator.

PROCEDURE

- Ensure that the group 5 (**GP5**) clock is selected in the Clock Generation section. Selection is done with the help of switch **S1**. Observe the corresponding LED indication.
- Select the transmitter clock of frequency **32 KHz**, using switch **S2** and the selected clock is indicated on the corresponding LED indication in the Clock Generation section.
- Connect **500Hz** sine wave having amplitude of 2Vpp using pot **P4** to **IN15** post of the integrator1 section.
- Connect **OUT9** post of the Integrator 1 section to **IN13** post and **TXCLK** post to **CLK3** post of the Digital Sampler section.
- Keep switch **S5** in sigma delta position.
- Connect **OUT 8** post of the Digital Sampler to **IN19** post of Integrator 2.
- Keep the switch **S6** of the Integrator 2 to High position.
- Connect **OUT14** post of Integrator 2 to **IN14** post of the Digital Sampler.
- Observe the modulated output at **OUT8** post of the digital sampler.
- Adjust Pot **P8** and observe the integrated output at the **OUT14** post integrator2 section.
- Connect the **OUT8** post of the Digital Sampler to **IN25** post of the Demodulator section.
- Connect the **OUT21** of the Demodulator section to the **IN33** post of the 2nd order LPF.
- Connect **OUT30** post of the 2nd order LPF to **IN34** post of the 4th order LPF.
- Observe the reconstructed signal through the 2nd order LPF and the 4th order LPF.
- Repeat the above procedure for different input signals and clock frequencies.
- Repeat the above procedure for DC as input signal and set the clock frequency to 128 KHz or 68 KHz.

OBSERVATION

Sampling clock **TXCLK**

Input Signal

Integrator 1 output **OUT 9**

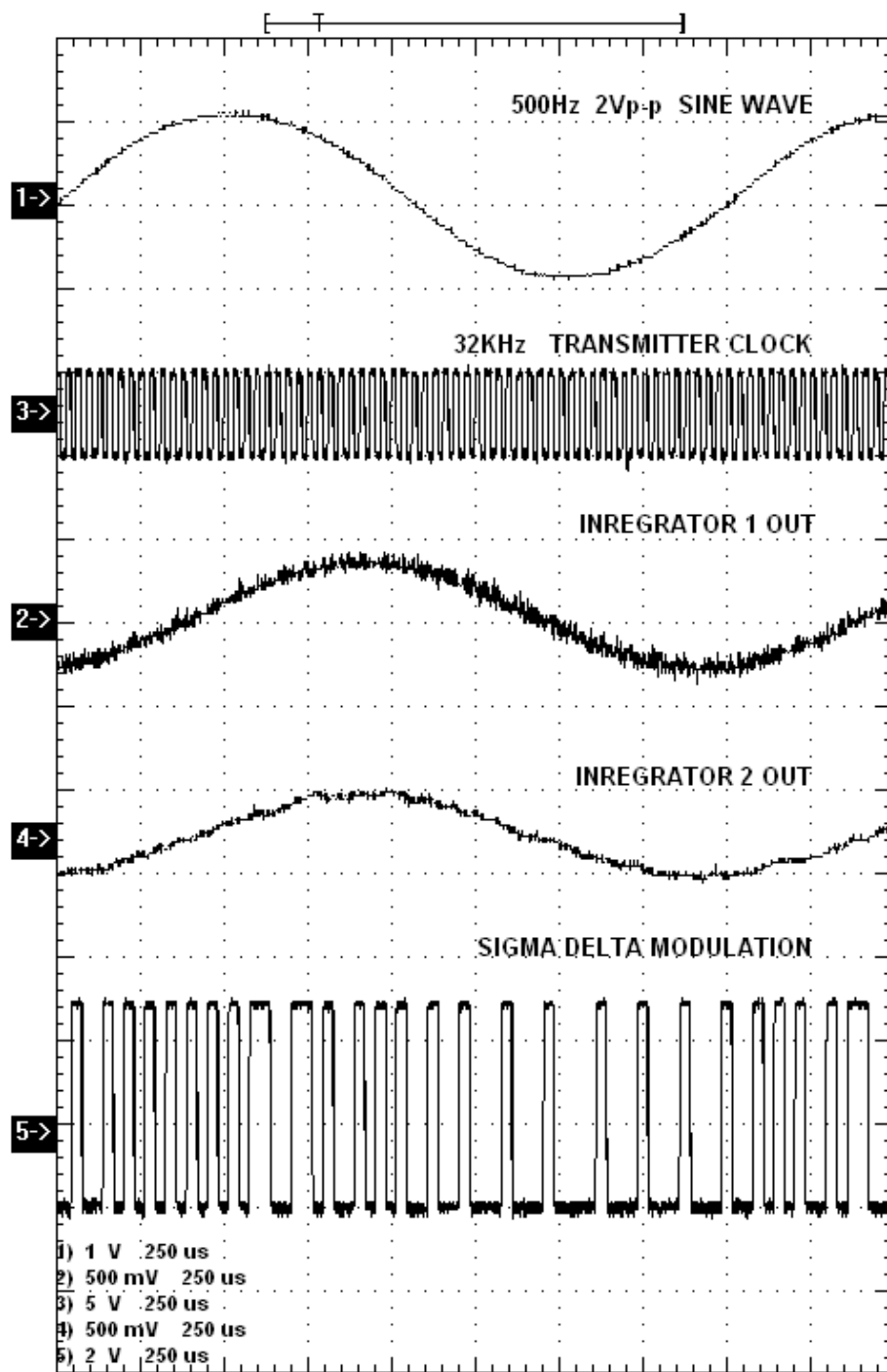
Digital sampler Output **OUT 8**

Integrator 2 output **OUT 14**

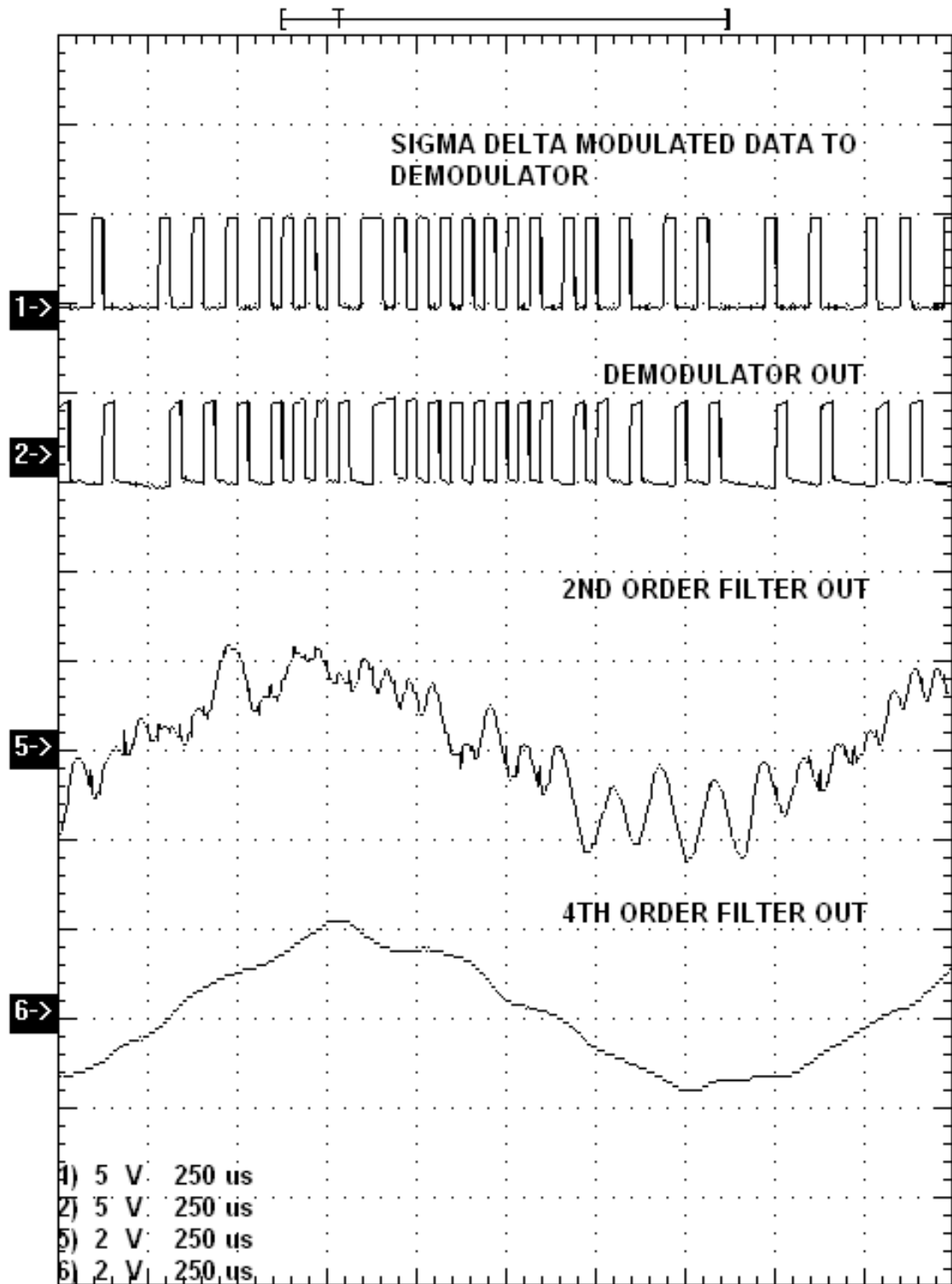
Filter Output **OUT 31**

CONCLUSION

The name sigma delta modulator comes from putting the integrator (Sigma) in front of the delta modulator. Sometimes the sigma delta modulator is referred to as an interpolative coder. The quantization noise characteristics (noise performance) of such a coder is frequency dependent in contrast to delta modulation

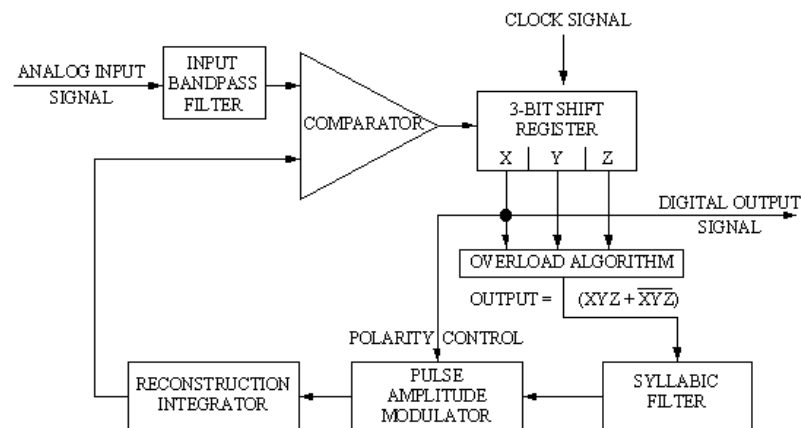
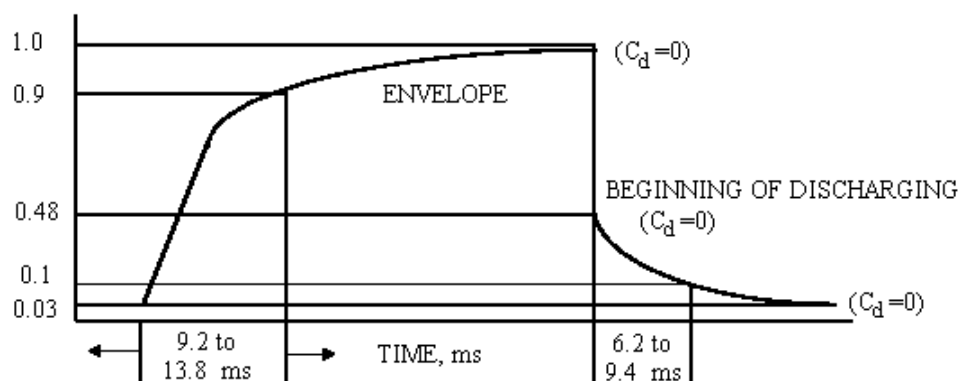
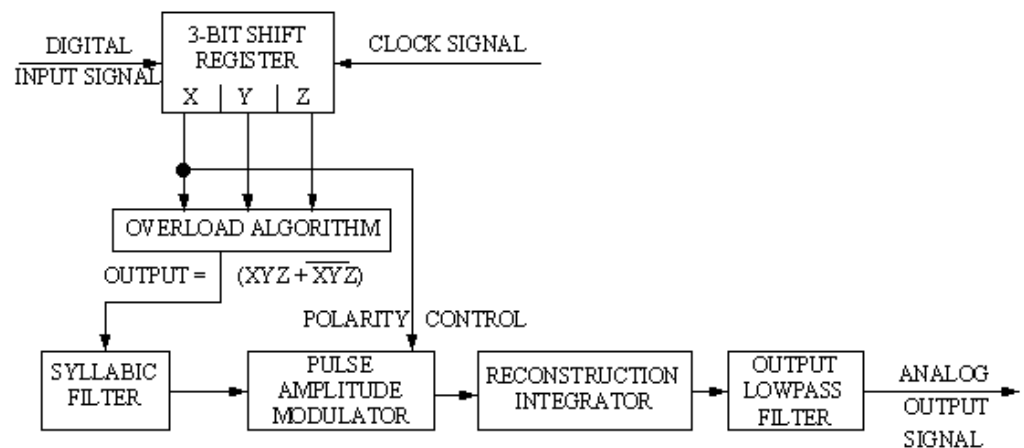


Sigma Delta Modulation



Sigma Delta Demodulation

**EXPERIMENT
NO. 11**

Typical CVSD encoder:Typical CVSD decoder:

NOTES:

1. C_d is duty cycle.
2. Figure is half of the envelope.

EXPERIMENT NO.11

NAME

Study Of Continuously Variable Slope Detector Technique

OBJECTIVES

- A. Study of Continuously Variable Slope Detector Modulation And Demodulation
- B. Study Of Companding System

THEORY

A. Study Of Continuously Variable Slope Detector Modulation And Demodulation

The Continuously Variable Slope Delta (CVSD) modulation is a nonlinear, sampled data, feedback system which accepts a band-limited analog signal and encodes it into binary form for transmission through a digital channel. At the receiver, the binary signal is decoded into a close approximation of the original analog signal.

A typical CVSD converter consists of an encoder and a decoder.

The analog input signal of the CVSD encoder is band-limited by the input band, pass filter. The CVSD encoder compares the band-limited analog input signal with an analog feedback approximation signal generated at the reconstruction integrator output. The digital output signal of the encoder is the output of the first register in the "run-of-three" counter. The digital output signal is transmitted at the clock (sample) rate and will equal "1" if the analog input signal is greater than or equal to the analog feedback signal at the instant of sampling. For this value of the digital output signal, the pulse amplitude modulator (PAM) applies a positive feedback pulse to the reconstruction integrator; otherwise, a negative pulse is applied. This function is accomplished by the polarity control signal, which is equal to the digital encoder output signal. The amplitude of the feedback pulse is derived by means of a 3-bit shift register, logic sensing for overload, and a syllabic low pass filter. When a string of three consecutive ONES or ZEROS appears at the digital output, a discrete voltage level is applied to the syllabic filter and the positive feedback pulse amplitude increases until the overload string is broken. In such an event, ground potential is fed to the filter by the overload algorithm, forcing a decrease in the amplitude of the slope voltage out of the syllabic filter. The encoder and decoder have identical characteristics except for the comparator and filter functions.

The CVSD decoder consists of an input band pass filter, shift register, overload algorithm, syllabic filter, PAM and reconstruction integrator used in the encoder and an output low-pass filter. The decoder performs the inverse function of the encoder and regenerates speech by passing the analog output signal of the reconstruction integrator through the low-pass filter. Other characteristics optimize the CVSD modulation technique for voice signals. These characteristics include the following:

- a. Changes in the slope of the analog input signal determine the step-size changes of the digital output signal.

- b. The feedback loop is adaptive to the extent that the loop provides continuous or smoothly incremental changes in step size.
- c. Companding is performed at a syllabic rate to extend the dynamic range of the analog input signal.
- d. The reconstruction integrator is of the exponential (leaky) type to reduce the effects of digital errors.

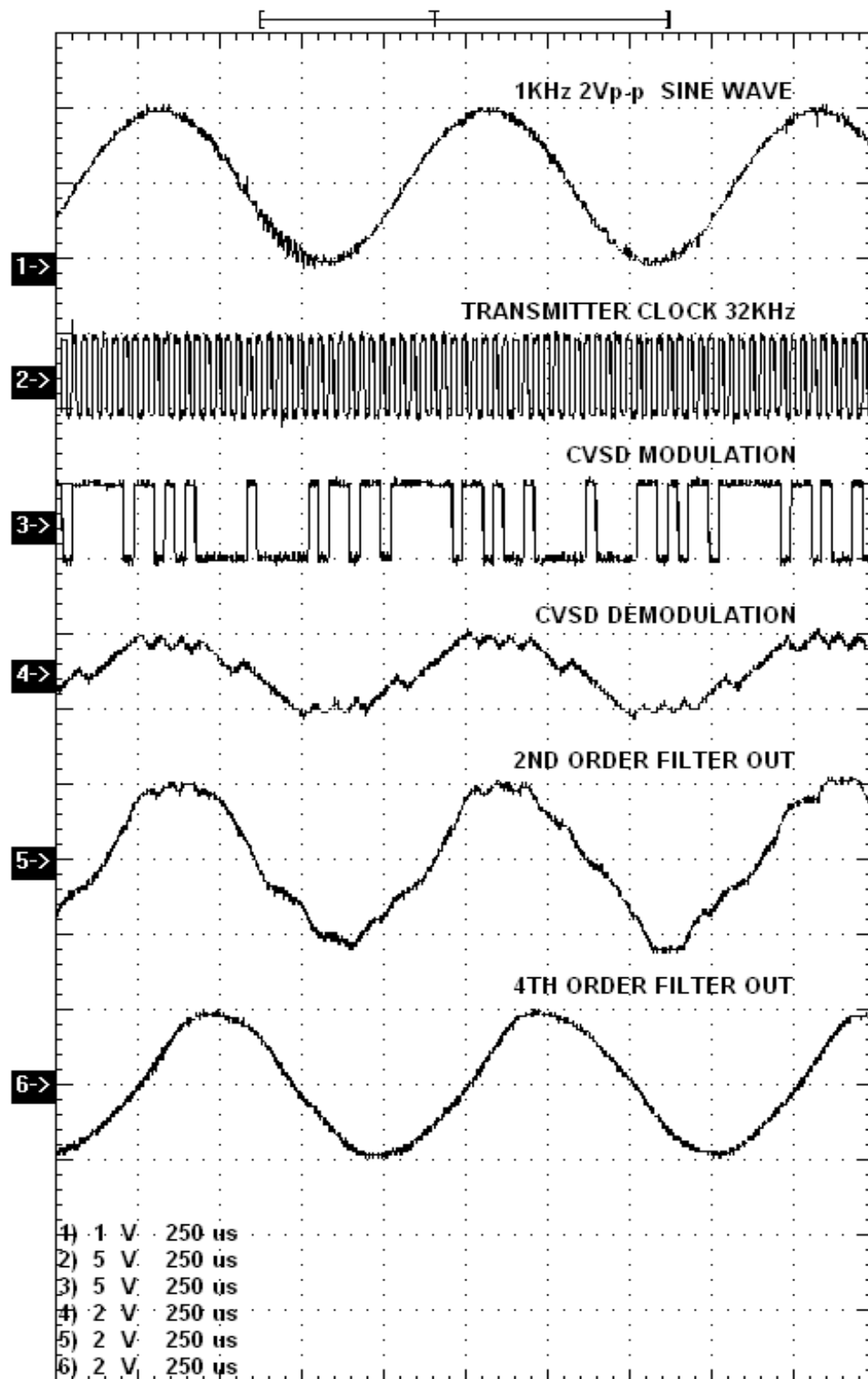
Continuously variable slope delta modulation (CVSD or CVSDM) is a voice coding method. It is a delta modulation with variable step size. CVSD encodes at 1 bit per sample, so that audio sampled at 16kHz is compressed to 16kbit/s. It is a lossy codec; the output of the decoder does not match the input to the encoder.

The encoder maintains a reference sample and a step size. Each input sample is compared to the reference sample. If the input sample is larger, the encoder emits a 1 bit and adds the step size to the reference sample. If the input sample is smaller, the encoder emits a 0 bit and subtracts the step size from the reference sample. The encoder also keeps the previous N bits of output ($N = 3$ or $N = 4$ are very common) to determine adjustments to the step size; if the previous N bits are all 1's or 0's, the step size is doubled. Otherwise, the step size is halved. The step size is adjusted for every input sample processed.

The decoder reverses this process, starting with the reference sample, and adding or subtracting the step size according to the bit stream. The sequence of adjusted reference samples are the reconstructed waveform, and the step size is doubled or halved according to the same all-1's-or-0's logic as in the encoder.

Adaptation of step size allows one to avoid slope overload (step of quantization increases when the signal rapidly changes) and decreases granular noise when the signal is constant (decrease of step of quantization).

CVSD is sometimes called a compromise between simplicity/low bit rate/quality. Bit rates are 9.6 to 128 Kbit/s.



CVSD Modulation And Demodulation

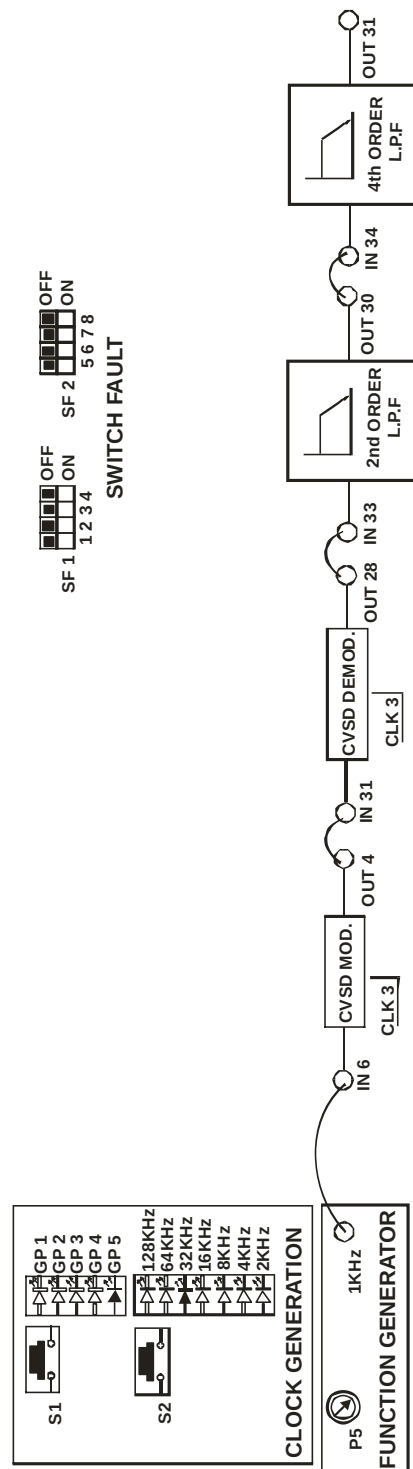


FIG 11A.BLOCK DIAGRAM FOR CONTINUOUSLY VARIABLE SLOPE DETECTOR MODULATION AND DEMODULATION

EQUIPMENTS

DCS-B kit
Connecting Chords
Power supply
20 MHz Dual Trace Oscilloscope
Power connection cables

A. Study Of Continuously Variable Slope Detector Modulation And Demodulation

PROCEDURE

- Ensure that the group 5 (**GP5**) clock is selected in the Clock Generation section. Selection is done with the help of switch **S1**. Observe the corresponding LED indication.
- Select the transmitter clock of frequency **32 KHz** using switches **S2** and the selected clock is indicated on the corresponding LED indication in the Clock Generation section.
- Connect **1 KHz** sine wave having amplitude of 2Vpp using pot **P5** to **IN6** post of the CVSD mod. section.
- Connect TXCLK post to the **CLK3** post of the CVSD mod. section.
- Observe the CVSD modulated signal at the **OUT4** post of the CVSD mod. section.
- Connect **OUT4** post of the CVSD mod. section to **IN31** post of the CVSD demod. Section.
- Connect **OUT28** post of the CVSD demod. Section to **IN33** post of 2nd order LPF.
- Connect **OUT30** post of the 2nd order LPF to **IN34** post of the 4th order LPF.
- Observe the reconstructed signal through the 2nd order LPF and the 4th order LPF.
- Repeat the above procedure for different input signals and clock frequencies.

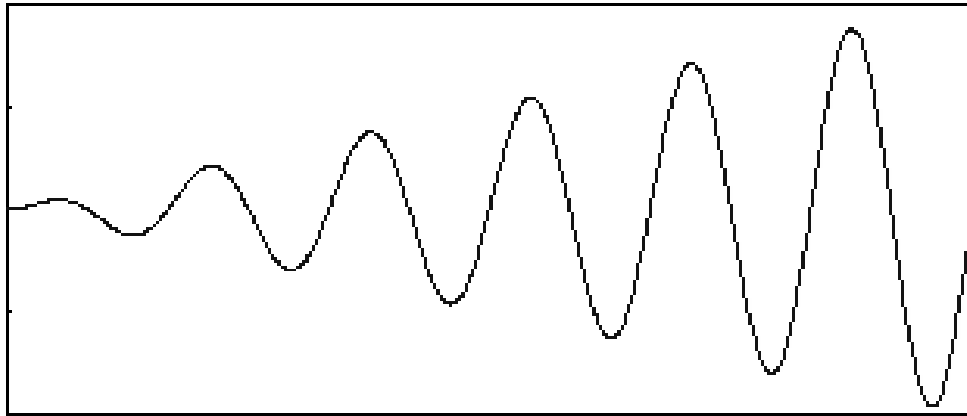
OBSERVATION

Sampling clock **TXCLK**
Input Signal
CVSD Modulator Output **OUT 4**
CVSD Demodulator Output **OUT 28**
Filter Output **OUT 31**

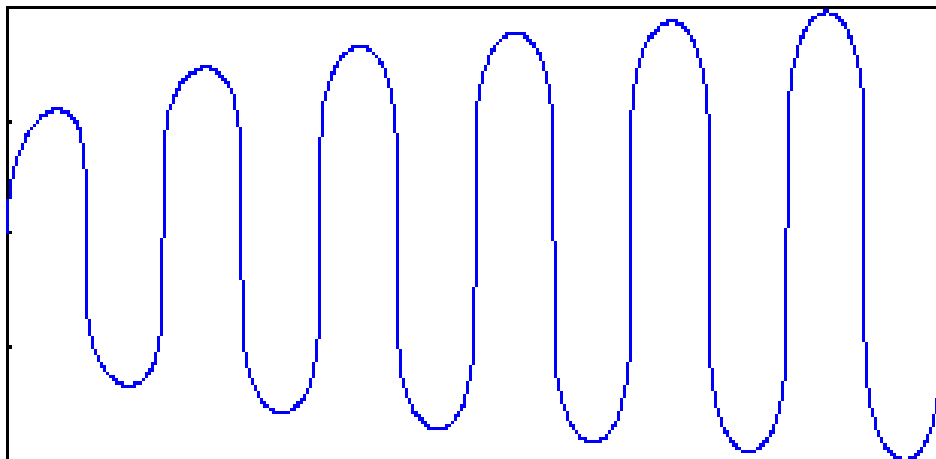
CONCLUSION

Continuously Variable Slope Delta modulation (CVSD) is differential waveform quantization techniques. it employ two level quantizes (one bit). CVSD is basically DM with an adaptive quantizer.

Applying adaptive techniques to a DM quantizer allows for continuous step size adjustment. By adjusting the quantization step size, the coder is able to represent low amplitude signals with greater accuracy (where it is needed) without sacrificing performance on large amplitude signals



A waveform before companding



A waveform after the compression stage of non-linear companding

11B. Study of Companding System

THEORY

In telecommunication, signal processing and thermodynamics, companding (occasionally called compansion) is a method of reducing the effects of a channel with limited dynamic range. It is a combination of compressing and expanding (the word companding therefore is a portmanteau word), and is a variant of audio level compression.

The compression used in audio recording depends on a variable-gain amplifier. Therefore it is a locally linear process (linear for short regions, but not globally). Companding is non-linear and takes place in the same way at all points in time. The dynamic range of a signal is compressed before transmission and is expanded to the original value at the receiver. Companding is also used in professional wireless microphones to improve the dynamic range of the microphone (the dynamic range of the microphone itself is higher than the dynamic range of the radio transmission).

The use of companding allows signals with a large dynamic range to be transmitted over facilities that have a smaller dynamic range capability. Companding reduces the noise and crosstalk levels at the receiver.

This is sometimes used in digital systems, compressing before input to an analog-to-digital converter, and then expanding after a digital-to-analog converter. This is equivalent to using a non-linear ADC. This is also used in digital file formats for better signal-to-noise ratio at lower bit rates. For example, a linearly encoded 16-bit PCM signal can be converted to an 8-bit WAV or AU file while maintaining a decent signal-to-noise ratio by compressing before the transition to 8-bit and expanding after a conversion back to 16-bit. This is effectively a form of lossy audio data compression.

A process called COMPANDING, a word formed by process of compressing and expanding, can materially improve the dynamic range. Companding is the process of compressing the signal amplitudes before transmission and expanding the compressed signal amplitudes after reception.

The dynamic range of Delta Modulation system is limited. The maximum input level is limited by slope overload and minimum input level is limited by quantization noise. The limited dynamic range of delta modulation can be increased by the use of Companding.

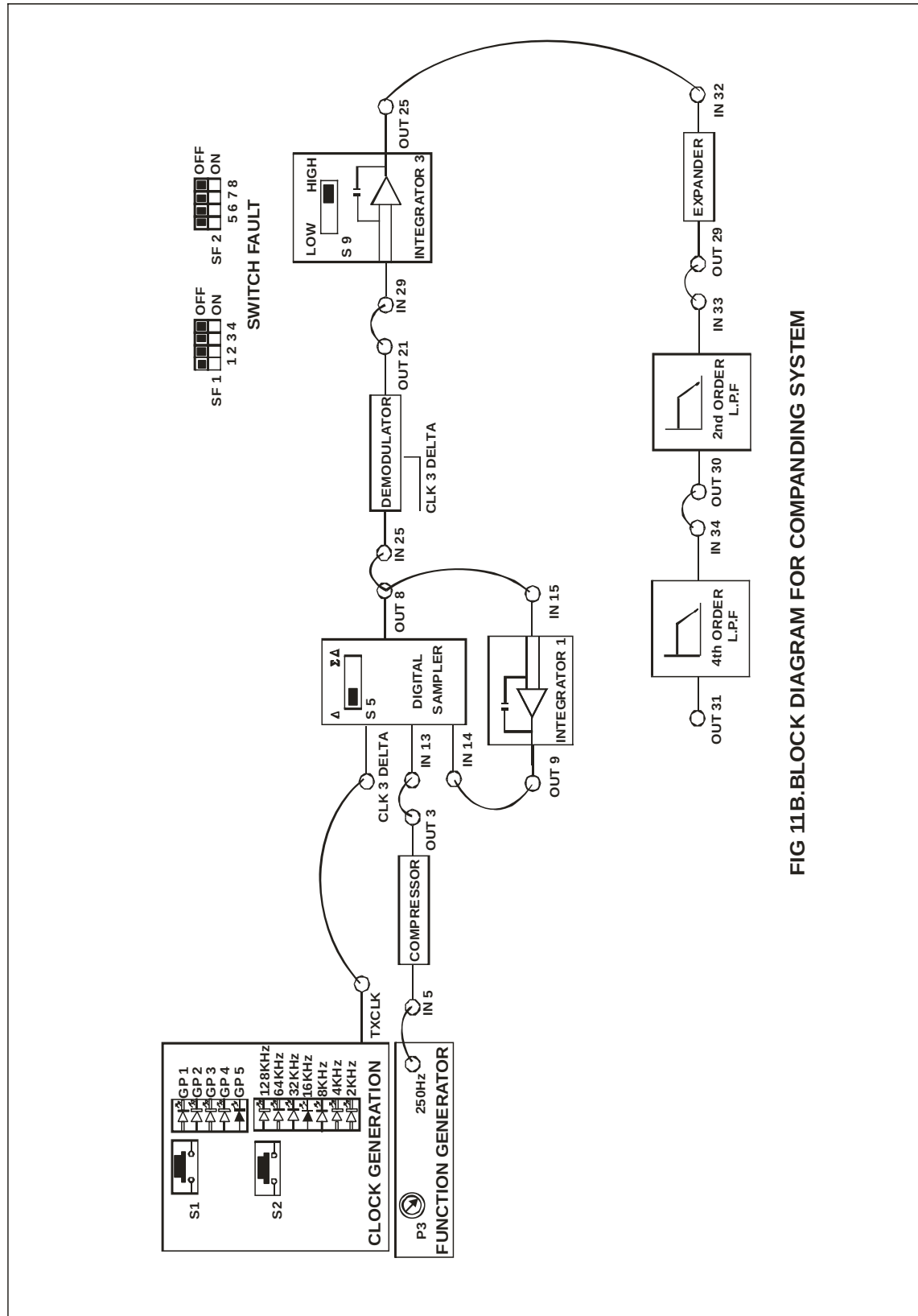


FIG 11B.BLOCK DIAGRAM FOR COMPANDING SYSTEM

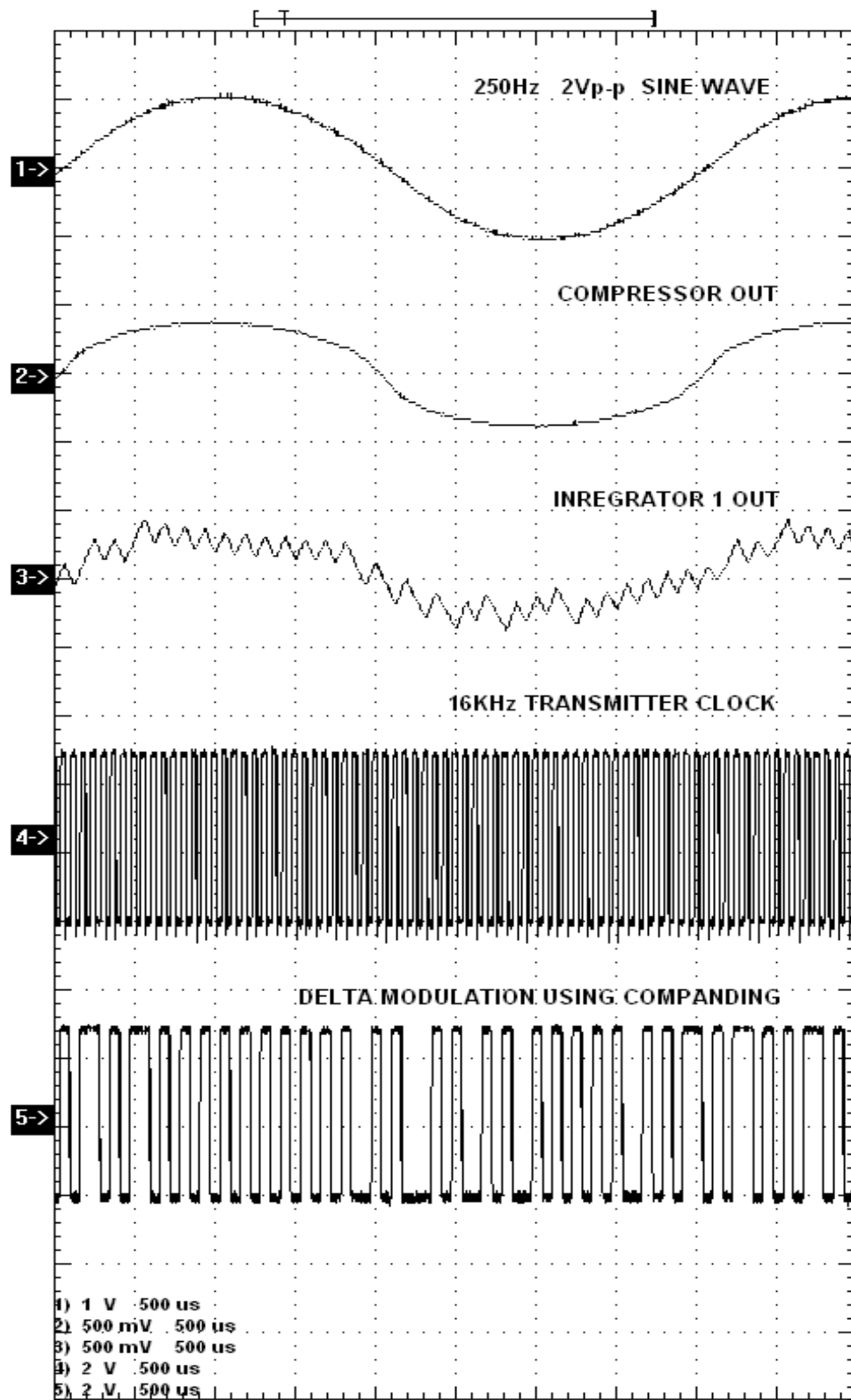
EQUIPMENTS

DCS-B kit
Connecting Chords
Power supply
20 MHz Dual Trace Oscilloscope
Power connection cables

11B. Study of Companding System

PROCEDURE

- Ensure that the group 5 (**GP5**) clock is selected in the Clock Generation section. Selection is done with the help of switch **S1**. Observe the corresponding LED indication.
- Select the transmitter clocks of frequency 16 KHz using switch **S2** and the selected clock is indicated on the corresponding LED indication in the Clock Generation section.
- Connect **250Hz** sine wave having amplitude of 2Vpp using pot **P3** to **IN5** post of the Compressor section.
- Connect **OUT3** post of the compressor section to **IN13** post and **TXCLK** post to **CLK3 DELTA** post of the Digital Sampler section.
- Keep switch **S5** in delta position.
- Connect **OUT 8** post of the Digital Sampler to **IN15** post of Integrator 1 section.
- Connect **OUT9** post of Integrator 1 to **IN14** post of the Digital Sampler.
- Observe the modulated output at **OUT8** post of the Digital Sampler.
- Connect the **OUT8** post of the Digital Sampler to **IN25** post of the Demodulator section.
- Keep switch **S9** towards High position.
- Connect the **OUT21** of the Demodulator section to the **IN29** post of the Integrator 3 section.
- Connect **OUT25** post of the Integrator 3 to the **IN32** post of the Expander section.
- Connect **OUT29** post of the Expander section to the **IN33** post of the 2nd order LPF.
- Connect **OUT30** post of the 2nd order LPF to **IN34** post of the 4th order LPF.
- Observe the reconstructed signal through the 2nd order LPF and the 4th order LPF.
- Repeat the above procedure for different input signals and clock frequencies.



Delta Modulation With Companding

OBSERVATION

Sampling clock **TXCLK**

Signal Input

Compressor Output **OUT 3**

Integrator 1 output **OUT 9**

Digital sampler Output **OUT 8**

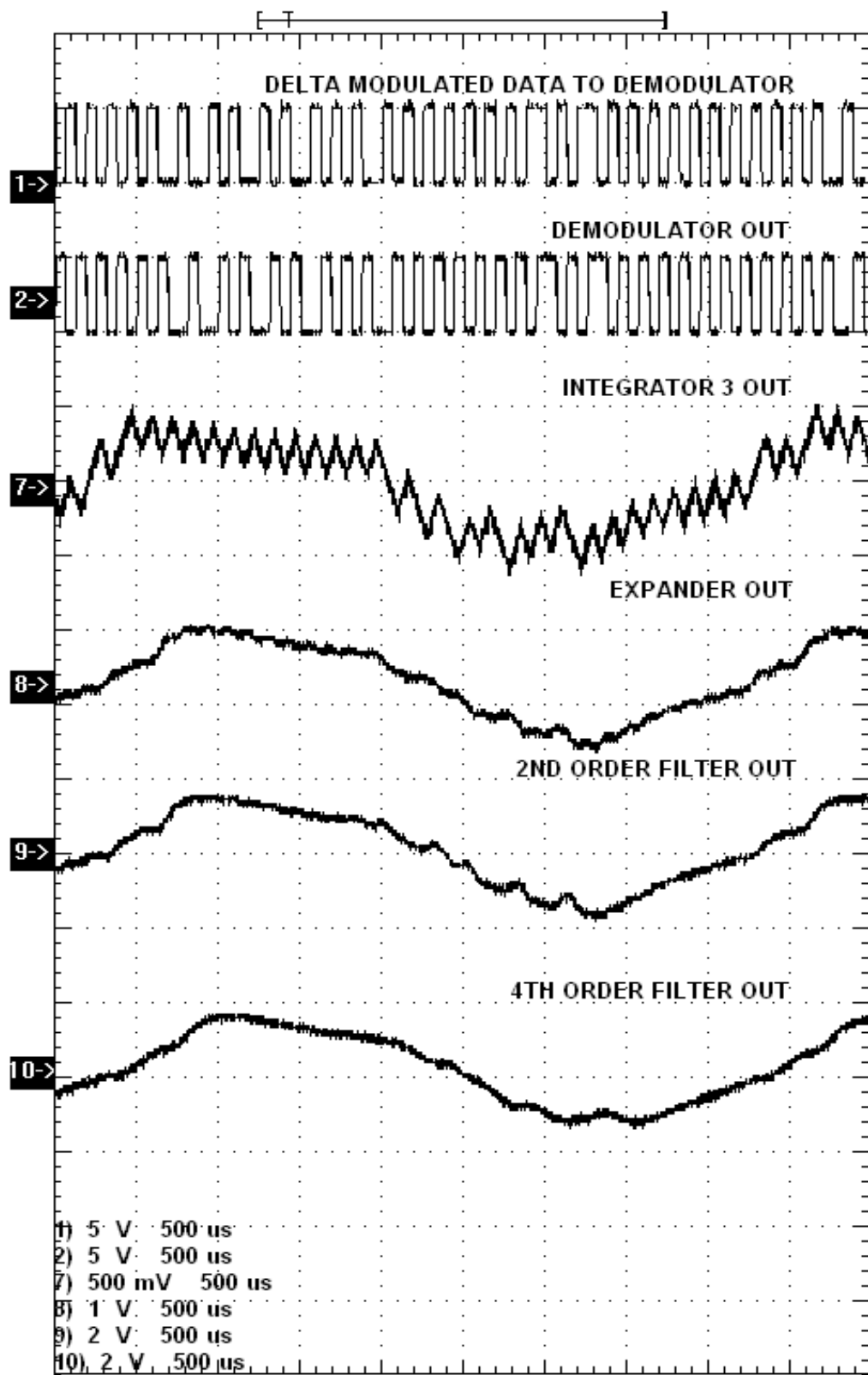
Integrator 3 output **OUT 25**

Expander Output **OUT 29**

Filter Output **OUT 31**

CONCLUSION

An adaptive delta modulation system is disclosed which features an improved digital compandor which may be used with a digital integrating network. The compandor executes a companding algorithm having a nonlinear characteristic to provide a rapid increase in step size to follow the fast rising start in input signals so as to minimize any slope overload distortions and to provide a small increase or decrease in step size .



DELTA DEMODULATION WITH EXPANDING

**EXPERIMENT
NO. 12**

EXPERIMENT NO.12

NAME

To study the Pulse Width Modulation

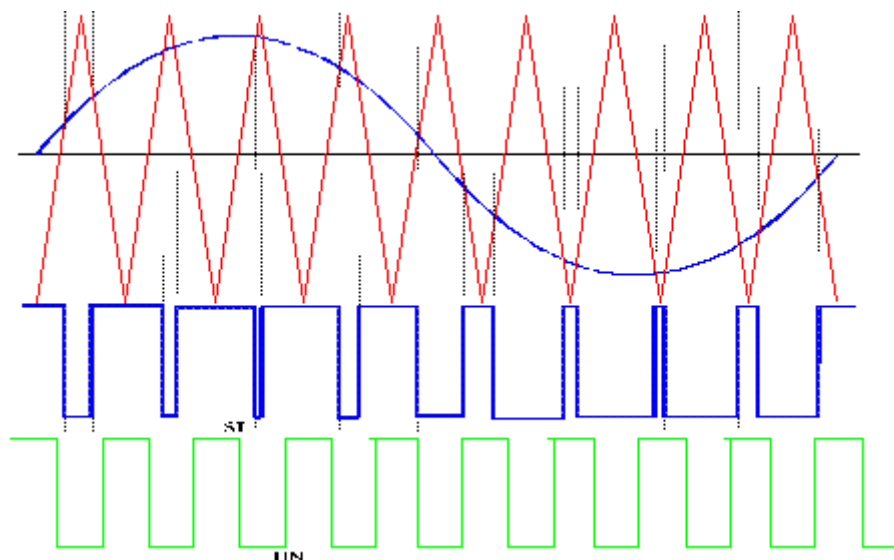
OBJECTIVES

- 12A. To study the Pulse Width Modulation
- 12B. To study the Pulse position Modulation

THEORY

Pulse Width Modulation (PWM) Basics

There are many forms of modulation used for communicating information. When a high frequency signal has amplitude varied in response to a lower frequency signal we have AM (amplitude modulation). When the signal frequency is varied in response to the modulating signal, we have FM (frequency modulation). These signals are used for radio modulation because the high frequency carrier signal is needed for efficient radiation of the signal. When communication by pulses was introduced, the amplitude, frequency and pulse width became possible modulation options. In many power electronic converters where the output voltage can be one of the two values, the only option is modulation of average conduction time.

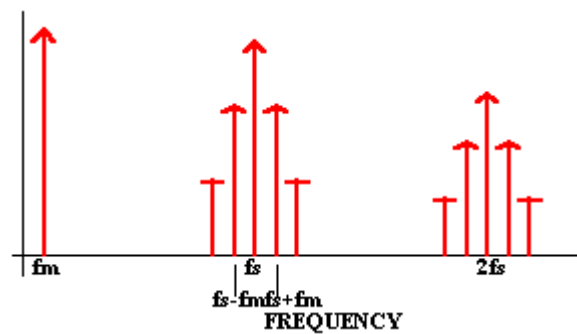


Unmodulated, sine modulated pulses

1. Linear Modulation:

The simplest modulation to interpret is where the average ON time of the pulses varies proportionally with the modulating signal. The advantage of linear processing for this application lies in the ease of de-modulation. The modulating signal can be recovered from the PWM by low pass filtering. For a single low frequency sine wave as modulating signal,

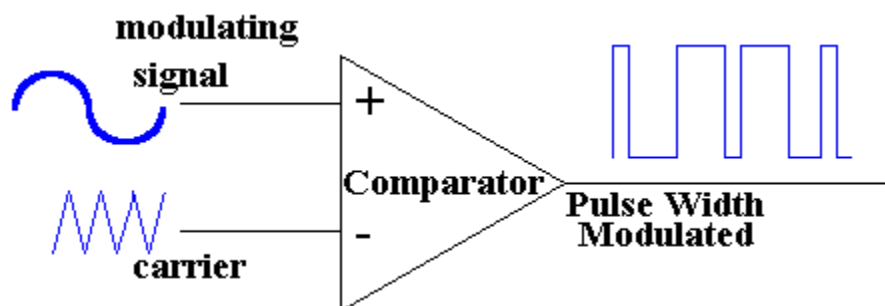
modulating the width of a fixed frequency (f_s) pulse train, the spectra is as shown in Fig 2. Clearly, a low pass filter can extract the modulating component f_m .



Spectra of PWM

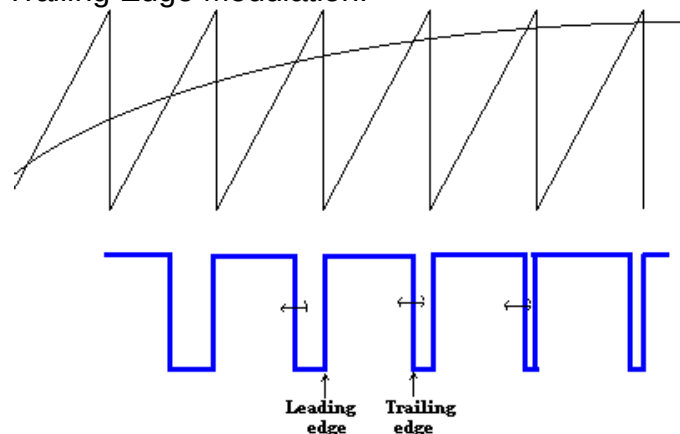
2.Sawtooth PWM:

The simplest analog form of generating fixed frequency PWM is by comparison with a linear slope waveform such as a saw tooth. As seen in Fig 2 the output signal goes high when the sine wave is higher than the saw tooth. This is implemented using a comparator whose output voltage goes to logic HIGH when the input is greater than the other.



Sine Saw tooth PWM

Other signals with straight edges can be used for modulation. A rising ramp carrier will generate PWM with Trailing Edge Modulation.

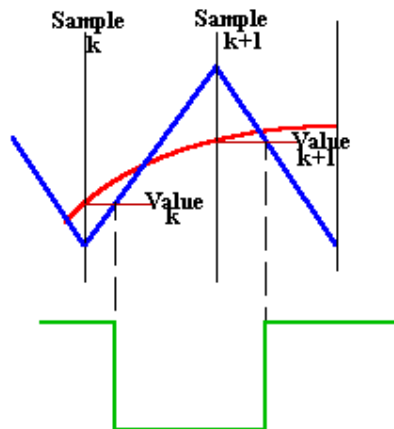


Trailing Edge Modulation

It is easier to have an integrator with a reset to generate the ramp as seen in Fig 4 but the modulation is inferior to double edge modulation.

3. Regular Sampled PWM:

The scheme illustrated above generates a switching edge at the instant of crossing of the sine wave and the triangle. This is an easy scheme to implement using analog electronics, but suffers the imprecision and drift of all analog computation as well as has difficulties in generating multiple edges when the signal has even a small-added noise. Many modulators are now implemented digitally but there is difficulty in computing the precise intercept of the modulating wave and the carrier. Regular sampled PWM makes the width of the pulse proportional to the value of the modulating signal, at the beginning of the carrier period. In Fig 5 the intercept of the sample values with the triangle determine the edges of the Pulses. For a saw tooth wave of frequency f_s the samples are at $2f_s$.

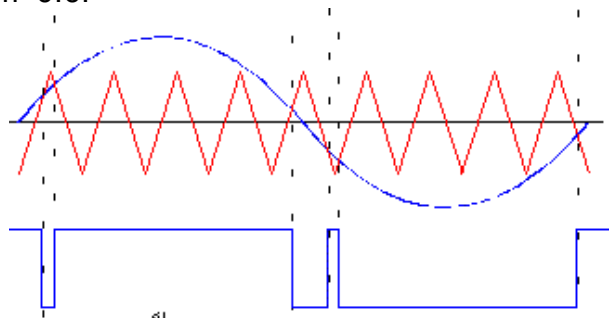


Regular Sampled PWM

There are many ways to generate a Pulse Width Modulated signal other than fixed frequency sine saw tooth. For three phase systems, the modulation of a Voltage Source Inverter can generate a PWM signal for each phase lag by comparison of the desired output voltage waveform for each phase, with the same saw tooth. One alternative, which is easier to implement in any computer and which gives a larger MODULATION DEPTH is using SPACE VECTOR MODULATION.

4. Modulation Depth:

For a single phase inverter, modulated by a sine-saw tooth comparison, if we compare a sine wave of magnitude from -2 to +2 with a triangle from -1 to +1 the linear relation between the input signal and the average output signal will be lost. Once the sine wave reaches the peak of the triangle the pulses will be of maximum width and the modulation will then saturate. The Modulation depth is the ratio of the current signal to the case when saturation is just starting. Thus, sine wave of peak 1.2 compared with a triangle with peak 2.0 will have a modulation depth of $m=0.6$.



Saturated Pulse Width Modulation:**Pulse Width Modulation**

Pulse Width Modulation refers to a method of carrying information on a train of pulses, the information being encoded in the width of the pulses.

In applications to motion control, it is not exactly information we are encoding, but a method of controlling power in motors without (significant) loss.

There are several schemes to accomplish this technique. One is to switch voltage on and off, and let the current recirculation through diodes when the transistors have switched off. Another technique is to switch voltage polarity back and forth with a full-bridge switch arrangement, with 4 transistors. This technique may have better linearity, since it can go right down to an effective 0% duty cycle by having the positive and negative voltage periods precisely equal. On/Off techniques may have trouble going down extremely close to 0% duty cycles and may jitter between minimum duty cycles of positive and negative polarity. In battery systems, PWM is the most effective way to achieve a constant voltage for battery charging by switching the system controller's power devices on and off. The generation of exact working PWM circuitry is complicated, but it is extremely conceptually important since there is good reason to believe that neurons transmit information using PWM spike trains.

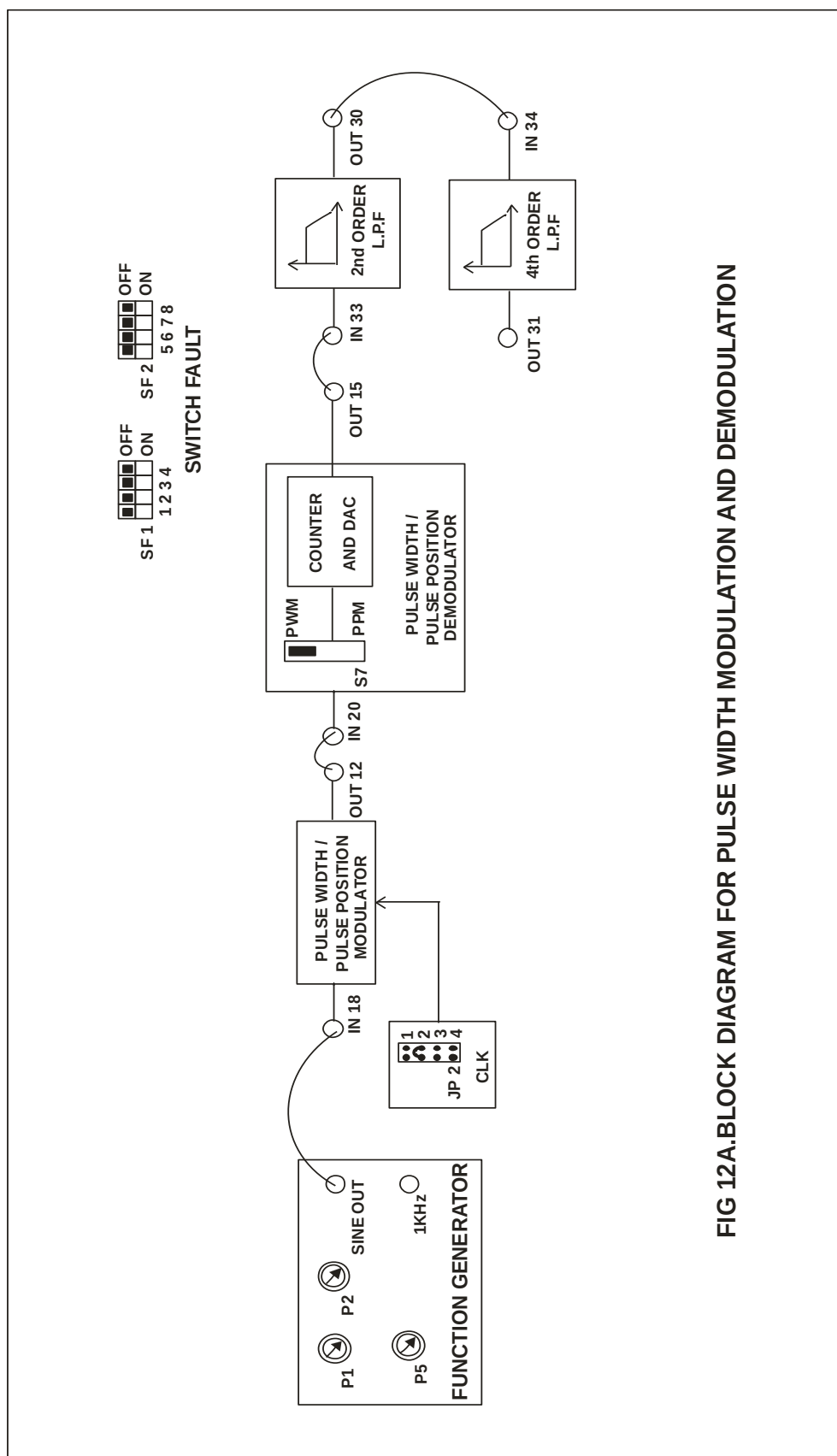


FIG 12A.BLOCK DIAGRAM FOR PULSE WIDTH MODULATION AND DEMODULATION

EQUIPMENTS

DCS-B kit
Connecting Chords
Power supply
20 MHz Dual Trace Oscilloscope
Power connection cables

NOTE: Due to high frequency we cannot observe the variation in the PWM signal. The following experiment is conducted to observe the pulse width variation.

12A.To Study The Pulse Width Modulation

PROCEDURE

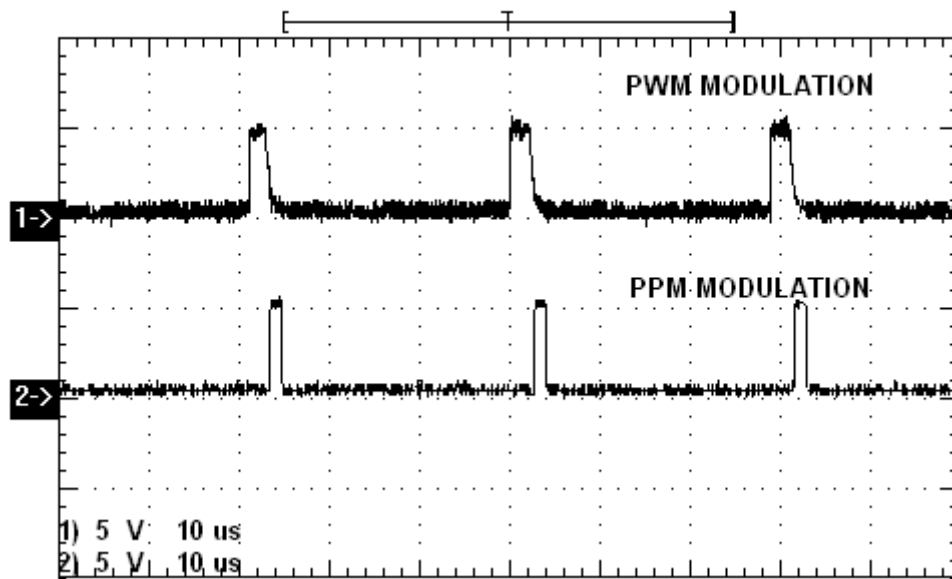
- Connect a low frequency sine wave from **SINE OUT** post having amplitude of 1Vpp using pot **P2** from the Function generator section to the **IN18** post of the PWM/PPM section.
- Keep jumper **JP2** on 2nd position.
- Observe the variation in the width of the carrier at the **OUT12** post of the PWM/PPM section, vary the frequency of input sine wave from 1 to 30 Hz using pot **P1** and observe the variation.
- Now connect **1 KHz** sine wave having amplitude of 1Vpp using pot **P5** from the Function generator section to the **IN18** post of the PWM/PPM section. Also, observe the counter outputs at their corresponding test points (**TP9** to **TP16**).
- Observe the pulse width modulated output at **OUT12** post of the PWM/PPM section.
- Connect **OUT12** post of the PWM/PPM to the **IN20** post of the PWM/PPM Demodulator section.
- Keep switch **S7** to PWM position.
- Observe the pulse width demodulated output at **OUT15** post of the PWM/PPM demodulator section.
- Connect **OUT15** post of the PWM/PPM demodulator section to the **IN33** post of the 2nd order LPF.
- Connect **OUT30** post of 2nd order LPF to **IN 34** of 4th order LPF
- Observe the recovered signal at the **OUT31** post of the 4th order LPF.
- Repeat the experiment for different input signals and sampling clocks by changing the position of the jumper **JP2**.

OBSERVATION

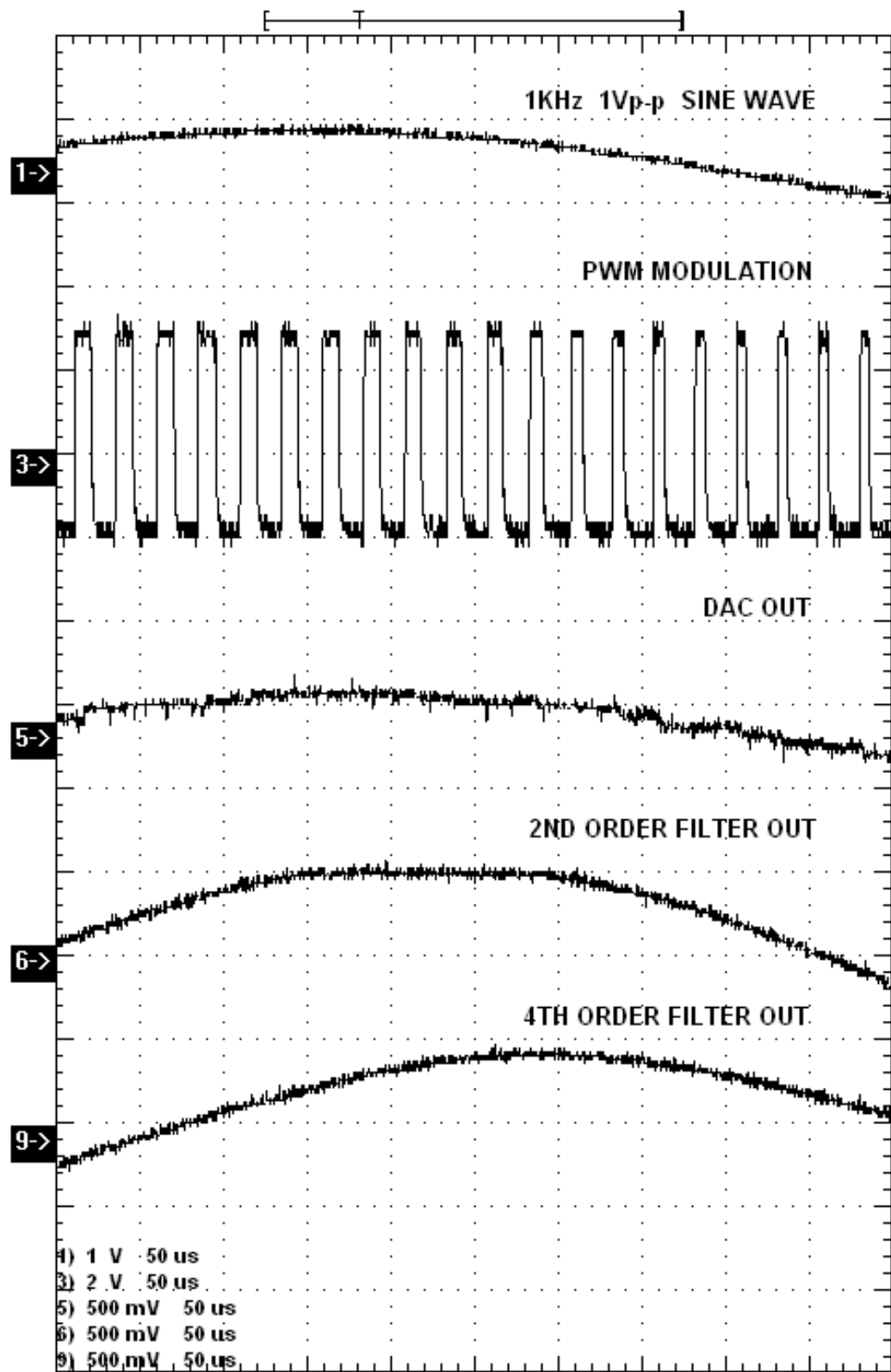
- Input Signal
- PWM Modulated Signal **Out 12**
- DAC Output **Out 15**
- Filter Output **Out 31**

CONCLUSION

PWM modulation is technique in which width of the carrier is modulated in accordance with the amplitude of the modulating signal

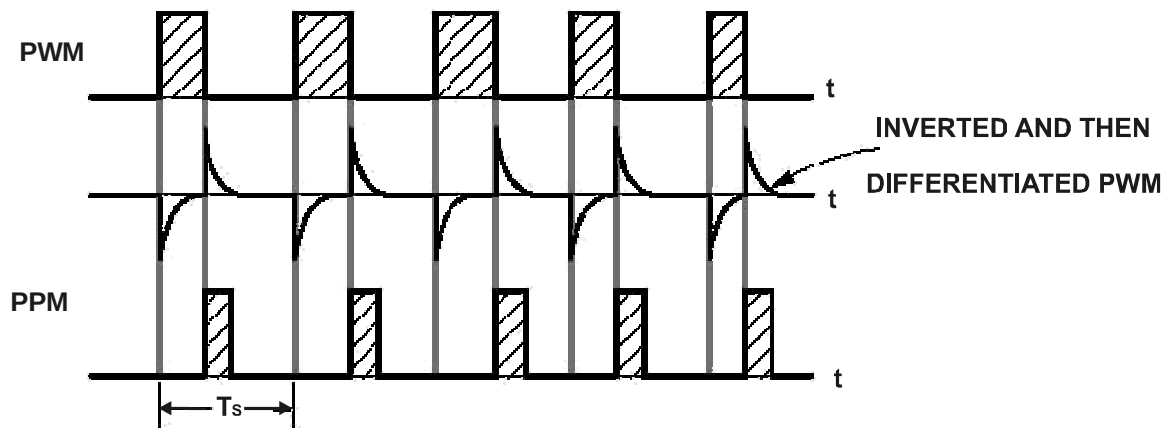


PWM/PPM using low frequency



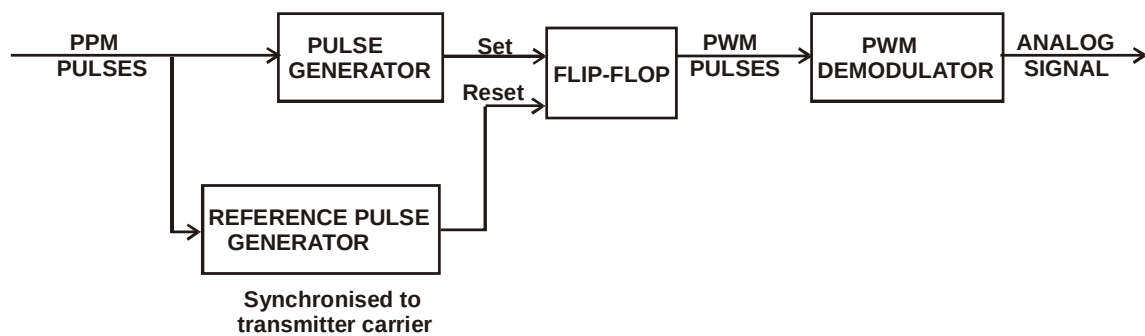
PWM Modulation And Demodulation

PPM GENERATION



PPM Demodulator

The demodulation can be processed by the generation of fixed period reference pulses from the incoming PPM pulse. These pulses set a flip-flop and the flip-flop is reset by the received PPM pulses, thereby at the output of the flip-flop a waveform is generated which can be demodulated by a PWM demodulator.



RECEPTION OF PPM PULSES (DEMODULATION)

12B. Study of Pulse Position Modulation And Demodulation

THEORY

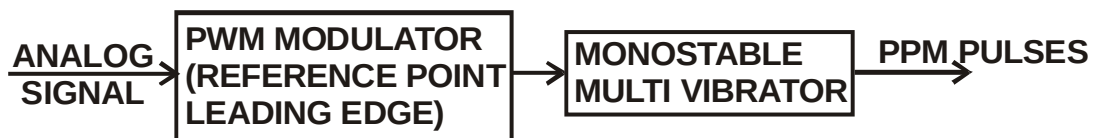
Pulse Position Modulation (PPM)

In this technique of modulation, the position of TTL pulse is changed on time scale, according to the variation of input, modulating signal amplitude. The pulse positions are directly proportional to the instantaneous values of the modulating signal.

The amplitude and width of the pulse is kept constant in the system. The position of each pulse, in relation to the position of a recurrent reference pulse, is varied by each instantaneous sampled value of the modulating wave. PPM has the advantage of requiring constant transmitter power since the pulses are of constant amplitude and duration. It is widely used but has a big disadvantage that it needs synchronization between the transmitter and the receiver.

PPM Modulator

For generating a PPM pulse, a PWM pulse modulator can be made to trigger a monostable multivibrator from the negative going edge of the PWM pulses. Thereby, producing a pulse of fixed height and width, at the negative going edge of the PWM pulse.



GENERATION OF PPM PULSES

EQUIPMENTS

DCS-B kit
Connecting Chords
Power supply
20 MHz Dual Trace Oscilloscope
Power connection cables

NOTE: Due to high frequency, we cannot observe the variation in the PPM signal. The following experiment is conducted to observe the pulse POSITION variation

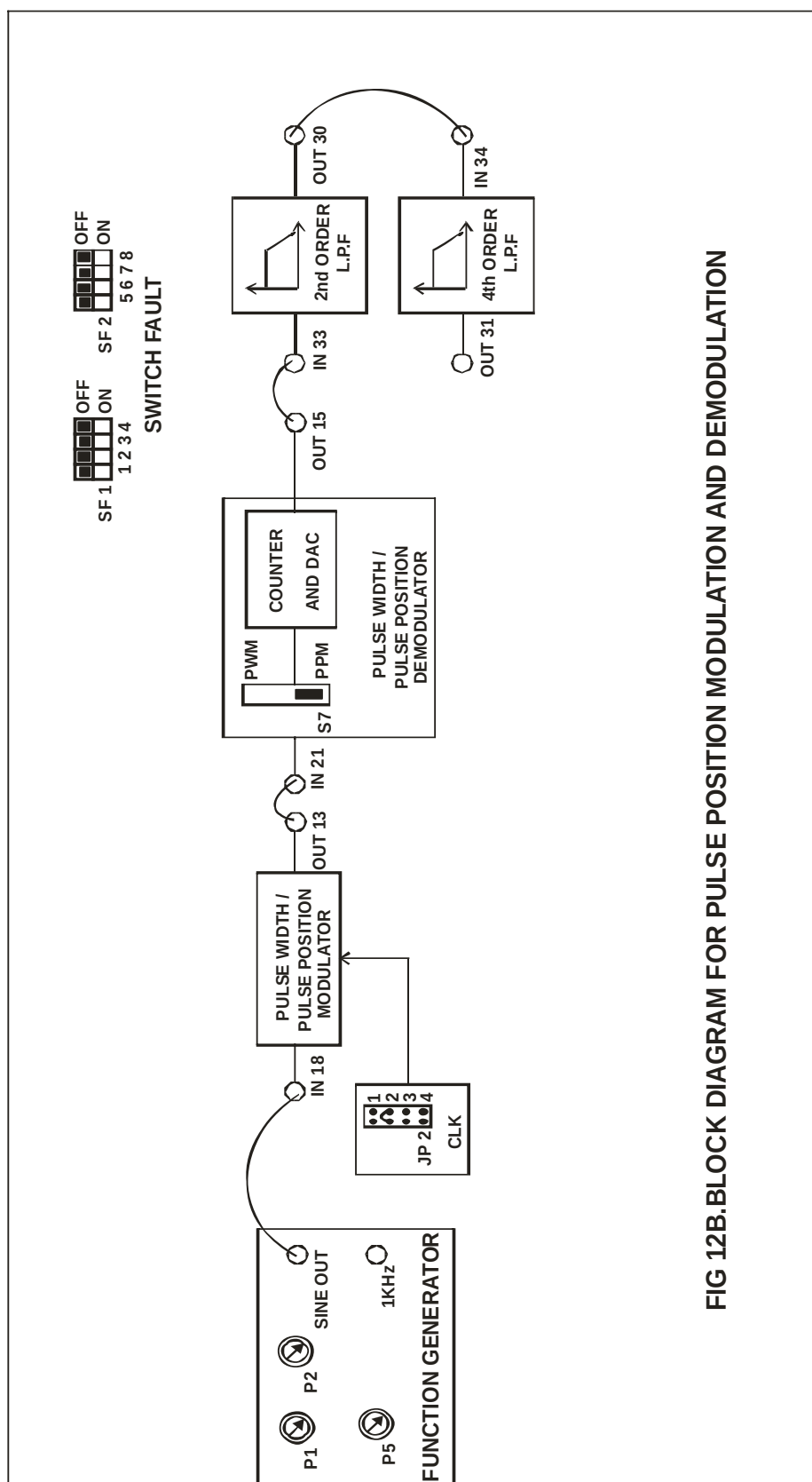
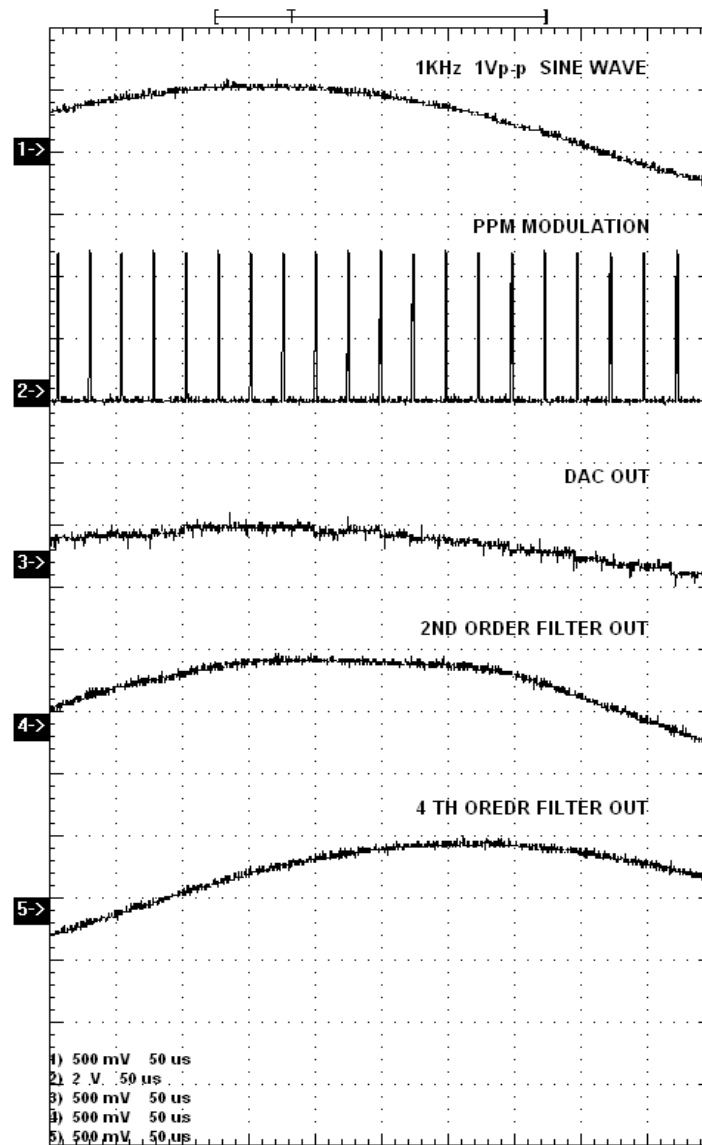


FIG 12B.BLOCK DIAGRAM FOR PULSE POSITION MODULATION AND DEMODULATION

PPM Modulation and Demodulation



12C. To Study Pulse Position Modulation

PROCEDURE

- Connect a low frequency sine wave from **SINE OUT** post having amplitude of 1Vpp, using pot **P2** from the Function generator section to the **IN18** post of the PWM/PPM section.
- Keep jumper **JP2** on the 2nd position.
- Observe the variation in the position of the carrier at the **OUT13** post of the PWM/PPM section. Change the frequency of input sine wave from 1 to 30 Hz using pot **P1** and observe the variation.
- Now connect **1 KHz** sine wave having amplitude of 1Vpp, using pot **P5** to the **IN18** post of the PWM/PPM section.

- Observe the pulse position modulated output at **OUT13** post of the PWM/PPM section. Also, observe the counter outputs at their corresponding test points (**TP9** to **TP16**).
- Connect **OUT13** post of the PWM/PPM to the **IN20** post of the PWM/PPM Demodulator section.
- Keep switch **S7** to PPM position.
- Observe the pulse width demodulated output at **OUT15** post of the PWM/PPM Demodulator section.
- Connect **OUT15** post of the PWM/PPM demodulator section to the **IN33** post of the 2nd order LPF.
- Connect **OUT30** post of 2nd order LPF to **IN 34** of 4th order LPF
- Observe the recovered signal at the **OUT31** post of the 4th order LPF.
- Repeat the experiment for different input signals and sampling clocks by changing the position of the jumper **JP2**.

OBSERVATIONS

Input Signal

PPM Modulated Signal **Out 13**

DAC Output **Out 15**

Filter Output **Out 31**

CONCLUSION

PPM modulation is technique in which Position of the carrier is modulated in accordance with the amplitude of the modulating signal. In PPM the width of carrier remains same, only position of pulse varies on time scale.

**EXPERIMENT
NO. 13**

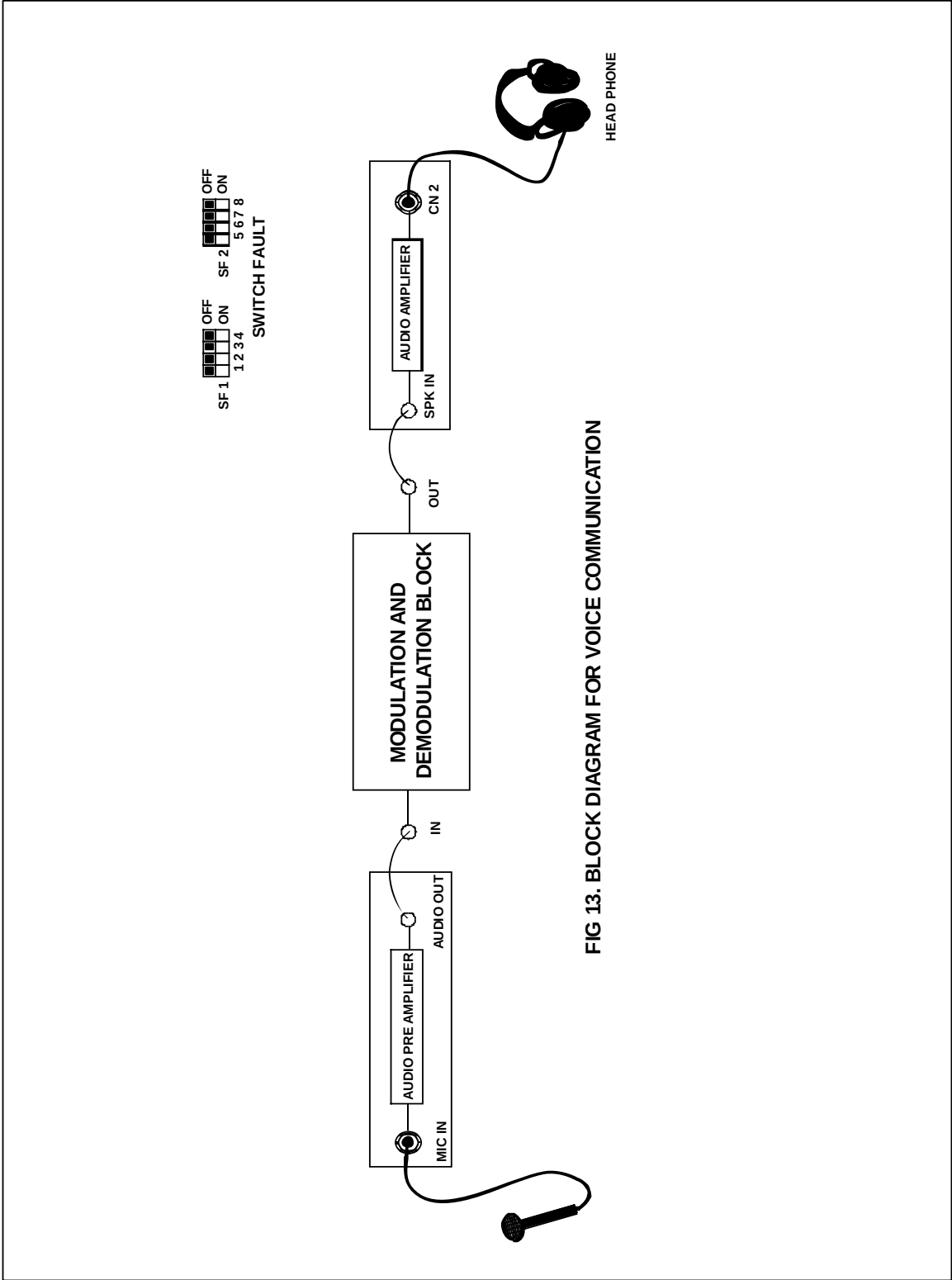


FIG 13. BLOCK DIAGRAM FOR VOICE COMMUNICATION

EXPERIMENT NO.13

NAME

Modulation Techniques Using Audio Signal.

OBJECTIVES

This experiment is performed to study the application of various modulation techniques using audio signal

THEORY

In DCS-B kit, we have provision for Microphone connection. And, in AUDIO OUTPUT KIT, we have provision for Speaker connection. Audio Pre-Amplifier section will amplify the voice signal from Microphone and feed it to the power connection port. Analog converted voice signal is amplified and is given to the Speaker in Audio Amplifier kit. LM386 is used for audio amplification purpose.

EQUIPMENTS

DCS-B kit
Connecting Chords
Power supply
20 MHz Dual Trace Oscilloscope
Microphone
Speaker
Power connection cables

PROCEDURE

- Connect the external microphone to the **MIC IN** post.
- Observe the voice signal at **AUDIO OUT** post.
- Connect **AUDIO OUT** post to any one of the analog modulation techniques experiment as an input signal.
- Demodulate the voice signal using the corresponding demodulation technique.
- Connect the demodulated voice signal to the filter and observe the recovered voice signal at the given filter out post.
- Connect the out post of the filter to the **SPK IN** post.
- Connect the speaker at the **CN2** post.
- Observe the quality of reproduction with different analog modulation techniques.

OBSERVATION

Electrical voice signal **AUDIO OUT**
Reproduction of voice signal on speaker

